

THE ELEMENTS OF ACCOUNTANCY



*Being a Plain Account of Recording, Classifying,
Summarising and Reporting*

A. JAIRA, N. TIMI

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A . J A I R A , N . T I M I

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*The book I need most and miss most, dedicated to
the person I need most and miss most. Naima Islam
Oyshee*

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TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

This book teaches THE ELEMENTS OF ACCOUNTANCY one idea at a time, from the very beginning.

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.

PREFACE

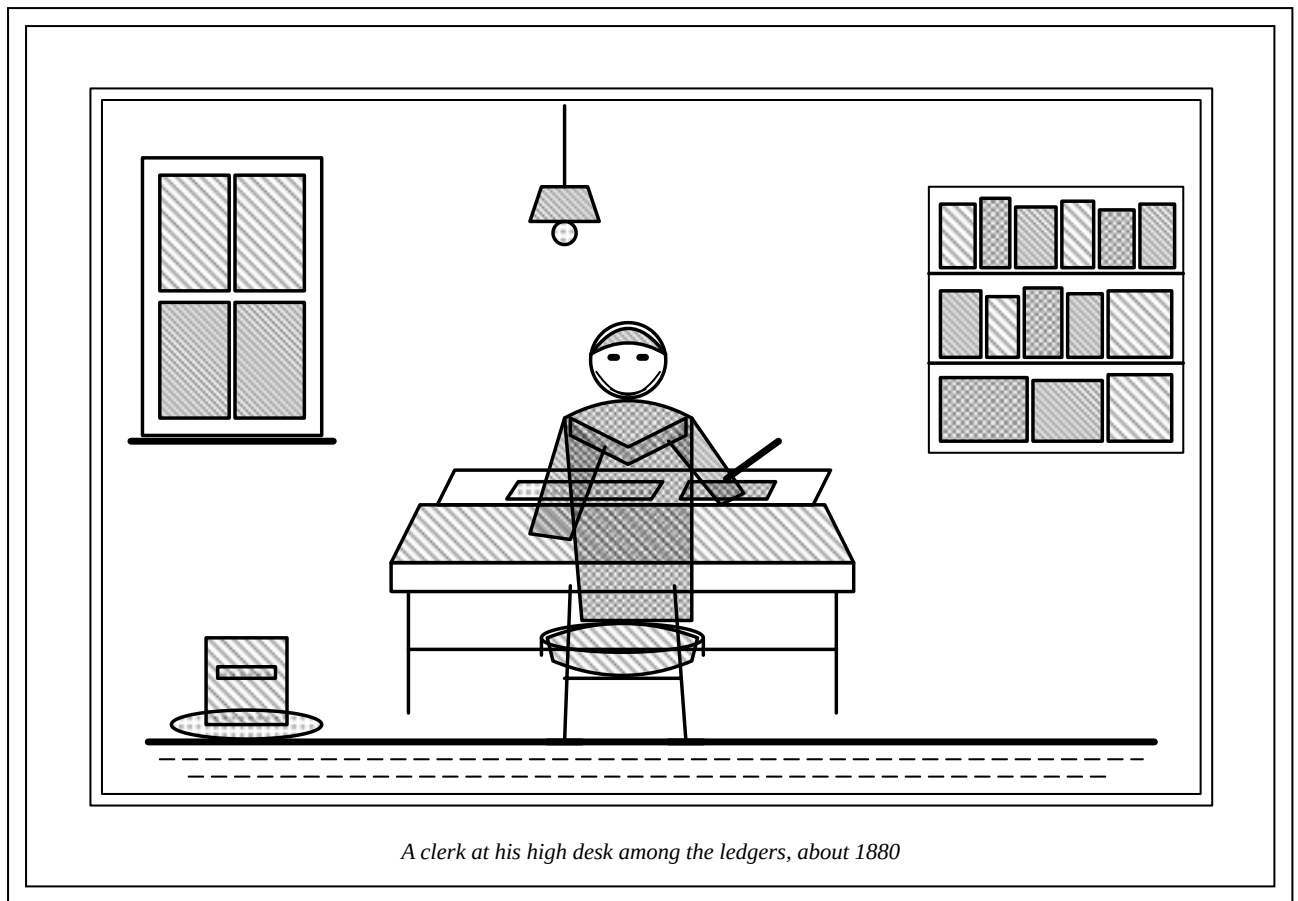
This treatise was written for the reader who has never studied accounts and wishes to begin properly. We have tried to say each thing once, plainly, and in its right place.

The book moves from the nature and purpose of accounting, through the recording and classifying of transactions, to the summarising of them and the reporting of the result. A reader who takes the chapters in order will find that each rests upon the one before.

Where a word has a special meaning, it is explained at its first use and gathered into the glossary at the end. The questions at the close of each chapter are answered at the back of the book; we advise the reader to attempt them first.

If this book spares a single student an hour of needless confusion, its labour is repaid.

A. Jaira, N. Timi



CHAPTER I

Business Activity and Companies

Who does the work, who owns the shop, and who needs to know?



This chapter shows what businesses do: some sell work, some sell goods bought from others, and some make what they sell. It then explains who may own a business, from one person to many shareholders, and why every owner needs written records.

YOU WILL NEED

Nothing from earlier pages, because this is the first chapter. You need only your own experience of buying, selling and saving money.

A shop can be busy all day and still leave its owner unsure of the result.

The cash in the drawer does not say whether the day was profitable.

This chapter asks what businesses do, who owns them, and why they keep records.

The book marks its key words, notes and questions in the same way every time.

Picture a bookshop on a busy street in Dhaka. From morning until evening, customers enter, choose a book, pay, and leave. At closing time the owner empties the cash drawer onto the desk and counts the notes. The pile looks large. Was it a good day? The owner hesitates, because the cash alone cannot answer the question.

Consider what the owner would need to know. Some of the cash may be payment for a book that has not yet been delivered. Some books on the shelves may have been bought from publishers who have not yet been paid. The rent for the month must still be found from somewhere. Even part of the pile may be borrowed from a relative last week. A drawer of cash is a fact. A good day is a judgement, and a judgement needs more facts.

This chapter begins to teach you how such judgements are made. First we ask what kinds of work businesses perform, because a bookshop, a tailor and a bakery have different things to count. Next we ask who owns a business, because one owner and a thousand owners need different things from the same records. Last we ask why records are kept at all. The answer gives you the reason for the rest of the book.

A word about the way this book is built. A new key word appears in bold, and its meaning follows at once. A short note in the side column repeats that meaning. Each section ends with a paragraph that begins *Before moving on*. Where an idea is new and hard, a flag tells you to go slowly, and a *Pause and check* question lets you test yourself before you continue. Many students find the first chapters slow, because every word is new. That is normal, and the pace is meant to help you.

* * *

1.1 WHAT BUSINESSES DO

GO SLOWLY HERE.

A business supplies something people want and receives money in return.

Business. Word in use. In daily life, it can mean any occupation or concern. In accounting, it is an organisation that supplies goods or services to earn a profit.

Every business activity has three parts: obtain, work and deliver.

Begin with something you may have seen yourself. A woman sells tea from a small stall beside a bus stop. She buys leaves and milk, boils them, pours the cups, and takes payment. Each action belongs to one pattern: she gives people something they want, and she receives money in return. We call that pattern business activity, the work an organisation performs to supply goods or services to customers for money. The word *business* has two meanings, so notice which one this book uses.¹

Look closer at the tea seller, because her day shows three parts that every business activity contains. First, she obtains what she needs: leaves, milk, fuel and a place to stand. Second, she performs work on them: she boils the mixture and pours the cups. Third, she delivers the result to customers and collects their money. A larger business has the same three parts, with more people and larger amounts. Meghna Binding Service, for example, obtains materials and the time of its workers, performs the work of repair and binding, and delivers the finished

¹ The word *business* is also used for any private matter or occupation. This book uses it only for an organisation that supplies goods or services to earn a profit.

books to schools for a fee. When you meet a new business, ask these three questions in turn. The answers tell you what the business must count.

Why does business activity exist? Nobody can grow all their food, make all their clothes and print all their books. So people specialise: one person does a task well and exchanges the result for what others do well. Profit is the reason a business continues. Profit is the money left after the costs of the business are paid.² The tea seller earns profit only when the money from customers is larger than the cost of leaves, milk, fuel and her own time.

Where does business activity fail? It fails when the costs are larger than the money received. The difference is then a loss, not a profit. A loss is not a disgrace, but a business that suffers losses for long cannot continue. This is why the owner must learn the result, and why the result must be learned from facts and not from feeling.

Different people see the same business differently. The owner asks whether the business rewards the money and effort put into it. A manager, who directs the daily work for the owner, asks whether the work is well organised. An employee, who works for a wage, asks whether the wage will be paid. A customer asks whether the goods are good and the price fair. Every one of these questions depends on facts about the business.

Businesses differ in what the customer receives. We use three groups, and the first test is simple: does the customer receive work, goods that the business bought, or goods that the business made? Each group needs different records, so each group has its own name. We shall meet the groups one at a time, starting with the one that is easiest to see.

A **service business** earns money by doing work for customers, not by selling goods. A hairdresser, a tailor who stitches the customer's own cloth, and a teacher are examples. In this book, Meghna Binding Service is a service business: it repairs and binds books for schools. The customer leaves with a repaired book, but the business was paid for the work.

A **merchandising business** buys finished goods from others and sells them again at a higher price. A bookshop is the clearest example: it buys books from publishers and sells the same books to readers. It does not change the books. It earns its profit from the gap between the price it pays and the price it receives.

PAUSE AND CHECK 1.P1 A shop buys sacks of rice and sells the rice by the kilogram. Which kind of business is it, and why?

A **manufacturing business** makes goods from materials and then sells them. A bakery turns flour, sugar and fuel into bread. A factory turns timber into chairs. The business holds three things that the other two kinds do not: materials waiting to be used, goods half finished, and goods ready to sell. Later chapters will give these three names.

The table below compares the three kinds. Read it one row at a time: the first column names the kind, and the other columns say what the customer receives, what the business must count, and one example. The comparison matters because the thing a business must count decides what its records look like.

Businesses exist because no one can make everything that he or she needs.

Business activity can fail, and failure is called a loss.

Owners, managers, employees and customers each look at a business differently.

Businesses differ in what the customer finally receives.

A service business is paid for work, not for goods.

Service business. A business that earns money by doing work for customers rather than by selling goods.

A merchandising business buys goods and sells the same goods again.

Merchandising business. A business that buys finished goods and sells them again at a higher price, such as a bookshop.

A manufacturing business changes materials into goods and sells them.

Manufacturing business. A business that makes goods from materials and then sells them.

The table sets the three kinds of business side by side.

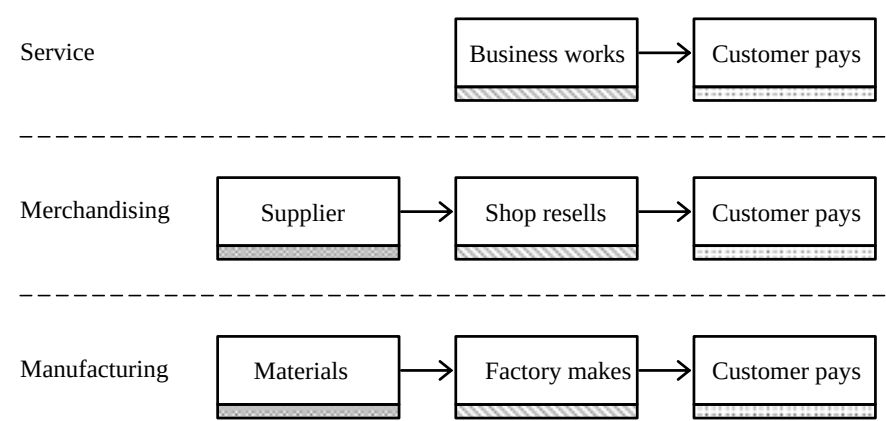
² Profit has a precise meaning in accounting, and later chapters teach it. For now, the plain meaning is enough.

THE THREE KINDS OF BUSINESS

KIND	THE CUSTOMER RECEIVES	THE BUSINESS MUST COUNT	EXAMPLE
Service	Work	Fees earned and the cost of doing the work	Meghna Binding Service
Merchandising	Goods bought from others	Goods bought, goods sold, and the price gap	Meghna and Sons, a bookshop
Manufacturing	Goods the business made	Materials, goods half finished, goods finished	A bakery that bakes its own bread

The figure shows each kind as a chain from the source to the customer.

The figure shows each kind as a chain. Read each row from left to right. A box names who holds the work or the goods, and an arrow shows the direction in which they travel toward the customer. A service chain is short, a merchandising chain has a supplier before the shop, and a manufacturing chain has the factory in the middle.



Strip under each box: lines = the business, dots = the customer, crosses = the source.

FIG. 1.1. Three kinds of business drawn as chains from source to customer

The kind of business decides which facts its owner must watch.

Why does the kind matter? Because it tells the owner where to look. In a service business, the owner watches the fees earned and the cost of the workers' time. In a merchandising business, the owner watches how many goods were bought, how many were sold and what remains on the shelves. In a manufacturing business, the owner watches all of these and also the materials and goods still being made. More things to count means more records.

Many real businesses combine two kinds, and that is allowed.

Real businesses often combine kinds. A bakery that also sells tea bought from a wholesaler is manufacturing and merchandising together. A shop that repairs what it sells is merchandising and service together. The three groups are tools for thinking, not boxes into which every firm must fit. When a firm mixes kinds, we ask which kind brings in most of its money, and we describe the other kinds as additions.

The recap gathers the section into three sentences.

Before moving on. Business activity is work done to supply goods or services for money, and profit is what remains after costs. A service business sells work, a merchandising business sells goods it bought, and a manufacturing business sells goods it made. You should now be able to say which of the three describes a bookshop, a hairdresser and a bakery, and why.

* * *

1.2 WHO OWNS THE BUSINESS

Every business belongs to someone. The owner provides the money and effort to begin, and receives the profit or bears the loss. Owners come in three main shapes: one person, a few people, and many people. Each shape gives the business different duties, because other people lend money, sell goods or buy shares according to who stands behind it.

The first shape is the sole proprietor, a single person who owns the business and answers for it. The tea seller beside the bus stop is a sole proprietor. So is the owner of Meghna and Sons, Nurul Meghna, who has traded in Dhanmondi since 4 April 1996. The business is simple to begin and simple to direct. Its weakness is that all the money, work and risk rest on one person.

The second shape is the partnership: a business owned by two or more people who agree to run it together and share its profit. For example, two friends might open a bookshop and divide the profit between them. Partnerships bring more money and more skills than one person can supply. They also bring a new question, which is who receives how much. Chapter 18, 'Partnership Accounts', teaches how partners' records answer it.

The third shape is the **company**. A company is a business that the law treats as a person separate from its owners, with its ownership divided into equal parts called shares. A person can buy a share, so a company can have hundreds of owners. The idea is strange at first. The company can own goods and owe money in its own name, even though the people who own it remain separate.³

A **shareholder** is a person who owns one or more shares of a company. The proposal for Meghna and Sons Limited illustrates this: it names 410,000 shares of Tk 10 each. Suppose a person held 41,000 of those shares. That person would own one tenth of the company, because 41,000 is one tenth of 410,000. More shares give a larger part of the ownership.

PAUSE AND CHECK 1.P2 A company has 100 shares, and one person holds 25 of them. What part of the company does this person own?

Consider the three shapes from the viewpoint of an outsider who thinks of lending money. For a sole proprietor, the lender deals with one person. For a partnership, the lender deals with several people who agreed together. For a company, the lender deals with the company itself, and the shareholders may be many strangers. As the number of owners grows, more people need clear, written facts, because they cannot watch the business every day.

The company shape has one more feature that matters for records. Because a company can issue shares, it can collect small sums from many people and gather a large total. This is how very large businesses, such as ships and railways in earlier centuries, found money that no single owner possessed. The cost is that shareholders, who may live far away, rely on the records to learn what their business is doing.

Do not conclude that the company is the best shape. A company has more rules to follow, more reports to prepare and more people to inform. A small family shop may reasonably remain a sole proprietor for many years. The right shape depends on how much money

Every business belongs to someone, and the kind of owner changes the records.

A sole proprietor is the one person who owns and answers for the business.

A partnership shares ownership among two or more people by agreement.

A company is a business that the law treats as a person of its own.

Company. A business that the law treats as a person separate from its owners, with ownership divided into shares.

A shareholder is an owner of a company through its shares.

Shareholder. A person who owns one or more shares of a company.

The three shapes differ in who owns, who decides and who must be told.

A company's shares let many people own a small part of one business.

The company shape is not always better; each shape has a use.

³ The rules for forming a company differ from country to country. In Bangladesh, companies are formed under the Companies Act 1994; other countries have their own laws.

A sole proprietor keeps control, which is why many small shops remain one.

Advanced note: a shareholder's risk is limited, and a beginner may defer it.

The recap gathers the section into three sentences.

A record is a written note that lets anyone check what happened.

Record. Word in use. In daily life, it can mean the best result ever achieved, as in a sports record. In accounting, it is a written note of an event in the business, such as a sale or a payment.

Memory is weak, and other people cannot see inside it.

Many decisions depend on records, and each decision has its own reader.

A useful record is dated, complete and open to checking.

Records are a basis for judgement, not a guarantee of success.

is needed, how many people share the work, and how many outsiders must be told.

Why would an owner choose to remain a sole proprietor? Control is the main reason. One person decides what to buy, what to charge and when to open, and no agreement or vote slows the decision. The profit also belongs to one person. For a small family shop with modest needs, these benefits often outweigh the extra money that more owners might bring.

Advanced note, which you may defer. One more feature explains why companies are popular. In most countries, a shareholder who has fully paid for his or her shares cannot be made to pay the company's debts from personal wealth. This protection is called limited liability. It makes people willing to buy shares, because the most they can lose is the amount they paid. The rules differ by country, so check the law of the place where the company is formed. Chapter 19, 'Company Accounts: Shares', returns to shares.

Before moving on. A sole proprietor is one owner, a partnership is two or more owners by agreement, and a company is a separate legal person owned through shares. A shareholder owns one or more shares. You should now be able to say who owns each of three businesses, and why a company with many shareholders needs written facts.

* * *

1.3 WHY RECORDS ARE NEEDED

Return to the shop owner and the pile of notes. What would turn the pile into an answer? The owner needs a written note of each sale, each purchase and each payment, made when the event happened. Such written notes are the records of a business. The word has two meanings, so notice that this book uses the second. Records are the memory of the business, written down so that it does not depend on any one person.

Why not trust memory? A busy owner forgets a small payment, remembers a large one wrongly, and cannot show either to anyone else. A lender, a tax officer or a partner will not accept "I believe so". A written record can be checked, copied, filed and examined years later. It also protects honest people, because it shows that an account was kept fairly.

Several decisions depend on records. The owner decides whether to hire another worker, buy more books or begin a new workshop. A lender decides whether a business can repay a loan. The tax authority decides how much tax is due. A person thinking of buying shares decides whether the company is sound. For Meghna and Sons, the proposal to form a company in 2026 will need records, because strangers will be asked to trust it. Chapter 2, 'What Accounting Is', describes these readers more fully.

What makes a record useful? Three qualities matter. It is dated, so that anyone can place the event in time. It is complete, so that a sale or payment is never left out because it seemed small. It is open to checking, so that a second person can follow it to the original paper, such as a receipt. A note on a scrap of paper, undated and unsigned, may be true, but nobody can prove it. The later chapters teach methods that give every record these qualities.

Records have limits. They do not make a business succeed; they only show what has happened, so that better choices become possible. Poor records are worse than none when they mislead. For that

reason the later chapters teach not only how to write records but how to check them.

Before moving on. A record is a written note of an event in the business, kept so that facts do not depend on memory. Owners, lenders, tax officers and shareholders each make decisions that need those records. You should now be able to say why a busy owner cannot rely on memory, and name two people who need the records.

The recap gathers the section into three sentences.

* * *

WORKED CASE: CLASSIFYING THE MEGHNA BUSINESSES

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhanmondi, Dhaka, owned by Nurul Meghna and trading since 4 April 1996. In 2025 it sold books worth Tk 14,400,000, which later chapters will call its net sales (Chapter 10, 'Accounts of a Merchandising Company'). The shop buys books and sells them. On 1 January 2026 a separate workshop, Meghna Binding Service, began to repair and bind books for schools, with its own set of books. A proposal exists to form a company in 2026 with 410,000 shares of Tk 10 each, a total of Tk 4,100,000.

The case file gives the facts about three Meghna businesses and one proposal.

Question. What kind of business is each of the two Meghna businesses? Who owns the shop today? If the company is formed, what is the total value of the shares at their face price of Tk 10? Use only the facts in the case file.

The question asks for the kind, the shape of ownership and the sum.

Reasoning. Begin with the shop. The case file says it buys books and sells them. It does not change the books, and it is not paid for work. This matches the definition of a merchandising business. Next, the workshop repairs and binds books for schools. The schools pay for work, and the workshop does not resell goods it bought. This matches the definition of a service business.

The shop buys finished books and resells them, so it is a merchandising business.

Now consider ownership. The case file names one owner, Nurul Meghna, so the shop today has the shape of a sole proprietor. The proposal would change the shape, because a company has shares and shareholders. To find the total value, multiply the number of shares by the price of one share: $410,000 \text{ shares} \times \text{Tk } 10 = \text{Tk } 4,100,000$. A buyer of any of these shares would become a shareholder.

One named owner makes the shop a sole proprietor today; the proposal would change that.

Conclusion. The shop is a merchandising business owned by one person, Nurul Meghna. The workshop is a service business. If the company is formed, its 410,000 shares have a total face price of Tk 4,100,000.

The conclusion gives the three answers in plain sentences.

Check. Multiply again from the other direction: $\text{Tk } 4,100,000 \div \text{Tk } 10 = 410,000 \text{ shares}$, which agrees with the proposal. Test each kind against its definition: goods bought and resold give merchandising, and work done for payment gives service. Notice that the sales figure of Tk 14,400,000 was not needed to decide the kind, because the kind depends on what the business does, not on how much it earns.

The check repeats the arithmetic and the definitions.

* * *

SUMMARY

Business activity is work done to supply goods or services for money. A service business sells work, a merchandising business sells goods it bought, and a manufacturing business sells goods it made. A business may be owned by one person, by partners or by shareholders of a company, which the law treats as a separate

The summary restates the chapter in a few lines.

person. Written records let owners and outsiders check the facts and make decisions.

Four short problems test whether the kinds of business and ownership are secure.

THINK IT THROUGH

Meghna Binding Service repairs and binds books for schools. The four problems below move from a familiar case, to the same case reshaped, to a new setting, and then to a case that combines earlier ideas.

1.T1 The workshop earns money from binding work. Why does it keep no records of goods bought to sell again?

1.T2 A hairdresser begins to sell bottles of shampoo bought from a wholesaler. Which kinds of business does the salon now combine, and which kind brings in most of its money if shampoo is only a small part of its income?

1.T3 A village club collects a monthly fee from its members and spends it on a reading room. Is the club a business in the sense used in this chapter? Give a reason.

1.T4 A bakery owned by three friends bakes bread and also sells tea that it buys from a wholesaler. State the kinds of business it combines, the shape of its ownership, and one person who would need its records.

WHERE PEOPLE GO WRONG

Cash in the drawer is not the same as profit.

A mistaken belief: "The cash in the drawer is the profit." *Correction:* cash is only money held. Some of it may be owed to suppliers or may belong to customers who paid in advance. Profit is what remains after all the costs of the period are paid.

A service business still needs records, though it holds no goods for sale.

A mistaken belief: "A service business has nothing to record." *Correction:* it holds no goods for resale, but it earns fees, pays wages and rent, and owes money. These all need written notes.

A company and its owners are different persons in law.

A mistaken belief: "A company and its owner are the same." *Correction:* a company is a separate legal person. It can own goods and owe money in its own name, and its shareholders remain separate people.

More owners do not by themselves make a business better.

A mistaken belief: "A business with more owners is stronger." *Correction:* more owners bring more money and more opinions, and more people to inform. Strength depends on the work the business does and the care with which it is run.

The three kinds of business are tools for thinking, not strict boxes.

A mistaken belief: "Every firm belongs to exactly one kind." *Correction:* many firms mix kinds. We name the kind that earns most of the money and treat the others as additions.

IN HISTORY

A royal charter of 1600 created an early company that served its investors.

On 31 December 1600, Queen Elizabeth I granted a royal charter to a group of London merchants trading to the East Indies. The charter gave the group the sole right to trade there for fifteen years, and its aim was to compete with the Dutch merchants who then controlled much of the spice trade. Thomas Smythe became the first governor of the English East India Company.

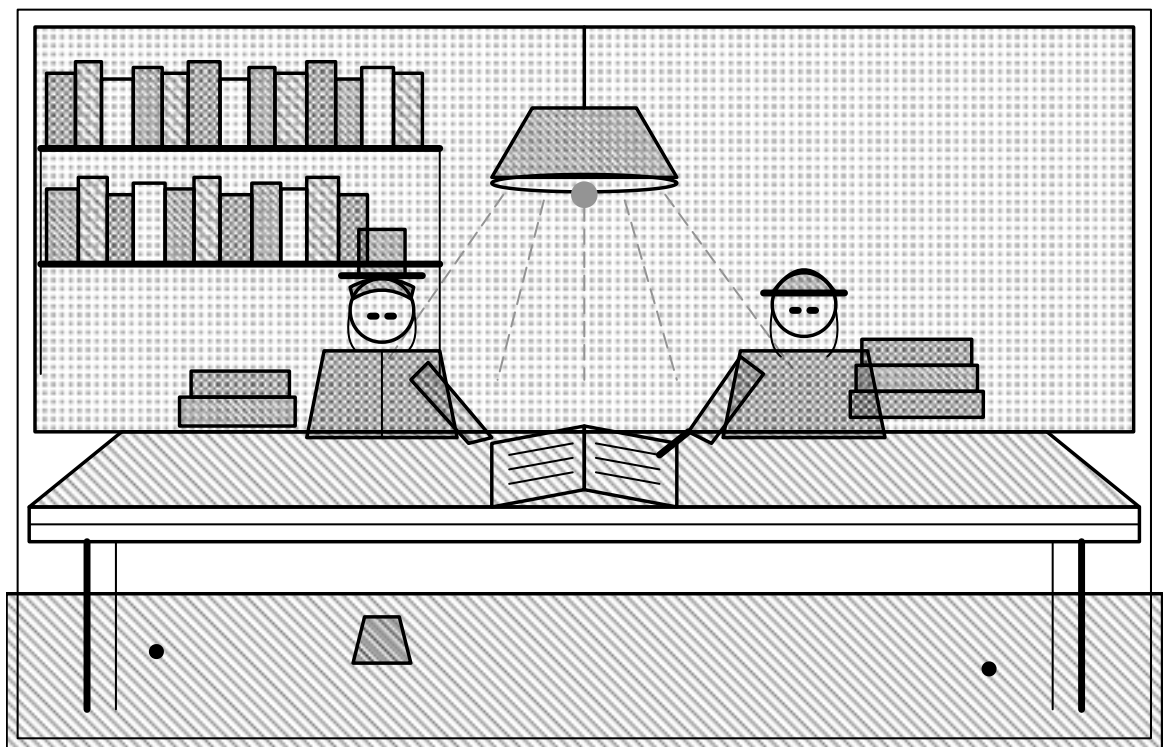
The company's voyages were paid for by investors, who expected to learn the result.

Voyages across the ocean were costly and dangerous, so no single merchant paid for them. The Company raised money by selling shares to merchants, and these investors expected an account of what became of their money. In this way the Company shows early the link between a company, its shareholders and its

records. The later political history of the Company, which includes the governing of large parts of South Asia, belongs to history books and is not told here.

REVIEW QUESTIONS

- 1.1 A tailor stitches clothes from cloth that customers bring. A grocer buys rice and sells it again. A furniture workshop builds chairs from timber and sells them. State the kind of business each one is.
- 1.2 List four people or groups who would want to see the records of a bakery, and say what each wants to learn.
- 1.3 In the proposal for Meghna and Sons Limited, there are 410,000 shares of Tk 10 each. Calculate their total face price, and say how many shares would be one tenth of the company.
- 1.4 In one sentence each, explain what a shareholder is and how a merchandising business differs from a manufacturing business.
- 1.5 Meghna Binding Service began on 1 January 2026 as a separate set of books from the shop. Name its kind of business, and give one reason why the owner may want the shop and the workshop recorded apart.



Auditors examining ledgers by lamplight

CHAPTER 11

What Accounting Is

Who needs to know how a shop is doing, and why should they believe the answer?



Accounting turns the daily events of a business into reports that other people can read and trust. This chapter explains what accounting is, which people use its reports, and which basic rules keep its numbers fair. It is the map for the rest of the book.

YOU WILL NEED

Chapter 1 'Business Activity and Companies', section 1.1 'What Businesses Do'; Chapter 1 'Business Activity and Companies', section 1.3 'Why Records Are Needed'.

A family notebook can answer several questions for several readers.

The shop in the running case has five readers and one set of records.

Five readers ask five different questions, which creates a puzzle for one set of records.

Think of a family that writes every payment in one notebook: rent, food, school fees, bus fares. At the end of the month, the mother opens the notebook and asks one question: did we save anything? The same notebook can answer other questions. The landlord may want proof that the rent was paid. A bank, before lending money for a repair, may want to know whether the family can repay it. There is one notebook, but there are three readers and three questions. The notebook is useful only if every reader believes what it says.

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhanmondi, Dhaka, owned by Nurul Meghna. Five parties read its records for 2025. Nurul reads the profit of Tk 750,000. The bank reads the loan of Tk 600,000. The publishers, who supply the books, read the Tk 1,150,000 owed to them. The tax authority reads the profit return. The staff read the wages of Tk 2,940,000.

Notice that the five readers do not ask the same question. Nurul asks whether the shop earned a fair reward for the work put into it. The bank asks whether the loan will be repaid. The publishers ask whether they will be paid for their books. The tax authority asks whether the profit has been reported honestly. The staff ask whether the shop is secure enough to keep them employed. If each reader demanded a private set of records, the sets would disagree, and nobody would trust any of them. How can one business keep one set of records that serves all five readers? And why should any of them believe it? This chapter answers both questions. It also states the purpose of the whole book, because every later chapter teaches one step of the process described here.

* * *

2.1 THE IDEA OF ACCOUNTING

A market seller already performs the three actions of accounting.

GO SLOWLY HERE.

Picture a woman who sells vegetables at a market stall. Each morning she buys sacks of produce. During the day she sells by the kilogram. In the evening she counts the money in her tin and writes three numbers in a notebook: what she spent, what she received, and what is left. She has done three things without naming them. She has noticed events, written them down, and drawn a conclusion. Accounting is these same actions, performed in an orderly way for many readers.

Accounting means recording, sorting and reporting the money effects of events.

Accounting. The process of recording, sorting and reporting the money effects of a business's events to people who need them.

The name for the whole process is **accounting**. Accounting is the process of recording, sorting and reporting the money effects of a business's events to people who need them. An event is anything that changes the money position of a business, such as buying books, selling a novel or paying a wage. Recording means writing the event down. Sorting means placing like events together, so that all sales stand in one place. Reporting means presenting the sorted totals to a reader in a form the reader can use. When Meghna and Sons sells a novel, accounting records the sale, sorts it with the other sales, and reports the total.

You may know the word *account* from a bank, where it means a record of the money you hold there. Accountants use the word in a related but different way. In their sense, an account is a named record of one kind of item, such as Cash or Sales. Chapter 4, 'Accounts, Debits and Credits', teaches accounts in detail. For now, remember only that accounting is built from such named records.

The figure that follows shows the whole process on one page. Read the top row from left to right. Each box is a stage, and each arrow shows the direction in which information moves. The first box holds the events. The second holds the records made from them. The third holds the reports built from the records. The lines that leave the third box lead to the readers in the bottom row, who are the subject of the next section.

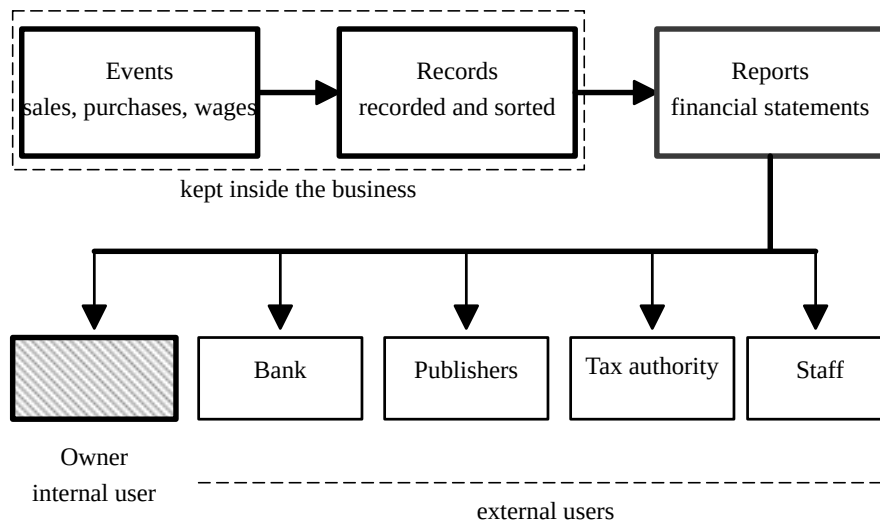


FIG. 2.1. Accounting as a flow from events to reports to users

People often use the words accounting and bookkeeping as if they meant the same thing, but they do not. **Bookkeeping** is the recording part of accounting: writing down each transaction in the right book. A transaction is one event that can be measured in money, such as the sale of one novel. The bookkeeper records. The accountant also sorts, checks and reports, and decides how a doubtful item should be treated. An everyday bridge helps here. The person who writes each shopping payment in the family notebook is keeping books. The person who studies the notebook to plan next month's spending is doing accounting.

Accounting divides into branches, and the readers decide the division. Think first of Nurul, who wants a weekly report to decide which books to order. Think next of the bank, which wants a formal report once a year. These two readers need different reports, prepared in different ways. That difference gives us the two main branches of accounting.

Financial accounting is accounting that prepares reports for outsiders, such as lenders, owners and tax authorities. An owner who takes no part in managing the business is also an outsider. The bank's yearly report and the tax authority's profit return come from financial accounting. Its reports follow written rules, which section 2.3 explains, so that outsiders can compare one business with another.

The word account has an everyday meaning and a special accounting meaning.

Account. Word in use. In daily life, a record of money in a bank. In accounting, a named record of one kind of item, such as Cash or Sales.

A flow chart shows events becoming records, then reports, then decisions by readers.

Bookkeeping is the recording stage; accounting is the whole process.

Bookkeeping. The recording part of accounting: writing down each transaction in the right book.

Accounting divides into branches according to the readers it serves.

Financial accounting serves outsiders, and its reports follow written rules.

Financial accounting. Accounting that prepares reports for outsiders, such as lenders, owners and tax authorities.

Management accounting serves managers, and its reports follow no outside rule.

Management accounting.

Accounting that prepares reports inside the business to help managers plan and control.

A short review of the ideas in this section.

A sentence the reader can test against the section.

One question sorts the readers into an inside group and an outside group.

Each external user wants to know something different about the business.

GO SLOWLY HERE.

Staff are a borderline group, and this book treats them as external users.

A table lays out each reader beside the question that reader asks.

Management accounting is accounting that prepares reports inside the business to help managers plan and control. Nurul's weekly report on which books sell fastest is management accounting. No outside rule governs such a report; the manager designs it to answer whatever question needs an answer. The chapters that follow concentrate on financial accounting. Management accounting is named here so that you know it exists.

PAUSE AND CHECK 2.P2 A school employs a clerk who writes each fee payment into the fee book. Is the clerk's work bookkeeping or accounting? Give one reason.

Before moving on. Accounting is the process of recording, sorting and reporting the money effects of events to people who need them. Bookkeeping is one part of that process, the recording part. Financial accounting serves outsiders by written rules, and management accounting serves managers without outside rules.

You should now be able to say: accounting is the whole journey from an event to a report, and bookkeeping is only its first stage.

* * *

2.2 WHO USES ACCOUNTING INFORMATION

Return to the five readers of the shop's records. Is there a way to sort them? One question gives a useful sorting: does this reader help to run the business from day to day? Nurul does. The bank, the publishers and the tax authority do not. Readers who help to run the business are called internal users, and those who do not are called external users. Internal users are people inside the business, such as managers and owner-managers, who use information to plan and control. External users are people outside it who have a claim on it or an interest in it.

Take the external users one at a time, and imagine each person's point of view. A bank that has lent money asks whether the shop can repay the loan on the agreed dates. A publisher that has delivered books not yet paid for asks whether the shop will pay what it owes. A tax authority asks whether the profit has been reported honestly, because the tax due depends on it. An owner who does not manage the business asks whether the profit is worth the money invested. A customer rarely reads accounting reports at all, but cares whether the shop will still be open next year to order a special book.

Staff are a harder case, and many students find them confusing, because staff work inside the shop yet do not run it. In this book we treat them as external users. They read finished reports, not the working records, and they cannot change what the reports say. Their question is whether the wages of Tk 2,940,000 recorded for them are safe and whether the shop will continue to employ them. Other books may place employees among the internal users, so the grouping is a convention, not a law of nature.

The table that follows gathers the five readers of the case. Read each row from left to right: the reader, the group, the question that reader asks, and the figure from the case file that answers part of that question. For example, the row for the bank says that the bank is an external user, that it asks whether the loan can be repaid, and that the loan of Tk 600,000 is the figure that matters to it.

WHAT EACH READER OF THE SHOP'S RECORDS ASKS

READER	GROUP	QUESTION ASKED	FIGURE FROM THE CASE
Nurul	Internal	Did the shop earn enough profit?	Profit Tk 750,000
The bank	External	Can the loan be repaid?	Loan Tk 600,000
The publishers	External	Will the shop pay for the books?	Owed Tk 1,150,000
The tax authority	External	Is the profit honestly reported?	Profit return
Staff	External (see text)	Are wages and jobs safe?	Wages Tk 2,940,000

Now the puzzle from the start of the chapter can be partly answered. The five questions differ, but the facts behind them are the same: what the shop earned, what it owes, what it holds. One set of records can serve all five readers when every reader receives reports built from the same events and the same rules. Section 2.3 explains the rules. Notice also the viewpoint of the owner, who is both a reader and a person who must answer to the others: Nurul needs honest reports precisely because the bank, the publishers and the tax authority will test them.

Remember that Meghna and Sons is a merchandising business: it buys books and sells them again. A service business, such as a tailor who sells stitching, and a manufacturing business, such as a printing works that makes books, have the same kinds of reader. The figures differ, but the questions are alike. A lender always asks about repayment, a supplier about payment, and a tax authority about honest profit.

Before moving on. Internal users run the business and use information to plan and control. External users stand outside it and ask about repayment, payment, honest profit, return on money invested, or security of work. One set of records serves them all when the reports share the same facts and rules.

You should now be able to say: an internal user helps to run the business, an external user does not, and each external user asks a different question of the same records.

* * *

2.3 RULES AND CONCEPTS

Suppose two bookshops each report a profit. The first records its books at the price it paid for them. The second records them at the price it hopes to receive one day. The two profits cannot be compared, and a bank reading both reports would not know which to believe. To prevent this, accountants agree on rules before the records are made. This section presents the rules that give accounting its trustworthiness: written standards, then three basic ideas about what is being recorded, then the reports that result.

One set of records can answer many questions because the questions share the same facts.

The same five kinds of reader exist whatever kind of business is studied.

Recall: merchandising business, service business, manufacturing business, Chapter 1, section 1.1.

A short review of the readers and their questions.

A sentence the reader can test against the section.

Reports can be trusted only if every business follows the same rules.

An accounting standard is a written rule that makes businesses report alike.

Accounting standard. A written rule about how to record and report items, so that different businesses report in the same way.

Standard. Word in use. In daily life, a level of quality, as in a high standard of service. In accounting, a written rule about how to record and report items.

The first basic idea keeps the shop's money apart from the owner's own money.

GO SLOWLY HERE.

The business entity rule treats the business as separate from its owner.

Business entity. The rule that a business is treated as separate from its owner when records are kept.

The business entity rule is an accounting rule, which may differ from the law.

A short review of the first basic idea.

The second basic idea assumes that the business will continue.

Going concern. The assumption that a business will continue to operate for the foreseeable future.

The going concern assumption fails when a business is about to close.

An **accounting standard** is a written rule about how to record and report items, so that different businesses report in the same way. For example, a standard may say how the cost of goods held for sale must be measured. In daily speech, the word *standard* means a level of quality, as in "a high standard of service". Here it means a rule, not a level of quality. Which standards apply depends on the place. Many countries, including Bangladesh, use the International Financial Reporting Standards, while others use national standards of their own.¹

Begin with a household example. A father who runs a small shop takes money from the till to pay the family electricity bill. If the shop's records show that payment as a shop cost, the shop will look less profitable than it is. The bank, reading the report, will form a false picture of the shop. The remedy is to keep two sets of money, the family's and the shop's, and to write only the shop's events in the shop's records.

This remedy is the rule of the **business entity**. The business entity rule says that a business is treated as separate from its owner when records are kept. Nurul's household spending does not appear in the records of Meghna and Sons. The rule also applies between two businesses of one owner. Meghna Binding Service, a book-binding workshop that began on 1 January 2026, keeps its own separate set of books, even though Nurul owns both businesses.

The rule has an exception that causes confusion. In law, a sole trader, meaning one person who owns a business alone, is often not a separate person from the business. A company, in contrast, is a separate person in law. The accounting rule does not wait for the law: it treats every business as separate from its owner, whether or not the law agrees. The rule serves the readers, who ask about the business and not about the family.

PAUSE AND CHECK 2.P1 Nurul takes a book from the shop shelf as a gift for a relative. Using the business entity rule, explain why the shop's records should show this event.

Before moving on. The business entity rule keeps the shop's records free of the owner's private spending, and it applies between two businesses of one owner. You should now be able to say: the shop's records describe the shop, not the family.

Consider a farmer who plants fruit trees. The farmer expects to pick fruit for many years, so the trees are valued by what they will bear, not by the timber they would give if cut down today. Accounting makes a similar choice. The **going concern** assumption is that a business will continue to operate for the foreseeable future. Under this assumption, the shelves and counter of the bookshop are treated as items that the shop will use for years, not as items that must be sold tomorrow.

The assumption has a clear limit. If a business is about to close, it will not use its shelves for years; it will sell them. In that case the records must show what the items would fetch when sold, and the going concern assumption no longer holds. Staff and lenders watch for this danger, because it changes the meaning of every number in the report.

¹ The International Financial Reporting Standards are published by the International Accounting Standards Board, which replaced the committee described in the history section on 1 April 2001.

Suppose the shop buys a display shelf. A year later, a neighbour says the shelf would sell for more than the shop paid. Which number should the records show? The rule of **historical cost** is the rule of recording an asset at the price paid for it, not at its present value. The word *asset* means something of value that the business owns, such as a shelf or a book; Chapter 3, 'The Accounting Equation', studies assets properly.

The reason for the rule is trust. The price paid is a fact that an invoice or a receipt can prove. A present value is an opinion, and two honest people may hold different opinions. A bank would prefer the fact. There is an exception: some accounting standards allow certain items to be shown at present value, and later chapters show how the cost of an item is lowered over time. This book first teaches the plain rule of historical cost, because it is the base on which the exceptions stand.

Before moving on. Business entity keeps the owner's money apart from the shop's. Going concern assumes the shop will continue. Historical cost records items at the price paid. You should now be able to say: each idea protects the reader from a misleading number.

The reports that result from these rules are called financial statements. A **financial statement** is a formal report of a business's results or position, prepared at the end of a period. A period may be a month, a quarter or a year. The yearly reports of Meghna and Sons for 2025 are financial statements. They carry the profit of Tk 750,000 that Nurul, the bank and the tax authority read.

Three statements are most common: the income statement, which shows profit; the statement of owner's equity, which shows what the owner's share has become; and the balance sheet, which lists what the business owns and owes on one day.² Chapter 9, 'Financial Statements and Closing Entries', teaches them. Notice how the whole process now closes into a circle. Events are recorded, recording follows the standards and the three ideas, and the statements carry the result to the readers.

IN SHORT

Accounting records, sorts and reports money events. Standards, the business entity rule, going concern and historical cost make the reports trustworthy for their readers.

Before moving on. Accounting standards make businesses report alike. Three ideas decide what is recorded: separate entity, continued operation and price paid. Financial statements deliver the result. You should now be able to say: readers can trust accounting reports because rules, not moods, decide what the reports contain.

* * *

WORKED CASE: MATCHING USERS TO QUESTIONS

Case file, illustrative. Situation: the 2025 records of Meghna and Sons are read by Nurul (profit Tk 750,000); the bank (loan Tk 600,000); publishers (owed Tk 1,150,000); the tax authority (profit return); and staff (wages Tk 2,940,000).

² Under the International Financial Reporting Standards, the balance sheet is called the statement of financial position. This book uses the name balance sheet.

The third basic idea records items at the price actually paid.

Historical cost. The rule of recording an asset at the price paid for it, not at its present value.

Historical cost is reliable because the price paid can be proved.

A short review of the three basic ideas.

Financial statements are the formal reports at the end of the process.

Financial statement. A formal report of a business's results or position, prepared at the end of a period.

The main statements are previewed here and taught in a later chapter.

A short review of the section.

A worked case applies the chapter's ideas to the shop's five readers.

The question asks for a reader, a branch and a rule for each party.

Reasoning begins with the bank and the publishers, who are outsiders with money at stake.

Reasoning continues with the tax authority and the staff.

Nurul needs both kinds of accounting.

The conclusion states the match and the check confirms it is complete.

The chapter's main ideas in a few lines.

Bookkeeping and accounting are different things.

Only managers read accounting information.

The business entity rule is not the same as company law.

Question. For each reader, which question does the reader ask, which branch of accounting serves the reader, and which rule makes the answer trustworthy?

Reasoning. Take the bank first. It is an external user, so financial accounting serves it. It asks whether the loan of Tk 600,000 can be repaid. The answer is trustworthy because accounting standards make the shop report in the same way as other borrowers. Take the publishers next. They are external users. They ask whether the shop will pay the Tk 1,150,000 owed to them. The business entity rule matters here, because the debt belongs to the shop and not to Nurul's household.

The tax authority is an external user. It asks whether the profit return is honest, and standards give it a common method to test. The staff, treated here as external users, ask whether the wages of Tk 2,940,000 are safe and whether the shop will go on. The going concern assumption is the rule that matters to them, because it states that the shop is expected to continue.

Nurul is the internal user. Nurul reads the profit of Tk 750,000, which comes from financial accounting. Nurul also needs management reports, such as the weekly report on fast-selling books, to decide what to order. Historical cost supports the profit figure, because every book is recorded at the price paid.

Conclusion. Five readers, five questions, one set of records. Four readers are served by financial accounting, and Nurul is served by both branches. **Check.** Count the readers: five in the case file, five in the answer. Each answer names one reader, one question and at least one rule. The wages figure, the loan figure and the profit figure all agree with the case file.

* * *

SUMMARY

Accounting is the process of recording, sorting and reporting the money effects of a business's events to people who need them. Bookkeeping is its recording part. Financial accounting serves outsiders and follows written standards; management accounting serves managers. Internal users run the business; external users do not. Four ideas keep reports trustworthy: accounting standards, the business entity rule, the going concern assumption and historical cost. Financial statements carry the result to the readers.

THINK IT THROUGH

2.T1 Nurul pays a household bill from the shop's till. Using the business entity rule and the point of view of the bank, explain why a shop's money and a home's money must be kept apart.

WHERE PEOPLE GO WRONG

Many students think that bookkeeping and accounting are the same. They are not. Bookkeeping is the recording part. Accounting also sorts and reports, and it serves readers who need decisions.

A common error is to think that only managers use accounting. In fact, banks, suppliers, tax authorities, owners and staff all read it. Each asks a different question of the same records.

Some students believe that the business entity rule applies only to companies. It applies to every business, including a

one-person shop, because accounting treats the business as separate whatever the law says.

A fourth confusion is to think that historical cost shows what an item is worth today. It does not. It shows the price paid, which can be proved. Present value may differ, and that is why the rule is a choice of reliability over currency.

Historical cost is not always the best measure, but it is the most reliable.

IN HISTORY

By the early 1970s, businesses in many countries sold goods and borrowed money across borders, yet each country kept its own accounting rules. A bank in one country could not easily compare a borrower's reports with those of a borrower in another. On 29 June 1973, professional accounting bodies from several countries formed the International Accounting Standards Committee in London. The idea came from Sir Henry Benson, a leading British accountant and former president of the Institute of Chartered Accountants in England and Wales. The bodies came from Australia, Canada, France, Germany, Japan, Mexico, the Netherlands, the United Kingdom and Ireland, and the United States.

The committee of 1973 first tried to give the world one set of accounting rules.

Its members worked part-time as standard setters while holding other jobs. They aimed to write standards in the public interest and to win acceptance for them across the world. The committee worked until 2000, and on 1 April 2001 a new body, the International Accounting Standards Board, replaced it and continued its work. The story shows what this chapter has argued: accounting is trustworthy when many people agree on the rules first.

The committee's standards were later taken over by a board.

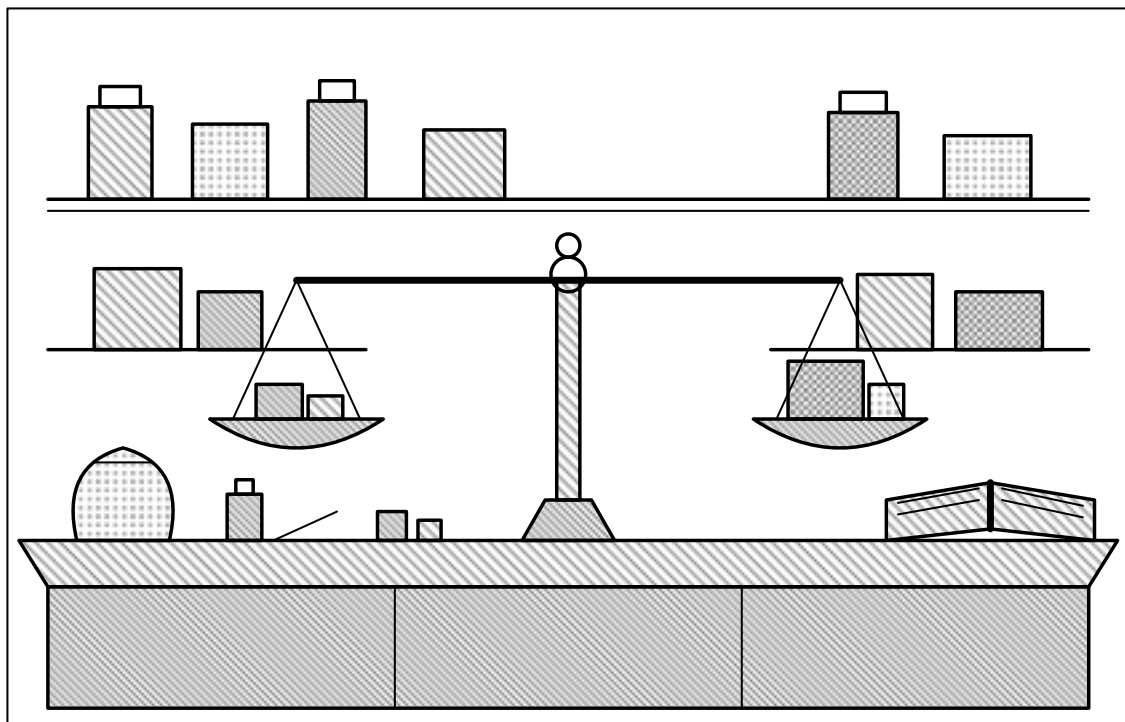
REVIEW QUESTIONS

2.1 Match each reader of Meghna and Sons to the question it asks: Nurul, the bank, the publishers, the tax authority and the staff.

2.2 Suppose the bank asks the shop for its records, but only the weekly book-sales report is shown. Which branch of accounting does the weekly report belong to, and why is it not enough for the bank?

2.3 A village tailor, who sells stitching, keeps her family's rent and food costs in her business book. Which rule is broken, and who is misled?

2.4 A shop is told it will close in three months. Explain what happens to the going concern assumption and to the way its shelves are recorded under historical cost.



A grocer's balance scale with weights on both pans, and a ledger beside it

CHAPTER III

The Accounting Equation

Why must what a business holds always match the claims on it?



This chapter teaches assets, liabilities and owner's equity, the three building blocks of every business record. It shows why they always balance, adds revenue, expenses and drawings, and tests the idea on seven January events in a small binding workshop.

YOU WILL NEED

Chapter 2, 'What Accounting Is', section 2.1 'The Idea of Accounting', for the meaning of accounting and bookkeeping; Chapter 2, section 2.3 'Rules and Concepts', for historical cost; and Appendix A1, 'Maths You Will Use', for moving a term across an equals sign.

A market stall holds Tk 100,000 of value, and someone must have a claim on all of it.

The uncle's claim and Rahim's claim together equal the whole value of the stall.

This chapter explains why what a business holds always equals the claims on it.

Picture a market stall on a busy street in Dhaka. Its owner, Rahim, counts what the stall holds: a folding table, a rack of umbrellas for sale, and cash in a tin. A careful friend values everything together at Tk 100,000. "Whose is all this?" the friend asks. Rahim answers: "Mostly mine. But my uncle lent me Tk 30,000 to buy the umbrellas, and he expects it back."

Now the friend has an answer. The uncle has a claim of Tk 30,000. The rest, Tk 70,000, belongs to Rahim. Add the two claims and the total is Tk 100,000 again. Try to find one taka of value in the stall that nobody has a claim on. You cannot. Every taka entered the stall in one of two ways: someone lent it, or Rahim put it in or earned it.

This small discovery is the subject of the whole chapter. A business holds things of value. The people who lent money, and the owner, hold claims on those things. The two totals must agree, in every business, in every country, on every day. Is that a law of nature, or only a habit of bookkeepers? By the end of the chapter you will see that it is a rule of arithmetic with a plain reason, and you will watch seven events in a small workshop obey it, one after another.

* * *

3.1 ASSETS, LIABILITIES AND EQUITY

Records begin with events that change a business's money, and these events have a name.

Transaction. An event that has a money effect on a business and must be recorded.

We follow one small workshop through this chapter, and its figures never change.

Bookkeeping, as Chapter 2 explained, means writing down what happens to the money of a business.¹ But which events deserve a place in the records? A business meets many events in a day: a telephone call, a promise, a visit from a customer. Only some of them change its money. A **transaction** is an event that has a money effect on a business and must be recorded. Buying shelving for Tk 120,000 is a transaction. A customer who says she may buy a book next month has not yet created one.

Case file, illustrative. Meghna Binding Service is a book-binding workshop that repairs and binds books for schools. It began on 1 January 2026 and keeps a separate set of books from the bookshop Meghna and Sons of Dhanmondi, Dhaka. This chapter uses the workshop's records for January 2026. Amounts are in taka (Tk), with international digit grouping.²

¹ Recall: accounting and bookkeeping, Chapter 2, 'What Accounting Is', section 2.1 'The Idea of Accounting'.

² In Bangladesh you may also see lakh and crore. Tk 500,000 is five lakh. This book writes every amount with commas after each three digits.

Begin with your own life. You may own a bicycle, a mobile telephone and Tk 2,000 in cash. Each has value to you, and you control each one. A business owns things of the same kind. An **asset** is something of value that a business owns or controls, such as cash, goods held for sale or equipment. The shelving that the workshop buys on 2 January 2026 is an asset, and so is the cash in its drawer. Goods held for sale have a name of their own, inventory, which Chapter 13, 'Inventory: Meaning and Measurement', section 13.1 'What Inventory Is' teaches in full. We use that word from here onward.

Now suppose you borrow Tk 1,500 from a friend to buy a second-hand textbook. The book is yours, but you owe your friend. A business owes money in the same way. A **liability** is an amount a business owes to others, such as a loan or an unpaid supplier bill. On 3 January 2026 the workshop buys books from a publisher for Tk 200,000 and promises to pay later. Until it pays, it has a liability. The amount owed to one supplier is called an account payable; Chapter 4, 'Accounts, Debits and Credits', section 4.1 'The Account' explains that name more fully. Some books say creditor for the same idea. This book says account payable.

Return to the bicycle. Suppose it is worth Tk 12,000, and Tk 4,000 of its price is still owed to your uncle. How much of the bicycle is really yours? Tk 12,000 minus Tk 4,000 leaves Tk 8,000. A business works in the same way. **Owner's equity** is the owner's claim on the business: what is left of the assets after the liabilities are paid. Other books call it capital, net worth or proprietorship. This book uses only owner's equity, so that one idea always has one name. At 31 January 2026 the workshop holds assets of Tk 565,000 and owes Tk 120,000, so its owner's equity is Tk 445,000.

Go back to Rahim's stall. It holds Tk 100,000. His uncle's claim is Tk 30,000 and his own is Tk 70,000. In words: what the business holds equals what it owes plus what the owner can claim. The **accounting equation** is the rule that assets always equal liabilities plus owner's equity. In symbols, $A = L + E$, where A stands for assets, L for liabilities and E for owner's equity. For the stall, $100,000 = 30,000 + 70,000$. The equation is true because each asset was paid for by someone: a lender, who gains a liability claim, or the owner, who gains an equity claim. No third source exists.

Start from your own belongings; a business owns things of the same kind.

GO SLOWLY HERE.

Asset. Word in use. In daily life, a useful quality or a helpful person, as in "she is an asset to the team". In accounting, an item of value that a business owns.

Asset. Something of value that a business owns or controls, such as cash, goods held for sale or equipment.

Money owed to others is the second building block.

Liability. An amount a business owes to others, such as a loan or an unpaid supplier bill.

The owner's share is what remains after the debts are taken away.

GO SLOWLY HERE.

Capital. Word in use. In daily life, a city where a government sits, or any money used to start something. In accounting, it often means the owner's money in the business; this book says owner's equity.

Owner's equity. The owner's claim on the business: what is left of the assets after the liabilities are paid.

Both sides of a business must agree, and this agreement has a name.

Accounting equation. The rule that assets always equal liabilities plus owner's equity.

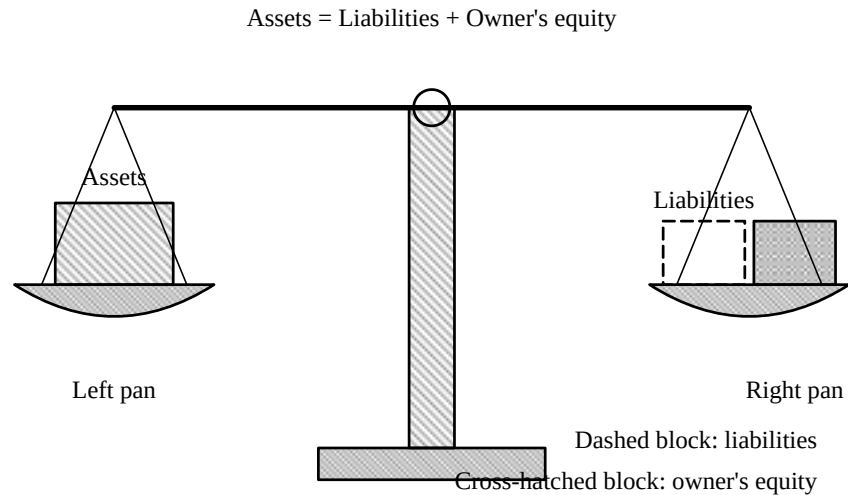


FIG. 3.1. An equation as a balanced scale: assets on one pan, liabilities and owner's equity on the other

Read the scale as an equals sign: the two pans carry the same total.

The equation can be rearranged to find any one of its three parts from the other two.

Worked example: owner's equity from assets and liabilities, with small numbers.

The equation measures recorded amounts, and equity can fall below zero.

Each person reads the same equation with a different question in mind.

Four ideas have been built so far, and each rests on the one before.

The picture shows the idea a second time. Read it from left to right. The left pan carries the assets. The right pan carries two blocks: the liabilities and the owner's equity. The beam is level because the totals are equal. If a block is added to one pan, a block of the same weight must be added to the other, or the scale tips.

Often you know two parts and need the third. Rearranging the equation gives $E = A - L$, and also $L = A - E$ (Appendix A1, section A1.3 'Rearranging an Equation', shows the move). Here is the first calculation, set out in the order used throughout this book.

Given. A business has assets of Tk 50,000 and liabilities of Tk 20,000. **Find.** The owner's equity. **Formula.** In words: owner's equity equals assets minus liabilities. In symbols: $E = A - L$. **Steps.** $E = 50,000 - 20,000 = 30,000$. **Answer.** The owner's equity is Tk 30,000. **Check.** Liabilities plus equity should equal assets: $20,000 + 30,000 = 50,000$. It does.

Two limits are worth knowing. First, the equation uses the prices written in the records. Chapter 2, section 2.3 'Rules and Concepts', taught historical cost: an asset is recorded at the price paid.³ Shelving bought for Tk 120,000 stays at Tk 120,000, even if a dealer would offer a different price today. Second, if liabilities exceed assets, then $E = A - L$ gives a negative number. The owner then has no claim left, and the business owes more than it holds. This is a warning sign, not an error in the equation.

Different people read the equation differently. A lender looks at A and L and asks whether the assets are large enough to repay the loan. The owner looks at E and asks how large the personal claim has become. A supplier looks at the account payable and asks whether the business can pay it.

PAUSE AND CHECK 3.P1 A workshop has assets of Tk 80,000 and liabilities of Tk 25,000. What is its owner's equity, and how do you check the result?

Before moving on. A transaction is a money event that must be recorded. Assets are what the business owns, liabilities are what it owes, and owner's equity is the owner's claim after debts. The three always obey $A = L + E$. You should now be able to say why the two sides of the equation can never differ.

³ Recall: historical cost, Chapter 2, 'What Accounting Is', section 2.3 'Rules and Concepts'.

3.2 REVENUE, EXPENSES AND DRAWINGS

On the first day the owner's claim is the money put in. Then the business starts to work. How can the owner's claim grow beyond what the owner invested? Think of a household. Your wealth rises when you receive a salary, falls when you pay the rent, and falls again when you give money to a relative. Business owners see the same three movements, and each has a name.

A business earns by selling goods or doing work. **Revenue** is money earned or becoming due from selling goods or doing services. On 10 January 2026 the workshop binds books for a school and receives Tk 40,000 in cash. That is revenue of Tk 40,000. Notice the words "or becoming due". A customer may take the work today and pay next month. The revenue is earned when the work is done, not when the cash arrives. Chapter 7, 'Adjusting the Accounts', section 7.1 'Why Adjust' returns to this timing.

Earning revenue costs something. An **expense** is a cost used up in earning revenue, such as rent or wages. On 15 January 2026 the workshop pays rent of Tk 60,000, and on 31 January it pays wages of Tk 35,000. Both are used up already: the rented room has served its month, and the worker has done the work. Two payments are not expenses, and many students confuse them. Buying shelving only changes one asset into another, and paying a supplier only reduces a liability.

The owner may also withdraw money. **Drawings** means money or goods the owner withdraws from the business for personal use. If Rahim takes Tk 500 from the tin to buy his family's dinner, that is drawings. It reduces the assets and the owner's claim, yet it did not help to earn any revenue, so it is not an expense. The workshop has no drawings in January 2026.

The four movements can now be joined. The owner's equity at any date equals the money the owner has put in (with any equity carried from earlier periods), plus revenue, minus expenses, minus drawings. In symbols, $E = I + R - X - D$, where I is the owner's investment, R is revenue, X is expenses and D is drawings. Put this into the equation and the expanded form appears: $A = L + (I + R - X - D)$. It tells you where the owner's claim came from and where it went.

Given. Karim's umbrella stall has an owner's investment of Tk 30,000, revenue of Tk 18,000, expenses of Tk 11,500, drawings of Tk 4,000 and liabilities of Tk 9,500. *Find.* The owner's equity and the total assets. *Formula.* Owner's equity equals investment plus revenue minus expenses minus drawings: $E = I + R - X - D$. Then $A = L + E$. *Steps.* $E = 30,000 + 18,000 - 11,500 - 4,000 = 32,500$. $A = 9,500 + 32,500 = 42,000$. *Answer.* Owner's equity is Tk 32,500 and total assets are Tk 42,000. *Check.* Change the order: $30,000 - 4,000 = 26,000$; $26,000 + 18,000 = 44,000$; $44,000 - 11,500 = 32,500$. The result is the same.

Case file, illustrative. Meghna Binding Service, January 2026: owner's investment Tk 500,000; revenue Tk 40,000; expenses Tk 60,000 (rent) plus Tk 35,000 (wages) = Tk 95,000; drawings none. Owner's equity = $500,000 + 40,000 - 95,000 - 0 = \text{Tk } 445,000$. This agrees with assets Tk 565,000 minus liabilities Tk 120,000.

Four slips appear again and again. Students subtract revenue and add expenses, which reverses the signs. They record drawings as an expense, which hides the owner's withdrawals. They mix units, for example thousands of taka with whole taka. And they treat revenue as cash received, although a customer may not have paid yet. Before any answer, ask which way each item moves the owner's claim.

The owner's claim changes after the first day, and three new ideas explain how.

Revenue is what a business earns by selling or by doing work.

Revenue. Money earned or becoming due from selling goods or doing services.

An expense is a cost used up while earning revenue.

Expense. A cost used up in earning revenue, such as rent or wages.

Drawings are the owner's own withdrawals, and they are not a cost of earning.

Drawings. Money or goods the owner withdraws from the business for personal use.

The owner's claim at any date is money put in, plus revenue, minus expenses and drawings.

Worked example: a messier stall with all four movements and a liability.

The workshop's January figures give the same owner's equity by a second route.

Four common slips: wrong sign, drawings as expense, mixed units, revenue as cash.

Three new items and one longer formula have been added to the equation.

Every transaction has matching effects, so the equation stays balanced.

Three questions analyse any transaction in the same order.

GO SLOWLY HERE.

Five patterns cover all seven events of the month.

Every row has two effects, which is the seed of a larger system.

Three questions, five patterns and one test close this section.

PAUSE AND CHECK 3.P2 The owner invested Tk 40,000. Revenue was Tk 15,000, expenses Tk 9,000 and drawings Tk 2,000. What is the owner's equity?

Before moving on. Revenue raises owner's equity, expenses lower it, and drawings lower it without being an expense. The expanded form is $E = I + R - X - D$. You should now be able to say which of the four movements raises the owner's claim and which three lower it.

3.3 TRANSACTIONS AND THE EQUATION

How can a business record hundreds of events and still keep both sides equal? The answer lies in how a transaction behaves. Each transaction changes the items of the equation, but always in matching ways: when one side changes, the other side changes by the same amount, or the change happens inside one side only.

Take any transaction and ask three questions in order. First, which items change, and by how much? Second, is each item an asset, a liability, owner's equity, revenue, an expense or drawings? Third, after the change, does $A = L + E$ still hold? If the third answer is no, an effect is missing or a sign is wrong. Begin with the smallest case. On 1 January 2026 the owner puts Tk 500,000 into the workshop. The asset cash rises by Tk 500,000, and owner's equity rises by Tk 500,000. Both sides are Tk 500,000. The equation holds.

The table below groups the effects into five patterns. Read a row from left to right: the first cell names the pattern, the second states what happens to the two sides, and the third gives an example from the workshop.

THE FIVE PATTERNS OF CHANGE IN THE EQUATION

PATTERN	EFFECT ON THE EQUATION	EXAMPLE FROM JANUARY 2026
One asset rises, another falls	Total assets unchanged	Shelving bought for cash
Asset rises, liability rises	Both sides rise equally	Inventory bought on credit
Asset rises, owner's equity rises	Both sides rise equally	Owner invests; binding fee earned
Asset falls, owner's equity falls	Both sides fall equally	Rent paid; wages paid
Asset falls, liability falls	Both sides fall equally	Publisher paid

Notice that every row contains at least two effects. No transaction changes only one item and leaves the rest alone. This fact is the seed of double entry, which Chapter 4, 'Accounts, Debits and Credits', section 4.3 'Debit and Credit' builds into a full method of recording. For now, keep the simpler tool: test $A = L + E$ after every transaction. A transaction that changes only the asset side, such as the shelving purchase, still balances, because one asset rises exactly as another falls.

Before moving on. Analyse any transaction with three questions: what changes, what kind of item it is, and whether $A = L + E$ still holds. Five patterns cover the January events. You should now be able to say why no transaction can leave the equation unbalanced.

WORKED EXAMPLE: SEVEN TRANSACTIONS OF JANUARY

Case file, illustrative. *Given.* Meghna Binding Service, January 2026: 1 January, the owner invests cash of Tk 500,000; 2 January, buys shelving for cash, Tk 120,000; 3 January, buys books (inventory) on credit from a publisher, Tk 200,000; 10 January, earns Tk 40,000 in cash for binding work; 15 January, pays rent, Tk 60,000; 20 January, pays the publisher Tk 80,000; 31 January, pays wages, Tk 35,000.

Find. Total assets, liabilities and owner's equity at 31 January 2026, and a check that they balance. *Formula.* $A = L + E$. Each row of the table below shows the effect of one event on A, L and E, in taka. *Steps.* The table lists each event with its effect. Assets rise or fall by the cash and other items affected, and equity changes only when revenue or an expense arises or the owner invests.

The seven events of the month are the data, and the task is to find the final position.

Find, formula and steps: classify each event, then total the effects.

EFFECTS OF THE SEVEN EVENTS ON THE EQUATION (TK)

DATE	EVENT	ASSETS	LIABILITIES	OWNER'S EQUITY
1 Jan	Owner invests cash	+500,000	0	+500,000
2 Jan	Shelving bought for cash	+120,000 and -120,000	0	0
3 Jan	Inventory bought on credit	+200,000	+200,000	0
10 Jan	Binding fee earned (revenue)	+40,000	0	+40,000
15 Jan	Rent paid (expense)	-60,000	0	-60,000
20 Jan	Publisher paid	-80,000	-80,000	0
31 Jan	Wages paid (expense)	-35,000	0	-35,000
Total		565,000	120,000	445,000

Event	Assets	Liabilities	Owner's equity
1 Jan owner invests	500,000	0	500,000
2 Jan shelving (cash)	500,000	0	500,000
3 Jan inventory (credit)	700,000	200,000	500,000
10 Jan binding fee	740,000	200,000	540,000
15 Jan rent paid	680,000	200,000	480,000
20 Jan publisher paid	600,000	120,000	480,000
31 Jan wages paid	565,000	120,000	445,000

Check at 31 January: $120,000 + 445,000 = 565,000$

FIG. 3.2. Totals of assets, liabilities and owner's equity after each of the seven events, in taka

Read the figure by rows, each showing the three totals after one event.

The answer is stated in a sentence, and two independent checks confirm it.

Three slips are common in a worked list like this one.

A long list is solved one row at a time and then totalled.

The chapter in a few lines: three items, one equation, and what moves it.

Owner's equity is a claim, and it is not the same as the owner's cash.

Paying a debt lowers a liability, and it does not create an expense.

A balanced equation does not prove that every figure is correct.

To read the second figure, take one row at a time. The three number columns show the totals of assets, liabilities and owner's equity after that event. Look at the row for 3 January: assets rise to 700,000, and liabilities to 200,000, with owner's equity still at 500,000. In every row, the first number equals the sum of the other two.

Answer. At 31 January 2026 the workshop has assets of Tk 565,000, liabilities of Tk 120,000 and owner's equity of Tk 445,000. The assets are cash Tk 245,000, shelving Tk 120,000 and inventory Tk 200,000. *Check.* Liabilities plus equity: $120,000 + 445,000 = 565,000$, which equals the assets. The cash can be tested alone: $500,000 - 120,000 + 40,000 - 60,000 - 80,000 - 35,000 = 245,000$. The expanded form agrees: $500,000 + 40,000 - 95,000 = 445,000$.

Students often match a rent payment to the liability column, when it should lower owner's equity. They also forget that the shelving appears twice, once rising and once falling. And they sometimes change owner's equity when the publisher is paid. Test each row against the three questions before totalling.

Before moving on. A long list of events is solved one row at a time, then totalled. Two checks, $A = L + E$ and the expanded form, give confidence. You should now be able to analyse a new set of events in the same way.

* * *

SUMMARY

Assets are what a business owns, liabilities are what it owes, and owner's equity is the owner's claim after debts. The accounting equation, $A = L + E$, holds for every business at every date. Revenue raises owner's equity, expenses lower it, and drawings lower it without being an expense, giving $E = I + R - X - D$. Every transaction has matching effects, so the equation stays balanced. At 31 January 2026, Meghna Binding Service has assets of Tk 565,000, liabilities of Tk 120,000 and owner's equity of Tk 445,000.

THINK IT THROUGH

3.T1 A transaction changes only the asset side of the equation. Give one example from this chapter and explain why both sides stay equal.

3.T2 A student says that paying the publisher Tk 80,000 on 20 January 2026 was an expense. Decide whether she is right, and explain what changed.

WHERE PEOPLE GO WRONG

Many readers think owner's equity is the cash the owner has in the business. It is not. Owner's equity is a claim, not an item. At 31 January 2026 the workshop's owner's equity is Tk 445,000, but its cash is only Tk 245,000. The claim covers the shelving and the inventory too, after the liability is removed.

Another common belief is that every payment is an expense. Paying the publisher Tk 80,000 lowered cash and the account payable by the same amount. Owner's equity stayed at Tk 480,000. Only costs used up in earning revenue, such as rent and wages, are expenses.

A third belief is that if $A = L + E$ balances, every figure must be right. This does not follow. Suppose the rent of Tk 60,000 were recorded as Tk 6,000. Cash and equity would both fall by Tk 6,000, and the equation would still balance, yet

the records would be wrong. Chapter 6, 'The Trial Balance', shows how such mistakes are sought.

IN HISTORY

In 1397 Giovanni di Bicci de' Medici (1360–1429), a banker of Florence in Italy, founded the Medici Bank. It grew into one of the most powerful banking houses of the fifteenth century, with branches in cities such as Rome, Venice, Geneva, Lyon, Bruges and London. Each branch was a separate company with its own name, owner's equity, managers and account books, under a head office in Florence.

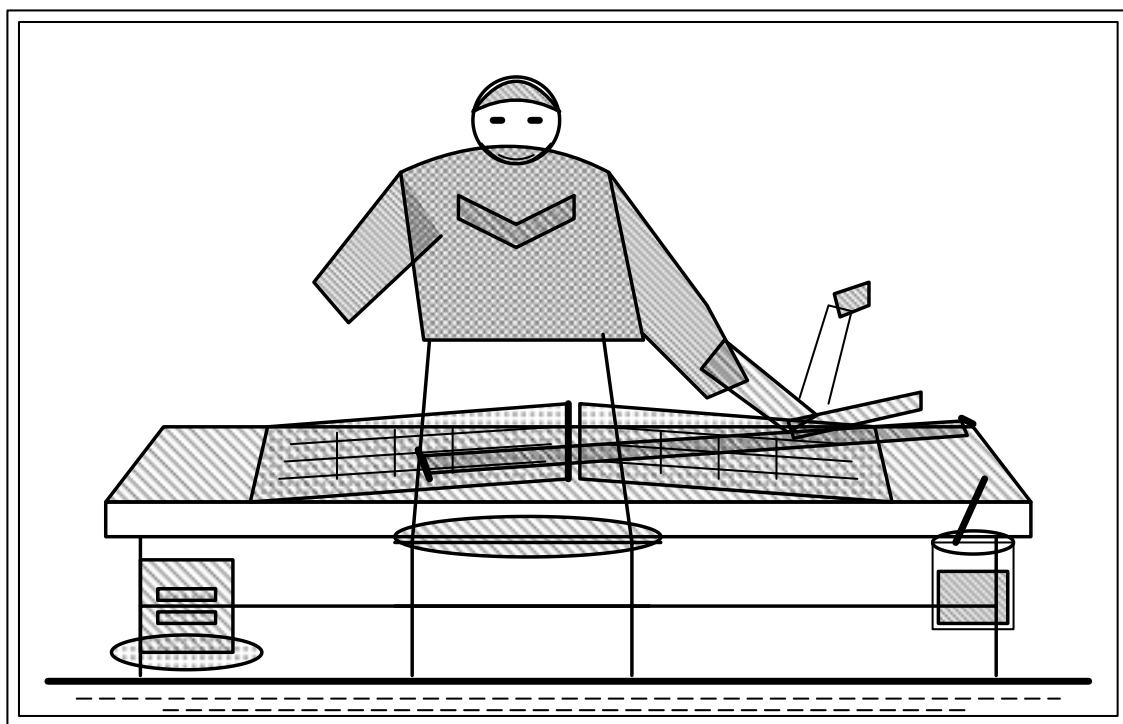
A branch of such a bank held coins, loans owed to it and goods, and it owed deposits and shares of profit to its partners. These are the two lists that the accounting equation says must match. A set of confidential ledgers, discovered in 1950, allowed the historian Raymond de Roover to reconstruct how the bank was organised and run. The bank collapsed in 1494, when the Medici were expelled from Florence. Its books remain a reminder that every business, old or new, must show what it holds and who has a claim on it.

A bank of the fifteenth century needed the same two lists that the equation joins.

Surviving ledgers let a modern historian reconstruct what the branches held and owed.

REVIEW QUESTIONS

- 3.1 Classify each item as an asset, a liability or owner's equity: (a) cash in the drawer, Tk 30,000; (b) a bill of Tk 12,000 owed to a paper supplier; (c) a binding press owned by the workshop; (d) a bank loan of Tk 90,000; (e) the owner's claim after all debts are paid.
- 3.2 A business has assets of Tk 240,000 and owner's equity of Tk 150,000. Find the liabilities.
- 3.3 At the start of a year a workshop's owner's equity was Tk 300,000. During the year the owner invested no more money. Revenue was Tk 410,000, expenses Tk 365,000 and drawings Tk 50,000. Find the owner's equity at the end of the year.
- 3.4 A workshop reports, in thousands of taka, assets of 1,250 and liabilities of 480. State its owner's equity in taka.
- 3.5 On 1 March a workshop starts with Tk 200,000 in cash from the owner. It then (a) buys equipment on credit for Tk 90,000; (b) earns Tk 30,000 in cash; (c) pays wages of Tk 18,000; (d) the owner withdraws Tk 10,000; (e) pays Tk 40,000 of the equipment debt. Find assets, liabilities and owner's equity after the fifth event, and check that they balance.



A clerk ruling a ledger page with a straight-edge and pen, about 1880

CHAPTER IV

Accounts, Debits and Credits

*How can one fact be written down so that it can be found,
added and checked?*



*This chapter teaches the account, the record that gathers every
change to one kind of item. It shows the six types of account and
the rule of debit and credit, called double entry, and applies the
rule to seven transactions of a small business.*

YOU WILL NEED

Chapter 3, 'The Accounting Equation', section 3.1 'Assets, Liabilities and Equity'; section 3.2 'Revenue, Expenses and Drawings'; and section 3.3 'Transactions and the Equation'. Also Appendix A1, 'Maths You Will Use', for adding and subtracting columns of figures.

A diary of events makes the simplest question hard to answer.

The equation of Chapter 3 works for a few events but not for thousands.

The chapter teaches the place, the six kinds and the rule that catches mistakes.

Imagine that you keep a diary for a small business. Each evening you write one line for every event: money received, a bill paid, books bought. After a month the diary holds dozens of lines. Then a lender asks one question: how much cash does the business hold today? To answer, you must read every line, decide which ones touched cash, and add and subtract them in the right order. If your answer is wrong, you cannot see where the mistake lies.

Chapter 3 gave one answer: change the accounting equation after each transaction. That method works well for seven events. It fails for seven thousand, because the table would need a column for every kind of item and a row for every event. A real business needs two things. It needs to keep each kind of item in one place, with every increase and decrease together. It also needs a rule that catches mistakes at once.

This chapter teaches both. First we meet the account, which is the place. Next we sort accounts into six types. Then we learn the rule of debit and credit, which links every entry to a second entry and so exposes many errors. The rule was written in a printed book more than five hundred years ago, and it has hardly changed since. Few ideas in business have lasted so long.

* * *

4.1 THE ACCOUNT

GO SLOWLY HERE.

A household notebook with one page for each item shows the idea of an account.

An account holds all the changes to one kind of item.

Account. A record that collects all the increases and decreases of one kind of item, such as Cash.

A chart of accounts is the list that stops duplicate or missing accounts.

Chart of accounts. A list of every account a business uses, with a code for each.

A T-account is a quick drawing of an account with two sides.

T-account. A simple account drawn as a letter T, with a left side and a right side.

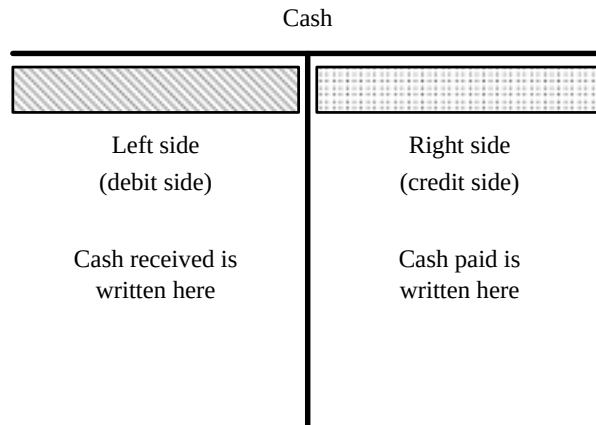
Think of a household exercise book. One page is headed *Rice*, another *Electricity*, a third *Money lent to a neighbour*. Each time something happens, you write it on the page where it belongs. To learn how much was spent on rice this month, you read one page and not the whole book. A business uses the same idea, with many more pages and with strict rules for writing on them.

An **account** is a record that collects all the increases and decreases of one kind of item. For example, the Cash account collects every receipt and every payment of cash, so it shows how much cash the business holds. The business keeps one account for each kind of item it must follow: one for Cash, one for shelving, one for money owed to a publisher, and so on.

As the number of accounts grows, the business needs a list, so that nobody opens a second Cash account by mistake. A **chart of accounts** is a list of every account a business uses, with a code for each. For example, a business might give the code 101 to Cash and the code 201 to Accounts Payable. Meghna Binding Service, the running case, uses only eight accounts, so its chart is short. A large company may use hundreds.

Before we use real ledger books, we need a quick way to draw an account. A **T-account** is a simple account drawn as a letter T, with a left side and a right side. The name of the account is written across the top, and each entry is written below it, on one side or the other. The figure shows a T-account for Cash. The horizontal line separates the name from the entries, the vertical line separates the two sides, and each

figure is one entry. Later in this chapter, the two sides receive their proper names.



The line under the name and the line down the middle make the letter T.

FIG. 4.1. A T-account for Cash, with its two sides labelled

Before moving on. An account collects all the changes to one kind of item, the chart of accounts lists every account in use, and a T-account is a quick drawing with a left side and a right side. You should now be able to say what the Cash account would show, and why a business keeps a separate account for each kind of item.

The recap gathers the section into three sentences.

* * *

4.2 SIX TYPES OF ACCOUNT

Chapter 3 sorted the items of a business into groups when it built the accounting equation.¹ Accounts follow the same groups, and there are six types. Why six? Because each type moves in its own way when something happens, and the rules of the next section depend on knowing the type. The table below lists them. Read each row from left to right: the type, what its accounts record, and examples from Meghna Binding Service.

Every account belongs to one of six types taken from the equation.

THE SIX TYPES OF ACCOUNT

TYPE	WHAT ITS ACCOUNTS RECORD	EXAMPLES
Asset	Things of value the business owns or controls	Cash, Shelving, Inventory
Liability	Amounts the business owes to others	Accounts Payable
Owner's capital	Value the owner has put into the business	Owner's Capital
Drawings	Value the owner takes out for personal use	Drawings
Revenue	Money earned or due from goods or services	Service Revenue
Expense	Costs used up in earning revenue	Rent, Wages

¹ Recall: asset, liability and equity, Chapter 3, 'The Accounting Equation', section 3.1 'Assets, Liabilities and Equity'; revenue, expense and drawings, section 3.2 'Revenue, Expenses and Drawings'.

Asset accounts record what is owned; liability accounts record what is owed.

The owner's capital account records money put in; drawings records money taken out.

Revenue accounts record what is earned; expense accounts record what is used up.

The recap gathers the section into three sentences.

An asset account records something of value that the business owns or controls, such as Cash, Shelving or Inventory. Here *inventory* means the books held for sale, a subject that Chapter 13, 'Inventory: Meaning and Measurement', teaches in full. A liability account records an amount owed to others. Accounts Payable, for example, holds the amounts owed to publishers. The owner can look at the first and see what the business has, and look at the second and see what it owes.

The owner's capital account records the value the owner has put into the business, and it is part of owner's equity. The drawings account records what the owner takes out for personal use. Why keep them apart? Because the owner should see both how much was put in and how much was taken out. A single combined account would hide the second figure behind the first.

A revenue account records money earned or due from selling goods or doing work. Service Revenue is the revenue account of Meghna Binding Service. An expense account records a cost used up in earning revenue, such as Rent or Wages. The business keeps a separate expense account for each cost, because an owner wishes to know the rent and the wages separately and not as one lump.

Before moving on. The six types are asset, liability, owner's capital, drawings, revenue and expense. Each account belongs to exactly one type, and its type decides how it behaves. You should now be able to name the type of Cash, Accounts Payable, Rent and Service Revenue.

* * *

4.3 DEBIT AND CREDIT

GO SLOWLY HERE.

Bank statements use the two words, but with a meaning that does not fit here.

Debit. Word in use. In daily life, a bank debits an account when it takes money out. In accounting, a debit is an entry on the left side of an account, whether the balance rises or falls.

Credit. Word in use. In daily life, credit means buying now and paying later, or a bank adding money to an account. In accounting, a credit is an entry on the right side of an account.

A debit is an entry on the left side of an account.

Debit. An entry on the left side of an account; it increases assets and expenses and decreases liabilities and equity.

A credit is an entry on the right side of an account.

Credit. An entry on the right side of an account; it increases liabilities, equity and revenue and decreases assets.

The sides of an account copy the sides of the accounting equation.

You have probably seen the words *debit* and *credit* on a bank statement. The bank says it has debited your account when it takes money out and credited it when it adds money. Shops also use *credit* for buying now and paying later. In accounting, the two words have a fixed and different meaning, and the bank's habit can mislead you. Put it aside for this chapter. The words come from Venice: *debit* from the Latin *debere*, to owe, and *credit* from *credere*, to believe.

A **debit** is an entry on the left side of an account. That is the whole meaning of the word: the left side. It does not mean good or bad, and it does not mean more or less. In the Cash T-account, a figure written on the left side is a debit to Cash. Whether a debit raises or lowers the balance depends on the type of account, as the next paragraphs show.

A **credit** is an entry on the right side of an account. Again, the word names a side and nothing else. A payment of cash is written on the right side of the Cash T-account, so we say that Cash is credited. The balance of an account is the difference between its two sides. Whether a credit raises or lowers that balance also depends on the type of account.

Why does an asset increase on the left? The reason lies in the accounting equation, which has two

sides.² Assets stand on the left of the equals sign, and liabilities and owner's equity stand on the right. The rule copies this layout. An increase in an item from the left side of the equation is recorded on the left of its account. An increase in an item from the right side is recorded on the right. Decreases are recorded on the opposite side from increases.

Three types need a second thought: revenue, expenses and drawings. They are parts of owner's equity. Revenue raises equity, so it sits with the right side. Expenses and drawings lower equity, so their increases take the opposite side, the left. The expanded equation shows the result. In words: assets plus drawings plus expenses equal liabilities plus owner's capital plus revenue. In symbols: $A + D + E = L + C + R$, where A is assets, D is drawings, E is expenses, L is liabilities, C is owner's capital and R is revenue.

Now the rule can be read straight from the equation. The three types on the left, which are assets, drawings and expenses, increase on the left, so they increase by debit. The three types on the right, which are liabilities, owner's capital and revenue, increase on the right, so they increase by credit. Every decrease is on the other side. The table gathers the rule. Read each row from the left: the type, the side of an increase, the side of a decrease, and the normal balance, which the next paragraph explains.

THE SIDE OF INCREASE AND DECREASE FOR EACH TYPE

TYPE	INCREASE	DECREASE	NORMAL BALANCE
Asset	Debit	Credit	Debit
Drawings	Debit	Credit	Debit
Expense	Debit	Credit	Debit
Liability	Credit	Debit	Credit
Owner's capital	Credit	Debit	Credit
Revenue	Credit	Debit	Credit

An account usually grows more than it shrinks, so its balance usually ends on the side of its increases. The **normal balance** is the side of an account, debit or credit, on which increases are recorded. Cash has a debit normal balance, because cash increases by debit. Accounts Payable has a credit normal balance. When a balance falls on the other side, it is a signal to look closer: an asset that ends on the credit side may mean an error, or a bank overdraft.

PAUSE AND CHECK 4.P1 Rent is an expense. On which side is its normal balance, and why?

The figure shows the same rule as a picture. The left column lists the three types that increase by debit, and the right column lists the three that increase by credit. A short bar under each name marks its normal balance. You can rebuild the whole picture from one fact: assets, drawings and expenses stand on the left, and the rest stand on the right.

Drawings and expenses lower equity, so they behave like assets; revenue behaves like equity.

The three types on the left increase by debit; the three on the right increase by credit.

The normal balance is the side on which an account usually ends.

Normal balance. The side of an account, debit or credit, on which increases are recorded and where its balance usually falls.

The figure shows each type with its side of increase and its normal balance.

² Recall: the equation in expanded form, Chapter 3, 'The Accounting Equation', section 3.2 'Revenue, Expenses and Drawings'.

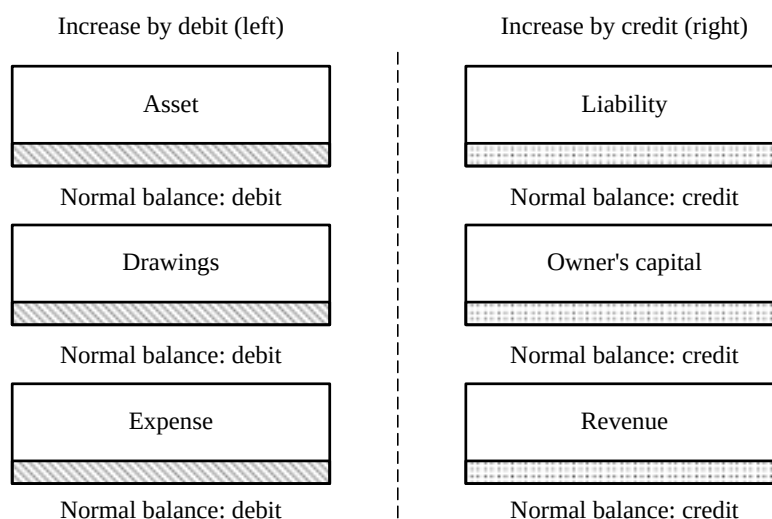


FIG. 4.2. Six account types, each with its side of increase and normal balance

Double entry means every transaction has two equal sides.

Double entry records each transaction twice, with equal debit and credit.

Double entry. The rule that every transaction is recorded twice, once as a debit and once as a credit of equal amount.

Equal entries keep the equation in balance and expose many mistakes.

The owner and the lender both gain from the check of equal totals.

Double entry has limits: equal totals do not prove that every entry is right.

Many students find this the hardest step, because it asks you to hold two ideas together. Start with a question. If all changes to an item sit in one account, how can we know that none was forgotten? Chapter 3 gives the clue. Every transaction left the accounting equation in balance, so every transaction had at least two effects. The method of recording should show both effects each time. That method is called double entry.

Double entry is the rule that every transaction is recorded twice, once as a debit and once as a credit of equal amount. Take the first transaction of Meghna Binding Service. On 1 January 2026 the owner invests cash of Tk 500,000. Cash is an asset and it increases, so Cash is debited Tk 500,000. Owner's Capital increases, so Owner's Capital is credited Tk 500,000. One fact, two entries, one amount.

PAUSE AND CHECK 4.P2 A business buys shelving for cash. Name the two accounts, and say which one is debited and which one credited.

Why does the rule help? Each entry changes the equation by the same amount on each side, so the equation stays in balance after every transaction. It follows that, at any moment, the total of all debit entries must equal the total of all credit entries. If they differ, an error exists, and the business knows it at once. This check is far better than the diary of the opening puzzle. The Venetian method was described in print by Luca Pacioli in 1494, and its essential rule is the one you have now learned.

Consider the viewpoints. The owner gains a quick way to find mistakes before they grow. A lender, who cannot watch the business, gains confidence that the figures were kept under a rule that exposes many errors. An employee who keeps the records gains protection, because a difference points to the mistake and not to the person.

Where does the rule fail? Equal totals show that debits and credits agree. They do not show that the right accounts were chosen. If rent is debited to Wages by mistake, the totals still agree. If a transaction is left out completely, both sides are missing and the totals still agree. Chapter 6, 'The Trial Balance', teaches how to test the totals and what the test cannot find. Also, some transactions affect three or more accounts. The rule still holds: the debits together equal the credits together. Chapter 5, 'The Journal and the Ledger', shows such entries.

Before moving on. A debit is the left side and a credit is the right side. Assets, drawings and expenses increase by debit, while liabilities, owner's capital and revenue increase by credit, and the normal balance of each type is the side of its increase. Under double entry every transaction has equal debits and credits. You should now be able to say, for any account, which side raises it and which side lowers it.

* * *

4.4 APPLYING THE RULES

Every transaction can be recorded by the same four questions, asked in order. First, which two accounts does the transaction affect? Second, what is the type of each? Third, does each account increase or decrease? Fourth, use the table to find the side: an increase in an asset is a debit, a decrease is a credit, and so on for each type. Finish by checking that the debit amount equals the credit amount. If you follow the questions in order, you do not need to memorise any transaction.

* * *

WORKED EXAMPLE: RECORDING THE JANUARY TRANSACTIONS

We apply the method twice. The first example uses small round numbers, so that the method is the only new thing. The second is the real case, with seven transactions. Each example follows the same order: Given, Find, Formula, Steps, Answer and Check.

Example 1. Given. A student begins a tea stall. (a) She puts Tk 1,000 of her own cash into the business. (b) She buys a stove for Tk 300 in cash. (c) She earns Tk 200 in cash by selling tea. *Find.* The debit and credit of each event, and the balance of each account.

Formula. In words: increases of assets, drawings and expenses are debits, increases of liabilities, owner's capital and revenue are credits, and each transaction has an equal debit and credit. In symbols, the rule comes from $A + D + E = L + C + R$.

Steps. (a) Cash is an asset and increases, so debit Cash 1,000. Owner's Capital increases, so credit Owner's Capital 1,000. (b) The stove is an asset and increases, so debit Equipment 300. Cash decreases, so credit Cash 300. (c) Cash increases, so debit Cash 200. Tea Revenue increases, so credit Tea Revenue 200. In the Cash T-account the left side holds 1,000 and 200, which total 1,200. The right side holds 300. The balance is $1,200 - 300 = 900$ on the left.

Answer. Cash has a debit balance of Tk 900. Equipment has a debit balance of Tk 300. Owner's Capital has a credit balance of Tk 1,000, and Tea Revenue has a credit balance of Tk 200.

Check. The debit balances total $900 + 300 = 1,200$, and the credit balances total $1,000 + 200 = 1,200$. They agree. In the equation, assets of Tk 1,200 equal owner's capital of Tk 1,000 plus revenue of Tk 200, with no liabilities.

Case file, illustrative. *Example 2. Given.* Meghna Binding Service, January 2026: 1 January, the owner invests cash of Tk 500,000. 2 January, it buys shelving for cash, Tk 120,000. 3 January, it buys books for sale on credit from a publisher, Tk 200,000. 10 January, it earns Tk 40,000 in cash for binding work.

The recap gathers the section into three sentences.

Four questions turn any transaction into a debit and a credit.

Two examples follow: a small one to learn the method, then the full case.

The given facts of the small example are three events of a tiny stall.

The formula says which side raises each type and that the amounts are equal.

Each event gets one debit and one credit of equal amount.

The answer states each balance and its side.

The check compares the two sides of the equation.

The case file gives the seven January transactions of Meghna Binding Service.

The find and the formula are the same as in the first example.

The table shows the debit and credit of each transaction.

15 January, it pays rent of Tk 60,000. 20 January, it pays Tk 80,000 to the publisher. 31 January, it pays wages of Tk 35,000.

Find. The debit and credit of each transaction, and the balance of each account. **Formula.** The same as in Example 1. Use the four questions of the last section for each line, and remember that Inventory is an asset and Accounts Payable is a liability.

Steps. The table below gives the result of the four questions for each transaction. Read each row from left to right: the date, the account debited, the account credited and the amount. Notice the sixth row. Paying the publisher lowers a liability, so Accounts Payable is debited, and it lowers an asset, so Cash is credited.

THE SEVEN TRANSACTIONS WITH THEIR DEBITS AND CREDITS

DATE	DEBIT	CREDIT	AMOUNT (TK)
1 January	Cash	Owner's Capital	500,000
2 January	Shelving	Cash	120,000
3 January	Inventory	Accounts Payable	200,000
10 January	Cash	Service Revenue	40,000
15 January	Rent	Cash	60,000
20 January	Accounts Payable	Cash	80,000
31 January	Wages	Cash	35,000

Adding each account's two sides gives its balance.

Now add the two sides of each account. Cash has debits of 500,000 and 40,000, which total 540,000, and credits of 120,000, 60,000, 80,000 and 35,000, which total 295,000. The difference is $540,000 - 295,000 = 245,000$ on the debit side. The other accounts are found in the same way, and the table below gives them all.

THE BALANCE OF EACH ACCOUNT AT 31 JANUARY 2026

ACCOUNT	DEBITS	CREDITS	BALANCE
Cash	540,000	295,000	245,000 debit
Shelving	120,000	0	120,000 debit
Inventory	200,000	0	200,000 debit
Accounts Payable	80,000	200,000	120,000 credit
Owner's Capital	0	500,000	500,000 credit
Service Revenue	0	40,000	40,000 credit
Rent	60,000	0	60,000 debit
Wages	35,000	0	35,000 debit

The answer lists the eight balances in one sentence.

Answer. At 31 January 2026, Cash has a debit balance of Tk 245,000, Shelving Tk 120,000 and Inventory Tk 200,000. Rent has a debit balance of Tk 60,000 and Wages Tk 35,000. Accounts Payable has a credit balance of Tk 120,000, Owner's Capital Tk 500,000 and Service Revenue Tk 40,000.

Two checks confirm the result: equal totals and the equation.

Check. First, the debit balances total $245,000 + 120,000 + 200,000 + 60,000 + 35,000 = 660,000$. The credit balances total $120,000 + 500,000 + 40,000 = 660,000$. They agree. Second, the debit entries of the seven rows total 1,035,000, and so do the credit entries. Third, the assets are $245,000 + 120,000 + 200,000 = 565,000$, which equals liabilities of 120,000 plus owner's equity of 445,000, found as $500,000 + 40,000 - 60,000 - 35,000$.

Common slips. Students often treat debit as meaning decrease, which is wrong for assets. They record an expense on the credit side because the cash leaves. They forget the second entry, so the totals differ. Or they read a liability as an asset and debit it when the business borrows. When an entry looks strange, return to the four questions.

Before moving on. The four questions are: which accounts, what types, increase or decrease, and which side. Each transaction gets one debit and one credit of equal amount. You should now be able to record any simple transaction without memorising it.

* * *

SUMMARY

An account collects all the changes to one kind of item, and the chart of accounts lists them all. There are six types: asset, liability, owner's capital, drawings, revenue and expense. A debit is the left side and a credit is the right side. Assets, drawings and expenses increase by debit, and liabilities, owner's capital and revenue increase by credit. Double entry records every transaction with an equal debit and credit, which keeps the equation in balance.

THINK IT THROUGH

4.T1 Explain, using the accounting equation, why Cash increases on the debit side and not on the credit side.

4.T2 The owner takes Tk 10,000 in cash from the business for personal use. Name the two accounts, say which is debited and which credited, and explain why the business does not record this as an expense.

WHERE PEOPLE GO WRONG

A mistaken belief: "Debit means less and credit means more." *Correction:* the words name the left side and the right side. Debits raise assets, expenses and drawings, but they lower liabilities, owner's capital and revenue.

A mistaken belief: "A credit is good news and a debit is bad news." *Correction:* neither word has a moral meaning. A debit to Cash is good news for the business, and a credit to Accounts Payable means the business owes more.

A mistaken belief: "Money going out is a credit, so every payment is a credit to the expense." *Correction:* the credit goes to Cash, which falls. The expense account increases, so it is debited.

A mistaken belief: "If debits equal credits, the books are correct." *Correction:* equal totals show agreement only. A wrong account, a missing transaction or a wrong amount on both sides still leaves the totals equal.

IN HISTORY

In 1494, in Venice, a printer issued a large book on mathematics by Luca Pacioli, a friar of the Franciscan order who also taught mathematics. Its title, *Summa de arithmetica, geometria, proportioni et proportionalita*, means a summary of arithmetic, geometry, proportions and proportionality. Within it is a section on bookkeeping, called *Particularis de*

Four common slips cause most wrong entries.

The recap gathers the section into three sentences.

The summary restates the chapter in a few lines.

Two problems test whether the rule can be explained and applied again.

Debit does not mean decrease, and credit does not mean increase.

Debit is not bad and credit is not good.

Expenses are debited even though money leaves the business.

Equal totals do not prove that every entry is right.

Luca Pacioli described the Venetian method of double entry in a printed book.

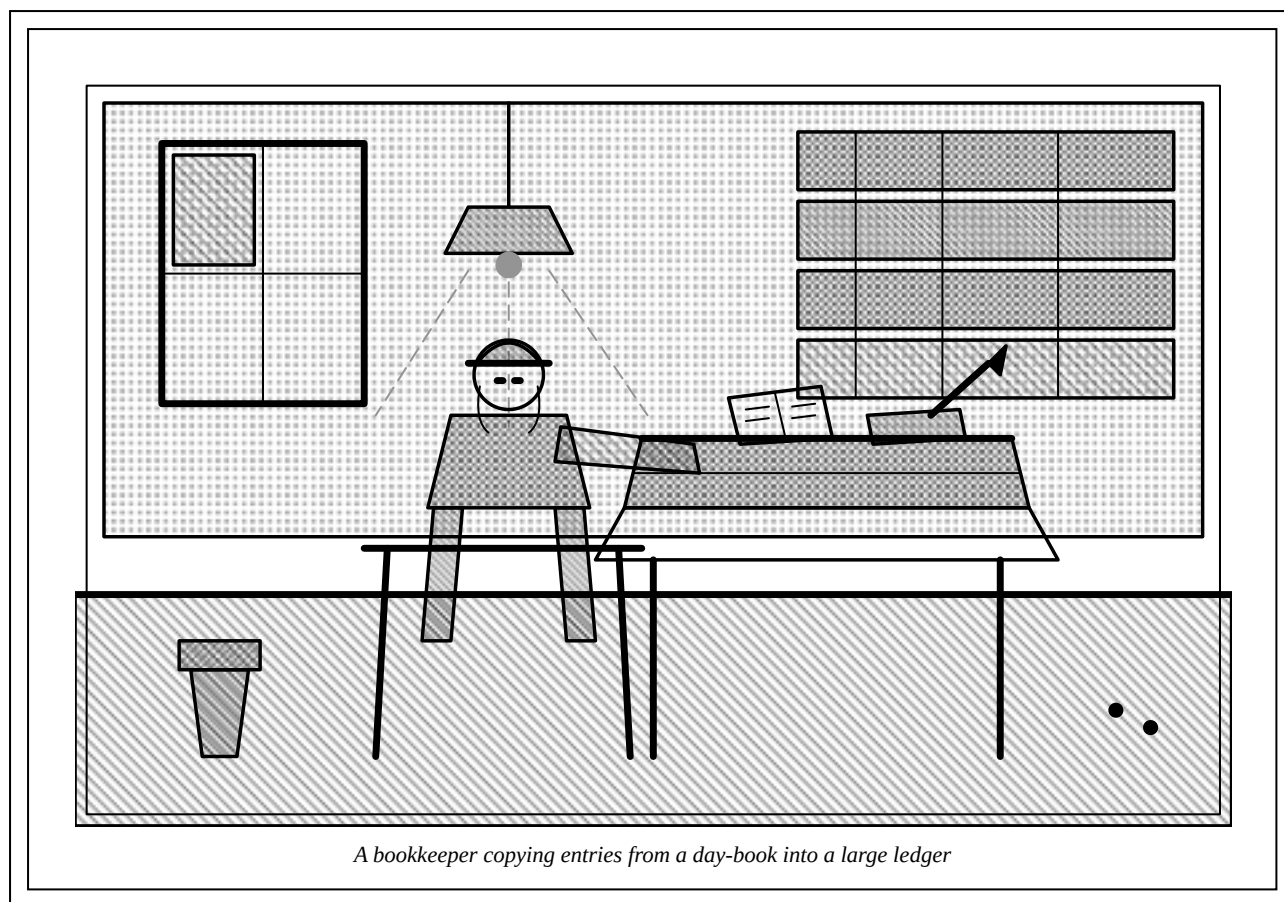
Pacioli recorded a practice that merchants already used; he did not invent it.

computis et scripturis, which describes how merchants of Venice kept their books.

Pacioli did not claim to have invented the method. He described what he called the method of Venice, which merchants had used for generations before the printing press made it widely available. His book was a textbook for students of mathematics and business. That is why it mattered: a method kept in merchants' habit became a method that anyone could read and teach.

REVIEW QUESTIONS

- 4.1 A small kiosk begins when its owner puts Tk 2,000 in cash into the business. It then buys a shelf for Tk 500 in cash and earns Tk 300 in cash from sales. Give the debit and the credit for each event, and state the balance of Cash.
- 4.2 A workshop has a Cash balance of Tk 50,000 on the debit side. It buys paper for Tk 18,000 on credit, pays Tk 7,500 to the supplier, earns Tk 26,400 in cash, pays wages of Tk 9,200 in cash, and the owner takes Tk 4,000 in cash for personal use. Give the entries, and find the balances of Cash and Accounts Payable.
- 4.3 A business borrows Tk 100,000 in cash from a bank, and later repays Tk 40,000 in cash. Give the entries for both events and the balance of the Bank Loan account.
- 4.4 A Cash account shows a balance of Tk 5,000 on the credit side. Say what this would normally mean, and name one way to find the cause.
- 4.5 For each account, state its type, the side that increases it, and its normal balance: Inventory, Accounts Payable, Drawings, Service Revenue, Wages.



CHAPTER V

The Journal and the Ledger

If every event is written down once, how can any fact be found again?



Every transaction leaves a paper trail: a source document, a journal entry and a ledger account. This chapter follows that trail step by step, names the steps of the accounting cycle, and works through the seven January 2026 transactions of Meghna Binding Service.

YOU WILL NEED

Chapter 4 'Accounts, Debits and Credits', section 4.2 'Six Types of Account';
 Chapter 4 'Accounts, Debits and Credits', section 4.3 'Debit and Credit';
 Appendix A1 'Maths You Will Use'.

A drawer of loose slips holds every fact but answers no question quickly.

Two books, one arranged by date and one by account, answer both questions.

Imagine a shopkeeper who writes every payment and every receipt on a loose slip of paper and drops the slip into a drawer. After a month the drawer holds hundreds of slips. A question arrives: how much did the shop spend on rent? The shopkeeper must read every slip to find the answer. A second question follows: what happened on the tenth of the month? Again every slip must be read. The slips contain all the facts, yet they cannot answer either question without a long search.

The puzzle is how to arrange the facts so that both kinds of question can be answered at once. Accountants found the answer long ago: write each event twice, in two different books. The first book lists events in the order in which they happened. The second collects the same events under the name of the thing they affect, such as Cash or Rent. This chapter explains both books, the path that leads from the first to the second, and the repeating series of steps to which they belong.

* * *

5.1 THE ACCOUNTING CYCLE

A baker's daily routine is a cycle, and accounting has a cycle too.

The accounting cycle is the series of steps repeated each period.

Accounting cycle. The series of steps repeated each period, from recording transactions to preparing closing records.

This book divides the cycle into eight steps, and three belong to this chapter.

Think of a baker's working day. The baker buys flour, mixes dough, bakes loaves, sells them, counts the money and cleans the ovens. Tomorrow the baker performs the same steps in the same order. A routine that repeats in this way is a cycle. Accounting has a routine of this kind. In each period a business performs the same ordered steps to turn its events into reports, and then it begins again.

The name for this routine is the **accounting cycle**. The accounting cycle is the series of steps repeated each period, from recording transactions to preparing closing records. A period is the stretch of time covered by one set of reports, such as a month or a year. When one cycle ends, the next begins with the first transaction of the new period. Meghna Binding Service, the book-binding workshop that began on 1 January 2026 with its own separate set of books, runs one cycle for each period it chooses to report.

The eight steps appear in the table that follows. Read each row from left to right: the number of the step, what happens in it, and where this book teaches it. Steps 1 to 3 are the subject of this chapter. The later steps are previewed here in one line each and taught in full later. A trial balance, for example, is a list of the balances of all the accounts, used to test that the records agree; Chapter 6 teaches it. Some books group the steps differently, but the order never changes.

THE EIGHT STEPS OF THE ACCOUNTING CYCLE

STEP	WHAT HAPPENS	WHERE TAUGHT
1	Collect and check source documents	This chapter
2	Record each transaction in the journal	This chapter
3	Post the journal to the ledger	This chapter
4	List the ledger balances in a trial balance	Chapter 6, 'The Trial Balance'
5	Adjust the accounts for changes not yet recorded	Chapter 7, 'Adjusting the Accounts'
6	Collect the figures on a worksheet	Chapter 8, 'The Worksheet'
7	Prepare the financial statements	Chapter 9, 'Financial Statements and Closing Entries'
8	Prepare the closing records	Chapter 9, 'Financial Statements and Closing Entries'

Every cycle begins with proof. A **source document** is paper or electronic proof of a transaction, such as an invoice, receipt or cheque stub. When the workshop buys books on credit from a publisher, the publisher's invoice is the source document. When the workshop pays rent, the landlord's receipt is the source document. Records begin from the document and never from memory, because a document can be shown to a bank, a tax authority or an auditor, who is a person that checks records for accuracy.

The idea has limits. Some events have no paper from an outsider: the owner's own investment may be proved only by a bank slip or a signed note. Other entries, which Chapter 7 describes, rest on a calculation made inside the business. A careful bookkeeper therefore asks one question of each source document before recording it: does this prove that an event happened, and does it show the date and the amount? If it does not, the entry waits.

The next figure shows the path that one transaction takes. Read it from left to right. The three boxes are the source document, the journal and the ledger. The arrow between the first two is labelled "recorded", and the arrow between the last two is labelled "posted". Below each box is a short note saying how that record is arranged. The example in the bottom row follows the publisher's invoice of 3 January.

A source document is the first proof that a transaction happened.

Source document. Paper or electronic proof of a transaction, such as an invoice, receipt or cheque stub.

Not every event has outside paper, and the bookkeeper must still act with care.

The figure that follows shows the path of one transaction through the first three steps.

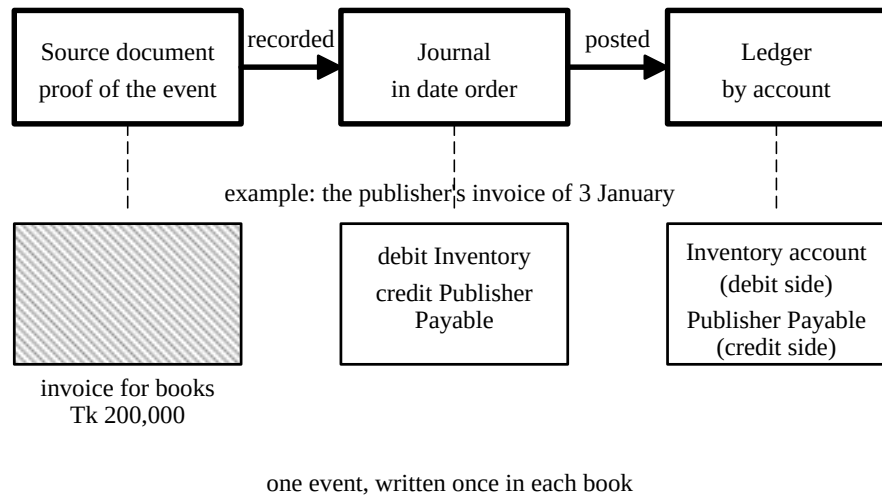


FIG. 5.1. The path from source document to journal to ledger

The ring shows all eight steps and the return to the beginning.

The second figure draws the whole cycle as a ring. Begin at the box marked 1 at the top and follow the arrows clockwise. Each box is one step. After step 8 the ring returns to step 1, which is why it is drawn as a circle and not as a line. The boxes with heavy outlines are the three steps taught in this chapter.

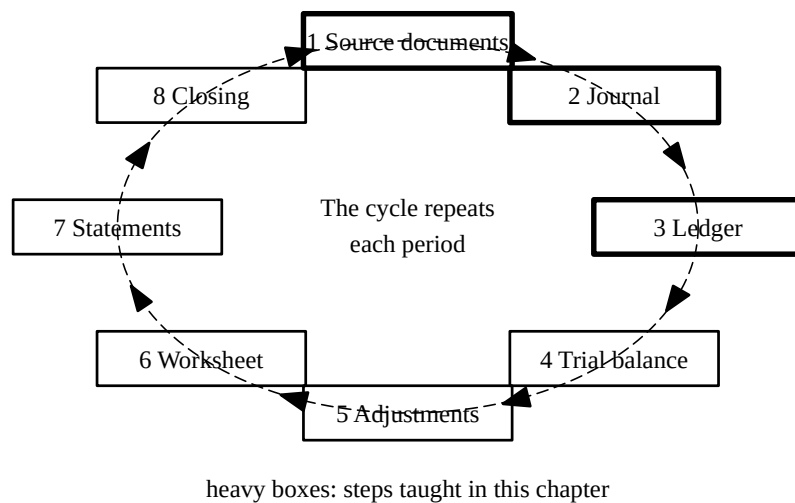


FIG. 5.2. The accounting cycle as a ring of steps

A short review of the cycle and of source documents.

Before moving on. The accounting cycle is the series of steps repeated each period. Steps 1 to 3 move a transaction from a source document to the journal and then to the ledger. A source document is the proof on which the whole record rests.

* * *

5.2 THE JOURNAL

One transaction on 1 January shows what the first book must hold.

On 1 January 2026, Nurul invests Tk 500,000 in cash in the workshop. What must be written down? Two accounts are touched. The account Cash rises, because the workshop now holds more cash. The account Owner's Equity rises, because the owner's share in the workshop has risen by the same

amount. Both facts belong to one event, so both must be written together in one place.

A reminder from Chapter 4 helps here. Each type of account increases on one side and decreases on the other. The table that follows lists the five types met in this chapter. Read a row from left to right: the type, the side on which it increases, and an example from the case. A decrease is always entered on the opposite side.

WHICH SIDE INCREASES EACH TYPE OF ACCOUNT

TYPE OF ACCOUNT	INCREASES WITH	EXAMPLE IN THE CASE
Asset	Debit	Cash
Liability	Credit	Publisher Payable
Owner's equity	Credit	Owner's Equity
Revenue	Credit	Service Revenue
Expense	Debit	Rent

The first book is called the journal. The **journal** is the book of first entry, in which transactions are recorded in date order with debits and credits. It is called the book of first entry because every transaction is written there before it is written anywhere else. The record of one transaction in the journal is a journal entry. A journal may be a bound volume, a ledger-style sheet or a computer file; the rules are the same.¹

A journal entry follows a fixed pattern. The columns are Date, Particulars, Folio, Debit and Credit. The date goes on the first line. The account to be debited is written first, with its amount in the Debit column. The account to be credited is written on the next line, indented a little to the right, with its amount in the Credit column. The Folio column stays empty until the entry is posted, a step that section 5.3 explains. The table that follows shows the entry for 1 January.

THE JOURNAL ENTRY FOR 1 JANUARY 2026

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
1 January 2026	Cash		500,000	
	Owner's Equity			500,000
	Narration: Owner invested cash to begin the workshop.			

The last line of the entry is a narration. A **narration** is a short note under a journal entry explaining the transaction. For example, "Owner invested cash to begin the workshop" tells a later reader why Cash was debited. A good narration names the event, not the accounts, and fits on one or two lines. It matters most to the

Recall: debit and credit, Chapter 4, section 4.3; account types, Chapter 4, section 4.2.

The table reminds the reader which side increases each type of account.

A word with two meanings appears in the name of the first book.

The journal is the book of first entry, kept in date order.

Book. Word in use. In daily life, a set of printed pages that you read. In accounting, a record in which transactions are written, such as the journal or the ledger.

Journal. The book of first entry, in which transactions are recorded in date order with debits and credits.

A journal entry has five columns and a fixed order of lines.

A narration records the reason for the entry in one line.

Narration. A short note under a journal entry explaining the transaction.

¹ Some businesses keep several journals, one for each frequent kind of transaction. Chapter 11, 'Special Journals', teaches them. This chapter uses the single general journal.

reader who was not present: an auditor, a new bookkeeper or the owner a year later.

PAUSE AND CHECK 5.P1 A workshop pays Tk 10,000 in cash for electricity. Which account is debited, which is credited, and what narration would you write?

Every entry must balance: total debits equal total credits.

A compound entry has more than one debit or more than one credit.

Compound entry. A journal entry that has more than one debit or more than one credit.

Every journal entry obeys one rule: the total of the debit amounts equals the total of the credit amounts. In the entry above, Tk 500,000 is debited and Tk 500,000 is credited. If the totals differ, the entry is wrong, and the error must be found before anything is posted. This rule is the safeguard that makes the records dependable.

Most entries touch two accounts, but some touch three or more. A **compound entry** is a journal entry that has more than one debit or more than one credit. Take a different workshop, not part of the case, that buys a binding press for Tk 30,000. It pays Tk 10,000 in cash and owes Tk 20,000 to the dealer. The press is debited for Tk 30,000. Cash is credited for Tk 10,000 and the dealer's payable account is credited for Tk 20,000. The debit total, Tk 30,000, equals the credit total, Tk 30,000, even though three accounts appear.

A COMPOUND ENTRY FOR A DIFFERENT WORKSHOP

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
5 March 2026	Equipment		30,000	
	Cash			10,000
	Equipment Dealer Payable			20,000
	Narration: Bought a binding press, part paid in cash, balance owed.			

A short review of the journal.

Before moving on. The journal is the book of first entry. Each entry has a date, a debit line, a credit line, and a narration. Debits equal credits in every entry. A compound entry has more than one debit or more than one credit, and it still balances.

* * *

5.3 THE LEDGER AND POSTING

The second book gathers every entry that touches one account.

The ledger holds every account, and posting fills it.

Ledger. The book containing all of a business's accounts, where the effects of journal entries are collected.

Posting. Copying the debits and credits from the journal to the right accounts in the ledger.

Return to the shopkeeper's question: how much was spent on rent? The journal lists events by date, so the rent payments lie scattered among all the other entries. A second book is needed, in which all the rent entries sit together on one page. Another page holds all the Cash entries, another all the Owner's Equity entries, and so on. Each page is one account.

The **ledger** is the book containing all of a business's accounts, where the effects of journal entries are collected. The ledger of the workshop will hold eight accounts: Cash, Shelving, Inventory, Publisher Payable, Owner's Equity, Service Revenue, Rent and Wages. The act of filling it is called posting. **Posting** is copying the debits and credits from the journal to the right accounts in the ledger. Posting changes no amount and invents no entry; it only moves what the journal already holds.

The method has four steps, and the same four serve every entry. First, find the account that was debited and enter on its debit side the date, the name of the other account, the journal page, and the amount. Second, find the account that was credited and enter on its credit side the date, the name of the other account, the journal page and the amount. Third, repeat for any further lines of the entry. Fourth, write the ledger page of each account in the Folio column of the journal. The tables that follow show the postings of the 1 January entry, with the journal on page 1 and the accounts on pages L1 and L5.

THE 1 JANUARY POSTING IN THE CASH ACCOUNT (LEDGER PAGE L1)

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
1 January 2026	Owner's Equity	J1	500,000	

THE 1 JANUARY POSTING IN THE OWNER'S EQUITY ACCOUNT (LEDGER PAGE L5)

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
1 January 2026	Cash	J1		500,000

Folio references are small numbers that link the two books. In the ledger, "J1" says that the amount came from journal page 1. In the journal, "L1" written beside Cash says that the amount was posted to ledger page L1. With these links a reader can start from any figure in either book and find the other half of the record in seconds. A folio entry in the journal also shows that the line has been posted, which prevents posting the same line twice.²

Posting leaves each account with many amounts on its two sides. To learn how much the account holds, we find its balance. Balancing an account means totalling both sides and finding the difference. Many students find this step confusing, because the arithmetic is easy but the layout is not. The method is as follows. Total the debit side. Total the credit side. Subtract the smaller total from the larger. The result is the balance, and it belongs to the side with the larger total.

Formula. In words: the balance equals the larger side total minus the smaller side total. In symbols: $B = L - S$, where B is the balance, L is the larger total and S is the smaller total. For a tiny example with made-up figures, suppose an account has debits of Tk 200 and Tk 100, and a credit of Tk 120. Then $L = 300$ on the debit side and $S = 120$, so $B = 300 - 120 = \text{Tk } 180$, a debit balance. **Check.** Add the balance to the smaller side: $120 + 180 = 300$, which equals the larger total, so the two sides now agree.

Before moving on. The ledger collects the journal's entries under the name of each account. Posting copies each debit and each credit to its account and records the folio. A balance is the larger side total less the smaller, placed on the larger side.

* * *

WORKED EXAMPLE: THE JANUARY TRANSACTIONS

Case file, illustrative. Given: Meghna Binding Service, a separate set of books, January 2026. 1 January: owner invests cash Tk 500,000. 2 January: buys

Posting follows four short steps for every journal entry.

The folio columns link each journal line to its ledger account and back.

Folio. Word in use. In daily life, a page number or a large sheet of paper. In accounting, a reference number linking a journal line to a ledger page, or the reverse.

After posting, an account is balanced to show how much it now holds.

The formula for a balance is the larger total less the smaller total.

A short review of the ledger, posting and balancing.

The case file lists seven transactions to be journalised, posted and balanced.

² Accounting software posts automatically, but the logic is the same: each journal line reaches the account it names, once.

shelving for cash Tk 120,000. 3 January: buys book inventory on credit from a publisher Tk 200,000. 10 January: earns Tk 40,000 cash for binding work. 15 January: pays rent Tk 60,000. 20 January: pays Tk 80,000 to the publisher. 31 January: pays wages Tk 35,000.

The task is three jobs: journalise, post, and balance Cash.

The method reuses the debit rules and the balance formula.

Steps 1 and 2 journalise the first four transactions.

Find. The journal entry for each transaction, with narrations; the postings to the ledger accounts Cash, Shelving, Inventory, Publisher Payable, Owner's Equity, Service Revenue, Rent and Wages; and the balance of Cash.

Formula. For each transaction: debit the account that gains (assets, expenses) or loses (liabilities, owner's equity, revenue) value, credit the other, and make the totals equal. For the Cash balance: $B = L - S$, where L is the larger side total and S the smaller.

Steps. Step 1: journalise. The first table covers 1 to 10 January.

JOURNAL ENTRIES FOR 1 TO 10 JANUARY 2026

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
1 January 2026	Cash	L1	500,000	
	Owner's Equity	L5		500,000
	Narration: Owner invested cash to begin the workshop.			
2 January 2026	Shelving	L2	120,000	
	Cash	L1		120,000
	Narration: Bought shelving for cash.			
3 January 2026	Inventory	L3	200,000	
	Publisher Payable	L4		200,000
	Narration: Bought book inventory on credit from a publisher.			
10 January 2026	Cash	L1	40,000	
	Service Revenue	L6		40,000
	Narration: Received cash for binding work.			

The second table covers the last three transactions.

The second table covers 15 to 31 January. In every entry the debit total equals the credit total.

JOURNAL ENTRIES FOR 15 TO 31 JANUARY 2026

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
15 January 2026	Rent	L7	60,000	
	Cash	L1		60,000
	Narration: Paid the month's rent.			
20 January 2026	Publisher Payable	L4	80,000	
	Cash	L1		80,000
	Narration: Paid part of the amount owed to the publisher.			
31 January 2026	Wages	L8	35,000	
	Cash	L1		35,000
	Narration: Paid wages for January.			

Step 2: post. Each of the fourteen lines above is copied to its account. The Cash account receives six postings, and the table that follows shows them with the totals of each side.

Step 3 posts every line, and the Cash account shows six postings.

THE CASH ACCOUNT AFTER POSTING (LEDGER PAGE L1)

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
1 January 2026	Owner's Equity	J1	500,000	
2 January 2026	Shelving	J1		120,000
10 January 2026	Service Revenue	J1	40,000	
15 January 2026	Rent	J1		60,000
20 January 2026	Publisher Payable	J1		80,000
31 January 2026	Wages	J1		35,000
Totals			540,000	295,000

Step 3: balance Cash. Debit total: $500,000 + 40,000 = \text{Tk } 540,000$. Credit total: $120,000 + 60,000 + 80,000 + 35,000 = \text{Tk } 295,000$. The larger total is the debit side, so $B = 540,000 - 295,000 = \text{Tk } 245,000$, a debit balance. The other accounts are balanced in the same way, and the table that follows lists every balance.

Step 4 balances Cash line by line, then the other accounts.

THE BALANCE OF EVERY LEDGER ACCOUNT AT 31 JANUARY 2026

ACCOUNT	DEBIT TOTAL (TK)	CREDIT TOTAL (TK)	BALANCE (TK)
Cash	540,000	295,000	245,000 debit
Shelving	120,000		120,000 debit
Inventory	200,000		200,000 debit
Publisher Payable	80,000	200,000	120,000 credit
Owner's Equity		500,000	500,000 credit
Service Revenue		40,000	40,000 credit
Rent	60,000		60,000 debit
Wages	35,000		35,000 debit

The answer and the check agree with the case file.

Answer. The Cash account closes with a debit balance of Tk 245,000, and the other balances are those of the table above. *Check.* The debit balances total $245,000 + 120,000 + 200,000 + 60,000 + 35,000 = \text{Tk } 660,000$. The credit balances total $120,000 + 500,000 + 40,000 = \text{Tk } 660,000$. The two totals are equal, which is what correct journalising and posting must produce. Chapter 6 shows how this test is used.

Common slips arise from the side, the folio and the date.

Common slips are easy to name. A student may post a debit to the credit side of the account, which reverses the amount. A student may forget to enter the folio, so the line seems unposted and is posted twice. A student may total the wrong side or subtract in the wrong order and obtain a negative balance, a sure sign of a slip. Check the side, the folio and the subtraction, in that order.

* * *

SUMMARY

The chapter's main ideas in a few lines.

A transaction starts with a source document, is recorded in the journal in date order with debits and credits, and is then posted to the ledger, which holds one account for each kind of item. These are steps 1 to 3 of the eight-step accounting cycle, which repeats each period. A narration explains each entry, and a compound entry has more than one debit or credit. A ledger account is balanced by subtracting the smaller side total from the larger.

THINK IT THROUGH

5.T1 A clerk posts the 20 January entry to the Cash account but forgets to post it to Publisher Payable. What balance will Publisher Payable then show, and by how much will the credit balances exceed the debit balances?

WHERE PEOPLE GO WRONG

The journal and the ledger arrange the same facts in different orders.

Many students think the journal and the ledger hold different information. They hold the same transactions. The journal arranges them by date; the ledger arranges them by account.

Posting copies a line; it does not create or change one.

Another error is to alter an amount while posting. Posting only copies. If an amount looks wrong, the correction belongs in the journal, with its own entry and narration.

A third slip is to place a debit on the credit side of the account because the other account's name was written there. The Particulars column names the other account; the amount goes on the side of the account whose page you are on.

A fourth confusion treats the balance as the sum of both sides. It is the difference between them, placed on the larger side.

A debit goes to the debit side of its own account, not to the other account's side.

A balance is a difference, not a total.

IN HISTORY

Francesco Datini was a merchant of Prato, a town near Florence in Tuscany, Italy. He was born around 1335 and died on 16 August 1410. He traded in cloth and many other goods, and kept offices in several countries. After his death, his papers stayed in his house in Prato. In 1870, workers altering the house found them stored in sacks in a disused stairwell.

The find included some 150,000 letters and more than 500 account books and ledgers. Historians still use them to study how merchants of the fourteenth century recorded debts, payments and goods. The survival of so many books, six centuries later, shows the aim of this chapter: a transaction written once in an orderly book can be found again by someone who never met the writer.

A medieval merchant's surviving books show that recorded transactions endure.

The Datini archive shows how long a well-kept record can last.

REVIEW QUESTIONS

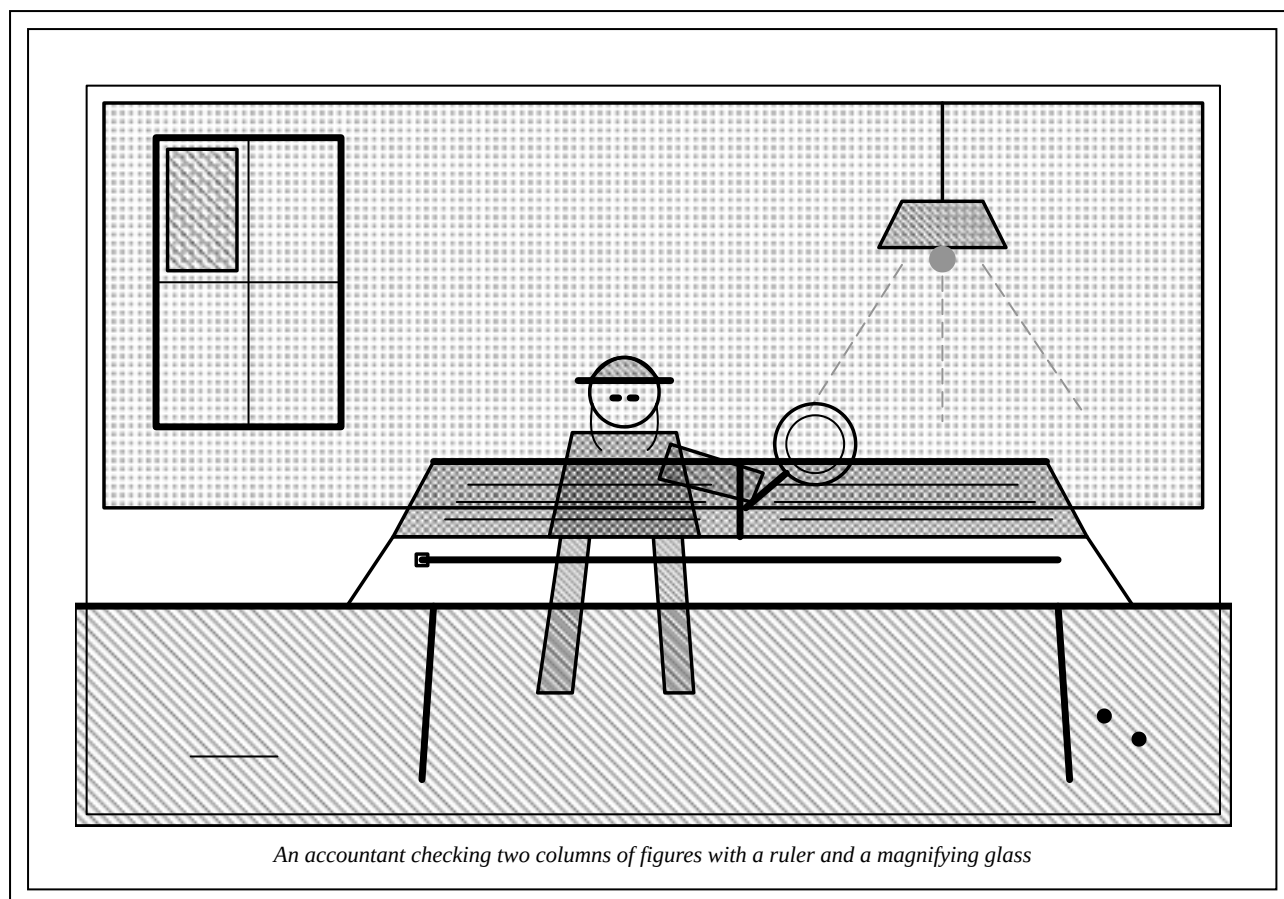
5.1 A workshop owner invests Tk 100,000 in cash. A week later the workshop pays Tk 20,000 in cash for shelving. Journalise both transactions, with narrations.

5.2 Post the two entries of question 5.1 to the Cash account and balance it.

5.3 A Cash account has debits of Tk 125,500 and Tk 48,250 and credits of Tk 30,750 and Tk 61,400. Find its balance and say on which side it falls.

5.4 On 1 February 2026 the Cash account of Meghna Binding Service starts with the balance of Tk 245,000 brought down. During February it receives Tk 52,000 and pays out Tk 118,500. Find the balance at 28 February 2026.

5.5 A different workshop buys a press for Tk 30,000, paying Tk 10,000 in cash and owing the rest to the dealer. Journalise the compound entry, post it to the three accounts, and state the balance of each.



An accountant checking two columns of figures with a ruler and a magnifying glass

CHAPTER VI

The Trial Balance

If every fact is written twice, can the books be tested against themselves?



Every entry in the books has equal debits and credits, so the whole ledger must carry the same equality. This chapter shows how to list the ledger balances, test the equality, and find and fix a difference, using the January 2026 books of Meghna Binding Service.

YOU WILL NEED

Chapter 5 'The Journal and the Ledger', section 5.3 'The Ledger and Posting';
Chapter 5 'The Journal and the Ledger', section 5.1 'The Accounting Cycle';
Appendix A1 'Maths You Will Use'.

A second, independent count tests the first, and the ledger offers such a pair.

After a month of posting, the bookkeeper needs a test that avoids rereading every line.

The ledger should keep the equality of debits and credits that every entry has.

Recall: ledger and posting, Chapter 5, section 5.3.

A tiny example with made-up figures shows the equality surviving the balancing.

Two everyday words take a special meaning in this chapter.

Balance. Word in use. In daily life, to keep steady, or the amount left in a bank account. In accounting, the difference between the two sides of an account.

Trial. Word in use. In daily life, a hearing in a court of law, or a test of something new. In accounting, a test of whether two totals are equal.

A shopkeeper closes for the evening and counts the coins in the till. Then the shopkeeper adds the till roll, the paper strip on which each sale was printed. If the two answers agree, the count is probably right. If they differ, something is wrong, and the shopkeeper knows to search. Neither answer proves anything alone, but two independent answers that agree give real comfort.

The books of a business offer a similar pair of answers. Every transaction was written with debits equal to credits. After a month of journalising and posting, a bookkeeper has hundreds of lines in many accounts. How can the bookkeeper learn, without rereading every line, whether a slip has occurred? And if a test says "wrong", how can the slip be found without starting again? This chapter answers both questions, and it also explains an uncomfortable truth: a test that says "right" does not prove that the books are perfect.

* * *

6.1 PREPARING A TRIAL BALANCE

Recall that every journal entry has debits equal to credits, and that posting copies each debit and each credit to its account. If both jobs are done correctly, one result must follow: the total of all debits in the ledger equals the total of all credits. Let D be the total of all debit amounts posted and C the total of all credit amounts. Because each entry has equal sides, $D = C$. Balancing an account removes the same amount from both of its sides, so the equality survives. The total of the debit balances therefore equals the total of the credit balances.

Here is a tiny example with made-up figures. Entry one debits Cash and credits Owner's Equity for Tk 100. Entry two debits Rent and credits Cash for Tk 30. The debit total D is $100 + 30 = 130$, and the credit total C is $100 + 30 = 130$. After balancing, Cash is a debit of $100 - 30 = 70$, Rent is a debit of 30, and Owner's Equity is a credit of 100. The debit balances total $70 + 30 = 100$, and the credit balance is 100. The equality holds.

Two words need care. In everyday speech, a *balance* may be the amount left in a bank account. In accounting, the balance of an account is the difference between its two sides, as Chapter 5 showed. The word *trial* in everyday speech may mean a hearing in court. Here it means only a test. The test asks one question and gives a plain answer, but it is not a verdict on the books.

The list that carries out the test has a name. A **trial balance** is a list of all ledger account balances, with debits and credits totalled, to test that they are equal. The list has three parts: the name of each account, its balance in the Debit column if the balance is a debit, and its balance in the Credit column if the balance is a credit. At the foot, each column is added. The trial balance is taken at one date, so its heading reads "as at" that date.

The preparation has five steps. First, balance every account in the ledger, using the method of Chapter 5. Second, write the heading: the name of the business, the words "Trial Balance", and the date. Third, list each account with its balance in the correct column. Fourth, add each column. Fifth, compare the two totals. If they are equal, the test is passed. If they differ, the books contain a slip that must be found before the next step of the cycle.

The accounts are usually listed in the order of the ledger: assets first, then liabilities, owner's equity, revenue and expenses. An account with no balance, such as a suspense account that has been cleared, may be left off the list or shown with a dash. The order is a convention that makes lists easy to read and to compare from month to month. It does not affect the test.

The next figure shows the layout of a trial balance. Read it from the top. The first row holds the headings. The rows below hold one account each, with its balance in one of the two columns. A ruled line separates the accounts from the totals, and the totals sit in the last row, where the eye can compare them. A trial balance is step 4 of the accounting cycle.

Trial balance as at 31 January 2026		
Account	Debit	Credit
Cash	245,000	
Shelving	120,000	
Inventory	200,000	
Publisher Payable		120,000
Owner's Equity		500,000
Service Revenue		40,000
Rent	60,000	
Wages	35,000	
Totals	660,000	660,000
equal totals: the test is passed (amounts in taka)		

FIG. 6.1. A trial balance as a two-column list with totals

The January list is an **unadjusted trial balance**: a trial balance taken before the period-end adjustments have been recorded. An adjustment is a change that brings the accounts up to date at the end of a period; Chapter 7, 'Adjusting the Accounts', teaches them. Every trial balance in this chapter is unadjusted, because no adjustment has yet been made.

Case file, illustrative. Given: Meghna Binding Service, trial balance as at 31 January 2026. Debit balances: Cash 245,000; Shelving 120,000; Inventory 200,000; Rent 60,000; Wages 35,000. Credit balances: Publisher Payable 120,000; Owner's Equity 500,000; Service Revenue 40,000. Find: the totals of the two columns, and whether they are equal.

A trial balance lists every ledger balance and compares the two column totals.

Trial balance. A list of all ledger account balances, with debits and credits totalled, to test that they are equal.

Preparing the list takes five steps, always in the same order.

Accounts are listed in a fixed order, and accounts with no balance are left out.

The figure that follows shows the layout, row by row.

Recall: accounting cycle, Chapter 5, section 5.1.

The trial balance in this chapter is taken before any period-end adjustment.

Unadjusted trial balance. A trial balance taken before the period-end adjustments have been recorded.

The January ledger is balanced first, then each balance is listed in its column.

The formula is a plain addition of each column, followed by a comparison.

Formula. In words: the debit total is the sum of all debit balances, the credit total is the sum of all credit balances, and the test is passed when the two are equal. In symbols: DT = sum of debit balances; CT = sum of credit balances; the test is $DT = CT$. **Steps.** $DT = 245,000 + 120,000 + 200,000 + 60,000 + 35,000 = \text{Tk } 660,000$. $CT = 120,000 + 500,000 + 40,000 = \text{Tk } 660,000$. The table that follows sets out the list.

TRIAL BALANCE OF MEGHNA BINDING SERVICE AS AT 31 JANUARY 2026

ACCOUNT	DEBIT (TK)	CREDIT (TK)
Cash	245,000	
Shelving	120,000	
Inventory	200,000	
Publisher Payable		120,000
Owner's Equity		500,000
Service Revenue		40,000
Rent	60,000	
Wages	35,000	
Totals	660,000	660,000

A trial balance is taken often, because later steps depend on it.

The test has a place in the cycle. Steps 5 to 8 of the accounting cycle build on the ledger balances, so an undetected slip at this point would travel into every later report. Taking the trial balance at the end of each month, or even each week, keeps the search for a slip short. In a small bookshop with few transactions, one list a month is enough; a busy business may take one daily.

The answer and the check show a passed test that still proves less than it seems.

Answer. Both columns total Tk 660,000, so the test is passed. **Check.** A second sum in a different order helps: debits, $35,000 + 60,000 + 200,000 + 120,000 + 245,000 = 660,000$. The same total appears. From the owner's viewpoint, the result is reassuring. From the viewpoint of a bank, it is a first sign of care, though not a proof that every figure is right, as section 6.2 explains.

Common slips in preparing the list are easy to avoid with a tick for each account.

Common slips are few. A balance may be written in the wrong column. An account that has a balance may be left out. A digit may be added wrongly. Tick each ledger account as it is listed, and compare the number of accounts on the list with the number in the ledger.

Different readers use the trial balance in different ways.

The trial balance serves several people inside the business. The bookkeeper uses it to learn whether posting went well. The owner, Nurul, uses it as a quick picture of what the accounts hold, though it is not a formal report. An employee who prepares the later steps of the cycle uses it as the starting list, because adjustments and statements are built from its figures. A bank usually asks for the formal reports rather than for this working paper.

PAUSE AND CHECK 6.P1 Explain in your own words why the debit total of a trial balance must equal the credit total when every journal entry and every posting is correct.

A short review of the trial balance and its preparation.

Before moving on. A trial balance lists all ledger balances in two columns and tests that the totals are equal. The test works because every entry has equal debits and credits. The five steps are: balance, head, list, add, compare.

* * *

6.2 WHAT THE TRIAL BALANCE DOES NOT SHOW

Suppose that the rent payment of 15 January had never been written in the journal. Rent would show nothing, and Cash would show Tk 305,000 instead of Tk 245,000. The debit total would be $305,000 + 120,000 + 200,000 + 35,000 =$ Tk 660,000. The credit total would still be Tk 660,000. The test is passed, yet the books are wrong. This example is the central lesson of the section: the test checks equality, not truth.

A framework helps. Divide errors into two groups. The first group changes only one side of the books, so the totals differ and the trial balance finds the error. Examples are a debit posted without its credit, an amount posted wrongly on one side only, a column added wrongly, and a balance listed in the wrong column. The second group changes both sides by the same amount, so the totals stay equal and the test says nothing.

The table that follows lists six kinds of error. Read a row from left to right: the kind of error, whether the totals differ, and an example from the January case. The example of the wrong account is a payment of wages debited to Rent. Rent would show Tk 95,000 and Wages nothing, but the debit total would remain Tk 660,000.

SIX KINDS OF ERROR AND THE TRIAL BALANCE

KIND OF ERROR	TOTALS DIFFER?	EXAMPLE FROM THE JANUARY CASE
One side of an entry not posted	Yes	The Rent debit of 60,000 is never posted; the debit total becomes 600,000
Wrong amount on one side only	Yes	Wages posted as 53,000 on the debit side only
Whole transaction left out	No	The 15 January rent entry is never recorded
Same wrong amount on both sides	No	Wages and Cash both posted as 53,000
Wrong account of the same type	No	Wages debited to Rent
Two errors that cancel	No	A debit too high by 10,000 and another too low by 10,000

The wages example in the table is a special slip. A **transposition error** is a mistake in which two digits of a number are written in the wrong order, such as 53 for 35. It is common because the eye reads quickly and the hand writes quickly. Posted on one side only, it makes the totals differ by a number that always divides by 9. The reason is short. A two-digit number with digits a and b equals $10a + b$. With the digits reversed it equals $10b + a$. The difference is $9a - 9b = 9(a - b)$, which is a multiple of 9. For 53 and 35, the difference is $18 = 9 \times 2$.

The property has a limit. A difference that divides by 9 suggests a transposition, but other slips can also give such a difference, so the sign is a lead and not a proof. The rule also holds when the digits stand in thousands, as in 53,000 for 35,000, where the difference is 18,000. Always confirm by looking at the source document.

The last row of the table needs a word of explanation. Suppose one debit is posted Tk 10,000 too high, and another debit, in a different entry, is posted Tk 10,000 too low. The debit total rises by 10,000 and falls by 10,000, so it ends

A balanced list can still hide an error, as one missing entry shows.

The test finds errors that put the totals out of line.

A table sets six kinds of error beside their effect on the totals.

One common slip in copying digits has a name and a useful property.

Transposition error. A mistake in which two digits of a number are written in the wrong order, such as 53 for 35.

A difference divisible by 9 suggests a transposition but does not prove one.

Two errors can hide each other, and so equal totals cannot rule out several errors.

A balanced list is a starting point for the auditor and a warning for the owner.

A short review of what the test finds and what it misses.

A difference of Tk 18,000 shows where a search for a slip begins.

The search moves from the cheapest check to the most costly.

Two further differences show how the second and third checks work.

Careful habits reduce the search.

Sometimes the search cannot finish before the reports are due.

Suspense account. A temporary account that holds a difference until its cause is found.

where it began. Such luck is rare, but it shows that equal totals cannot rule out several errors.

An auditor who reads a balanced trial balance treats it as a starting point and not as an end. The auditor tests source documents against the journal to find the errors that the totals cannot show. An owner who sees equal totals may feel safe, but a missing sale or a payment posted to the wrong account still spoils the reports. A careful bookkeeper therefore combines the test with other checks, such as confirming that every source document has been recorded.

Before moving on. The trial balance finds errors that change one side only. It cannot find a missing transaction, an equal error on both sides, a wrong account of the same type, or errors that cancel. A transposition error on one side gives a difference divisible by 9.

* * *

6.3 FINDING AND FIXING DIFFERENCES

Suppose the wages of 31 January were posted to the debit side of the Wages account as Tk 53,000, while the credit to Cash was posted correctly as Tk 35,000. The trial balance now shows a debit total of Tk 678,000 and a credit total of Tk 660,000. The difference is Tk 18,000. The bookkeeper holds one number and faces a long ledger. Where should the search begin?

Locating a difference follows a fixed order, from the cheapest check to the most costly. First, add both columns again, in a different order. Second, test whether the difference is divisible by 9, which points to a transposition error. Third, test whether the difference equals one account balance, which points to a posting that was missed, and whether half of the difference equals a balance, which points to a balance listed in the wrong column. Fourth, compare every posting with the journal and tick each line that agrees. Fifth, recalculate each balance. Stop as soon as the cause is found.

The checks have examples in the January ledger. Suppose the Rent debit of Tk 60,000 was never posted. The debit total would be Tk 600,000 against a credit total of Tk 660,000. The difference of Tk 60,000 equals one amount in the journal, so the bookkeeper would look for a missed posting of that size. Suppose instead that Service Revenue, a credit balance of Tk 40,000, were listed in the Debit column. The debit total would become Tk 700,000 and the credit total Tk 620,000. The difference of Tk 80,000 is twice Tk 40,000, so half of it points to a balance in the wrong column.

The best search is the one that is never needed. Careful bookkeepers check each entry before posting, tick each journal line when it has been posted, and take a trial balance often, for example at the end of each week. A difference found among a week of entries is far easier to trace than a difference found among a year of them.

When time runs short, the bookkeeper may open a suspense account. A **suspense account** is a temporary account that holds a difference until its cause is found. If the credits fall short, the difference is entered as a credit balance in the suspense account. If the debits fall short, it is entered as a debit balance. The two column totals then agree. This is an honest device only if it stays temporary: the account must be cleared as soon as the cause is found, and it must never remain in the final reports.

When the cause is found, the books are corrected by a new journal entry. The entry carries a narration and is posted like any other. The wrong figure is never rubbed out, because the books must stay readable to an auditor, who is a person that checks records for accuracy. A correcting entry usually has two parts. It removes the wrong amount from the account in which it was posted, and it clears the suspense account.

Before moving on. A difference is searched for in a fixed order: re-add, test for divisibility by 9, compare with one balance or half a balance, tick the postings, recalculate. A suspense account holds the difference for a short time. A correcting entry, never an erasure, fixes the error.

* * *

WORKED EXAMPLE: THE WAGES DIFFERENCE

Case file, illustrative. Given: the January 2026 ledger of Meghna Binding Service is correct except that the wages of 31 January were posted to the Wages account as Tk 53,000 instead of Tk 35,000. The credit to Cash was posted correctly. The correct trial balance totals Tk 660,000 on each side.

Find. The trial balance totals with the error, the difference, its probable cause, and the entries that clear it.

Formula. In words: the difference is the larger column total minus the smaller. In symbols: $d = DT - CT$, where DT is the debit total and CT is the credit total. A transposition of two digits gives $d = 9(a - b)$, where a and b are the two digits, in the units of their place.

Steps. Step 1: add the columns. The wrong debit balance for Wages is Tk 53,000, so $DT = 245,000 + 120,000 + 200,000 + 60,000 + 53,000 = \text{Tk } 678,000$. $CT = 120,000 + 500,000 + 40,000 = \text{Tk } 660,000$. Step 2: subtract. $d = 678,000 - 660,000 = \text{Tk } 18,000$, with the debits larger.

Step 3: test for 9. $18,000 \div 9 = 2,000$, so the difference divides by 9. Step 4: look for the digits. Two numbers that differ by 18 in the thousands are 53 and 35, which are a transposition of each other. The payroll record, which is the source document, shows Tk 35,000. The cause is therefore a transposition error in the debit posting to Wages.

Step 5: hold the difference. The credits are short by Tk 18,000, so a credit balance of that amount is entered in a suspense account. The table that follows shows the list.

A found error is corrected by a new journal entry and never by erasing.

A short review of the search, the suspense account and the correcting entry.

The case file gives an error to find and to fix.

The task is to find the difference, its cause and the correction.

The two formulas are a subtraction and a divisibility test.

Steps 1 and 2 find the totals and the difference.

Steps 3 and 4 test the difference and confirm the cause.

Step 5 shows the trial balance with the difference held in suspense.

TRIAL BALANCE AS AT 31 JANUARY 2026 WITH THE ERROR AND A SUSPENSE ACCOUNT

ACCOUNT	DEBIT (TK)	CREDIT (TK)
Cash	245,000	
Shelving	120,000	
Inventory	200,000	
Rent	60,000	
Wages	53,000	
Publisher Payable		120,000
Owner's Equity		500,000
Service Revenue		40,000
Suspense		18,000
Totals	678,000	678,000

Step 6 corrects the books with one journal entry.

Step 6: correct. Wages is too high by Tk 18,000, so Wages is credited by that amount. The suspense account is cleared by a debit of the same amount. The table that follows shows the journal entry.

THE CORRECTING ENTRY

DATE	PARTICULARS	FOLIO	DEBIT (TK)	CREDIT (TK)
31 January 2026	Suspense		18,000	
	Wages			18,000
	Narration: Corrected wages posted as 53,000 instead of 35,000, and cleared the suspense account.			

The answer and the check agree with the case file.

Answer. After posting, Wages = 53,000 – 18,000 = Tk 35,000, and the suspense account = 18,000 – 18,000 = nil. The totals of the correct trial balance are Tk 660,000 on each side. **Check.** Wages now equals the Tk 35,000 of the source document. The suspense account has no balance. The debit total and the credit total both equal the Tk 660,000 of section 6.1.

Common slips arise when the sign, the column or the unit is mixed up.

Common slips occur at this stage. A student may place the suspense balance on the wrong side, which doubles the difference instead of removing it. A student may correct the error by overwriting the figure in the Wages account, which hides the history. A student may forget to clear the suspense account, so that it stays in the books. Check the side, the entry and the final balance of Suspense, in that order.

* * *

SUMMARY

The chapter's main ideas in a few lines.

A trial balance lists all ledger balances in two columns and tests that the debit total equals the credit total. It is step 4 of the accounting cycle. Taken before adjustments, it is an unadjusted trial balance. The test finds errors that change one side only, and it cannot find omissions or equal errors on both sides. A transposition error gives a difference divisible by 9. A difference

may be held in a suspense account, and a correcting entry then removes the error.

THINK IT THROUGH

6.T1 Name two errors that leave a trial balance in balance. For each, give an example from the January ledger of Meghna Binding Service and say what would still be wrong.

WHERE PEOPLE GO WRONG

Many students believe that a balanced trial balance proves that the books are correct. It proves only that the debit balances and the credit balances are equal. Missing entries and equal errors on both sides remain hidden.

A second error is to assume that the difference equals the error. A transposition error of 53 for 35 makes an error of 18, which equals the difference only because the credit was right. A balance in the wrong column makes the difference twice the balance.

A third error is to leave the difference in the suspense account and carry on. The account is temporary. Its balance must be explained and cleared before reports are prepared.

A fourth confusion is to treat the trial balance as the balance sheet. The trial balance is a working list for testing the books. The balance sheet is a formal report taught in Chapter 9, 'Financial Statements and Closing Entries'.

A balanced trial balance does not prove the books are right.

The difference is not always the size of the error.

The suspense account is not a place to leave a problem.

A trial balance is a test and not a financial statement.

IN HISTORY

Simon Stevin was born in Bruges, in Flanders, in 1548 and died in 1620. He was a mathematician and engineer who served as tutor and adviser to Prince Maurice of Nassau, the military leader of the Dutch Republic. Stevin wrote on many subjects, among them bookkeeping. His book on the subject, usually called *Princely Bookkeeping*, was printed in Leiden in the first decade of the seventeenth century, with a French edition in 1608.

Stevin urged that the household and finances of a prince be recorded by the double-entry method that merchants already used, and he put the method into practice for Prince Maurice. The idea behind a trial balance is the same idea: the books of a business or a state should be tested against themselves at regular times, so that slips are found by method and not by chance.

A Flemish mathematician argued that a ruler's accounts also need method.

Stevin wanted a state to keep its books by the method of merchants.

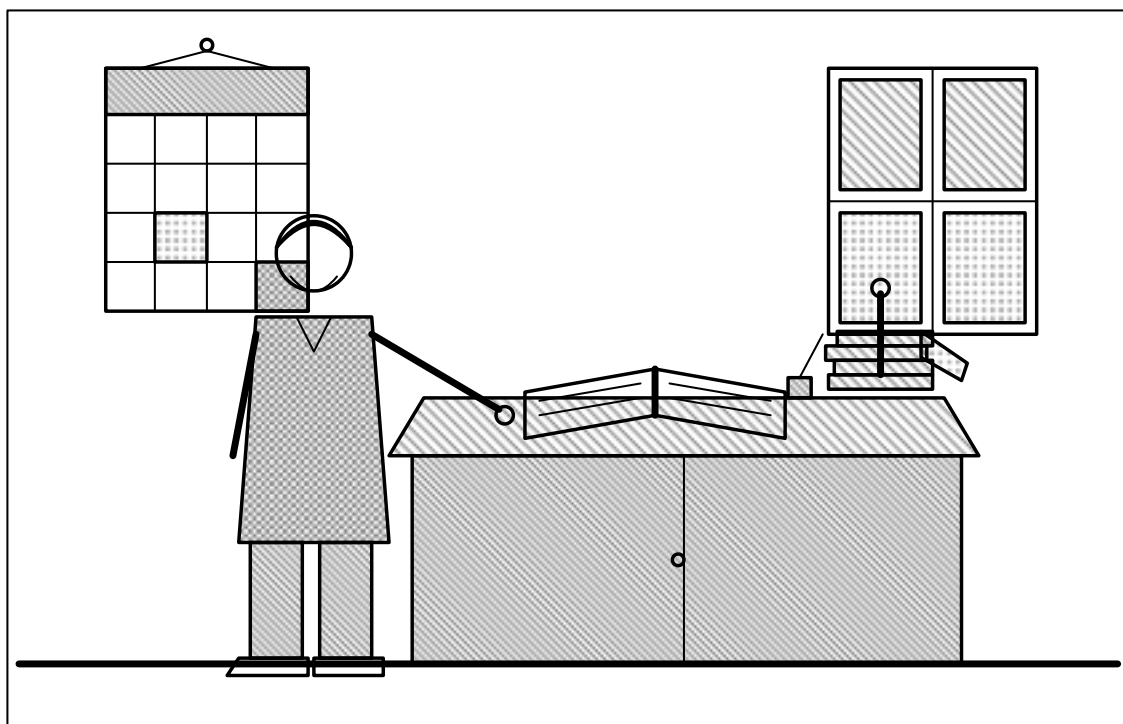
REVIEW QUESTIONS

6.1 Prepare a trial balance as at 31 March 2026 from these balances: Cash Tk 50,000 debit; Equipment Tk 30,000 debit; Owner's Equity Tk 80,000 credit. State whether the totals are equal.

6.2 Prepare a trial balance from these balances and state whether it is balanced: Cash 187,450 debit; Inventory 96,300 debit; Equipment 150,000 debit; Rent 42,000 debit; Wages 78,500 debit; Publisher Payable 64,250 credit; Owner's Equity 400,000 credit; Service Revenue 90,000 credit.

6.3 A trial balance shows a debit total of Tk 458,900 and a credit total of Tk 465,500. Find the difference. Is it divisible by 9? What does half of it suggest?

- 6.4 The credit total of a trial balance is Tk 7,200 larger than the debit total. State the suspense account balance that would be added and on which side.
- 6.5 Rent of Tk 18,000 was posted to the debit side of the Rent account as Tk 81,000, and the credit to Cash was correct. Find the difference, say on which side the suspense balance goes, and write the correcting entry with its narration.



A clerk at year-end with a wall calendar, an open ledger and a pile of unpaid bills

CHAPTER VII

Adjusting the Accounts

If money moves on one day and the benefit arrives on another, which day counts?



Records made as money moves do not always show what a period really earned and used. This chapter explains why accounts need adjustment at period end, teaches the two families of adjustment, deferrals and accruals, and shows how each is entered, using the year-end records of a small binding workshop.

YOU WILL NEED

Chapter 6, 'The Trial Balance', section 6.1 'Preparing a Trial Balance'; Chapter 4, 'Accounts, Debits and Credits', section 4.3 'Debit and Credit'; and Appendix A1, 'Maths You Will Use', for percentages and simple division.

Paying for a year of use on one day raises the question of how much belongs to each month.

A business meets the same puzzle, and its records must choose one answer.

The records are true as a diary of events, but incomplete as a measure of the period.

Reports cover a fixed span of time, which must be named before anything can be measured.

GO SLOWLY HERE.

Period. Word in use. In daily life, a lesson at school or the full stop at the end of a sentence. In accounting, a span of time such as a month or a year.

Accounting period. The span of time, such as a month or year, for which results are measured and reported.

Imagine that you pay Tk 12,000 on 1 October for a bus pass that lasts twelve months. By 31 December you have used it for three months. A friend asks: "How much did the pass cost you this year?" Your wallet says Tk 12,000, because that is the amount that left it. Your journeys say Tk 3,000, because that is the amount you have used. Both answers are honest. They answer two different questions.

Now give the puzzle to a business. On 1 October 2026 the book-binding workshop Meghna Binding Service pays Tk 36,000 for twelve months of insurance. On 31 December 2026 its records still show the whole Tk 36,000 as paid. In the same records nothing appears for the wages of the last days of December, which the workers have earned but not yet received. The first item makes the year look dearer than it was. The second makes it look cheaper. A lender who read such records would form a wrong picture in both directions.

The records are not false. Each payment did happen, and each was written down correctly on its day. They are incomplete as a measure of one period of time. Closing that gap is the work of this chapter: a small set of corrections, made once at the end of each period, each needing only a few lines of arithmetic. Many students find this chapter difficult at first, because a ledger that balances looks finished. It is not finished, and you will soon see why.

* * *

7.1 WHY ADJUST

A school reports on your work term by term, not once for your whole life. A business reports in the same way. An **accounting period** is the span of time, such as a month or year, for which results are measured and reported. Meghna Binding Service uses the calendar year 2026, from 1 January to 31 December. Some businesses also report every month or every quarter. The shorter the period, the more often the records need correction, because more items fall across the edge of the period.¹

¹ Reporting periods of different lengths are taught in Chapter 20, 'The Whole Cycle and Reporting Periods', section 20.1 'Fiscal Year and Interim Periods'.

There are two ways to decide which period an item belongs to. The first follows the cash: record an item when money moves. The second follows the activity: record an item when the work is done or the benefit is received. The second way is the **accrual basis of accounting**: recording revenue when earned and expenses when incurred, whether or not cash has moved. An expense is incurred when the cost is used up, not when it is paid. A worker who labours on 28 December 2026 has earned wages in December, even if the pay arrives on 5 January 2027. On the accrual basis those wages belong to 2026. From here onward this book uses the accrual basis.

Think about fairness. If the workshop earns fees all year, the report for the year should carry the cost of the room, the labour and the insurance that made those fees possible. The **matching principle** is the rule that expenses are recorded in the same period as the revenue they helped to earn. Insurance for the months from October to December belongs beside the revenue of those months. Insurance for the following nine months belongs beside the revenue of 2027. The accrual basis says when to count an item; matching says where to place its cost.

PAUSE AND CHECK 7.P1 A workshop pays Tk 24,000 on 1 November for twelve months of rent. How much of this cost belongs to a year ending on 31 December, and which principle decides it?

The ordinary entries of the year, taught in Chapter 5, are made when something happens: cash is paid, a bill arrives, a customer is billed. Two problems follow. Some items are recorded too early or too late for the period: a payment may cover many months, and a fee may arrive before the work. Other items are never recorded, because no event pushed anyone to write them: wages that are earned but unpaid, and work that is done but not yet billed. So at the end of each period the accounts must be checked and corrected. The corrections fall into two families. In a deferral, cash moves first and the effect is put off to later. In an accrual, the effect comes first and the cash moves later. Chapter 8, 'The Worksheet', and Chapter 9, 'Financial Statements and Closing Entries', use the corrected figures.

The table below is a trial balance.² It was taken on 31 December 2026 before any correction. Read each row as an account name and its balance. The debit total and the credit total both equal Tk 1,240,000.

Revenue and expenses belong to the period of the activity, not the period of the cash.

GO SLOWLY HERE.

Accrual. Word in use. In daily life, a gradual build-up, such as interest growing in a savings account. In accounting, an amount earned or used up that cash has not yet settled.

Accrual basis of accounting. Recording revenue when earned and expenses when incurred, whether or not cash has moved.

Costs are placed beside the revenue they helped to earn.

Matching principle. The rule that expenses are recorded in the same period as the revenue they helped to earn.

At period end some items are wrongly timed and others are missing.

GO SLOWLY HERE.

The year-end figures of the workshop show what needs correction.

² Recall: trial balance, Chapter 6, 'The Trial Balance', section 6.1 'Preparing a Trial Balance'.

UNADJUSTED TRIAL BALANCE, MEGHNA BINDING SERVICE, 31 DECEMBER 2026 (TK)

ACCOUNT	DEBIT	CREDIT
Cash	150,000	
Accounts Receivable	90,000	
Supplies	48,000	
Prepaid Insurance	36,000	
Equipment	360,000	
Drawings	80,000	
Wages	330,000	
Rent	120,000	
Utilities	26,000	
Accounts Payable		60,000
Unearned Revenue		24,000
Capital		400,000
Service Revenue		756,000
Total	1,240,000	1,240,000

The uncorrected figures give a profit that is too high by Tk 96,000.

Different readers of the records are harmed in different ways by a wrong profit.

The accrual basis is the basis used here, though other bases exist.

The section's four ideas are the period, the accrual basis, matching, and the need to correct.

In a deferral, cash moves first and the effect arrives in later periods.

Case file, illustrative. Meghna Binding Service, 31 December 2026. Before correction, revenue is Tk 756,000 and the expenses are wages Tk 330,000, rent Tk 120,000 and utilities Tk 26,000, a total of Tk 476,000. The profit, meaning revenue minus expenses, appears to be Tk 280,000. After the six corrections of this chapter, revenue is Tk 788,000 and expenses are Tk 604,000, so profit is Tk 184,000. Chapter 8, 'The Worksheet', section 8.3 'Worked Worksheet', calls this amount net income.

Each reader of the records would be misled by the first figure. The owner might withdraw money in the belief that Tk 280,000 was earned. A lender might judge the business safer than it is. A tax authority would assess tax on the wrong amount. An employee might read the high profit as a reason to expect higher pay. Correct timing protects all of these people.

The accrual basis is not the only method in use. Some very small businesses, in some countries, keep records on the cash basis and may be allowed to report tax that way. Rules differ from country to country, and this book describes the accrual basis because it is the basis behind the statements studied here. Even on the accrual basis, judgement is needed: a very small amount may be left uncorrected if leaving it would mislead nobody.

Before moving on. An accounting period is the span of time being reported. The accrual basis places revenue and expenses in the period of the activity, and matching places each cost beside the revenue it helped to earn. Because ordinary entries follow cash and events, corrections are needed at period end. You should now be able to say why a ledger that balances can still give the wrong profit.

7.2 DEFERRALS

Return to the bus pass. The money left your wallet on 1 October, but the benefit arrives month by month. An accountant calls the correction for such a case a deferral, because part of the effect is put off to later periods. This section covers three deferrals of the workshop and one related correction for equipment. Each follows one pattern: find how much of the cash has already been

used up, move that part into an expense or into revenue, and leave the rest where it is.

On 1 October 2026 the workshop pays Tk 36,000 for twelve months of insurance. On that day nothing has been used up, and the whole payment is an asset: a right to be protected for a year. A **prepaid expense** is a cost paid in advance that is an asset until it is used up. Insurance, rent and subscriptions are common examples. Each day, a small part is used up and becomes an expense. At the end of the period, the entry moves the used part from the asset to an expense account.

A cost paid in advance is an asset until it is used up.

GO SLOWLY HERE.

Prepaid expense. A cost paid in advance that is an asset until it is used up.

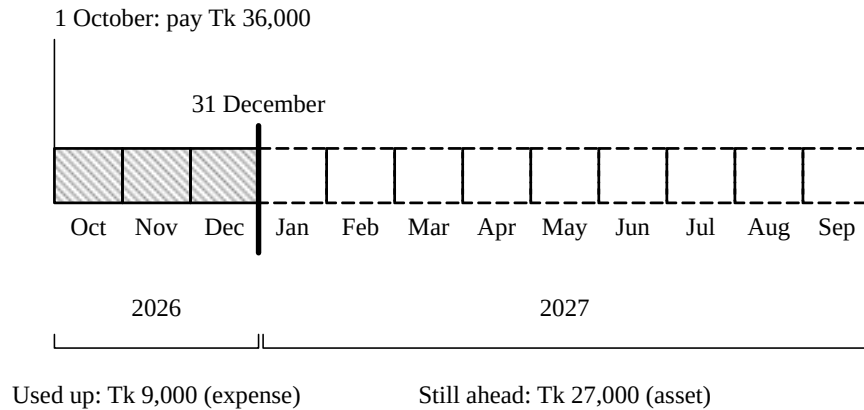


FIG. 7.1. Insurance paid on 1 October 2026 for twelve months: the part used by 31 December and the part still to come

To read the figure, follow the bar from left to right. Each box is one month. The hatched boxes, October to December 2026, are used up in the period. The dashed boxes, January to September 2027, are still ahead. The vertical line marks the end of the period. Each box stands for Tk 3,000.

The bar shows twelve months of cover, split at 31 December.

Given. Insurance of Tk 36,000 paid on 1 October 2026 for twelve months; the period ends on 31 December 2026. *Find.* The insurance expense for 2026 and the prepaid expense that remains. *Formula.* In words: monthly cost equals the amount paid divided by the months covered; expense equals monthly cost times months used; remainder equals amount paid minus expense. In symbols: $m = P \div n$; $X = m \times k$; $R = P - X$, where P is the amount paid, n the months covered and k the months used. *Steps.* $m = 36,000 \div 12 = 3,000$. The months used are October, November and December, so $k = 3$. $X = 3,000 \times 3 = 9,000$. $R = 36,000 - 9,000 = 27,000$. *Entry:* debit Insurance Expense 9,000; credit Prepaid Insurance 9,000. *Answer.* The insurance expense for 2026 is Tk 9,000, and Tk 27,000 remains as a prepaid expense. *Check.* $9,000 + 27,000 = 36,000$. Also, $27,000 \div 3,000 = 9$, the nine months still ahead.

Worked example: the insurance adjustment, step by step.

Some assets are used up without a clear daily rate. The workshop buys paper, thread and glue during the year, and records the cost in an asset account called Supplies. These are items bought to be used in the work, not goods held for sale. Nobody records each sheet used. So at year end the staff count what remains, and the difference between what was bought and what remains is the amount used. This is a prepaid cost, too, but the count, not a monthly rate, fixes the amount.

Supplies are counted at the end, and the difference is the amount used.

Worked example: supplies used during 2026.

Money received in advance is a debt of work until the work is done.

Unearned revenue. Money received before work is done, held as a liability until the work is performed.

Worked example: the part of the advance that has been earned.

Equipment is paid for once but serves for years, so its cost is spread.

Worked example: one year of depreciation on the equipment.

Four corrections have been made; the pattern is to find the used part and move it.

In an accrual the effect comes first and the cash moves later.

Given. The Supplies account shows Tk 48,000 before adjustment; a count on 31 December 2026 values the supplies on hand at Tk 15,000. *Find.* The supplies used during 2026. *Formula.* In words: supplies used equal the supplies account balance minus supplies on hand. *Steps.* $48,000 - 15,000 = 33,000$. Entry: debit Supplies Expense 33,000; credit Supplies 33,000. *Answer.* Supplies used in 2026 are Tk 33,000, and the Supplies account is left at Tk 15,000. *Check.* $33,000 + 15,000 = 48,000$, the original balance.

Now turn the idea around. Suppose a school pays a magazine publisher for a year of issues in advance. The publisher has the money, but it still owes twelve issues. The money is not yet earned. For the workshop, **unearned revenue** is money received before work is done, held as a liability until the work is performed. On 1 October 2026 a school pays Tk 24,000 in advance for binding work; the trial balance shows this as a liability of Tk 24,000. The work is done evenly at Tk 4,000 a month from 1 October. By 31 December three months of the work are complete, so part of the liability has become revenue.

Given. Unearned Revenue of Tk 24,000; work worth Tk 4,000 a month, started on 1 October 2026; period ends on 31 December 2026. *Find.* The revenue earned in 2026 and the liability that remains. *Formula.* In words: revenue earned equals monthly work times months done; liability remaining equals the advance minus revenue earned. *Steps.* $4,000 \times 3 = 12,000$. $24,000 - 12,000 = 12,000$. Entry: debit Unearned Revenue 12,000; credit Service Revenue 12,000. *Answer.* Tk 12,000 is earned in 2026, and Tk 12,000 stays as unearned revenue. *Check.* The whole advance covers $24,000 \div 4,000 = 6$ months, and 3 months are done, so half remains.

Equipment follows the same spirit. The workshop paid Tk 360,000 for equipment that will serve for about five years. It would be unfair to charge the whole cost to the first year. The method that spreads the cost is called depreciation, taught in Chapter 16, 'Depreciation', section 16.1 'Why Depreciate'. For now, use the simplest rule: divide the cost, less any amount expected at the end, by the years of service. The charge is recorded by debiting Depreciation Expense and crediting an account called Accumulated Depreciation, explained in Chapter 16, section 16.4 'Recording and Disposal'.

Given. Equipment cost Tk 360,000; life 5 years; no residual value, which means nothing is expected to be received at the end. *Find.* The depreciation for 2026. *Formula.* In words: yearly charge equals cost divided by years of life. In symbols: $D = C \div N$. *Steps.* $D = 360,000 \div 5 = 72,000$. Entry: debit Depreciation Expense 72,000; credit Accumulated Depreciation 72,000. *Answer.* The depreciation for 2026 is Tk 72,000. *Check.* $72,000 \times 5 = 360,000$, the whole cost after five years.

Before moving on. A prepaid expense is an asset until used, and the used part moves to an expense. Supplies are counted, and the difference is the amount used. Unearned revenue is a liability until the work is done, and the earned part moves to revenue. Depreciation spreads the cost of equipment over its life. You should now be able to say why cash may leave on one day while the cost belongs to several periods.

7.3 ACCRUALS

You use your telephone in December. The bill arrives in January, and you pay it in January. Yet the calls were made in December, and a fair record of December would include them. An accrual is the opposite of a deferral: the activity has

happened, but the cash has not yet moved, and nothing has yet been written down. Two accruals concern the workshop, one for a cost and one for revenue.

The workshop pays its workers on the fifth day of each month. On 31 December 2026 the workers have worked for several days since their last pay, and Tk 14,000 is owed to them. Nothing has been written down, because no money has moved. An **accrued expense** is a cost already used up but not yet paid, shown as a liability. The workers' labour has been used in earning December's revenue, so the matching principle puts its cost in December. The liability is real: the workshop owes the workers Tk 14,000.

Given. The Wages account shows Tk 330,000 before adjustment; wages of Tk 14,000 have been earned by workers but not paid. *Find.* The wages expense for 2026 and the liability at 31 December. *Formula.* In words: wages expense equals wages recorded plus wages owed. *Steps.* $330,000 + 14,000 = 344,000$. *Entry:* debit Wages 14,000; credit Wages Payable 14,000. *Answer.* The wages expense for 2026 is Tk 344,000, and the workshop owes Tk 14,000 in Wages Payable. *Check.* The new liability, Tk 14,000, equals the increase in the expense, so the equation still balances.

Now consider the other side. A freelance designer finishes a job in the last week of December, but sends the invoice in January. The work is done, so the revenue belongs to December. In the workshop's case, binding work worth Tk 20,000 was finished for a school before 31 December, and the invoice has not yet been prepared. **Accrued revenue** is revenue already earned but not yet received or billed, shown as an asset. The asset is the right to be paid. An amount owed to the business by a customer is called an account receivable; some books say debtor. This book says account receivable.

Given. Binding work worth Tk 20,000 is complete on 31 December 2026 but not billed. Service Revenue stands at Tk 756,000 before this item and before the item for unearned revenue. *Find.* The entry for the unbilled work and its effect on Accounts Receivable. *Formula.* In words: revenue earned equals the value of work done; the customer's debt is added to Accounts Receivable. *Steps.* *Entry:* debit Accounts Receivable 20,000; credit Service Revenue 20,000. Accounts Receivable becomes $90,000 + 20,000 = 110,000$. *Answer.* Revenue rises by Tk 20,000, and Accounts Receivable becomes Tk 110,000. *Check.* Assets rise by 20,000 and owner's equity rises by 20,000 through revenue, so the equation balances.

PAUSE AND CHECK 7.P2 At year end a workshop owes its workers Tk 6,500 and has finished work worth Tk 11,000 that it has not yet billed. State the two amounts, and say which is an asset and which is a liability.

Before moving on. An accrued expense is a cost used up but unpaid, and it creates a liability. Accrued revenue is revenue earned but unbilled, and it creates an asset. In both, the cash moves later. You should now be able to say how an accrual differs from a deferral.

A cost already used up but not yet paid is recorded as a debt.

GO SLOWLY HERE.

Accrued expense. A cost already used up but not yet paid, shown as a liability.

Worked example: wages owed at the end of the year.

Revenue already earned but not yet billed is recorded as an asset.

Accrued revenue. Revenue already earned but not yet received or billed, shown as an asset.

Worked example: work done but not yet billed.

Both accruals record an effect before any cash moves.

7.4 RECORDING ADJUSTING ENTRIES

Before listing the entries, hold the whole plan in one picture. The figure arranges the corrections in a square. The columns show whether cash moved before the effect (a deferral) or after it (an accrual). The rows show whether the item is on the cost side or the revenue side.

One picture holds the four kinds of correction, arranged by what moved first.

	Cash moves first (deferral)	Cash moves later (accrual)
Cost side	Prepaid expense an asset Insurance: Tk 9,000 used	Accrued expense a liability Wages owed: Tk 14,000
Revenue side	Unearned revenue a liability Advance fee: Tk 12,000 earned	Accrued revenue an asset Unbilled work: Tk 20,000

Supplies used and depreciation follow the deferral pattern.

FIG. 7.2. The four kinds of adjustment, arranged by when cash moves and by cost or revenue

An adjusting entry corrects one account for the period and changes no cash.

Adjusting entry. A journal entry made at period end to bring an account to its correct balance for the period.

The six corrections of the workshop, each with its two accounts and its reason.

Each correction is written as a journal entry, as in Chapter 5. An **adjusting entry** is a journal entry made at period end to bring an account to its correct balance for the period. It is dated on the last day of the period, here 31 December 2026, and carries a short narration. Three habits apply to every adjusting entry. It is made after the unadjusted trial balance. It changes at least one revenue or expense account and at least one asset or liability account. And it does not involve Cash, because the adjustment is about timing, and the cash has already moved or will move later.³

The table lists the six adjustments for Meghna Binding Service. Read each row from left to right: the reference letter, the account debited, the account credited, the amount in taka, and the narration that explains the entry.

ADJUSTING ENTRIES OF MEGHNA BINDING SERVICE, 31 DECEMBER 2026 (TK)

REF	DEBIT	CREDIT	AMOUNT	NARRATION
(a)	Supplies Expense	Supplies	33,000	Supplies used in 2026
(b)	Insurance Expense	Prepaid Insurance	9,000	Insurance expired, 1 October to 31 December
(c)	Depreciation Expense	Accumulated Depreciation	72,000	One year of depreciation on equipment
(d)	Unearned Revenue	Service Revenue	12,000	Advance fees earned in 2026
(e)	Wages	Wages Payable	14,000	Wages earned but unpaid
(f)	Accounts Receivable	Service Revenue	20,000	Work done but not billed

Common slips: forgetting that part of the original amount remains, and mixing up accounts.

Several slips occur in this work. Students record the whole payment as an expense, forgetting that part of it remains an asset or a liability. They adjust for the full year when only part of the year has passed. They debit Cash by habit, although no cash moves. And they reverse the two accounts, for example by crediting Wages when the wages are owed. A quick test helps: after each entry, ask whether the asset or liability account now shows a sensible figure.

³ Recall: debit and credit, Chapter 4, 'Accounts, Debits and Credits', section 4.3 'Debit and Credit'.

Before moving on. An adjusting entry is dated on the last day of the period, carries a narration and involves no cash. The workshop's six entries cover supplies, insurance, depreciation, advance fees, wages owed and unbilled work. You should now be able to write any of the six from its data.

Six entries, one date and no cash complete the section.

WORKED EXAMPLE: FROM UNADJUSTED TO ADJUSTED BALANCES

Case file, illustrative. *Given.* The unadjusted trial balance of Meghna Binding Service at 31 December 2026 (see the table in the first section), totalling Tk 1,240,000 on each side, and the six adjustments: (a) supplies on hand Tk 15,000, so supplies expense Tk 33,000; (b) insurance for 12 months paid on 1 October 2026, Tk 3,000 a month, expired Tk 9,000; (c) equipment depreciation Tk 72,000 (5 years, no residual value); (d) unearned revenue earned Tk 12,000 (Tk 4,000 a month from 1 October); (e) wages owed Tk 14,000; (f) revenue earned but not billed Tk 20,000.

The unadjusted balances and the six adjustments are the data, and the task is to find the new balances.

Find. The adjusted balance of every account that changes, the profit after adjustment, and a check that debits equal credits. *Formula.* Adjusted balance equals unadjusted balance plus or minus the adjustment. Profit equals revenue minus expenses. *Steps.* Apply each of the six entries to its two accounts, as in the next table. An asset or expense rises on a debit and falls on a credit; a liability or revenue rises on a credit and falls on a debit.

Find, formula and steps: apply each entry to its two accounts, and total the effect on profit.

BALANCES OF THE ACCOUNTS CHANGED BY THE SIX ADJUSTMENTS (TK)

ACCOUNT	SIDE	BEFORE	CHANGE	AFTER
Accounts Receivable	Debit	90,000	+20,000	110,000
Supplies	Debit	48,000	-33,000	15,000
Prepaid Insurance	Debit	36,000	-9,000	27,000
Accumulated Depreciation	Credit	0	+72,000	72,000
Wages Payable	Credit	0	+14,000	14,000
Unearned Revenue	Credit	24,000	-12,000	12,000
Service Revenue	Credit	756,000	+32,000	788,000
Wages	Debit	330,000	+14,000	344,000
Supplies Expense	Debit	0	+33,000	33,000
Insurance Expense	Debit	0	+9,000	9,000
Depreciation Expense	Debit	0	+72,000	72,000

Service Revenue rises by $12,000 + 20,000 = 32,000$, from 756,000 to 788,000. The expenses are wages 344,000, rent 120,000, utilities 26,000, supplies expense 33,000, insurance expense 9,000 and depreciation expense 72,000, which total 604,000. The profit is therefore $788,000 - 604,000 = 184,000$.

The expenses and revenue after adjustment give the true profit of the year.

Answer. After the six adjustments the profit for 2026 is Tk 184,000, not the Tk 280,000 that the unadjusted figures suggested. Tk 96,000 of the apparent profit had not really been earned. *Check.* The unchanged accounts are Cash 150,000, Equipment 360,000, Drawings 80,000, Rent 120,000 and Utilities 26,000 on the debit side, and Accounts Payable 60,000 and Capital 400,000 on the credit side. The debit side totals 1,346,000 and the credit side totals 1,346,000. Also, expenses rise by 128,000 and revenue by 32,000, so profit falls by 96,000: $280,000 - 96,000 = 184,000$.

The answer and the two checks show that the books still balance.

Three slips are common when the table is built.

The worked case joins the earlier ideas into one routine.

The chapter in a few lines: periods, the accrual basis, matching and six corrections.

Adjusting entries do not correct mistakes; they update the records for the passing of time.

A prepaid cost is an asset when paid, not an expense.

Money received in advance is not yet revenue.

Depreciation does not set money aside.

A company that failed in 2001 shows what can happen when reports hide the true position.

Students often forget one side of an entry, so the totals differ by one amount. They carry a figure to the wrong side, such as treating Accumulated Depreciation as a debit. And they replace the old balance instead of changing it, for example writing 33,000 for Supplies instead of 48,000 less 33,000. Compare each total with the check before moving on.

Before moving on. The worked case applies all six adjustments to the unadjusted balances and shows that profit falls from Tk 280,000 to Tk 184,000. You should now be able to carry a trial balance through its adjustments and test the result.

* * *

SUMMARY

Results are measured for an accounting period. The accrual basis places revenue and expenses in the period of the activity, and the matching principle sets each cost beside the revenue it helped to earn. Because ordinary entries follow cash, the accounts need adjustment at period end. A prepaid expense is an asset that is used up, unearned revenue is a liability that is earned, an accrued expense is a cost owed, and accrued revenue is revenue owed to the business. Each correction is an adjusting entry dated on the last day of the period.

THINK IT THROUGH

7.T1 Suppose Meghna Binding Service made none of the six adjustments. State the profit that its records would show for 2026, and say who might be misled and how.

7.T2 Explain why an adjusting entry does not involve Cash, using the insurance example.

WHERE PEOPLE GO WRONG

Many readers believe that an adjusting entry corrects a mistake. It does not. The original entries were right on the day they were made. Adjusting entries bring the accounts up to date for time that has passed or work that has been done.

Another belief is that a cost is an expense on the day it is paid. When the workshop pays Tk 36,000 for a year of insurance, nothing is used up on that day, so the payment is an asset. Only the part used up becomes an expense.

A third belief is that money received is always revenue. The advance of Tk 24,000 from a school was a liability on the day it arrived, because the work was still owed. Only the work done, Tk 12,000, became revenue by 31 December. Revenue follows the work, not the cash.

A fourth belief is that depreciation puts cash aside for replacing equipment. It does not. It spreads a cost already paid across the years of use, and no cash moves when it is recorded.

IN HISTORY

Enron Corporation was an energy company based in Houston, Texas, in the United States. It filed for bankruptcy on 2 December 2001. Investigations showed that its reports had kept large debts out of sight, partly by moving them into separate arrangements. Its collapse cost investors and employees heavily.

In 2002 the United States passed the Sarbanes-Oxley Act, which tightened the rules on company reporting and auditing. It applies to companies in the United States; other countries have their own laws. The story shows why correct and complete accounts matter: a report that leaves out what a business owes gives its readers a false picture.

A new United States law followed in 2002.

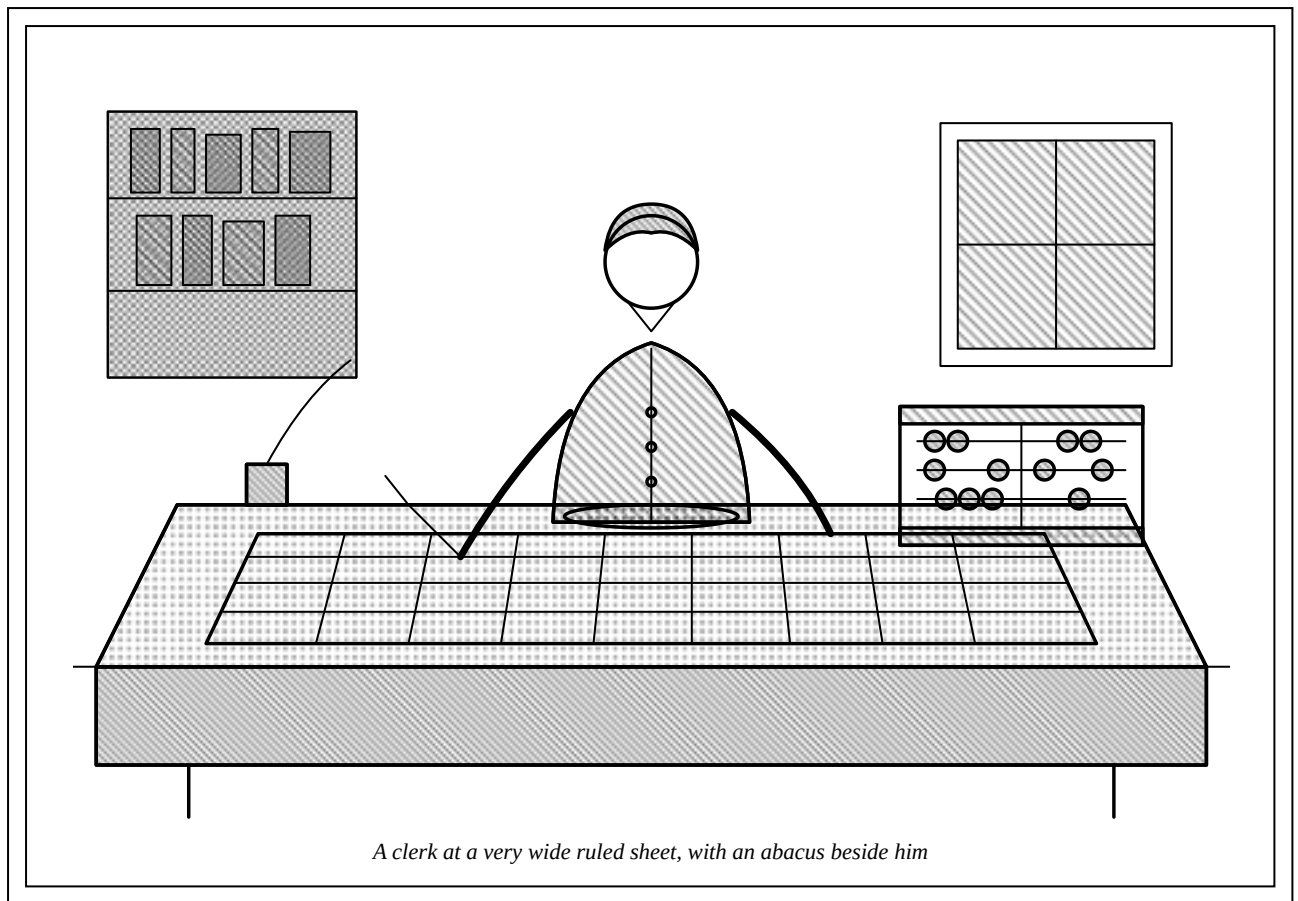
REVIEW QUESTIONS

7.1 On 1 March a workshop pays Tk 6,000 for six months of insurance. Find the insurance expense and the prepaid expense at 30 June of the same year.

7.2 The Supplies account shows Tk 52,500 before adjustment. A count on the last day of the year shows supplies on hand of Tk 18,200. Find the supplies expense and write the adjusting entry.

7.3 On 1 November a workshop receives Tk 18,000 for a school's binding work, to be done evenly over six months. Find the revenue earned and the unearned revenue at 31 December.

7.4 Equipment cost Tk 240,000, has a life of 8 years and no residual value. At year end wages of Tk 9,000 are owed, and work worth Tk 15,000 has been done but not billed. Write the three adjusting entries and state the effect on profit.



CHAPTER VIII

The Worksheet

How can an accountant test every change on paper before touching a single record?



A large ruled sheet carries figures from the trial balance through the adjustments to the statements. This chapter builds its ten columns one pair at a time, extends each balance, and teaches the checks that reveal a hidden mistake.

YOU WILL NEED

Chapter 7 'Adjusting the Accounts', section 7.4 'Recording Adjusting Entries'; Chapter 6 'The Trial Balance', section 6.1 'Preparing a Trial Balance'; Appendix A1 'Maths You Will Use'.

A year-end clerk holds correct balances that are not yet ready to use.

Changing the records directly, and drafting the statements early, are both risky.

The question of the chapter: how can every change be tested on paper first?

GO SLOWLY HERE.

A family budget on a rough sheet shows the idea before the subject.

A worksheet is a working paper that carries figures from the trial balance to the statements.

Sheet. Word in use. In daily life, a flat piece of paper or cloth. In accounting, the large ruled page, or its computer copy, on which a worksheet is drawn.

Worksheet. A working paper with columns that moves figures from the trial balance through adjustments to the statements.

Recall: adjusting entries, Chapter 7, section 7.4 'Recording Adjusting Entries'.

A worksheet has three jobs: to gather, to test and to check.

It is the evening of 31 December, and a clerk in a small office is holding the trial balance. It lists eighteen accounts, and the debit total equals the credit total. Yet the balances are not ready to use. Some supplies have been consumed, some wages are owed, and the equipment has lost part of its value. Six changes must be made before anyone can learn what the business gained or lost during the year.

The clerk could write each change into the ledger at once. But suppose the third change contains an error. The ledger now holds a wrong figure, and correcting it leaves crossings-out on a record that other people must trust. The clerk could also draft the final statements straight away. But the statements need the changed figures, so a late discovery means writing every statement a second time.

So here is the question for this chapter. How can an accountant try every change, see every result and find every error, without writing in any permanent record? Think about a difficult sum at home. You take a spare sheet of paper, write the figures in columns and work there, and only then write the answer neatly. Accountants do the same. Their spare sheet has a name, and it has an exact layout. By the end of this chapter you will be able to build it, complete it and check it.

8.1 WHY USE A WORKSHEET

Before meeting the subject idea, consider a family that is planning a wedding. On a large sheet of paper, the family lists every cost. Beside the list is a column headed "first estimate". Next to it is a column headed "changes", where the family enters a rise in the hall price and a cousin who has cancelled. A third column shows the new estimate. Only when the whole sheet is complete does the family write the final budget on a clean page. The large sheet is rough, and mistakes on it cost little. The clean page is the one that other people will read.

The accountant's method follows the same pattern, and the rough sheet has a name. A **worksheet** is a working paper with columns that moves figures from the trial balance through the adjustments to the statements. For example, a worksheet may show Supplies as Tk 1,000 in one column and Tk 400 in a later column, so that a reader sees the figure before and after a change. Notice the word "working". The worksheet is where the work is done. It is not where the answer is published. The family's first estimate matches the trial balance, the changes match the adjustments, and the clean page matches the statements.

A worksheet does three jobs, and each one answers a worry from the opening puzzle. The first job is to gather. All the account balances stand on one page, in one order, so the clerk does not turn between ledger pages. The second job is to test. The clerk can enter the six changes on the worksheet and see their effect before recording any of them in the journal. The third job is to check. Equal totals at each stage show whether the arithmetic is sound,

and the last columns show the result of the period before any statement is drafted.

Here is the idea in the smallest possible numbers. Supplies show Tk 1,000 in the trial balance, and Tk 600 of them have been used. On the worksheet we enter Tk 600 as a credit to Supplies, and we see at once that Tk 400 remain. Nothing in the ledger has changed. If we had made an arithmetic slip, we would correct it on the worksheet and begin again at no cost.

Different people value these jobs in different ways. The owner wants to know the result of the year early, before the statements are ready. The clerk wants to avoid rewriting, because a worksheet error is corrected on the sheet itself, while a ledger error needs a formal correction. A bank or a customer never sees the worksheet at all, because it is private working paper.

A worksheet is not a statement. This difference is easy to forget, because the worksheet holds the same figures that later appear in the statements. A financial statement is a formal report for outside readers. In this book we prepare two of them, and Chapter 9, 'Financial Statements and Closing Entries', teaches them fully. For now, here are short previews in plain words. The income statement lists the revenue and the expenses of a period. The balance sheet lists what the business owns and owes on one date. The worksheet only prepares the figures that these two statements need.

Because the worksheet is working paper, it changes no account. The ledger keeps its old balances until the adjusting entries are recorded in the journal and posted, as Chapter 7, section 7.4 'Recording Adjusting Entries' explains. A worksheet is also optional: a small business may omit it, and a computer program may perform the same task without showing it. Finally, a worksheet has limits. Equal totals show that debits equal credits, but they cannot show that an amount sits in the right account. A cost entered in the wrong account leaves the totals equal. A clerk must never hide a difference in a suspense account merely to make the totals agree.

Before moving on. A worksheet is a working paper with columns. It gathers balances, tests the adjustments and checks the arithmetic, and it is private. It is not a statement, and it changes no ledger account. Equal totals do not prove that every amount is in the right account. *You should now be able to say why an accountant uses a worksheet before the statements, and why it changes no record.*

The same idea in numbers: a Tk 600 change is tested without touching the ledger.

Owner, manager, clerk and outsider value the worksheet in different ways.

GO SLOWLY HERE.

The worksheet is private working paper; it is not a financial statement.

The worksheet changes no record, and it has limits.

The recap restates the three jobs, the private status and the limits.

8.2 BUILDING THE TEN COLUMNS

We now build a worksheet one pair of columns at a time. To keep the arithmetic small, we use a repair stall that exists only for teaching; it is not part of the running case. At the end of its year, the stall's trial balance shows these debit balances: Cash Tk 5,000, Supplies Tk 1,000, Equipment Tk 6,000 and Wages Tk 2,000. It shows these credit balances: Capital Tk 8,000 and Repair Revenue Tk 6,000. Each side totals Tk 14,000. Three changes are needed. Supplies worth Tk 600 have been used, wages of Tk 300 are owed, and the equipment has lost Tk 500 of its value.

Before filling any column, look at the layout. The worksheet has one column for account titles and then five pairs of columns, which makes ten numbered columns. Each pair has a debit column and a credit column. The figure below shows the layout. To read it, begin at the left and move right. Each box names a pair of columns, and the numbers 1 to 10 show the order of the columns. The

A tiny repair stall gives clean numbers for learning the method.

The figure shows five pairs of columns, read from left to right.

The rule for step 3 has two parts. If the balance and the adjustment stand on the same side, add them. Wages is a debit of Tk 2,000, and adjustment (b) adds a debit of Tk 300, so the adjusted balance is a debit of Tk 2,300. If they stand on opposite sides, subtract the smaller from the larger, and place the answer on the larger side. Supplies is a debit of Tk 1,000 and a credit of Tk 600, so the answer is a debit of Tk 400. A row with only an adjustment, such as Wages Payable, keeps that figure on its own side.

The combining rule: add figures on the same side, subtract figures on opposite sides.

The table below shows the stall's worksheet through column 6. Read each row from left to right. For Supplies, Tk 1,000 is a debit in column 1. The letter (a) beside Tk 600 marks a credit in column 4. The result, Tk 400, is a debit in column 5. The bottom row shows that every pair of columns has equal totals, and the adjusted trial balance totals Tk 14,800 on each side.

The table shows columns 1 to 6 for the stall; read each row from left to right.

STALL WORKSHEET, COLUMNS 1 TO 6 (TAKA)

ACCOUNT	TRIAL BALANCE DR	TRIAL BALANCE CR	ADJUSTMENTS DR	ADJUSTMENTS CR	ADJUSTED DR	ADJUSTED CR
Cash	5,000				5,000	
Supplies	1,000			(a) 600	400	
Equipment	6,000				6,000	
Capital		8,000				8,000
Repair Revenue		6,000				6,000
Wages	2,000		(b) 300		2,300	
Supplies Expense			(a) 600		600	
Depreciation Expense			(c) 500		500	
Wages Payable				(b) 300		300
Accumulated Depreciation				(c) 500		500
Totals	14,000	14,000	1,400	1,400	14,800	14,800

PAUSE AND CHECK 8.P1 In the stall's worksheet, what is the adjusted balance of Supplies, and on which side does it stand?

In one line. The adjusted trial balance combines columns 1 to 4 row by row, and its two totals must agree before we move on.

In one line: the adjusted trial balance is the trial balance after the changes, with equal totals.

The fourth and fifth pairs are the statement columns. In step 4, every adjusted balance is copied to one column only, and the kind of account decides which. Revenue and expense accounts go to the income statement columns, which are 7 and 8. Asset, liability and owner's equity accounts go to the balance sheet columns, which are 9 and 10. Within each pair, a debit balance stays a debit and a credit balance stays a credit.

GO SLOWLY HERE.

Step 4 sends each adjusted balance to exactly one statement column.

The table below shows the destination of each kind of account. Read a row from left to right: the kind of account, the side on which its balance normally stands, and the columns that receive it.

The table shows where each kind of account is entered.

WHERE EACH KIND OF ACCOUNT IS ENTERED

KIND OF ACCOUNT	NORMAL SIDE	STATEMENT COLUMNS
Asset, for example Cash	Debit	Balance sheet, debit
Accumulated Depreciation	Credit	Balance sheet, credit
Liability, for example Wages Payable	Credit	Balance sheet, credit
Capital	Credit	Balance sheet, credit
Drawings	Debit	Balance sheet, debit
Revenue	Credit	Income statement, credit
Expense	Debit	Income statement, debit

Two accounts cause most confusion, and some offices use fewer columns.

The recap restates the five pairs and the four steps.

Two rows need care. Accumulated Depreciation is not a liability; it is a credit balance that reduces the Equipment on the balance sheet, so it goes to the balance sheet credit column. Drawings is a debit balance, but it is not an expense, because it records money taken out by the owner. It reduces owner's equity, so it goes to the balance sheet debit column. Finally, some offices use fewer columns than ten.¹ We use ten because the adjusted trial balance then has its own pair of totals, and each pair is easier to check.

Before moving on. The worksheet has five pairs of columns: trial balance, adjustments, adjusted trial balance, income statement and balance sheet. Step 1 copies the trial balance. Step 2 enters the adjustments. Step 3 combines them into the adjusted trial balance. Step 4 sends each balance to one statement column. *You should now be able to say what each pair of columns holds and which accounts go to the income statement columns.*

* * *

WORKED WORKSHEET

We finish the stall first, with small numbers, then complete the running case.

Given and Find: the stall's adjusted balances, and the statement columns to complete.

Extend. Word in use. In daily life, to make something longer or to allow more time. In accounting, to copy an adjusted balance into the correct statement column.

Extending a balance means copying it to one statement column on its own side.

We now complete the worksheet. This section has three tasks: to extend each balance, to total the columns and find the result of the year, and to check the worksheet. We begin with the stall again, because its figures are small, and then we turn to the running case. Take each task slowly, and notice that none of them needs a new idea. They need only care.

Given. The stall's adjusted trial balance from section 8.2. The debits are Cash 5,000, Supplies 400, Equipment 6,000, Wages 2,300, Supplies Expense 600 and Depreciation Expense 500. The credits are Capital 8,000, Repair Revenue 6,000, Wages Payable 300 and Accumulated Depreciation 500. All figures are in taka. *Find.* The income statement columns, the balance sheet columns and the result of the year.

To extend a balance needs no calculation, only care. Wages 2,300, Supplies Expense 600 and Depreciation Expense 500 are expenses, so each is a debit in column 7. Repair Revenue 6,000 is revenue, so it is a credit in column 8. Cash 5,000, Supplies 400 and Equipment 6,000 are assets, so they are debits in column 9. Capital 8,000, Wages Payable 300 and Accumulated Depreciation 500 are credits in column 10. Every adjusted balance appears once, and only once.

¹ Some offices use eight columns, leaving out the adjusted trial balance pair and extending each balance straight from the first two pairs. The method is shorter, but an error is harder to locate.

Now we total columns 7 and 8. The difference between them is the result of the year, and it has a name. **Net income** is the amount by which total revenue exceeds total expenses for a period. For example, if the stall earns Tk 6,000 and its expenses are Tk 3,400, its net income is Tk 2,600.

Formula. In words: net income equals total revenue minus total expenses. In symbols, $NI = R - E$. Here NI stands for net income, R stands for total revenue and E stands for total expenses.

Steps. Column 7 total: $E = \text{Tk } 2,300 + \text{Tk } 600 + \text{Tk } 500 = \text{Tk } 3,400$. Column 8 total: $R = \text{Tk } 6,000$. Then $NI = \text{Tk } 6,000 - \text{Tk } 3,400 = \text{Tk } 2,600$. *Answer.* The stall's net income is Tk 2,600. *Check.* Enter Tk 2,600 under column 7, on the debit side, so that column 7 totals Tk 6,000 like column 8. Enter Tk 2,600 under column 10, on the credit side. Column 9 totals Tk 11,400, and column 10 totals $\text{Tk } 8,800 + \text{Tk } 2,600 = \text{Tk } 11,400$.

Why does Tk 2,600 appear twice, on opposite sides? In the income columns, the debit side is short by Tk 2,600, so the figure balances that pair. In the balance sheet columns, net income increases owner's equity, and owner's equity has a credit balance, so the figure belongs on the credit side. One amount, two columns, opposite sides.

PAUSE AND CHECK 8.P2 The stall's income columns total Tk 3,400 (debit) and Tk 6,000 (credit). On which side of the income columns is the balancing figure placed, and why?

A period can also end badly. A **net loss** is the amount by which total expenses exceed total revenue for a period. For example, if the stall earned Tk 3,000 in a slow quarter and its expenses were Tk 3,400, its net loss would be Tk 400. The formula reverses: $NL = E - R$. The balancing figure now goes to the credit side of the income columns and to the debit side of the balance sheet columns, because a loss reduces owner's equity.

A worksheet is checked three times. First, the totals of columns 1 and 2, of columns 3 and 4, and of columns 5 and 6 must agree. Second, after the balancing figure is added, columns 7 and 8 must agree, and columns 9 and 10 must agree. Third, the balancing figure under the income columns must equal the one under the balance sheet columns, on opposite sides. These checks catch arithmetic slips and extension errors. They do not catch an amount placed in the wrong account, because the totals still agree.

Case file, illustrative. Meghna and Sons uses the dataset of the Chapter 7 case. After the adjustments, the balances are, in taka: Cash 150,000; Accounts Receivable 110,000; Supplies 15,000; Prepaid Insurance 27,000; Equipment 360,000; Accumulated Depreciation 72,000; Accounts Payable 60,000; Wages Payable 14,000; Unearned Revenue 12,000; Capital 400,000; Drawings 80,000; Service Revenue 788,000; Wages 344,000; Rent 120,000; Utilities 26,000; Supplies Expense 33,000; Insurance Expense 9,000; Depreciation Expense 72,000. The file records total expenses of 604,000 and net income of $788,000 - 604,000 = \text{Tk } 184,000$.

The first four columns of this worksheet hold the unadjusted balances and the adjusting entries, which Chapter 7, section 7.4 'Recording Adjusting Entries' records. Here we begin at the adjusted trial balance, in columns 5 and 6, and extend every balance. Read each row from left to right. Cash is a debit of 150,000 in column 5, and it is extended to column 9. Service Revenue is a credit in column 6, and it is extended to column 8.

GO SLOWLY HERE.

Net income is the amount by which revenue exceeds expenses.

Net income. The amount by which total revenue exceeds total expenses for a period.

The formula for net income, in words and in symbols.

Steps, answer and check for the stall: net income is Tk 2,600.

The balancing figure goes to opposite sides because net income increases owner's equity.

Net loss is the opposite case, with expenses above revenue.

Net loss. The amount by which total expenses exceed total revenue for a period.

Three checks show whether a worksheet is sound, and each has limits.

The case file gives the adjusted balances of the running case.

The table extends each case balance, and the bottom rows give the totals.

MEGHNA AND SONS WORKSHEET, COLUMNS 5 TO 10 (TAKA)

ACCOUNT	ADJUSTED DR	ADJUSTED CR	INCOME DR	INCOME CR	BALANCE SHEET DR	BALANCE SHEET CR
Cash	150,000				150,000	
Accounts Receivable	110,000				110,000	
Supplies	15,000				15,000	
Prepaid Insurance	27,000				27,000	
Equipment	360,000				360,000	
Accumulated Depreciation		72,000				72,000
Accounts Payable		60,000				60,000
Wages Payable		14,000				14,000
Unearned Revenue		12,000				12,000
Capital		400,000				400,000
Drawings	80,000				80,000	
Service Revenue		788,000		788,000		
Wages	344,000		344,000			
Rent	120,000		120,000			
Utilities	26,000		26,000			
Supplies Expense	33,000		33,000			
Insurance Expense	9,000		9,000			
Depreciation Expense	72,000		72,000			
Totals	1,346,000	1,346,000	604,000	788,000	742,000	558,000
Net income			184,000			184,000
Final totals			788,000	788,000	742,000	742,000

Steps and answer: expenses Tk 604,000, net income Tk 184,000, both pairs balanced.

A second check with the accounting equation confirms the result.

Steps. Columns 5 and 6 each total Tk 1,346,000. Column 7 totals Tk 344,000 + Tk 120,000 + Tk 26,000 + Tk 33,000 + Tk 9,000 + Tk 72,000 = Tk 604,000. Column 8 holds Service Revenue of Tk 788,000. So NI = Tk 788,000 – Tk 604,000 = Tk 184,000, which we enter under column 7 and under column 10. Column 9 totals Tk 742,000. Column 10 totals Tk 72,000 + Tk 60,000 + Tk 14,000 + Tk 12,000 + Tk 400,000 = Tk 558,000 before net income, and Tk 742,000 after it. **Answer:** Net income is Tk 184,000, the same as the case file.

Check. The worksheet checks pass, and we can test the result a second way. Assets less Accumulated Depreciation are Tk 662,000 – Tk 72,000 = Tk 590,000. Liabilities are Tk 60,000 + Tk 14,000 + Tk 12,000 = Tk 86,000. Owner's equity is Tk 400,000 + Tk 184,000 – Tk 80,000 = Tk 504,000. Then Tk 86,000 + Tk 504,000 = Tk 590,000, so the accounting equation holds.

IN SHORT

A worksheet moves figures through five pairs of columns: trial balance, adjustments, adjusted trial balance, income statement and balance sheet. Net income is revenue minus expenses, and it is entered on the debit side of the income columns and the credit side of the balance sheet columns.

Many students find the last step confusing, because one amount appears in two places. These are the common slips. Some students extend a debit balance into a credit column. Some treat Drawings as an expense and extend it to the income columns. Some forget an account that came from an adjustment, such as Wages Payable. Some place a net loss on the same sides as a net income, which reverses the sign. Finally, some mix units, for example by writing 3,500 for a figure that is in thousands of taka. Name each slip before you begin, and you will notice it when it occurs.

Before moving on. To extend a balance, copy it to one statement column on its own side. Net income is revenue minus expenses, and net loss is expenses minus revenue. The balancing figure appears on opposite sides in the income columns and the balance sheet columns. Three checks test the worksheet, and none of them finds an amount in the wrong account. *You should now be able to say where net income is entered, and why it is entered twice.*

* * *

SUMMARY

A worksheet is a working paper with columns. It moves the figures of the trial balance through the adjustments, the adjusted trial balance and the statement columns, without changing any ledger account. Net income is total revenue minus total expenses, and net loss is the reverse. The balancing figure is entered twice, on opposite sides. Equal totals check the arithmetic, but they cannot show that an amount is in the right account.

THINK IT THROUGH

8.T1 A clerk extends the stall's balances. The balance sheet columns total Tk 11,400 (debit) and Tk 8,500 (credit) before net income. Net income of Tk 2,600 is added to the credit side. Find the difference between the two totals, name the stall account with the same balance, and say what you would check first.

8.T2 The stall's adjusted trial balance shows a debit total of Tk 14,530 and a credit total of Tk 14,800. Find the difference. Is it divisible by 9? What kind of error does this suggest, and which balance is the likely cause?

WHERE PEOPLE GO WRONG

Some students believe that the worksheet records the adjusting entries. It does not. The worksheet only shows the adjustments, so that they can be tested. The entries must still be recorded in the journal and posted to the ledger, or the ledger will keep its old balances.

Common slips: a wrong column, a wrong side, a forgotten account and a mixed-up sign.

The recap restates extending, finding net income or loss, and checking.

The chapter in a few lines: five pairs of columns, one result, three checks.

Misunderstanding: the worksheet records the adjusting entries in the ledger.

Misunderstanding: Drawings belong to the income statement columns.

Misunderstanding: net income is entered once, on the same side everywhere.

Misunderstanding: equal totals prove that the worksheet is correct.

In 1979 two Americans built an electronic version of the accountant's columnar worksheet.

The modern spreadsheet keeps the logic of the paper worksheet.

Some students extend Drawings to the income statement columns, because it looks like a cost. Drawings is money taken out by the owner. It is not an expense, and it reduces owner's equity. It belongs in the balance sheet debit column.

Some students enter net income on the debit side of both pairs of statement columns. The income columns need it on the debit side to balance, but the balance sheet columns need it on the credit side, because it increases owner's equity. A net loss reverses both sides.

Some students believe that equal totals prove a worksheet correct. They prove only that debits equal credits. A cost entered in the wrong account, or an amount left out of both sides, leaves the totals equal. The totals are a necessary test, not a sufficient one.

IN HISTORY

In 1979, Dan Bricklin and Bob Frankston wrote VisiCalc, the first electronic spreadsheet for the Apple II. Bricklin was then a student at Harvard Business School, in Massachusetts, in the United States. Frankston was a programmer. The Apple II was a small personal computer, sold to homes and offices. Their program was built from the accountant's columnar worksheet. It gave the computer the same columns and rows that an accountant draws on a large ruled sheet, and it recalculated the dependent figures whenever one figure changed.²

Today, the worksheet of this chapter often lives on a computer screen. A clerk who changes one adjustment sees every total change at once. Yet the logic has not changed. Figures still move from left to right through the same pairs of columns, and the same equal-total checks still apply. A machine calculates faster than a clerk, but it cannot tell whether an amount sits in the right account. That judgement remains with you.

REVIEW QUESTIONS

8.1 Clean numbers. A business has these unadjusted balances: Cash Tk 3,000, Prepaid Insurance Tk 1,200 and Wages Tk 3,000 (debits); Capital Tk 3,200 and Fees Revenue Tk 4,000 (credits). Insurance of Tk 300 has expired. Complete the ten-column worksheet and state the net income.

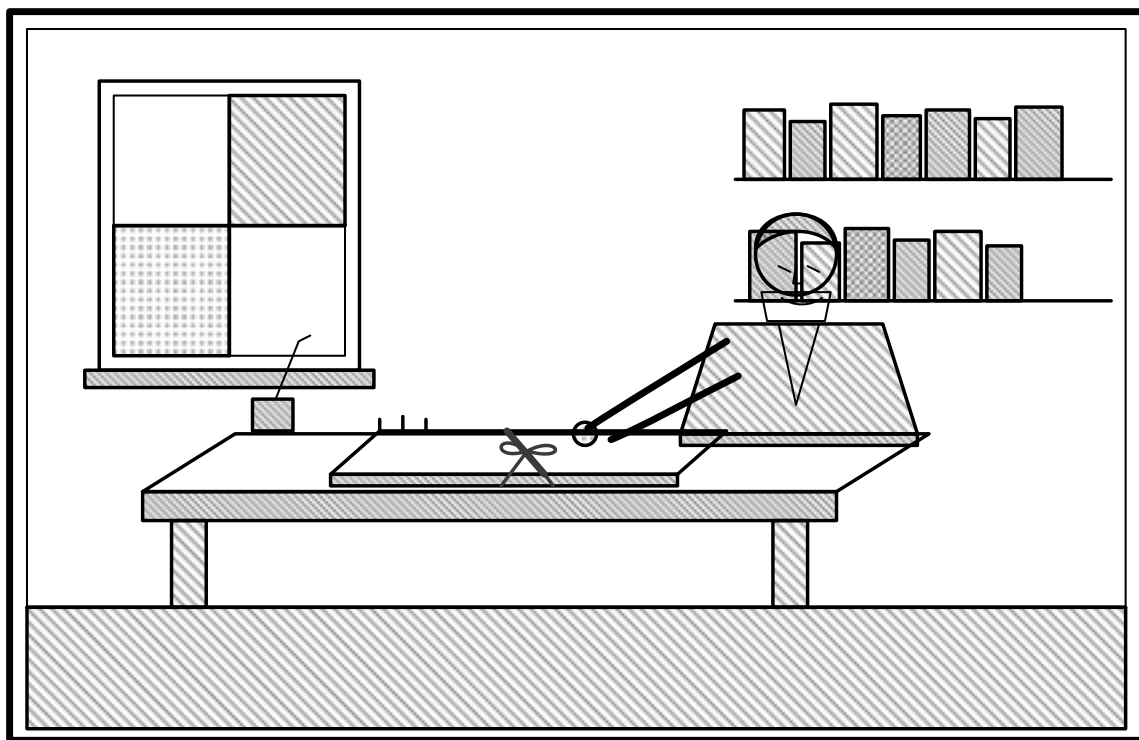
8.2 Realistic data. The unadjusted balances are, in taka: Cash 24,500; Accounts Receivable 18,300; Supplies 4,200; Equipment 48,000; Drawings 6,000; Wages 17,200; Rent 4,800; Utilities 2,000 (debits); Accounts Payable 9,600; Capital 54,000; Fees Revenue 61,400 (credits). Supplies on hand at the end are Tk 1,500; wages owed are Tk 1,300; depreciation of equipment is Tk 4,800. Complete the worksheet and state the net income.

8.3 Changed sign and unit. For one quarter, the income columns total 21,500 (debit) and 18,000 (credit), and the balance sheet columns, before the balancing figure, total 40,000 (debit) and 43,500 (credit). All figures are in thousands of taka. State the result, the column that receives it in each pair, and the amount in full taka.

² Software Arts was the company that Bricklin and Frankston formed to develop the program. A firm called Personal Software sold it.

8.4 Combined problem. The unadjusted balances are, in taka: Cash 9,000; Equipment 12,000; Prepaid Insurance 1,800; Wages 3,000 (debits); Capital 17,400; Unearned Revenue 2,400; Fees Revenue 6,000 (credits). Adjustments: (a) insurance of 900 has expired; (b) fees of 1,500 in Unearned Revenue are now earned; (c) depreciation is 1,200; (d) wages of 400 are owed. Complete the worksheet and state the net income.

8.5 Name two jobs that a worksheet does and a ledger alone does not. Then explain why the worksheet does not change any account.



A clerk seals a finished annual account into a folder with a ribbon.

CHAPTER IX

Financial Statements and Closing Entries

What must a business show at the end of the year, and what must it empty before the next year begins?



At the end of a period, the accounts must report their results and then prepare for a fresh start. This chapter teaches the three reports that present the results, the closing entries that reset the accounts, and the reversing entries that make the first payment of the new period easy to record.

This chapter rests on three earlier pieces of work.

Recall: worksheet, Chapter 8, 'The Worksheet', section 8.3 'Worked Worksheet'.

Recall: deferrals, Chapter 7, 'Adjusting the Accounts', section 7.2 'Deferrals'.

A market stall owner must answer two different questions on the last evening of the month.

The earnings belong to the month that ended; the tin and the debts continue.

Meghna Binding Service meets the same puzzle at the end of its first year.

The chapter answers the puzzle in three parts, then applies them to the workshop.

The word statement means a formal report of figures, not a spoken claim.

Statement. Word in use. In daily life, something said or written, such as a bank statement. In accounting, a formal report of figures for a period or for a date.

Here are the stall's small, round figures for the month.

The first statement answers: what did the business earn?

Income statement. A report of revenue and expenses for a period, ending in net income or net loss.

You will need three earlier pieces of work. Chapter 8, 'The Worksheet', section 8.3 'Worked Worksheet', showed how the adjusted figures are collected on one sheet. Chapter 7, 'Adjusting the Accounts', section 7.2 'Deferrals', explained the entries that move amounts between periods. Appendix A1, 'Maths You Will Use', holds the arithmetic.

A woman keeps a tea stall in a market. On the last evening of the month she empties the cash tin onto the table. She counts the takings, remembers what she paid for leaves, milk and sugar, and keeps apart the money she took home. Then she faces a question that every business must answer sooner or later. How much did the stall earn this month? And what does the stall own and owe tonight?

These are two different questions, and their answers behave differently. The earnings belong to the month that has ended. The cash tin, the debts and the owner's stake continue into the next month. On the first morning of the new month, the earnings must start again at nothing, while the cash tin must keep what it holds. How can one set of records do both?

Case file, illustrative. Meghna Binding Service, the separate set of books for the book-binding workshop that began on 1 January 2026, ends its first year on 31 December 2026. Its Revenue account then holds Tk 788,000. If nothing is done, the first sale of 2027 would be added to that balance, and the account could never show what the workshop earned in one year alone.

This chapter answers the puzzle in three parts. Section 9.1 teaches the three reports that present the results of a period. Section 9.2 teaches the closing entries, which reset the accounts that measure one period. Section 9.3 teaches reversing entries, a step that makes the first payment of a new period easier to record. A worked example with Meghna Binding Service follows, and the chapter ends with the history of the law that first required companies to publish a balance sheet.

* * *

9.1 THE THREE STATEMENTS

Return to the tea stall. At the end of the month the owner wants three answers. What did the stall earn after paying its costs? How did her own stake in the stall change? What does the stall own and owe tonight? Each answer needs a separate page, because each answers a different question. Accountants call each such page a statement, and this section teaches the three pages in the order in which they are prepared.

Here are her figures. She sold tea for Tk 5,000. She spent Tk 3,000 on leaves, milk, sugar and fuel. She took Tk 500 home for her family. When she started the stall, she had put in Tk 2,500. At the month's end the stall holds cash of Tk 3,700 and equipment worth Tk 1,300, and it owes its supplier Tk 1,000. We will use these figures for every statement in this section.

The first answer comes from the **income statement**. An income statement is a report of revenue and expenses for a period, ending in net income or net loss. Revenue is the amount a business earns by selling goods or services. An expense is a cost of earning that revenue. The stall's revenue is Tk 5,000 and its expenses are Tk 3,000, so its net income is Tk 2,000. If expenses were larger than revenue, the result would be a net loss.

The table below shows the income statement of the stall. Read it from the top downwards: revenue first, then expenses, and the result on the last line.

The table lists revenue first, expenses next and the result last.

INCOME STATEMENT OF THE TEA STALL FOR ONE MONTH

ITEM	TK
Revenue: tea sold	5,000
Expenses: leaves, milk, sugar and fuel	3,000
Net income	2,000

Every statement has a heading of three lines: the name of the business, the name of the statement, and the period or date. The income statement covers a period, so its third line reads, for example, *For the year ended 31 December 2026*. Without the period, a reader cannot judge whether Tk 2,000 is a good result for one week or a poor result for one year. Because the books follow the accrual basis of accounting, the statement reports revenue when it is earned and matches expenses to that revenue, as the matching principle requires.

Every statement has a three-line heading: who, which statement, and when.

The second answer comes from the **statement of owner's equity**. A statement of owner's equity is a report showing how the owner's capital changed during a period. Owner's equity is the owner's claim on the business. Some books call it capital, net worth or proprietorship; this book uses owner's equity for the idea, and Owner's Capital as the name of the account. The stall began with capital of Tk 2,500, and the statement shows how that figure became a new one.

The second statement answers: how did the owner's stake change?

Statement of owner's equity. A report showing how the owner's capital changed during a period.

The calculation has three parts. Ending capital equals starting capital, plus net income, minus drawings. Drawings are amounts that the owner withdraws from the business for personal use. For the stall, the steps are $\text{Tk } 2,500 + \text{Tk } 2,000 - \text{Tk } 500 = \text{Tk } 4,000$. A net loss would be subtracted instead of added, and any extra money that the owner invests during the period would be added.

Ending capital equals starting capital plus net income minus drawings.

The third answer comes from the **balance sheet**. A balance sheet is a report of assets, liabilities and owner's equity at one date. Assets are things of value that the business owns. Liabilities are amounts that the business owes. The report is called a balance sheet because its two sides must be equal: assets equal liabilities plus owner's equity. Some countries and standards use the title statement of financial position for the same report; this book uses balance sheet.¹

The third statement answers: what does the business own and owe on one date?

Balance sheet. A report of assets, liabilities and owner's equity at one date.

The table below shows the stall's balance sheet. The left part lists what the stall owns, and the lower part lists what it owes and what belongs to the owner. The two totals must agree.

The balance sheet for the stall shows that both sides equal Tk 5,000.

BALANCE SHEET OF THE TEA STALL AT THE END OF THE MONTH

ITEM	TK
Cash	3,700
Equipment	1,300
Total assets	5,000
Accounts payable (owed to the supplier)	1,000
Owner's Capital	4,000
Total liabilities and owner's equity	5,000

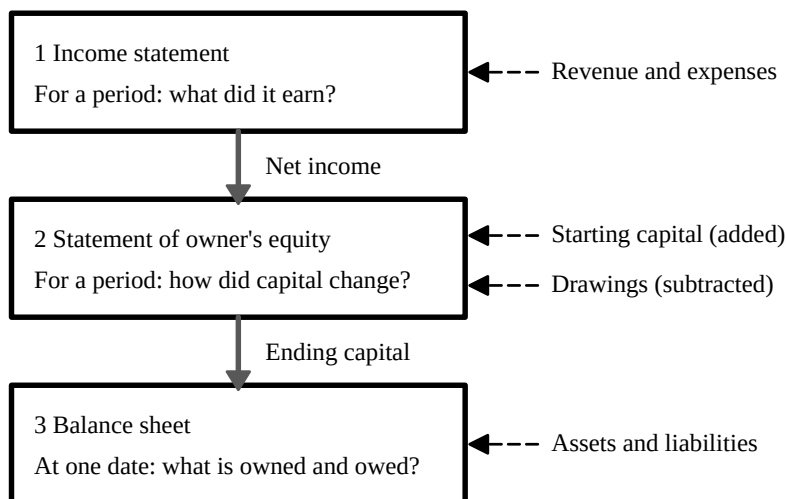
¹ The International Financial Reporting Standards use the title statement of financial position. The contents are the same as in a balance sheet. Which title is required depends on the law and standards of the country and on the kind of business.

The balance sheet shows one date, so its heading says at, not for.

The three statements are prepared in a fixed order because each feeds the next.

Notice the difference. The balance sheet shows a single date, so its third line reads *At 31 December 2026*. The other two statements cover a period.

The order matters, because each statement supplies a number to the next. The income statement is prepared first, because it produces net income. The statement of owner's equity comes second, because it needs net income and produces ending capital. The balance sheet comes last, because it needs ending capital. The figure below shows the links. Each box is one statement. Each arrow shows a number moving from one box to the next, and the words on the arrow name that number.



Solid line: a result passed on. Dashed line: data brought in.

FIG. 9.1. *The three statements and the numbers that link them*

Owners, lenders and managers read the three statements for different reasons.

Before moving on: each statement answers one question, and each feeds the next.

The owner reads the income statement to judge the work. A bank that lends money reads the balance sheet to see whether assets cover liabilities.

Before moving on. The income statement reports revenue, expenses and net income for a period. The statement of owner's equity reports how capital changed, using net income and drawings. The balance sheet reports assets, liabilities and owner's equity at one date. Prepare them in that order, and give each the right third line of heading.

* * *

9.2 CLOSING THE BOOKS

Some accounts must reset to zero at period end and others must not.

Look again at the tea stall's books after the month's results are known. Its Revenue account holds Tk 5,000, its Expenses account holds Tk 3,000, and its Drawings account holds Tk 500. Next month the owner wants the results of the new month only. If the old figures stay, each new sale is added to Tk 5,000, and no one can separate the two months. The books must be reset, but not every account may be reset. The Cash account must keep its Tk 3,700. Which accounts may be emptied, and which must not?

Accountants answer by dividing accounts into two groups. A **temporary account** is an account for revenue, expense or drawings, closed to zero at the end of each period. Revenue and Expenses are examples. A **permanent account** is an account for assets, liabilities or capital, whose balance carries forward to the next period. Cash and Accounts Payable are examples. A temporary account measures one period only. A permanent account shows the position of the business and has no end date.

The table below sorts the common account types. Read each row from left to right: the kind of account, its group, and what happens to its balance at the end of the period.

WHICH ACCOUNTS ARE TEMPORARY AND WHICH ARE PERMANENT

KIND OF ACCOUNT	GROUP	AT THE END OF THE PERIOD
Revenue	Temporary	Closed to zero
Expense	Temporary	Closed to zero
Drawings	Temporary	Closed to zero
Asset	Permanent	Balance carries forward
Liability	Permanent	Balance carries forward
Owner's Capital	Permanent	Updated, then carried forward

The tool that resets a temporary account is the closing entry. A **closing entry** is a journal entry at period end that transfers a temporary account's balance to capital. Suppose the stall's Revenue account has a credit balance of Tk 5,000. A closing entry debits Revenue for Tk 5,000. That debit cancels the credit, and the account stands at zero. The same amount must be credited somewhere, and that place is capital, or an account that leads to capital.

Most books use a helper account for this purpose. The **income summary** is a temporary account used to collect revenue and expenses before the net result goes to capital. Think of it as a small table where the month's revenue and expenses are placed side by side, so that the difference is easy to see. After the net result is moved to capital, the income summary itself stands at zero.

Closing takes four steps, and each step is one journal entry. The steps are performed in this order, after all adjusting entries have been recorded.

Step 1. Close each revenue account: debit it and credit Income Summary. Step 2. Close each expense account: credit it and debit Income Summary. Step 3. Close Income Summary to Owner's Capital: a credit balance means net income, which is credited to capital. Step 4. Close Drawings to Owner's Capital: credit Drawings and debit capital.

The table below shows the four entries for the tea stall. The stall has several expense accounts; their total is used here to keep the table short. In a real set of books, each expense account is credited separately.

Accounts that measure one period are temporary; accounts that show position are permanent.

Temporary account. An account for revenue, expense or drawings, closed to zero at the end of each period.

Permanent account. An account for assets, liabilities or capital, whose balance carries forward to the next period.

The table sorts the account types into temporary and permanent.

The word closing in accounting means moving a balance to capital so that an account starts again at zero.

Closing. Word in use. In daily life, shutting a shop for the night. In accounting, moving the balance of a temporary account to capital so that the account starts the new period at zero.

Closing entry. A journal entry at period end that transfers a temporary account's balance to capital.

A helper account collects revenue and expenses before the net result moves to capital.

Income summary. A temporary account used to collect revenue and expenses before the net result goes to capital.

Closing takes four steps, each one a journal entry, and the order is fixed.

The four steps close revenue, expenses, the income summary and drawings.

The table shows the four closing entries for the tea stall.

THE FOUR CLOSING ENTRIES OF THE TEA STALL

STEP	ACCOUNT	DEBIT (TK)	CREDIT (TK)
1	Revenue	5,000	
1	Income Summary		5,000
2	Income Summary	3,000	
2	Expenses (all accounts)		3,000
3	Income Summary	2,000	
3	Owner's Capital		2,000
4	Owner's Capital	500	
4	Drawings		500

After step 2, the income summary holds the net result: credits of revenue less debits of expenses.

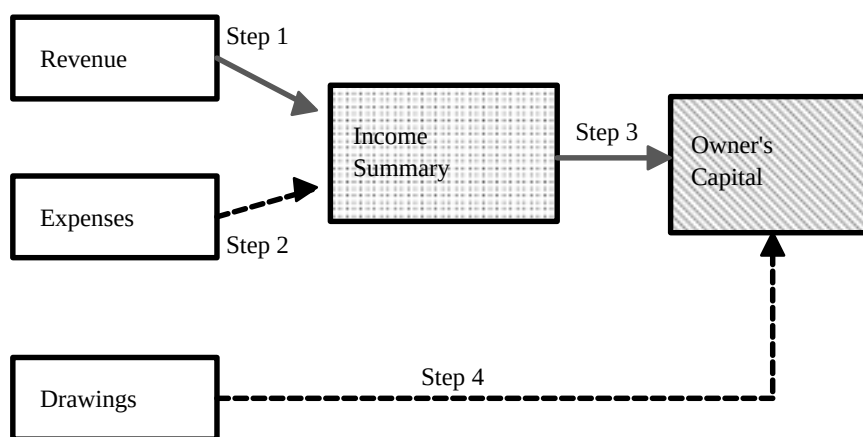
Follow the income summary through the steps. After step 1 it holds a credit of Tk 5,000. After step 2 it also holds a debit of Tk 3,000, so its balance is a credit of Tk 2,000. That balance is the net income. Step 3 debits Income Summary for Tk 2,000 and moves the amount to capital, and the account is left at zero. The rule is: a credit balance is closed by a debit of the same amount, and a debit balance is closed by a credit of the same amount.

PAUSE AND CHECK 9.P1 A shop has revenue of Tk 800 and expenses of Tk 500. After steps 1 and 2, what balance does Income Summary hold, and is it a debit or a credit?

The figure shows the four closing entries as arrows that end in Owner's Capital.

The figure below shows the same four entries as arrows. Revenue, Expenses and Drawings sit on the left. Income Summary sits in the middle. Owner's Capital sits on the right, and every path ends there. Each arrow is labelled with its step number. Where an arrow points shows where a balance moves; the line style tells the kind of balance: a solid line for a credit and a dashed line for a debit.

Revenue has a credit balance; expenses and drawings have debit balances.



Solid line: credit balance closed. Dashed line: debit balance closed.

FIG. 9.2. Closing entries as arrows into capital

Closing never erases records; it only resets the accounts that measure one period.

Three points often cause confusion. First, closing does not erase history. The journal keeps every entry, and the income statement keeps every figure. Second, drawings are not an expense. They do not pass through the income summary, because they are the owner's withdrawal and not a cost of earning revenue. Third, some books close revenue and

expenses directly to capital and skip the income summary.² The final capital is the same.

Permanent accounts are never closed; their balances continue. If the result of the period is a net loss, the income summary has a debit balance after step 2. Step 3 then credits Income Summary and debits Owner's Capital, which reduces capital.

Permanent accounts are never closed, and a net loss reverses the direction of step 3.

IN SHORT

Temporary accounts measure one period and return to zero through closing entries. Permanent accounts show the position and carry their balances forward.

Before moving on. A temporary account is closed to zero at the end of each period; a permanent account carries its balance forward. The four closing entries move revenue, expenses and drawings into Owner's Capital, using the income summary as a helper for the net result. After closing, only permanent accounts hold balances.

Before moving on: closing resets four kinds of balance and leaves the permanent accounts alone.

* * *

9.3 REVERSING ENTRIES

Case file, illustrative. At 31 December 2026, Meghna Binding Service owes its workers Tk 14,000 for wages earned in 2026 but not yet paid. The adjusting entry of that date debited Wages Expense and credited Wages Payable for Tk 14,000. On 5 January 2027 the workshop pays wages of Tk 28,000. Half of this payment settles the old debt, and half pays for work done in 2027. How should the payment be recorded?

A payment that covers two periods creates extra work for the person who records it.

The difficulty is the split. If the bookkeeper records the full Tk 28,000 as an expense, the Tk 14,000 of 2026 is counted twice: once by the adjusting entry, and again by the payment. To avoid this, the bookkeeper must remember to debit Wages Payable and Wages Expense for Tk 14,000 each. With many such payments, memory is not a safe method.

Without a reversal, the bookkeeper must split the payment by hand, and a slip counts the same wages twice.

The remedy is the **reversing entry**. A reversing entry is an entry on the first day of a new period that cancels an earlier accrual so that routine payment can be recorded in the usual way. Here the reversing entry is the exact opposite of the adjusting entry: it debits Wages Payable and credits Wages Expense for Tk 14,000. After it, the payment on 5 January needs no thought. The bookkeeper debits Wages Expense and credits Cash for the full amount.

A reversing entry cancels the earlier accrual so that payment can be recorded in the usual way.

Reversing entry. An entry on the first day of a new period that cancels an earlier accrual so that routine payment can be recorded in the usual way.

The table below shows the two entries in 2027. Read the rows by date: the reversal first, then the payment.

The table shows the reversal and the payment, with their dates.

² This shorter method gives the same ending capital. It is used less often in teaching, because the income summary shows the net result in one place before it moves to capital.

REVERSAL AND PAYMENT OF WAGES, MEGHNA BINDING SERVICE, 2027

DATE	ACCOUNT	DEBIT (TK)	CREDIT (TK)
1 January 2027	Wages Payable	14,000	
1 January 2027	Wages Expense		14,000
5 January 2027	Wages Expense	28,000	
5 January 2027	Cash		28,000

After both entries, the expense account holds only the wages that belong to 2027.

Reverse accruals that routine payments will settle; most deferrals need no reversal.

Reversing is a convenience, not a requirement, and the result is the same either way.

The reversal changes nothing in the old period's statements and corrects the new period's expense.

Before moving on: reverse accruals on the first day of the new period; the payment then follows the usual routine.

Now examine the accounts. Wages Payable holds a credit of Tk 14,000 from December and a debit of Tk 14,000 from the reversal, so it stands at zero. Wages Expense holds a credit of Tk 14,000 from the reversal and a debit of Tk 28,000 from the payment, so its balance is a debit of Tk 14,000. That is exactly the wages for work done in 2027. The old debt is gone, the new expense is correct, and the payment was recorded without any splitting.

The next question is when to reverse. The usual answer is: reverse an accrual that a later routine payment or receipt will settle. Wages payable is the classic case. Interest owed or interest earned but not yet received is another. Deferrals are different. In the deferral entries of Chapter 7, section 7.2, the original payment or receipt was first recorded in an asset or liability account, such as Prepaid Insurance or Unearned Revenue. The adjusting entry already left each balance correct, so a reversal is not needed.

Reversing is a convenience and never a requirement. A bookkeeper who splits the payment correctly by hand reaches the same balances. Two rules keep the method safe. Date the reversal on the first day of the new period, after the old period has been closed and its statements prepared. And never reverse a closing entry, because closing entries must stay in place for the period that ended.

The effect on the next period is easy to state. The balance sheet at 31 December 2026 still shows wages payable of Tk 14,000, because the reversal is dated 2027. The income statement for 2027 shows only the wages that belong to 2027. If the bookkeeper forgets the reversal and records the full payment, the 2027 wages are overstated by Tk 14,000.

Before moving on. A reversing entry cancels an earlier accrual on the first day of a new period. It is used for accruals that routine payments will settle, and it is optional. After the reversal, the full payment is recorded in the usual way, and the expense account shows only the new period's share.

* * *

A WORKED EXAMPLE: STATEMENTS AND CLOSING FOR MEGHNA BINDING SERVICE

Here is the data for the workshop's first year, used in every step below.

Case file, illustrative. Meghna Binding Service, statements for 2026. Revenue Tk 788,000. Expenses Tk 604,000. Capital at 1 January 2026 Tk 400,000. Drawings for the year Tk 80,000. Assets at 31 December 2026: cash Tk 150,000, accounts receivable Tk 110,000, supplies Tk 15,000, prepaid insurance Tk 27,000, equipment (net) Tk 288,000. Liabilities: accounts payable Tk 60,000, wages payable Tk 14,000, unearned revenue Tk 12,000.

The first calculation finds net income from the income statement.

Case file, illustrative. Calculation 1: net income. *Given:* revenue Tk 788,000; expenses Tk 604,000. *Find:* net income for 2026. *Formula:* net income equals revenue minus expenses; in symbols, $N = R - E$, where N is net income, R is revenue and E is expenses. *Steps:* Tk 788,000 – Tk 604,000 = Tk 184,000.

Answer: the net income of the workshop for the year ended 31 December 2026 is Tk 184,000. *Check:* revenue is larger than expenses, so the result must be a net income; and Tk 184,000 is about 23 per cent of Tk 788,000, a reasonable share.

Case file, illustrative. Calculation 2: capital at the end of the year. *Given:* capital at the start Tk 400,000; net income Tk 184,000; drawings Tk 80,000. *Find:* capital at 31 December 2026. *Formula:* ending capital equals starting capital plus net income minus drawings; in symbols, $C_1 = C_0 + N - D$. *Steps:* Tk 400,000 + Tk 184,000 = Tk 584,000; then Tk 584,000 – Tk 80,000 = Tk 504,000. *Answer:* ending capital is Tk 504,000. *Check:* the owner added net income and withdrew less than that, so capital must have grown: Tk 504,000 is larger than Tk 400,000.

Case file, illustrative. Calculation 3: the balance sheet. *Given:* the assets, liabilities and ending capital above. *Find:* total assets, total liabilities, and whether the two sides agree. *Formula:* assets equal liabilities plus owner's equity; in symbols, $A = L + E$. *Steps:* assets are Tk 150,000 + Tk 110,000 + Tk 15,000 + Tk 27,000 + Tk 288,000 = Tk 590,000. Liabilities are Tk 60,000 + Tk 14,000 + Tk 12,000 = Tk 86,000. Then $L + E = Tk 86,000 + Tk 504,000 = Tk 590,000$. *Answer:* the balance sheet has total assets of Tk 590,000, and both sides agree. *Check:* the equity of Tk 504,000 also equals assets less liabilities: Tk 590,000 – Tk 86,000.

The table below sets the balance sheet out in full. Read it from the top: assets, then liabilities, then owner's equity, and last the two equal totals.

BALANCE SHEET OF MEGHNA BINDING SERVICE AT 31 DECEMBER 2026

ITEM	TK
Cash	150,000
Accounts receivable	110,000
Supplies	15,000
Prepaid insurance	27,000
Equipment (net)	288,000
Total assets	590,000
Accounts payable	60,000
Wages payable	14,000
Unearned revenue	12,000
Total liabilities	86,000
Owner's Capital	504,000
Total liabilities and owner's equity	590,000

The books are now closed. The table below shows the four closing entries of 31 December 2026. Revenue and expenses are shown as totals, because the case file gives totals only.

The second calculation finds ending capital for the statement of owner's equity.

The third calculation totals assets and liabilities and tests the balance sheet.

The table presents the workshop's balance sheet at 31 December 2026.

The four closing entries, dated 31 December 2026, use the same figures.

CLOSING ENTRIES OF MEGHNA BINDING SERVICE, 31 DECEMBER 2026

STEP	ACCOUNT	DEBIT (TK)	CREDIT (TK)
1	Revenue	788,000	
1	Income Summary		788,000
2	Income Summary	604,000	
2	Expenses (all accounts)		604,000
3	Income Summary	184,000	
3	Owner's Capital		184,000
4	Owner's Capital	80,000	
4	Drawings		80,000

The check: Income Summary ends at zero, and capital ends at the figure in the statement.

Check the closing. After step 1, Income Summary holds a credit of Tk 788,000. After step 2 it holds Tk 788,000 – Tk 604,000 = a credit of Tk 184,000. Step 3 removes it, so the account is at zero. Capital is Tk 400,000 + Tk 184,000 = Tk 584,000 after step 3 and Tk 584,000 – Tk 80,000 = Tk 504,000 after step 4. This equals the balance sheet figure. On 1 January 2027, the reversal and the payment of 5 January from section 9.3 leave Wages Expense with a debit of Tk 14,000, the share that belongs to 2027.

Four slips are common when students perform these steps.

Four slips are common. The first is a wrong sign: adding drawings to capital instead of subtracting them. The second is a mixed-up period: heading the balance sheet *For the year ended* instead of *At*. The third is a forgotten liability, such as Tk 12,000 of unearned revenue, which makes the two sides disagree. The fourth is closing Drawings through the income summary, which understates net income in the report.

* * *

SUMMARY

The summary lists the five ideas of the chapter.

The income statement, the statement of owner's equity and the balance sheet are prepared in that order, each feeding the next. After they are prepared, closing entries move the balances of temporary accounts into Owner's Capital through the income summary, and permanent accounts carry their balances forward. On the first day of the new period, a reversing entry cancels an accrual so that the routine payment is recorded in the usual way. For Meghna Binding Service, the year produced net income of Tk 184,000 and ending capital of Tk 504,000.

THINK IT THROUGH

9.11 Suppose Meghna Binding Service did not close its Revenue account on 31 December 2026, and then recorded Tk 900,000 of revenue during 2027. What balance would the account show at the end of 2027, and why would that figure mislead a reader?

9.12 Explain in your own words why the balance sheet cannot be prepared before the statement of owner's equity.

WHERE PEOPLE GO WRONG

Some students close every account, including Cash. This is wrong. Only revenue, expense and drawings accounts are closed. Assets, liabilities and capital show the position of the business and must carry their balances forward.

Some students charge drawings to the income summary with the expenses. Drawings are the owner's withdrawal, not a cost of earning revenue. They are closed directly to Owner's Capital, and they never reduce net income.

Some students think the books are wrong without a reversal. They are not. A reversal only makes later recording easier. Deferrals recorded in asset or liability accounts usually need no reversal at all.

Mistake one: closing permanent accounts, which would erase the position of the business.

Mistake two: treating drawings as an expense.

Mistake four: believing that reversing entries are always required.

IN HISTORY

Before 1844, a company in the United Kingdom could usually gain a separate legal existence only by royal charter or by a private Act of Parliament. In 1844 Parliament passed the Joint Stock Companies Act, which received royal assent on 5 September 1844 and came fully into force on 1 November 1844. A company could now register with an official called the Registrar of Joint Stock Companies. The Act also placed duties on registered companies. They had to keep books of account, and their directors had to prepare a "full and fair" balance sheet. Auditors then examined it, and a copy went to the Registrar for the public to inspect. Parliament later replaced the Act with the Joint Stock Companies Act 1856. Today the law of each country sets how a company reports to its owners.

The Joint Stock Companies Act 1844 made a balance sheet a legal duty of registered companies.

REVIEW QUESTIONS

9.1 State the question that each of the three statements answers, and name the third line of each heading.

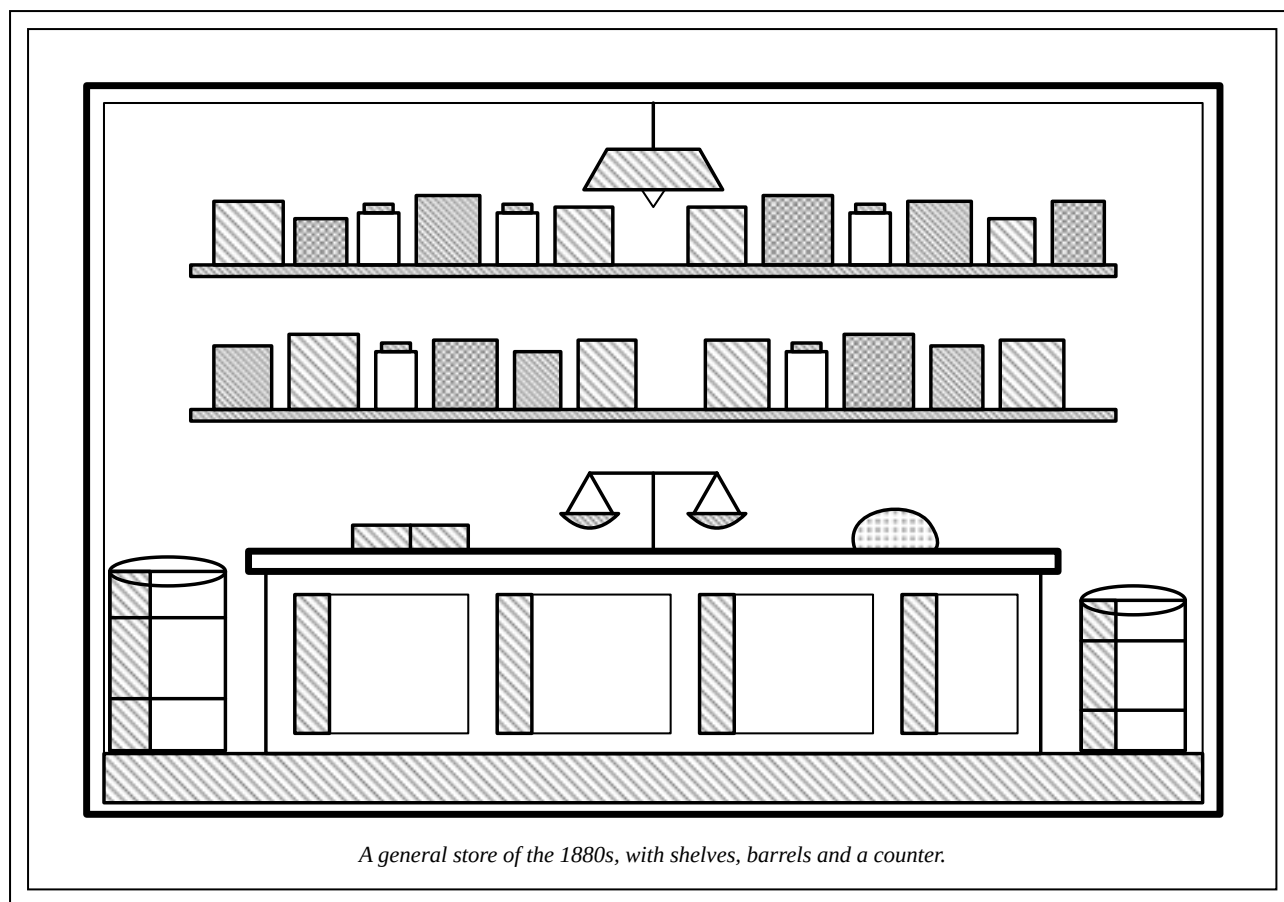
9.2 A small shop has this adjusted trial balance at 31 December 2026 (in Tk): Cash 6,000; Accounts Receivable 2,000; Equipment 9,000; Drawings 1,000; Expenses 7,000 (debits); Accounts Payable 3,000; Owner's Capital 10,000; Revenue 12,000 (credits). Prepare the three statements.

9.3 Write the four closing entries for the shop in question 9.2.

9.4 Sort these accounts into temporary and permanent: Revenue, Prepaid Insurance, Wages Expense, Drawings, Wages Payable, Owner's Capital.

9.5 Wages of Tk 6,000 were earned but unpaid at 31 December 2026, and Tk 8,000 was paid on 4 January 2027. Write the reversing entry and the payment, and state the wages expense of 2027.

9.6 A business has revenue of Tk 9,000 and expenses of Tk 11,000 for the year. Write the third closing entry and say what happens to capital.



CHAPTER X

Accounts of a Merchandising Company

If a shop buys goods all year, why is its cost of goods not the amount it spent?



A business that buys goods and sells them again needs accounts that a service business does not. This chapter teaches how sales, purchases and returns are recorded, how the income statement and balance sheet are arranged in classes, and how mistakes in the books are classified and corrected.

This chapter builds on two sections of Chapter 9.

Recall: the three statements, Chapter 9, 'Financial Statements and Closing Entries', section 9.1 'The Three Statements'.

Recall: closing the books, Chapter 9, 'Financial Statements and Closing Entries', section 9.2 'Closing the Books'.

A bookseller who still holds unsold books cannot find profit by subtracting all purchases.

A business that buys goods to sell again must separate goods sold from goods still held.

The running case shows the same gap between the goods bought and the cost of the goods sold.

The chapter has four sections and one worked example.

The word sales means revenue from goods sold, not a price reduction event.

Sales. Word in use. In daily life, the act of selling, or a time when prices are lowered. In accounting, the revenue earned from goods sold to customers.

A sales return is goods sent back; an allowance is a price reduction for a faulty item.

Net sales is the revenue the business really keeps after returns and allowances.

Net sales. Sales less sales returns and allowances; the revenue a merchandising business really keeps.

You will need two sections of Chapter 9, 'Financial Statements and Closing Entries': section 9.1 'The Three Statements' and section 9.2 'Closing the Books'. You will also use Appendix A1, 'Maths You Will Use', for percentages.

A bookseller buys 100 books at Tk 100 each and sells 80 of them at Tk 150 each. What profit did the bookseller make? A hasty answer subtracts the cost of all 100 books from the takings: Tk 12,000 – Tk 10,000 = Tk 2,000. That answer is wrong, and it is wrong in a useful way. Twenty books are still on the shelf. They have not been lost or used, and they are still worth Tk 2,000 to the bookseller. Only the cost of the 80 books that were sold, Tk 8,000, belongs against the takings, so the profit is Tk 4,000.

A business that buys goods in order to sell them again is called a merchandising business. Its main difficulty is the one in the puzzle: at any date, some of the goods bought have been sold, and some have not. The accounts of such a business must separate the two.

Case file, illustrative. Meghna and Sons, the family bookshop in Dhanmondi, Dhaka, bought books costing Tk 10,200,000 during 2025, before returns. Yet its cost of goods sold for 2025 is Tk 9,800,000. The difference is explained by opening inventory, purchase returns and closing inventory, as this chapter shows.

This chapter has four sections. Section 10.1 teaches sales, purchases, returns and the cost of goods sold. Section 10.2 arranges these in a classified income statement. Section 10.3 arranges the balance sheet in the same way. Section 10.4 teaches the kinds of error that can occur in the books and how each is corrected. A worked example with Meghna and Sons follows.

* * *

10.1 SALES, PURCHASES AND RETURNS

Think of a stationery seller with a small stall. In one month she sells goods for Tk 10,000. That figure is her sales. Then a customer returns notebooks worth Tk 500 because the pages are loose, and she gives another customer a price reduction of Tk 200 because a pen is scratched. Does she still have Tk 10,000 of sales? She does not, because she did not keep all of that money. Each of these cases has a name, and the accounts record them separately.

A sales return is goods that a customer sends back, and the business refunds or cancels the amount. A sales allowance is a reduction in the price that a customer agrees to pay because the goods are faulty. Both are recorded in their own accounts, and not by changing the original sale. The business can then see how often goods come back.

The amount that remains is called **net sales**. Net sales means sales less sales returns and allowances, the revenue a merchandising business really keeps. For the stationery seller, net sales are Tk 10,000 less Tk 500 less Tk 200. This figure, and not the larger one, is the first line of the income statement.

Given: sales Tk 10,000; sales returns Tk 500; sales allowances Tk 200. *Find:* net sales. *Formula:* net sales equal sales minus sales returns minus sales allowances; in symbols, $NS = S - R - A$. *Steps:* Tk 10,000 – Tk 500 = Tk 9,500; then Tk 9,500 – Tk 200 = Tk 9,300. *Answer:* net sales are Tk 9,300. *Check:* net sales must be smaller than sales, and Tk 9,300 is.

On the buying side the same pattern appears. The Purchases account records goods bought for resale. It does not record shop fittings or supplies, because those are not sold. A purchase return is goods that the business sends back to a supplier, recorded in its own account. For the stall, goods bought during the month cost Tk 6,000, and notebooks worth Tk 300 were sent back to the supplier. Purchases less purchase returns give net purchases of Tk 5,700.

Now we can solve the opening puzzle. The **cost of goods sold** is the cost of the goods that were sold during the period. The business does not record the cost of each sale as it happens. It counts the goods that remain at the end of the period, and from that count it calculates the cost of the goods sold.¹ Goods that remain are called inventory, and they are an asset, not an expense, until they are sold.

The formula has four parts. Opening inventory is the inventory held at the start of the period. Add purchases and subtract purchase returns. The result is the goods available for sale. Then subtract closing inventory, the inventory still held at the end. In symbols, $C = O + P - R - E$, where C is cost of goods sold, O is opening inventory, P is purchases, R is purchase returns and E is closing inventory.

Given: opening inventory Tk 2,000; purchases Tk 6,000; purchase returns Tk 300; closing inventory Tk 1,700. *Find:* cost of goods sold. *Formula:* $C = O + P - R - E$. *Steps:* Tk 2,000 + Tk 6,000 = Tk 8,000; Tk 8,000 – Tk 300 = Tk 7,700; Tk 7,700 – Tk 1,700 = Tk 6,000. *Answer:* the cost of goods sold is Tk 6,000. *Check:* goods available for sale were Tk 7,700, and Tk 1,700 of them remain, so Tk 6,000 must have been sold.

PAUSE AND CHECK 10.P1 A shop starts the month with inventory of Tk 500, buys goods for Tk 2,000, and counts Tk 700 of inventory at the end. What is the cost of goods sold?

Case file, illustrative. In 2025, Meghna and Sons began with inventory of Tk 3,300,000 and bought books for Tk 10,200,000. It sent books worth Tk 200,000 back to suppliers and counted inventory of Tk 3,500,000 at the end. So Tk 3,300,000 + Tk 10,200,000 – Tk 200,000 – Tk 3,500,000 = Tk 9,800,000. The bookshop bought Tk 10,200,000 of books, yet the cost of the books it sold was Tk 9,800,000.

Each person with an interest sees something different here. The owner wants to know how much revenue the business keeps. A manager watches the returns, because a high figure suggests faulty goods or poor service. A customer sees only the refund, but the refund is the reason the business records the return. Returns also show the limit of the method: if inventory is counted wrongly, the cost of goods sold is wrong by the same amount.

Three slips are common. If the closing count is forgotten, the cost of goods sold is too large and profit is too small. If purchase returns are forgotten, the same happens. If fittings are placed in the Purchases account, the cost of goods sold is too large, because fittings are not goods for resale.

The first calculation finds net sales for the stall.

Purchases are goods bought for resale, and purchase returns are goods sent back to the supplier.

Cost of goods sold takes the goods available for sale and removes the goods still held.

Cost of goods sold. The cost of the goods that were sold during the period.

The formula starts with opening inventory, adds net purchases and removes closing inventory.

The second calculation finds the cost of goods sold for the stall.

The bookshop's 2025 figures resolve the opening puzzle.

Owners, managers and customers each care about returns for a different reason.

Three common slips: forgetting the closing count, forgetting returns, and counting fittings as purchases.

¹ Some businesses update their inventory records with every sale and so do not need to calculate the cost of goods sold at the end. This chapter uses the count at the end of the period.

Before moving on: net sales and cost of goods sold both remove what the business did not really sell.

Before moving on. Net sales are sales less returns and allowances. The cost of goods sold equals opening inventory plus purchases less purchase returns less closing inventory. Unsold goods remain an asset until they are sold.

* * *

10.2 CLASSIFIED INCOME STATEMENT

A classified income statement groups items so that the reader can see the shop's two kinds of result.

Gross profit is the first result: net sales less the cost of the goods sold.

Gross profit. Net sales less cost of goods sold, before operating expenses are deducted.

Return to the stationery seller. Her net sales are Tk 9,300 and her cost of goods sold is Tk 6,000. She also paid for advertising, delivery, rent and wages. If all these costs were placed in one long list, she could not tell whether her trading with customers or her running costs were the problem. A classified income statement answers by grouping items into classes. It first shows the result of buying and selling goods, and then the cost of running the business.

The first result is the **gross profit**. Gross profit is net sales less cost of goods sold, before operating expenses are deducted. For the stationery seller, Tk 9,300 – Tk 6,000 = Tk 3,300. The figure below shows gross profit as the gap between two bars. The left bar is net sales, and the right bar is the cost of goods sold. The height of each bar shows the amount, and the gap at the top of the right bar is the gross profit.

Height of a bar shows the amount.

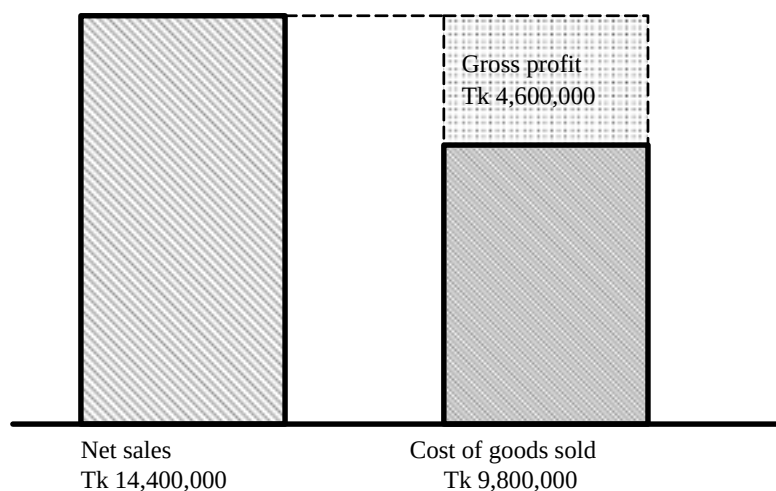


FIG. 10.1. Gross profit as the gap between net sales and cost of goods sold

Operating expenses are the costs of running the business, apart from the cost of goods.

Operating expense. A cost of running the business, such as wages or rent, apart from the cost of goods sold.

The third calculation finds the stall's profit.

The second class contains the operating expenses. An **operating expense** is a cost of running the business, such as wages or rent, apart from the cost of goods sold. Most statements divide operating expenses into two groups. Selling expenses are costs of finding customers and delivering goods, such as advertising. Administrative expenses are costs of managing the business as a whole, such as office rent and salaries. The groups show where the money goes.

Given: gross profit Tk 3,300; selling expenses Tk 900; administrative expenses Tk 1,200. Find: profit. Formula: profit equals gross profit minus operating expenses; in symbols, $P = G - (V + D)$, where G is gross profit, V is selling expenses and D is administrative expenses. Steps: Tk 900 + Tk 1,200 = Tk 2,100; then Tk 3,300 – Tk 2,100 = Tk 1,200. Answer: profit is Tk 1,200.² Check: profit must be smaller than gross profit, and Tk 1,200 is.

² In this chapter the case file calls the final result profit. It is the same figure that Chapter 9 calls net income.

Percentages make a comparison possible. The stationery seller's gross profit is Tk 3,300 on net sales of Tk 9,300, which is about 35 per cent. A shop with a higher percentage keeps more from each taka of sales. Appendix A1 shows how to calculate a percentage. Two shops can have the same gross profit and different profit, because one of them spends more on running costs. That is the reason the statement separates the two classes.

The next figure shows the layout. The six boxes are the lines of the statement from top to bottom. The two heavy boxes are the results: gross profit after the first three lines, and profit at the foot. The brackets on the right show the two parts: the goods, and the running costs.

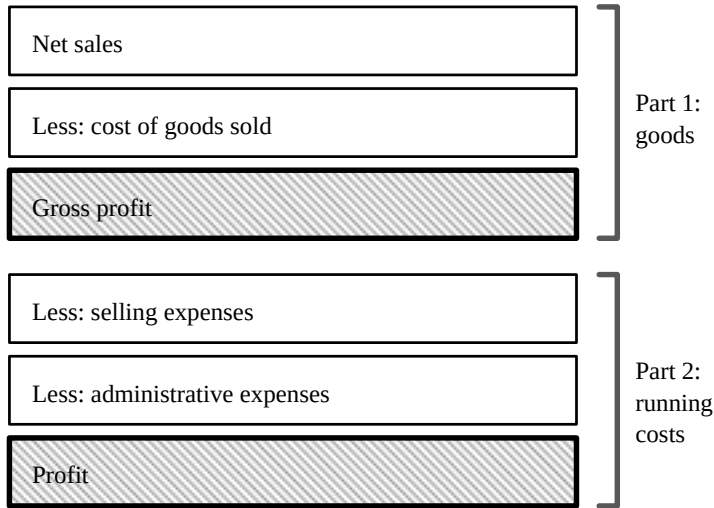


FIG. 10.2. The layout of a classified income statement

Case file, illustrative. The table below shows the 2025 income statement of Meghna and Sons. Read it from top to bottom: net sales, the build-up of the cost of goods sold, gross profit, the two groups of operating expenses, and profit.

CLASSIFIED INCOME STATEMENT OF MEGHNA AND SONS, YEAR ENDED 31 DECEMBER 2025 (TK)

ITEM	TK
Net sales	14,400,000
Opening inventory	3,300,000
Add: purchases	10,200,000
Less: purchase returns	(200,000)
Less: closing inventory	(3,500,000)
Cost of goods sold	9,800,000
Gross profit	4,600,000
Selling expenses	1,990,000
Administrative expenses	1,860,000
Total operating expenses	3,850,000
Profit	750,000

Before moving on. Gross profit is net sales less cost of goods sold. Operating expenses are the costs of running the business and are grouped as selling and administrative. Profit is gross profit less operating

A percentage of net sales lets the reader compare gross profit across years or shops.

The figure shows the order of the lines in a classified income statement.

Meghna and Sons' 2025 statement puts its figures in this layout.

Before moving on: the statement shows gross profit first, then operating expenses, then profit.

expenses. The classes tell the reader where the business gains and where it spends.

* * *

10.3 CLASSIFIED BALANCE SHEET

The word current in accounting means within one year.

Current. Word in use. In daily life, happening now. In accounting, expected to be used, collected or paid within one year.

Current assets are turned into cash or used up within a year.

Current asset. An asset expected to be turned into cash or used up within one year.

Current liabilities are debts that must be paid within a year.

Current liability. A debt that must be paid within one year.

Non-current items last longer than a year, and owner's equity forms its own group.

Meghna and Sons' balance sheet at 31 December 2025 follows the classes.

Suppose you run a shop and wish to know what money you can count on in the next twelve months, and what you must pay in the same time. A single list of assets and liabilities will not tell you, because a shop counter and a coin in the till are both assets, yet only one of them can pay a bill next month. The classified balance sheet solves this by separating items that are near to cash, or near to payment, from those that are not.

A **current asset** is an asset expected to be turned into cash or used up within one year. Cash is the plainest example. Accounts receivable, the amounts that customers owe, will be collected soon. Inventory will be sold. Prepaid expenses, such as insurance paid in advance, will be used up. Current assets are listed in the order in which they come closest to cash.

A **current liability** is a debt that must be paid within one year. Accounts payable, the amounts the business owes to suppliers, are the usual example. A long loan can also contribute a current part: the instalments due in the next twelve months are shown as current, and the rest is shown separately.

Items that last or fall due after more than one year are non-current. A non-current asset, such as shop fittings, serves the business for several years. A non-current liability, such as a long-term loan, is paid after more than a year. Owner's equity is shown in a group of its own. Fittings are shown net, meaning after the fall in value that Chapter 16, 'Depreciation', teaches.

Case file, illustrative. The table below shows the balance sheet of Meghna and Sons at 31 December 2025. Read it in three parts: assets first, then liabilities, then owner's equity. The two totals at the foot must agree.

CLASSIFIED BALANCE SHEET OF MEGHNA AND SONS AT 31 DECEMBER 2025 (TK)

ITEM	TK
Cash	420,000
Accounts receivable	280,000
Inventory	3,500,000
Prepaid expenses	60,000
Total current assets	4,260,000
Fittings (net)	600,000
Total assets	4,860,000
Accounts payable	1,150,000
Loan due within one year	120,000
Total current liabilities	1,270,000
Long-term loan	480,000
Owner's capital (2,860,000 + 750,000 – 500,000)	3,110,000
Total liabilities and owner's equity	4,860,000

The capital line comes from the statement of owner's equity of Chapter 9.

The capital figure is not new. It is the result of the statement of owner's equity: Tk 2,860,000 at the start, plus the profit of Tk 750,000, less drawings of Tk

500,000. The three statements of Chapter 9 therefore link here to the classified layout.

PAUSE AND CHECK 10.P2 A loan of Tk 90,000 is repayable in four equal yearly instalments of Tk 22,500, the first due in six months. Which amount is a current liability?

Different readers use these classes. A bank that lends money to Meghna and Sons compares current assets of Tk 4,260,000 with current liabilities of Tk 1,270,000. The current assets are about three times as large, which suggests that the shop can pay its debts of the coming year. The owner reads the same figures to decide how much to withdraw. One limit must be remembered: inventory is a current asset, but it becomes cash only if customers buy it.

Before moving on. A current asset is turned into cash or used up within one year. A current liability must be paid within one year. Items beyond one year are non-current, and owner's equity forms its own group.

* * *

10.4 CORRECTION OF ERRORS

The trial balance is the list that tests whether total debits equal total credits. Many students believe that when it agrees, the books must be right. They are not always right. Some mistakes leave the totals equal. Accountants name four kinds, and each has its own correction.

An **error of omission** is a mistake in which a transaction is left out of the books completely. A credit sale of Tk 800 that is never entered is an example. Neither side is recorded, so the trial balance still agrees.

An **error of commission** is a mistake in which an entry is made in the right class of account but the wrong account, or with a wrong amount. Tk 700 paid for electricity and debited to Telephone Expense is an example. Both are expenses, so the class is right and the account is wrong.

An **error of principle** is a mistake in which an entry is made in the wrong class of account, such as an expense recorded as an asset. Tk 40,000 spent on a delivery van and debited to Repairs Expense is an example. A van is an asset, and the entry placed it among the expenses.

A **compensating error** is two or more mistakes that cancel each other so that the trial balance still agrees. Suppose Wages is debited Tk 300 too much and Rent is debited Tk 300 too little. The debit total is correct, though both accounts are wrong.

The remedy for each is a correcting entry, which is a new journal entry that fixes the error without erasing the old record. The table below shows one correction for each kind. Read each row as the entry that would put the books right.

A lender compares what is due soon with what can be turned into cash soon.

Before moving on: current items fall within a year; non-current items last longer.

An agreeing trial balance does not prove that the books are free of error.

An error of omission leaves a transaction out of the books completely.

Error of omission. A mistake in which a transaction is left out of the books completely.

An error of commission uses the wrong account of the right class, or a wrong amount.

Error of commission. A mistake in which an entry is made in the right class of account but the wrong account, or with a wrong amount.

An error of principle puts an entry in the wrong class of account.

Error of principle. A mistake in which an entry is made in the wrong class of account, such as an expense recorded as an asset.

A compensating error is two mistakes that cancel each other in the trial balance.

Compensating error. Two or more mistakes that cancel each other so that the trial balance still agrees.

A correcting entry fixes an error with a new journal entry and erases nothing.

CORRECTING ENTRIES FOR THE FOUR KINDS OF ERROR

KIND OF ERROR	DEBIT	CREDIT	TK
Omission: unrecorded credit sale	Accounts Receivable	Sales	800
Commission: electricity in Telephone Expense	Electricity Expense	Telephone Expense	700
Principle: van in Repairs Expense	Vehicles	Repairs Expense	40,000
Compensating: Wages too high, Rent too low	Rent Expense	Wages Expense	300

A bookkeeper finds omission, principle and compensating errors by checking, not by the trial balance.

Before moving on: four kinds of error, and each is fixed by a new entry.

How are such errors found? A disagreeing trial balance warns of a one-sided mistake, and the bookkeeper searches for it. Errors of omission, principle and compensating errors do not disturb the totals, so only careful checks reveal them: comparing records with invoices, reviewing the accounts, and asking whether each entry used the right class. A good bookkeeper never relies on the trial balance alone. Because the books are permanent records, corrections are made by a new entry, dated and explained, so that the history stays visible.

Before moving on. An error of omission leaves out a whole transaction. An error of commission uses the wrong account of the right class, or a wrong amount. An error of principle uses the wrong class. A compensating error is two mistakes that cancel. Each is corrected by a new journal entry.

* * *

A WORKED EXAMPLE: MEGHNA AND SONS FOR 2025

The data for the bookshop's 2025 statements is given first.

Case file, illustrative. Meghna and Sons, 2025. Net sales Tk 14,400,000. Opening inventory Tk 3,300,000; purchases Tk 10,200,000; purchase returns Tk 200,000; closing inventory Tk 3,500,000. Selling expenses Tk 1,990,000; administrative expenses Tk 1,860,000. Capital at the start Tk 2,860,000; drawings Tk 500,000.

The first calculation finds the cost of goods sold and gross profit.

Case file, illustrative. Calculation 1. *Given:* the inventory and purchase figures above. *Find:* cost of goods sold and gross profit. *Formula:* $C = O + P - R - E$, then gross profit = net sales – C. *Steps:* Tk 3,300,000 + Tk 10,200,000 = Tk 13,500,000; less Tk 200,000 = Tk 13,300,000; less Tk 3,500,000 = Tk 9,800,000. Then Tk 14,400,000 – Tk 9,800,000 = Tk 4,600,000. *Answer:* cost of goods sold is Tk 9,800,000 and gross profit is Tk 4,600,000. *Check:* gross profit is about 32 per cent of net sales, a reasonable share for a bookshop.

The second calculation finds operating expenses and profit.

Case file, illustrative. Calculation 2. *Given:* selling expenses Tk 1,990,000; administrative expenses Tk 1,860,000; gross profit Tk 4,600,000. *Find:* operating expenses and profit. *Formula:* profit = gross profit – operating expenses. *Steps:* Tk 1,990,000 + Tk 1,860,000 = Tk 3,850,000; then Tk 4,600,000 – Tk 3,850,000 = Tk 750,000. *Answer:* operating expenses are Tk 3,850,000 and profit is Tk 750,000. *Check:* profit is smaller than gross profit.

The third calculation finds capital and tests the balance sheet.

Case file, illustrative. Calculation 3. *Given:* capital at the start Tk 2,860,000; profit Tk 750,000; drawings Tk 500,000; current assets Tk 4,260,000; fittings Tk 600,000; current liabilities Tk 1,270,000; long-term loan Tk 480,000. *Find:* ending capital, and whether assets equal liabilities plus owner's equity. *Formula:* ending capital = start + profit – drawings; $A = L + E$. *Steps:* Tk 2,860,000 + Tk 750,000 = Tk 3,610,000; less Tk 500,000 = Tk 3,110,000. Assets: Tk 4,260,000

+ Tk 600,000 = Tk 4,860,000. Liabilities: Tk 1,270,000 + Tk 480,000 = Tk 1,750,000; with capital, Tk 1,750,000 + Tk 3,110,000 = Tk 4,860,000. *Answer:* ending capital is Tk 3,110,000, and both sides total Tk 4,860,000. *Check:* Tk 4,860,000 – Tk 1,750,000 = Tk 3,110,000.

Now change one figure to see how the method responds. Suppose purchase returns had been Tk 300,000 instead of Tk 200,000. Then cost of goods sold would be Tk 3,300,000 + Tk 10,200,000 – Tk 300,000 – Tk 3,500,000 = Tk 9,700,000, and gross profit would be Tk 14,400,000 – Tk 9,700,000 = Tk 4,700,000. Profit would rise by Tk 100,000 to Tk 850,000, and capital would rise by the same amount. A larger return to suppliers lowers the cost of goods sold.

Three slips are common. The first is forgetting the closing inventory, which would give a cost of goods sold of Tk 13,300,000 and a gross profit of only Tk 1,100,000. The second is placing the loan due within one year among the non-current items. The third is adding drawings to capital instead of subtracting them.

Changing one figure shows how the statements respond.

Three slips are common in these calculations.

* * *

SUMMARY

A merchandising business records sales, returns, purchases and inventory. Net sales are sales less returns and allowances. The cost of goods sold is opening inventory plus purchases less purchase returns less closing inventory. Gross profit is net sales less cost of goods sold, and profit is gross profit less operating expenses. The balance sheet separates current from non-current items. Four kinds of error are corrected by new entries.

The summary gathers the chapter in a few lines.

THINK IT THROUGH

10.T1 Explain why a compensating error does not stop the trial balance from agreeing, and give an example of your own.

10.T2 A bookshop forgets to count its closing inventory of Tk 2,000 when it calculates its cost of goods sold. What happens to the cost of goods sold, to gross profit and to the balance sheet?

WHERE PEOPLE GO WRONG

Some students treat everything bought as the cost of goods sold. Only the goods that were sold count. Goods that remain are inventory, an asset, until they are sold.

Some students start from sales and forget the returns. Net sales are always smaller than sales when goods come back or prices are reduced.

Some students place a whole long loan below the current items. The installment due within one year is current, and only the rest is non-current.

Some students stop checking when the trial balance agrees. Omission, principle and compensating errors can all leave it in balance.

Mistake one: treating all purchases as the cost of goods sold.

Mistake two: ignoring returns and allowances.

Mistake three: treating a long loan as wholly non-current.

Mistake four: believing that an agreeing trial balance means no errors.

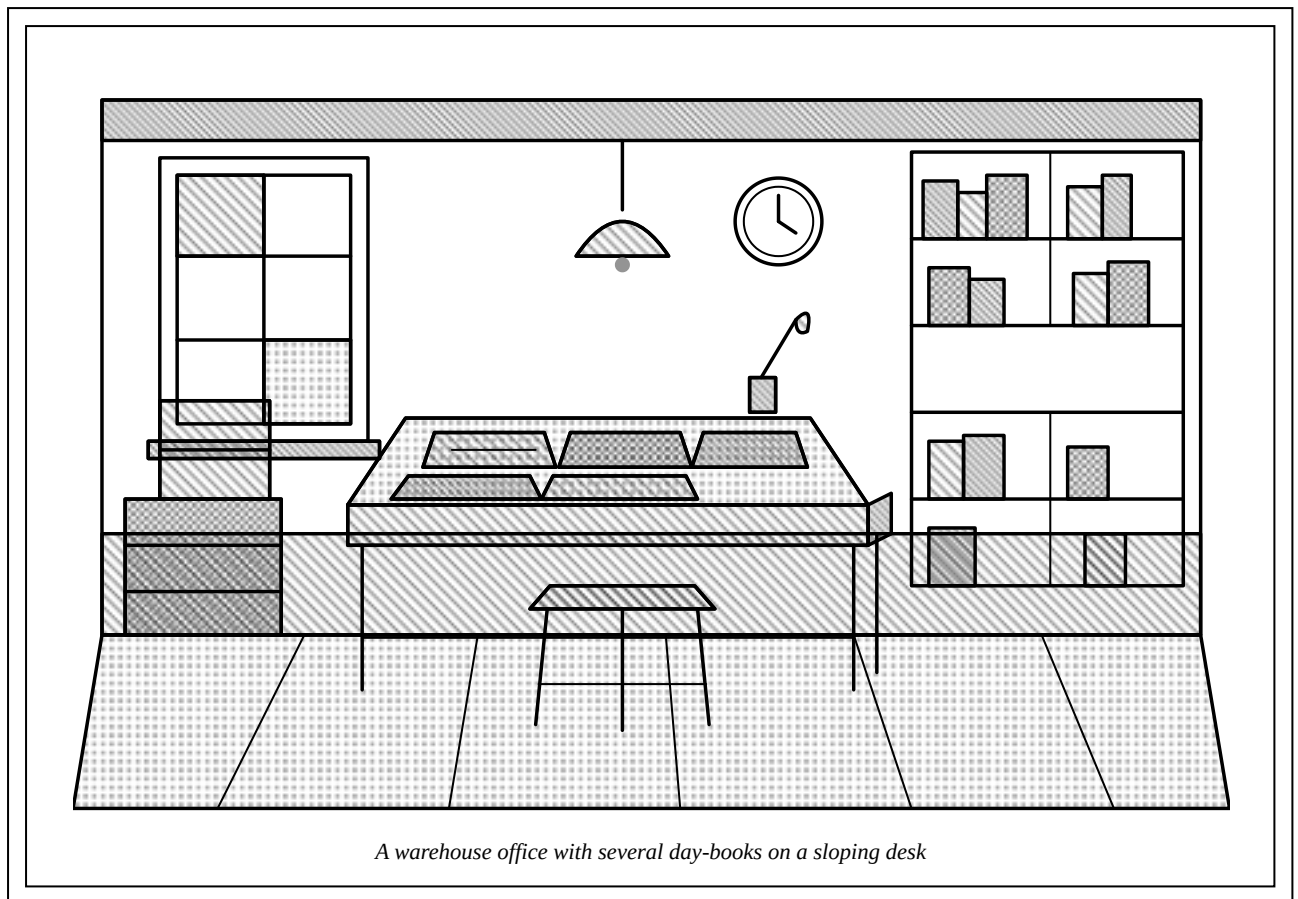
Frank Woolworth failed with his first store in 1879 and succeeded with his second.

IN HISTORY

Frank Winfield Woolworth was an American merchant, born in 1852. In 1879 he opened his first store in Utica, a city in New York State in the United States. It sold goods at a fixed price of five cents, and it failed within months. Later the same year he opened a second store in Lancaster, Pennsylvania, which succeeded and added goods at ten cents. A shop of this kind lives on the gap between the cost of its goods and the price at which they sell. A fixed price means that every customer pays the same amount for the same item. Woolworth's chain grew from that second store, and the gap between net sales and cost of goods sold, the gross profit, was the figure on which it depended.

REVIEW QUESTIONS

- 10.1 A shop has sales of Tk 5,000, sales returns of Tk 300 and sales allowances of Tk 100. Find net sales.
- 10.2 Prepare a classified income statement from this data (Tk): sales 80,000; sales returns 2,000; opening inventory 12,000; purchases 50,000; purchase returns 1,000; closing inventory 14,000; selling expenses 6,000; administrative expenses 7,000.
- 10.3 Prepare a classified balance sheet from this data (Tk): cash 3,000; accounts receivable 4,000; inventory 14,000; prepaid expenses 1,000; equipment (net) 20,000; accounts payable 5,000; loan due within one year 2,000; long-term loan 10,000; owner's equity 25,000.
- 10.4 Name the kind of error in each case: (a) a sale of Tk 900 never recorded; (b) a delivery van of Tk 40,000 debited to Repairs Expense; (c) electricity of Tk 700 debited to Telephone Expense; (d) Wages debited Tk 300 too much and Rent debited Tk 300 too little.
- 10.5 Write the correcting entries for errors (b) and (c) of question 10.4.
- 10.6 Classify each as a current asset, a non-current asset, a current liability or a non-current liability: Inventory; shop fittings; accounts payable; a loan due in four years; a loan due in six months.



CHAPTER XI

Special Journals

How can a shop record hundreds of credit sales without writing the same words hundreds of times?



This chapter shows how a business with many repeated transactions saves time and avoids error. You will meet the sales journal and the purchases journal, learn how their totals reach the ledger, and see how a control account checks the individual accounts of customers.

YOU WILL NEED

You will need the idea of the journal from Chapter 5 'The Journal and the Ledger', section 5.2 'The Journal', and the idea of posting from Chapter 5 'The Journal and the Ledger', section 5.3 'The Ledger and Posting'. Appendix A1 'Maths You Will Use' covers rounding and units, which help when you check totals.

*A busy shop faces a puzzle:
hundreds of similar sales, each
needing the same careful record.*

*Repeating the same entry many
times costs time and invites
mistakes.*

*The answer is to separate the
kinds of transaction and to give
each kind its own book.*

*This chapter builds that idea in
three steps and then adds a check
against error.*

Picture a stall in a market that sells tea and biscuits. Most customers pay at once. Three regular customers, however, take their goods now and promise to pay at the end of the month. Each of these sales must be recorded. The owner has learned that every transaction begins in the journal, so each credit sale receives its own entry: a date, the account to debit, the account to credit, the amount and a short explanation.

Now picture the stall growing into a large shop. It makes two hundred credit sales in one month. Two hundred journal entries must be written, and in each one the same two account names appear: the account receivable and the sales account. Then every entry must be posted to the ledger, which means two hundred postings to the sales account alone. A tired clerk might skip one entry, write a figure wrongly, or post the debit and forget the credit. Many students find this point puzzling, because the method works well for ten entries. The difficulty appears only at two hundred.

Think about how the shop could save effort. Every credit sale has the same shape. The debit is always the account receivable, and the credit is always sales. Only three things change from one sale to the next: the date, the customer and the amount. Suppose the shop prints those headings across a page and writes one line for each sale. The repeated words disappear. At the end of the month the clerk totals the amount column once and posts a single total to each of the two accounts. Two hundred postings shrink to two.

Section 11.1 names the idea and explains why a single journal reaches its limit. Section 11.2 shows the sales journal and the purchases journal and how their totals reach the ledger. Section 11.3 introduces a main account that sums many customer accounts, and a method to prove that the two agree. Our running example is a bookshop in Dhanmondi, Dhaka, where credit sales and credit purchases are frequent.

* * *

11.1 WHY SPECIAL JOURNALS

*The single journal records every
kind of transaction, so it repeats
the same words again and again.*

*Recall: the journal, Chapter 5,
'The Journal and the Ledger',
section 5.2 'The Journal'.*

Journal. Word in use. In daily life, a diary or a magazine on one subject. In accounting, a book where each transaction is first recorded, in date order.

In Chapter 5 you learned that the journal is the first book of record. Each transaction, supported by a source document such as an invoice, is written there in date order before it reaches the ledger. A journal of this kind accepts any transaction: a purchase of furniture, a payment of rent, a loan from a bank. For that reason we shall call it the general journal in this chapter. Its strength is flexibility. Its weakness appears when one kind of transaction occurs again and again.

The first limit is repetition. Each credit sale needs its own entry, and each entry names the same two accounts. The second limit is that only one person can write in one book at a time, so a second clerk must wait. The third limit is posting. Each entry is posted separately to two ledger accounts, so the number of postings is twice the number of entries. Errors also grow, and an error among hundreds of similar entries is hard to find.

The table compares the work in the general journal with the work in a book kept for credit sales alone. The first row counts the ledger postings to the two main accounts, the receivable account and the sales account. A row is read from left to right: the task, the work for three sales, then the work for two hundred sales. For the general journal the count is twice the number of sales, because each sale touches two accounts. For a book of credit sales alone the count is two, because only the total is posted.

POSTINGS TO THE TWO MAIN ACCOUNTS WHEN SALES ARE RECORDED ON CREDIT

METHOD	THREE SALES	TWO HUNDRED SALES
General journal, posting every entry	6 postings	400 postings
Book for credit sales, posting the total	2 postings	2 postings

A **special journal** is a journal used for only one kind of repeated transaction, such as credit sales. It names the two accounts once, in its heading, and records only the changing parts on each line. Return to the tea stall. Its three credit sales of Tk 100, Tk 200 and Tk 300 take three short lines in a special journal. In the general journal they would take three full entries, each with two account names and an explanation.

Consider the benefits from several viewpoints. The owner receives records sooner and at lower cost. A manager can give the book of credit sales to one clerk and the book of credit purchases to another, and both clerks work at the same time. An employee writes fewer words and makes fewer slips. A customer, who may query a bill, is helped because every sale sits on a line that is easy to find. The plain rule is this: when many transactions of one kind occur, give them a book of their own.

Special journals also have exceptions. A special journal for credit sales does not record cash sales, because no account receivable arises when the customer pays at once; Chapter 12 'The Cash Book and Bank Reconciliation' describes the book used for cash. It does not record the sale of a delivery van, because a van is not goods held for sale. Adjusting and correcting entries also stay in the general journal. Finally, a very small business may have too few transactions to justify extra books, and one journal is then enough.

Before moving on. A special journal is a book for one repeated kind of transaction. It saves writing, shares work and reduces postings. The general journal remains for everything that does not fit a special journal. You should now be able to say why two hundred credit sales need two postings in a special journal and four hundred in the general journal.

* * *

The general journal has three limits, and each one grows with the number of transactions.

A small count shows how quickly the work grows, so read the table row by row.

A special journal groups transactions of one kind in one book, with columns for the parts that change.

Special journal. A journal used for only one kind of repeated transaction, such as credit sales, to speed up recording.

Special journals give four benefits: less writing, shared work, fewer postings and easier checking.

A special journal has limits, because it accepts only its own kind of transaction.

Recap: one book per repeated kind of transaction, with the rest staying in the general journal.

11.2 SALES AND PURCHASES JOURNALS

A small clean example shows the method before we meet the real records.

The sales journal records credit sales of goods, one line for each sale invoice.

Sales journal. A special journal that records only sales of goods on credit.

The case file shows three credit sales in November 2026, each with an invoice number.

To read the sales journal, follow each row across and then the amount column down.

Each line goes to the customer's own account; the total goes to the main ledger once.

We begin with the tea stall, because its numbers are small. The calculation follows the order used throughout this book. *Given.* Three credit sales, of Tk 100, Tk 200 and Tk 300. *Find.* The total to post to the sales account. *Formula.* The total equals the first amount plus the second plus the third. In symbols, $T = a_1 + a_2 + a_3$, where T is the total and a_1 , a_2 and a_3 are the three amounts. *Steps.* Tk 100 + Tk 200 = Tk 300. Then Tk 300 + Tk 300 = Tk 600. *Answer.* The total is Tk 600, posted once to sales. *Check.* Posting the three amounts one by one would also give Tk 600.

The **sales journal** is a special journal that records only sales of goods on credit. Its source document is the copy of the sales invoice that the business keeps. Each invoice becomes one line, and the usual columns are the date, the invoice number, the name of the customer and the amount.¹ The sales journal lists the credit sales of the business, nothing more.

Case file, illustrative. In November 2026, Meghna and Sons made three credit sales. On 4 November it sold goods to Dhanmondi Girls' School for Tk 85,000 (invoice 701). On 12 November it sold goods to Rose Valley School for Tk 120,000 (invoice 702). On 20 November it sold goods to Lalmatia College for Tk 64,000 (invoice 703). The total of credit sales for the month is Tk 269,000.

The table below is the sales journal for the case. Read each row from left to right: the date, the invoice number, the customer and the amount in taka. Then read the amount column downwards and total it. The heading does not name the two accounts for each line, because every line has the same debit and credit.

SALES JOURNAL OF MEGHNA AND SONS, NOVEMBER 2026

DATE	INVOICE	CUSTOMER	AMOUNT (TK)
4 November	701	Dhanmondi Girls' School	85,000
12 November	702	Rose Valley School	120,000
20 November	703	Lalmatia College	64,000
Total			269,000

The figure shows what happens next. Each line is posted to the account of the customer named on it, so that the business always knows how much each customer owes. At the end of the month, the total of Tk 269,000 is posted once to two main accounts: a debit to the account receivable and a credit to sales. Solid lines in the figure show the posting of each line. Dashed lines show the posting of the total.

¹ Some businesses add columns, for example for tax charged on sales. The rules differ from country to country, so this chapter leaves them out and shows only the main idea.

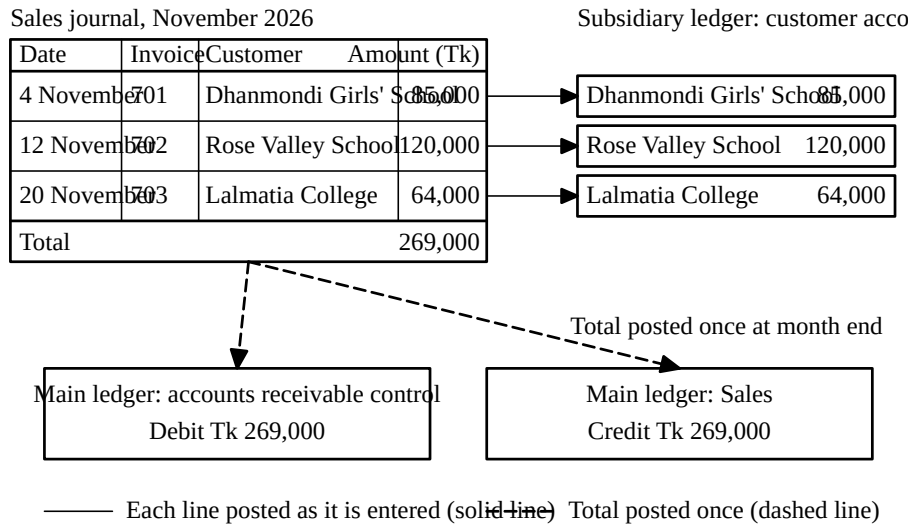


FIG. 11.1. Sales journal columns and posting to ledgers

The **purchases journal** is a special journal that records only purchases of goods on credit. Its source document is the invoice received from the supplier. The columns are the date, the name of the supplier and the amount. In November 2026, Meghna and Sons bought on credit as the next case paragraph records.

Case file, illustrative. On 3 November 2026, Meghna and Sons bought goods on credit from Ananda Publishers for Tk 150,000. On 9 November it bought goods on credit from Prothom Print for Tk 96,000. On 17 November it bought goods on credit from Jonaki Books for Tk 54,000. The total of credit purchases for the month is Tk 300,000.

The posting rule is the same in reverse. Each line is posted to the account of the supplier named on it. The total of Tk 300,000 is posted once to two main accounts: a debit to Purchases, an account that collects the cost of goods bought for resale, and a credit to the account payable. The debit and credit follow the rules of Chapter 4 'Accounts, Debits and Credits': a purchase of goods is a debit, and an increase in a debt owed is a credit. Debits and credits for the month are equal, at Tk 300,000 each.

Exceptions apply here too. A purchase paid in cash does not belong in the purchases journal, because no debt to a supplier arises. A purchase of equipment on credit, such as a new shelving unit, does not belong there either, because the business buys it to use and not to resell. Both go elsewhere: the first in the cash book of Chapter 12 'The Cash Book and Bank Reconciliation', the second in the general journal. A clerk therefore asks two questions of each invoice received. Was it bought on credit? Was it goods for resale? Only two answers of yes place it in the purchases journal.

Each posting should carry a reference. The journal line records the account to which it was posted, and the account records the journal and the date from which the figure came. This two-way trail costs a few seconds for each posting. It repays that time whenever a customer queries a bill or an auditor, who is a person who examines the records, asks where a figure originated.

Before moving on. The sales journal records credit sales, and the purchases journal records credit purchases. Each line is posted to one individual account, and each total is posted once to the main accounts. You should now be able to state which accounts the sales journal total of Tk 269,000 debits and credits.

The purchases journal is the matching book for goods bought on credit from suppliers.

Purchases journal. A special journal that records only purchases of goods on credit.

The case file lists three credit purchases, which total Tk 300,000.

Posting the totals gives each main account one entry for the month, and each supplier's account one entry per line.

Only credit purchases of goods for resale belong in the purchases journal.

A posting reference lets anyone trace a figure back to its source.

Recap: each sale or purchase has one line, and each total is posted once.

* * *

11.3 CONTROL ACCOUNTS

The individual accounts of customers need a home of their own, apart from the main ledger.

Subsidiary ledger. A ledger holding the individual accounts of customers or suppliers, supporting a main ledger account.

The main ledger keeps a single summary account for each subsidiary ledger.

Control. Word in use. In daily life, power to direct something. In accounting, a check that a total agrees with the sum of its parts.

Control account. A main ledger account that sums up the balances of all the accounts in a subsidiary ledger.

Pause and check that you can tell the two ledgers apart.

Agreeing totals means comparing the control account with the sum of the individual balances.

The case file gives the opening balance, which both records must match.

The calculation follows the standard order, and the check is the agreement itself.

Agreement does not prove that every figure is correct, only that no posting is missing or unequal.

A business with hundreds of customers cannot keep every customer account among its main accounts, because the main ledger would become huge and hard to read. It keeps them in a separate book. A **subsidiary ledger** is a ledger holding the individual accounts of customers or suppliers, supporting a main ledger account. One subsidiary ledger holds the accounts of customers. Another holds the accounts of suppliers. Meghna and Sons posts each line of the sales journal to the customer's account in the first of these.

A **control account** is a main ledger account that sums up the balances of all the accounts in a subsidiary ledger. The accounts receivable control account sums up every customer account. The accounts payable control account sums up every supplier account. Its name explains its job. The word control here means a check, not power over customers. If the control account and the subsidiary ledger are correct, they must tell the same story. A difference shows that an error exists somewhere.

PAUSE AND CHECK 11.P1 Which holds the balance of one named customer: the accounts receivable control account or the subsidiary ledger? Which holds the total owed by all customers?

Agreeing the totals is a short calculation, and we perform it with the case. The control account receives the total of the sales journal as a debit. The subsidiary ledger receives the same sales as individual debits, one for each customer. Neither side may be changed without the other. A student might ask why a business keeps both records. The answer is that each one checks the other.

Case file, illustrative. At 1 November 2026, the accounts receivable control account of Meghna and Sons had an opening balance of Tk 280,000. During November, credit sales posted from the sales journal total Tk 269,000.

Given. The opening balance is Tk 280,000. We assume that the customer accounts brought forward also total Tk 280,000, as a correct ledger requires. The November credit sales are Tk 85,000, Tk 120,000 and Tk 64,000. **Find.** The closing balance of the control account, and of the sum of customer accounts. **Formula.** Closing balance equals opening balance plus credit sales posted. In symbols, $C = O + S$, where C is the closing balance, O is the opening balance and S is the sales posted. **Steps.** Tk 85,000 + Tk 120,000 + Tk 64,000 = Tk 269,000. Then Tk 280,000 + Tk 269,000 = Tk 549,000. **Answer.** Both the control account and the sum of customer accounts show Tk 549,000. **Check.** The two totals are equal, so the posting is consistent.

Exceptions need care. Agreement shows that the control account and the customer accounts hold the same total. It does not prove that each customer account is correct, because an error could be posted to both sides. One example is a sale recorded under the wrong customer name. Two errors of equal and opposite size could also hide each other. In this chapter we ignore cash received from customers and goods returned, so the balance only rises; Chapter 12 shows how receipts reduce it.

The same method applies to suppliers. Meghna and Sons posts each line of the purchases journal to the supplier named on it: Tk 150,000 to Ananda Publishers, Tk 96,000 to Prothom Print and Tk 54,000 to Jonaki Books. These are credits, because they increase debts owed. The credit side of the accounts payable control account receives the total of Tk 300,000. The credits in the supplier accounts also total Tk 300,000, since $Tk\ 150,000 + Tk\ 96,000 = Tk\ 246,000$, and $Tk\ 246,000 + Tk\ 54,000 = Tk\ 300,000$. Only the new purchases are counted here, because the case gives no opening balance for suppliers.

Before moving on. A subsidiary ledger holds individual accounts. A control account in the main ledger sums them. To agree the totals, find the closing balance of the control account and compare it with the sum of the individual balances. You should now be able to say what a difference between the two totals tells you.

* * *

WORKED CASE: RECORDING AND AGREEING ONE MONTH

Situation. It is 30 November 2026. Meghna and Sons has recorded its credit sales on invoices 701, 702 and 703, with a total of Tk 269,000, and its credit purchases from three suppliers, with a total of Tk 300,000. The accounts receivable control account opened the month at Tk 280,000. *Question.* Which totals reach the main ledger, and do the control account and the customer accounts agree at month end?

Reasoning. First, total each journal. The sales journal total is $Tk\ 85,000 + Tk\ 120,000 + Tk\ 64,000$, which is Tk 269,000. The purchases journal total is $Tk\ 150,000 + Tk\ 96,000 + Tk\ 54,000$, which is Tk 300,000. Second, post each total once. For sales, debit the accounts receivable control account Tk 269,000 and credit Sales Tk 269,000. For purchases, debit Purchases Tk 300,000 and credit the accounts payable control account Tk 300,000. Third, post each line to the customer or supplier account named on it. Fourth, compare the control account with the sum of the customer accounts.

Conclusion. The accounts receivable control account is $Tk\ 280,000 + Tk\ 269,000$, which is Tk 549,000. The three customer accounts have received debits of Tk 85,000, Tk 120,000 and Tk 64,000, and, as assumed, the brought-forward customer balances total Tk 280,000. Their sum is also Tk 549,000. The totals agree. *Check.* Debits and credits posted from the two journals are Tk 569,000 each, as $Tk\ 269,000$ plus $Tk\ 300,000$ gives the same sum on both sides.

Posting the lines. The three customer accounts each receive a debit: Dhanmondi Girls' School Tk 85,000, Rose Valley School Tk 120,000 and Lalmatia College Tk 64,000. The three supplier accounts each receive a credit: Ananda Publishers Tk 150,000, Prothom Print Tk 96,000 and Jonaki Books Tk 54,000. Each posting names the journal that it came from, so any figure can be traced to its source document. This trail is the reason a business records a reference on every posting.

Common slips deserve a warning. The first is posting both the lines and the total to the same main account, which counts every sale twice. The second is posting the total to the wrong side, a credit where a debit belongs. The third is mixing periods, for example putting a sale of 2 December in the November total. Each slip is found by the same test: after posting, the control account and the individual accounts must agree.

The same method applies to suppliers, with the control account on the credit side.

Recap: one control account summarises one subsidiary ledger, and the two totals must agree.

This worked case joins the sales journal, the purchases journal and the control account in one month.

The reasoning starts with the totals and then moves to the check.

The conclusion is that the control account stands at Tk 549,000, matching the customer accounts.

Posting the lines to individual accounts is the step that makes later checking possible.

Three slips occur often in this work, and each one can be prevented.

* * *

The summary restates the whole chapter in a few lines.

SUMMARY

A special journal records one repeated kind of transaction in its own book. The sales journal holds credit sales, and the purchases journal holds credit purchases. Each line is posted to an individual account in a subsidiary ledger, and each total is posted once to the main ledger. A control account sums a subsidiary ledger, and the two totals must agree. For Meghna and Sons in November 2026, credit sales were Tk 269,000, credit purchases were Tk 300,000, and the accounts receivable control account closed at Tk 549,000.

THINK IT THROUGH

11.T1 A very small business makes five credit sales in a year. Explain why it may not need a sales journal, and say at what point the answer might change.

11.T2 A clerk posts every line of the sales journal to the customer accounts and then posts every line again to the accounts receivable control account. What goes wrong, and how would the agreement test reveal it?

11.T3 A clerk totals a sales journal as Tk 52,400, but the true total is Tk 54,200. The customer accounts have been posted correctly. Which test reveals the error, and what difference appears?

WHERE PEOPLE GO WRONG

Five common misunderstandings are stated here, each with its correction.

Some students believe that a special journal replaces the ledger. It does not. A special journal is a book of first record, like the general journal, and its entries still reach the ledger by posting. Others believe that all transactions of a business go into special journals. They do not: cash transactions, adjusting entries and correcting entries have other books. A third belief is that agreement of the control account with the customer accounts proves that all figures are right. It proves only that the two totals are equal. Last, students sometimes read the word control as a power over customers. In accounting it means a check. A fifth belief is that special journals need new rules for debit and credit. They do not: the rules of Chapter 4 'Accounts, Debits and Credits' apply unchanged, and only the method of recording is different.

IN HISTORY

A machine built for the census of 1890 started the mechanising of record-keeping.

In 1890 the United States counted its people with machines designed by Herman Hollerith, an American engineer and former employee of the Census Bureau. The count of 1880 had taken about seven years to tabulate. The Census Bureau then held a contest for a faster method, and Hollerith's system won.² His method placed the facts about each person on a punched card. Holes in the card completed an electrical circuit, and dials counted the totals. Hollerith had patented the machine in 1889, and the census of 1890 was the first large count to use it.

Clerks first transferred the facts from the census schedules onto cards, using a punch. The cards then passed through the tabulating machine, and dials recorded

The machines worked in two stages: first record each fact on a card, then total each kind of fact.

² The Census Bureau, the Smithsonian Institution and The Henry Ford museum each describe the contest and the machine, and their accounts agree on these points.

one count for each kind of fact. Notice that this is the two-stage method of this chapter: record each item on its own line or card, then total each kind once. In 1911 Hollerith's firm joined three other firms to form the Computing-Tabulating-Recording Company, which later became IBM.

A tabulating machine is not a special journal, but the two share one idea: group records of the same kind and total each group at once. In 1896 Hollerith incorporated the Tabulating Machine Company. In later decades, similar machines were adapted to bookkeeping, and the company became part of the business that grew into IBM. Mechanised records, like special journals, show that large numbers of similar transactions reward the effort of organising them.

The link with accounting is the idea of grouping like records and totalling them quickly.

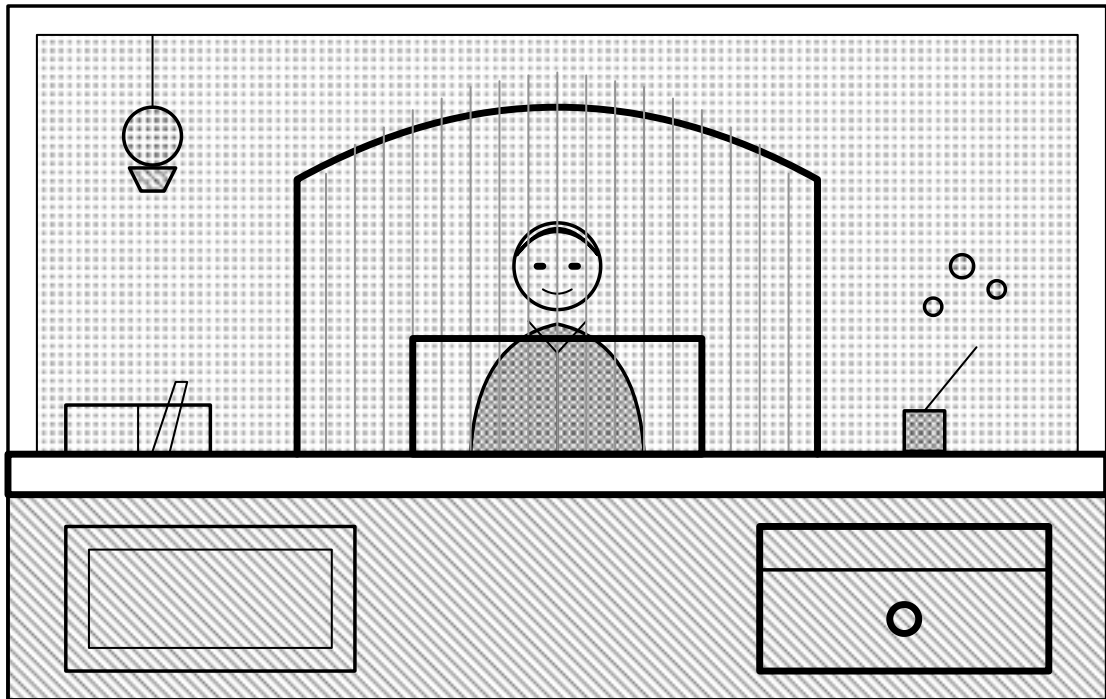
REVIEW QUESTIONS

11.1 A tea stall makes three credit sales: Tk 100, Tk 200 and Tk 300. Enter them in a sales journal, total the amount column, and state which two main accounts receive the total and on which side.

11.2 A different bookshop makes these credit sales in December 2026: 2 December, invoice 801, Greenfield School, Tk 42,500; 9 December, invoice 802, Padma College, Tk 18,750; 15 December, invoice 803, Greenfield School, Tk 31,200. Its accounts receivable control account opens at Tk 150,000. Enter the sales, total them, and find the closing balance of the control account.

11.3 In one week a sales journal totals Tk 126,400. The control account opened at Tk 310,000. After posting, the customer accounts show balances of Tk 120,000, Tk 95,400, Tk 150,000 and Tk 61,000. Do the totals agree? If not, state the difference.

11.4 A shop records three credit purchases: Tk 36,000, Tk 54,500 and Tk 29,500. The accounts payable control account opened at Tk 200,000. Total the purchases journal, state the debit and credit posted, and find the closing balance of the control account.



A bank counter in the age of ink and ledgers

CHAPTER XII

The Cash Book and Bank Reconciliation

*Two honest records of one account, and two different totals.
Which one is right?*



This chapter shows how a business records its cash and bank money in a cash book, and why its bank balance rarely matches the figure the bank reports. You learn the common reasons for a difference and how to prepare a bank reconciliation statement that explains every taka of it. You also learn which differences need a journal entry afterwards.

A small puzzle: two careful people record the same money and reach different totals.

Neither person is careless, so the difference needs an explanation, not a search for blame.

The same difference appears in the running case, and the chapter will explain all of it.

Imagine that you keep a small notebook for your savings account. Every time you put money in, you write the amount. Every time you take money out, you write that too. At the end of the month you add everything up and your notebook says Tk 5,000. Then the bank sends you its own list of what happened in the account. Its final figure is Tk 5,700. The difference is Tk 700.

Who is wrong? Perhaps the bank made a mistake. Perhaps you did. Many students expect this answer, and it is usually false. In most months both records are correct. They differ because the bank and you learn about each event at different times, and because some events reach one record before the other. Your task is not to decide which record is right. Your task is to explain, item by item, how one total becomes the other. If you can do that, you have proved that both records are sound, and you have found the true balance as well.

Case file, illustrative. On 31 October 2026 the cash book of Meghna and Sons showed a bank balance of Tk 612,000. The bank statement for the same date showed a balance of Tk 655,000. The difference is Tk 43,000. The case file lists seven items that explain this difference; by the end of the chapter each one has a place.

YOU WILL NEED

This chapter uses the idea of a special journal from Chapter 11 'Special Journals', section 11.1 'Why Special Journals'. It also uses the ledger and posting from Chapter 5 'The Journal and the Ledger', section 5.3 'The Ledger and Posting'. The arithmetic is addition, subtraction and checking two totals against each other; Appendix A1 'Maths You Will Use' holds a short reference.

* * *

12.1 THE CASH BOOK

Every business receives and pays money in two ways, and it needs one book that records both.

The cash book is the first record and the ledger account for cash and bank, in one place.

Recall: journal, Chapter 5 'The Journal and the Ledger', section 5.3 'The Ledger and Posting'.

Cash book. A book that serves as both journal and ledger for cash and bank receipts and payments.

The other half of every cash book entry still has to reach its own ledger account.

A business handles money in two places. Some money is in the drawer: notes and coins from customers, kept for small payments. The rest is in the bank account: cheques deposited, transfers received, and payments made by cheque. Think of your own purse and your own bank card. You know the amount in each, and you add them to know what you have. A business needs the same habit, kept with more care. It records both kinds of money in one book.

This book is called the **cash book**. A cash book is a book that serves as both journal and ledger for cash and bank receipts and payments. For example, when a customer pays Tk 500 in notes, you write the amount once in the cash book. Many students find this confusing, because they have learned that a journal and a ledger are two different books. Usually an entry is written in the journal first and posted to the ledger afterwards, which is two steps. For cash and bank, the cash book joins the two steps into one. This saves work, because a cash book is also the ledger account for Cash and the ledger account for Bank.

The cash book does not finish the job alone. Every transaction has two sides. If a customer pays Tk 500 in notes for books, the cash book records the money received. The sale itself must still reach the ledger account for sales. Likewise, if the customer was paying an account receivable, that account must be reduced. We call this the other side of the entry. The cash book holds one side, and posting carries the other side to the ledger.

A cash book page looks like a ledger account with two sides. The left side is the debit side and holds money received. The right side is the credit side and holds money paid. Each side is split into columns. A cash book with a Cash column and a Bank column on each side is called a two-column cash book. The Cash column follows the notes and coins in the drawer. The Bank column follows the money in the bank account. The table below shows four small entries and where each one is placed. Read each row from left to right: the event, the side, the column, and the reason.

FOUR ENTRIES IN A TWO-COLUMN CASH BOOK

EVENT	SIDE	COLUMN	REASON
---	---	---	---
Customer pays Tk 500 in notes	Debit	Cash	Money received into the drawer
Customer's cheque for Tk 2,000 deposited	Debit	Bank	Money received into the bank account
Rent of Tk 1,000 paid by cheque	Credit	Bank	Money paid out of the bank account
Tk 300 of notes paid for postage	Credit	Cash	Money paid out of the drawer

Sometimes the owner moves money between the drawer and the bank. Suppose the owner takes Tk 3,000 in notes from the drawer and deposits it. The Bank column receives money, so it is debited. The Cash column loses money, so it is credited. Both halves are in the same book. We call this a contra entry. The word *contra* is Latin for "against": the two halves face each other across the page. A contra entry has an important feature. It does not touch any other ledger account, because both sides are already in the cash book.

At the end of each period you total each column. On the debit side the total shows all money received. On the credit side the total shows all money paid. The difference is the balance. The Cash column balance should equal the notes and coins in the drawer. The Bank column balance is what the books say is in the bank. The bank reports its own figure, and the rest of this chapter compares the two. For the owner, this balance answers a daily question: how much can we spend today?

Before moving on. The cash book is a journal and a ledger in one. Its two sides hold receipts and payments, and its columns separate cash from bank. A contra entry moves money between the two columns and touches no other account. You should now be able to say where each of four events is written in a cash book.

12.2 WHY BANK AND BOOK DIFFER

The bank account has two keepers. The business keeps the Bank column of its cash book. The bank keeps its own record of the same account and sends the business a statement from time to time.

The page is divided into receipts on the left and payments on the right, with a column for each kind of money.

One kind of entry moves money between the two columns and needs no other ledger account.

Totalling a column gives its balance, and the Bank column balance is the figure we shall later compare with the bank.

Before moving on, we recall the three ideas of this section.

The bank keeps a second record of the same account, so two records must be compared.

Statement. Word in use. In daily life, something said or written, such as a statement to a newspaper. In banking, the list of an account's receipts, payments and balance that the bank sends to its customer.

A bank statement is a list of the account's movements, not a claim that someone has made.

The principle: a difference arises only when one record learns of an event before the other.

A deposit in transit is a receipt the business has recorded but the bank has not.

Deposit in transit. Money recorded as deposited in the books but not yet shown on the bank statement.

An outstanding cheque is a payment the business has recorded but the bank has not yet paid.

Outstanding cheque. A cheque written and recorded but not yet paid by the bank.

Clear. Word in use. In daily life, easy to understand or free of obstacles. In banking, a cheque clears when the bank has paid it and moved the money out of the writer's account.

A credit memo is the bank telling the business that it added money to the account.

Credit memo. A bank's notice that it has added money to the account, for example a collection or interest.

A debit memo is the bank telling the business that it took money out of the account.

Debit memo. A bank's notice that it has deducted money, for example a service charge.

A dishonoured cheque is a cheque the bank would not pay, so money recorded as received has gone.

Dishonoured cheque. A cheque that the bank refuses to pay, usually because the writer's account lacks funds.

A bank statement lists each receipt and payment the bank has recorded, with a balance after each. It is a source document, because it is written evidence that an event took place. It is also the bank's version of the account, so it is the natural thing to compare with the cash book.

Here is the principle that explains every difference. Each event in the account is recorded by two parties. If both record it in the same period, the records agree. If one records it earlier, the records differ until the other catches up. We can therefore sort the differences into two groups. In the first group, the business has recorded the event and the bank has not yet done so. The bank will catch up without any action from us, so no entry is needed in our books. In the second group, the bank has recorded the event and the business has not. The business must make an entry to catch up. Keep these two groups in mind, because they decide what we do with each item.

The first item in group one is the **deposit in transit**. A deposit in transit is money recorded as deposited in the books but not yet shown on the bank statement. For example, a shop banks Tk 400 after the bank has closed on the last day of the month. The shop records the receipt at once. The bank records it on the next working day. The statement therefore omits it, and the bank balance looks lower than the cash book balance by Tk 400.

The second item in group one is the **outstanding cheque**. An outstanding cheque is a cheque written and recorded but not yet paid by the bank. For example, you write a cheque for Tk 900 to a supplier and record the payment the same day. The supplier needs a few days to deposit it, and the banks need more days to clear it. Until it clears, the bank statement still shows the money in your account, so the bank balance looks higher than the cash book balance by Tk 900.

Now we move to group two, the events that the bank records first. The first is the **credit memo**. A credit memo is a bank's notice that it has added money to the account, for example a collection or interest. Suppose the bank collects a Tk 300 payment on the business's behalf, or pays Tk 300 of interest. The bank adds the money and informs the business with a credit memo. Why is it called a credit? The bank looks at the account from its own side. The money in your account is money the bank owes you, a liability of the bank. When the bank owes you more, it records a credit. For you, the memo increases the balance.

The opposite item is the **debit memo**. A debit memo is a bank's notice that it has deducted money, for example a service charge. Suppose the bank charges Tk 100 for running the account. It deducts the money and sends a debit memo. The bank owes you less, so it records a debit. For you, the memo decreases the balance. Remember the pairing: a credit memo adds to your balance and a debit memo subtracts from it. The names follow the bank's books, not yours. This is a common source of confusion, so check each memo against that rule.

The last item is the **dishonoured cheque**. A dishonoured cheque is a cheque that the bank refuses to pay, usually because the writer's account lacks funds. Suppose a customer pays you with a cheque for Tk 800, and you deposit it and record the receipt. Later the customer's bank refuses to pay, so your bank takes the Tk 800 back out of your account. Your cash book still shows the money, so it is Tk 800 too high. The customer also owes you again, so the account receivable is restored. The legal results of a dishonoured cheque differ

from country to country, so a business must check the law that applies where it trades.

One more situation fits group two. A customer may pay the bank account directly by transfer, with no cheque and no visit. The bank records the receipt at once. The business learns of it only from the statement. The same can happen for a payment out, such as a standing order to pay a bill. We treat a direct receipt like a credit memo and a direct payment like a debit memo.

The table below gathers the items. Read each row as a question: who records the event first, and what must the business do? The two items at the top need no entry, because the bank will catch up. The four at the bottom need an entry in the cash book.

WHY THE BANK STATEMENT AND THE CASH BOOK DIFFER

ITEM	RECORDED FIRST BY	IS AN ENTRY NEEDED IN OUR BOOKS?
---	---	---
Deposit in transit	Business	No; the bank catches up
Outstanding cheque	Business	No; the bank catches up
Credit memo, or direct receipt	Bank	Yes; we add to the cash book
Debit memo, or direct payment	Bank	Yes; we subtract from the cash book
Dishonoured cheque	Bank	Yes; we subtract and restore the account receivable

Case file, illustrative. On 31 October 2026 the items were as follows. Cash book bank balance: Tk 612,000. Bank statement balance: Tk 655,000. Deposit in transit: Tk 48,000. Outstanding cheques: no. 4417 for Tk 36,000 and no. 4421 for Tk 22,000. Bank service charge: Tk 1,500. Dishonoured customer cheque: Tk 8,000. Interest credited by the bank: Tk 2,500. School payment made directly to the bank and not yet in the books: Tk 40,000.

Before moving on. The bank statement and the cash book differ because one record learns of an event before the other. A deposit in transit and an outstanding cheque are known to the business first, and need no entry. A credit memo, a debit memo and a dishonoured cheque are known to the bank first, and need an entry. You should now be able to say, for any item, who knows first and what follows.

12.3 PREPARING THE RECONCILIATION

You now know the six causes of difference. The next question is how to arrange them so that nothing is forgotten. The answer is a working paper called the **bank reconciliation statement**.

A bank reconciliation statement is a statement that explains the difference between the bank balance in the cash book and the balance on the bank's statement. It has two halves. The first half starts from the bank statement balance and deals with the items the bank has not yet recorded. The second half starts from the cash book balance and deals with the items the business has not yet

Payments made straight into or out of the bank account are also learned from the statement.

A table sorts the items by who records first and whether an entry is needed.

The case file lists seven items at 31 October 2026, and each fits one row of the table.

Before moving on, we restate the principle of this section.

The reconciliation turns two different totals into two matching totals, with every step shown.

Bank reconciliation statement. A statement that explains the difference between the bank balance in the cash book and the balance on the bank's statement.

The statement is built in two halves, one starting from each record.

Each half has its own rule for adding and subtracting, based on the six items.

recorded. Each half ends with a figure, and we call these two figures the adjusted balances. If the work is correct, the adjusted balances are equal, and that common figure is the true amount in the bank.

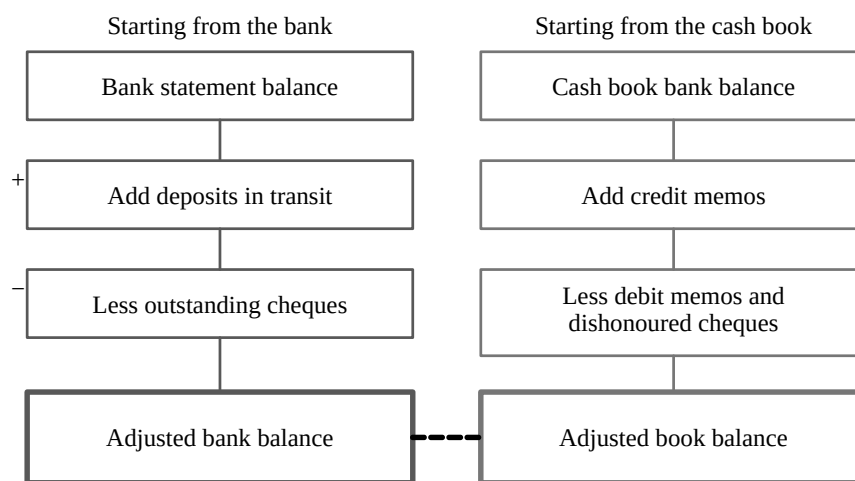
The rules follow from what you learned in section 12.2. In words: the adjusted bank balance is the bank statement balance plus deposits in transit, less outstanding cheques. The adjusted book balance is the cash book bank balance plus credit memos, less debit memos and dishonoured cheques. In symbols, let S be the bank statement balance, D the deposits in transit and C the outstanding cheques. Let B be the cash book bank balance, M the credit memos, N the debit memos and H the dishonoured cheques. Then the adjusted bank balance is $S + D - C$, and the adjusted book balance is $B + M - N - H$. A deposit in transit is added to the bank side because the bank will soon receive it. An outstanding cheque is subtracted because the bank will soon pay it.

IN SHORT

Adjusted bank balance = $S + D - C$. Adjusted book balance = $B + M - N - H$.
The two must be equal. If they are not, an item is missing or wrongly placed.

The figure shows the two halves side by side, ending in a single agreed balance.

The figure below shows the layout. Read it from top to bottom in each column. The left column begins with the bank statement balance and shows the two timing items. The right column begins with the cash book balance and shows the three items from the bank. Each column ends in a box, and a dashed line joins the two boxes to show that they must be equal.



The two adjusted balances must be equal

FIG. 12.1. Two-sided layout of a bank reconciliation

Four steps give a method that works for any month.

Follow this method. First, compare the cash book with the statement, line by line, and tick every pair that agrees. Second, list the unticked items from the cash book: these are deposits in transit and outstanding cheques. Third, list the unticked items from the statement: these are credit memos, debit memos, dishonoured cheques and direct payments. Fourth, calculate the two adjusted balances and compare them. If they differ, tick again and check the arithmetic before you change anything.

Now we apply the method to the puzzle at the start of the chapter. We use the tutor's pace: each step is announced.

Here is a first example with small, clean numbers, which returns to the puzzle that opened the chapter.

Given. The cash book shows a bank balance of Tk 5,000. The statement shows Tk 5,700. A deposit of Tk 400 is in transit. One cheque for Tk 900 is outstanding. The bank added a credit memo for Tk 300 and a debit memo for Tk 100.

Find. The adjusted balance on each side, and whether they agree.

Formula. Adjusted bank balance = $S + D - C$. Adjusted book balance = $B + M - N - H$, with no dishonoured cheque here, so $H = 0$.

FIRST EXAMPLE: SMALL NUMBERS

BANK STATEMENT SIDE	TK	CASH BOOK SIDE	TK
---	---	---	---
Statement balance	5,700	Cash book balance	5,000
Add deposit in transit	400	Add credit memo	300
Less outstanding cheque	900	Less debit memo	100
Adjusted bank balance	5,200	Adjusted book balance	5,200

Steps. Left: $5,700 + 400 = 6,100$, and $6,100 - 900 = 5,200$. Right: $5,000 + 300 = 5,300$, and $5,300 - 100 = 5,200$.

Answer. Both adjusted balances are Tk 5,200, so the two records agree and the true balance is Tk 5,200.

Check. The original difference was $5,700 - 5,000 = 700$. The bank side fell by 500 ($400 - 900$) and the book side rose by 200 ($300 - 100$), and $500 + 200 = 700$. The puzzle is solved: neither record was wrong.

PAUSE AND CHECK 12.P1 Suppose the adjusted bank balance is Tk 645,000 and the adjusted book balance is Tk 644,000. What should you do before you make any entry?

After the reconciliation, one question remains: which items need a journal entry? The reconciliation statement is not a book of account. It is a working paper, so writing it changes no ledger account. The deposit in transit and the outstanding cheque need no entry, because the bank will record them without our help. The credit memos, debit memos and dishonoured cheques need entries, because our books are missing them. Each entry is posted to the cash book on one side and to the matching ledger account on the other. After posting, the cash book balance equals the adjusted balance.

Before moving on. A bank reconciliation statement has two halves, one from the bank statement and one from the cash book. Timing items adjust the bank side. Items from the bank adjust the book side. Only the book side leads to journal entries. You should now be able to say why a deposit in transit needs no entry and a debit memo does.

WORKED EXAMPLE: THE OCTOBER RECONCILIATION

We now reconcile the seven items of the case file, with larger numbers and more varied items.

Case file, illustrative. Given for 31 October 2026: cash book bank balance Tk 612,000; bank statement balance Tk 655,000; deposit in transit Tk 48,000; outstanding cheques no. 4417 for Tk 36,000 and no. 4421 for Tk 22,000; bank service charge Tk 1,500; dishonoured customer cheque Tk 8,000; interest credited by the

The data of the small example.

The question to answer.

The two rules, in symbols.

Every line of arithmetic.

The result in a sentence, and a test of it.

A test that the result is sensible.

The reconciliation statement is a working paper, so only the book-side items lead to entries.

Recall: ledger and posting, Chapter 5 'The Journal and the Ledger', section 5.3 'The Ledger and Posting'.

Before moving on, we restate the method.

The case file for 31 October 2026 is reconciled in full, with every line shown.

The data for the October reconciliation.

The task is to find the adjusted balance on each side and to check that they agree.

bank Tk 2,500; school payment direct to the bank not yet in the books Tk 40,000.

Find. The adjusted bank balance and the adjusted book balance, and the entries that follow. The formula is the same as before: $S + D - C$ for the bank side, and $B + M - N - H$ for the book side. Here M is the interest of Tk 2,500 plus the school payment of Tk 40,000. N is the service charge of Tk 1,500. H is the dishonoured cheque of Tk 8,000.

OCTOBER RECONCILIATION, 31 OCTOBER 2026

BANK STATEMENT SIDE	TK	CASH BOOK SIDE	TK
---	---	---	---
Statement balance	655,000	Cash book balance	612,000
Add deposit in transit	48,000	Add interest credited	2,500
Less cheque no. 4417	36,000	Add school payment	40,000
Less cheque no. 4421	22,000	Less service charge	1,500
		Less dishonoured cheque	8,000
Adjusted bank balance	645,000	Adjusted book balance	645,000

The arithmetic is shown line by line so that you can repeat it by hand.

Steps. Bank side: $655,000 + 48,000 = 703,000$. The cheques total $36,000 + 22,000 = 58,000$. Then $703,000 - 58,000 = 645,000$. Book side: $612,000 + 2,500 = 614,500$. Then $614,500 + 40,000 = 654,500$. Then $654,500 - 1,500 = 653,000$. Then $653,000 - 8,000 = 645,000$.

The answer is a single balance, and a quick test confirms the work.

Answer. Both adjusted balances are Tk 645,000. This is the true bank balance, and it is the figure for the balance sheet. *Check.* The original difference was $655,000 - 612,000 = 43,000$. The bank side moved by $48,000 - 58,000 = -10,000$. The book side moved by $2,500 + 40,000 - 1,500 - 8,000 = +33,000$. And $10,000 + 33,000 = 43,000$, so every taka is explained.

Four items need journal entries, because the books of Meghna and Sons are missing them.

Now the entries. Four items came from the bank, so four entries are needed. Suppose the school owed Meghna and Sons Tk 40,000 for books supplied earlier. The table shows each entry.

ENTRIES AFTER THE OCTOBER RECONCILIATION

ITEM	DEBIT	CREDIT	TK
---	---	---	---
Interest credited	Bank	Interest Revenue	2,500
School payment	Bank	Account Receivable (the school)	40,000
Service charge	Bank Charges	Bank	1,500
Dishonoured cheque	Account Receivable (the customer)	Bank	8,000

After posting, the Bank column of the cash book equals the adjusted balance.

The net effect on the Bank account is $+2,500 + 40,000 - 1,500 - 8,000 = +33,000$. So the cash book balance changes from Tk 612,000 to Tk 645,000. The deposit in transit and the two cheques have no entry; they will appear on the statement in November, and the bank side will then agree without help.

Common slips: a wrong sign, a missed item, a misplaced half.

Three slips are common. The first is a wrong sign: a dishonoured cheque is subtracted from the cash book side, not added. The second is a missed item, usually the direct payment, because it is not in either record you started with. The third is adding an item to the wrong half; the table in section 12.2 shows which half each item belongs to.

SUMMARY

The cash book is a journal and a ledger in one, with a Cash column and a Bank column. The bank statement and the cash book differ because one learns of an event before the other. Deposits in transit and outstanding cheques are timing differences and need no entry. Credit memos, debit memos and dishonoured cheques come from the bank and need entries. A bank reconciliation statement brings both records to one adjusted balance.

The chapter's ideas in a few lines.

THINK IT THROUGH

12.T1 In the case file of 31 October 2026, which items need a journal entry and which do not? Give one reason for each group.

12.T2 Cheque no. 4417 for Tk 36,000 is still outstanding three months later. What should the business do, and what does the reconciliation show each month until then?

WHERE PEOPLE GO WRONG

Some students think the statement is always right. It is not. The statement lacks the business's deposits in transit and outstanding cheques, so it is also incomplete. Neither record is correct until both are adjusted. The adjusted balance is correct, and neither raw total is.

Belief: the bank statement is the correct record, so the cash book must be forced to match it.

Only the items that came from the bank need entries. A deposit in transit and an outstanding cheque are already in the cash book. Entering them again would record them twice.

Belief: every reconciling item needs a journal entry.

The names follow the bank's books. A credit memo means the bank owes you more, so your balance rises. A debit memo means the bank owes you less, so your balance falls.

Belief: a credit memo reduces the balance, because credit sounds like a reduction.

A dishonoured cheque is a loss of money that was recorded as received. It is subtracted from the cash book, and the customer's account receivable is restored.

Belief: a dishonoured cheque is a gain, because it is a cheque.

Agreement is strong evidence, not proof. Two errors of equal size in opposite directions could cancel. The agreement shows that the known items explain the difference.

Belief: an agreeing reconciliation proves there are no mistakes.

IN HISTORY

In the City of London, the old centre of banking in the English capital, banks once sent clerks from door to door to exchange the cheques they held against each other. From the 1770s the clerks began to meet each day in one place. Each clerk brought the cheques drawn on the other banks, and the clerks swapped them and settled only the differences, in cash. This daily meeting is the origin of cheque clearing. In the early nineteenth century a formal Bankers' Clearing House grew from it.

Cheque clearing began when bank clerks in London began meeting to exchange cheques.

This history matters for the reconciliation. A cheque must travel through the clearing process before the bank pays it, and the journey takes days. In those days the business has recorded the payment and the bank has not. That gap is the outstanding cheque.

The history explains why outstanding cheques exist at all.

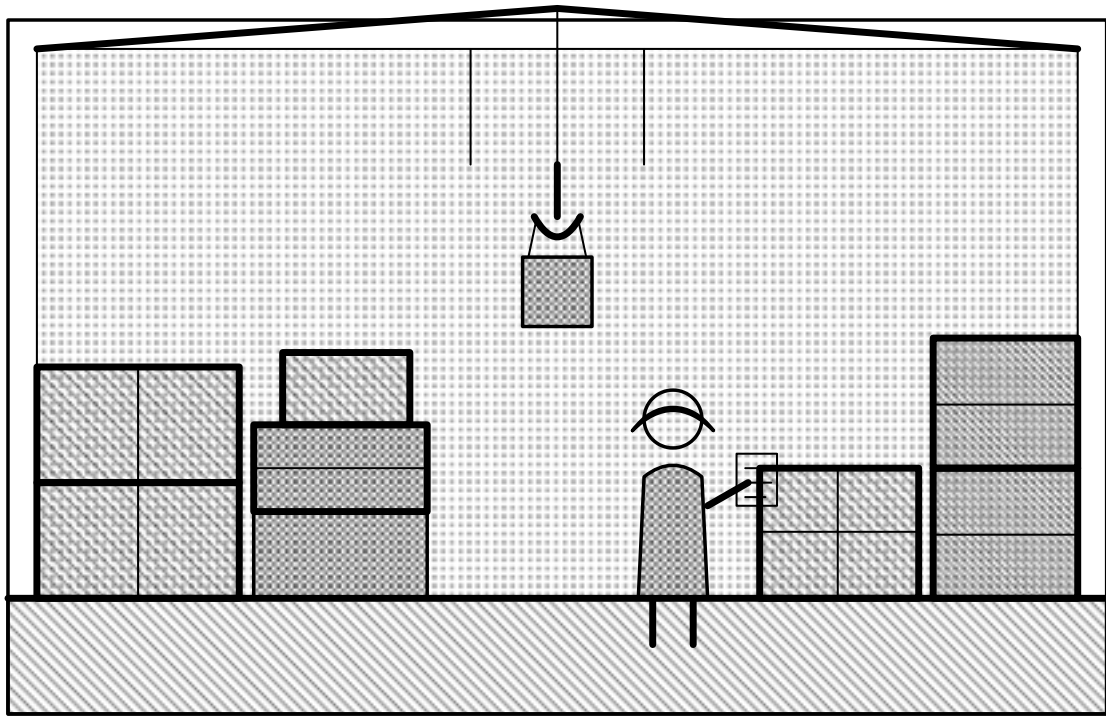
REVIEW QUESTIONS

12.1 The cash book shows a bank balance of Tk 3,000. The statement shows Tk 3,400. A deposit of Tk 700 is in transit. A cheque for Tk 300 is outstanding. The bank collected Tk 900 for the business (a credit memo) and charged Tk 100 (a debit memo). Prepare the reconciliation statement and state the adjusted balance.

12.2 The cash book shows Tk 95,050 and the statement shows Tk 91,250. Deposits in transit are Tk 12,000 and Tk 6,500. Outstanding cheques are no. 0215 for Tk 9,800 and no. 0219 for Tk 4,150. A credit memo shows a collection of Tk 5,000. A debit memo shows a service charge of Tk 750. A customer's cheque for Tk 3,500 was dishonoured. Prepare the reconciliation statement.

12.3 In a later month the statement shows Tk 18,400 and the cash book shows Tk 20,150. A deposit of Tk 5,000 is in transit. Cheques of Tk 3,200 and Tk 1,100 are outstanding. A service charge of Tk 250 and a dishonoured cheque of Tk 800 are not in the books. Say which balance is higher before adjusting, then reconcile.

12.4 The cash book shows Tk 40,000 and the statement shows Tk 47,000. A deposit of Tk 4,000 is in transit and a cheque for Tk 9,000 is outstanding. A customer paid Tk 3,000 directly into the bank. The bank credited interest of Tk 500 and charged Tk 1,500. Reconcile, write the journal entries, and state the figure for Bank on the balance sheet.



A warehouse in the age of hoists, crates and handwritten lists

CHAPTER XIII

Inventory: Meaning and Measurement

A shelf that is empty and a store room that is full: which mistake costs more?



This chapter explains the goods a business holds for sale and the materials it holds to make them. You learn the three kinds of inventory in a factory, why a business holds inventory at all, and what it costs to hold too little or too much. You then learn the two systems for recording inventory, how a physical count works, and how to reduce the value of goods that will sell for less than they cost.

A small puzzle: two bad mornings in a bookshop, one with too few goods and one with too many.

The puzzle is to find a sensible amount between none and too many, and then to record it honestly.

The running case gives a first number: the goods held by the bookshop at the end of 2025.

Imagine two bad mornings in a bookshop. On the first morning a student walks in and asks for a particular textbook. The shelf is empty. The student shrugs and walks to another shop, and the sale is lost for ever. On the second morning the owner opens the store room at the back and finds a thousand copies of a book that nobody has asked for in a year. The money that paid for those copies is not in the bank. It is locked in paper and dust.

Both mornings are mistakes, and they pull in opposite directions. If the owner buys more to avoid the first morning, the second morning comes closer. If the owner buys less to avoid the second, the first comes closer. How much should a business hold? And once it holds something, how should it count the goods and say what they are worth? Accountants answer the second question with care, because the figure appears twice: in the income statement, through cost of goods sold, and in the balance sheet, as an asset. A mistake in the count is a mistake in the profit.

Case file, illustrative. On 31 December 2025 Meghna and Sons held goods for sale with a total cost of Tk 3,500,000. The year's net sales were Tk 14,400,000 and its cost of goods sold was Tk 9,800,000. Two groups within the Tk 3,500,000 need special attention: 800 titles that have not sold for 18 months, and 120 copies of an old edition.

YOU WILL NEED

This chapter uses cost of goods sold from Chapter 10 'Accounts of a Merchandising Company', section 10.1 'Sales, Purchases and Returns'. It also uses the idea of current assets from Chapter 10 'Accounts of a Merchandising Company', section 10.3 'Classified Balance Sheet'. The arithmetic is multiplication, percentages and subtraction; Appendix A1 'Maths You Will Use' holds a short reference.

* * *

13.1 WHAT INVENTORY IS

A shop and a household both keep a supply of things for later use, and the business version has a name.

Inventory is the accounting name for goods held for sale and for materials held to make them.

Recall: current assets, Chapter 10, section 10.3 'Classified Balance Sheet'.

Inventory. Goods held for sale, or materials held to make goods for sale, in the ordinary course of business.

Begin at home. Most families keep a supply of rice, oil and soap in the kitchen. They do not eat the rice the moment they buy it. They keep it until it is needed, and when it runs low they buy more. A shopkeeper behaves in the same way, with one difference: the shopkeeper keeps the goods in order to sell them, not to use them.

In accounting this supply has a name: **inventory**. Inventory means goods held for sale, or materials held to make goods for sale, in the ordinary course of business. For example, the books on the shelves of Meghna and Sons are inventory, because the shop intends to sell them to customers. Inventory is an asset, because the business owns it and expects to receive money from it. It is a current asset, because the business expects to sell it within a short time, usually within one year.

Many students find the word "stock" confusing, because shopkeepers use it for goods and investors use it for shares in a company. Both uses are correct in their own settings. To avoid mixing them, this book uses inventory for goods and stock only for company shares. One phrase keeps the older word for tradition, and we meet it in section 13.3.

Inventory has a life in two steps. While the goods wait on the shelf, their cost is an asset on the balance sheet. When a customer buys them, the cost leaves the balance sheet and becomes the expense called cost of goods sold, which Chapter 10, section 10.1 'Sales, Purchases and Returns' explains. The sale itself is revenue. So the same Tk 100 of cost appears in two places in two different times: first as an asset, then as an expense. Gross profit is the revenue from the sale less this cost. If the cost is wrong, the profit is wrong.

What does not count? The test is the purpose. A cupboard in the shop is owned and useful, but the owner does not intend to sell it. It is equipment, not inventory. A box of carrier bags for customers is a supply used in the business, not a good held for sale. Books that belong to a publisher and are only stored in the shop do not count either, because the shop does not own them. Inventory is owned goods held for sale, or owned materials held to make goods for sale.

A shop such as a bookshop buys goods already finished and sells them in the same condition. It therefore holds one kind of inventory: finished goods bought for resale. A business that makes things holds three kinds, because making takes time and the product passes through stages. Imagine a book-binding workshop, such as the separate workshop Meghna Binding Service. It buys paper, board, thread and glue. It folds, sews and covers books over several days. It then holds finished volumes until a customer collects them. At any moment the workshop has materials waiting, books half-made, and books complete.

The first kind is **raw materials**. Raw materials are materials a manufacturer holds to turn into products. For example, the reels of paper and the sheets of board in the workshop are raw materials. They have not yet been changed in any way.

The second kind is **work in process**. Work in process means goods in a factory that have been started but are not yet finished. For example, a pile of printed sheets that are sewn but have no cover is work in process. Its cost includes the materials and the labour used so far.

The third kind is **finished goods**. Finished goods are completed products held for sale. For example, a bound and trimmed volume on the collection shelf is a finished good. A bookshop's inventory consists wholly of finished goods, bought ready-made from a publisher.

The figure below shows the chain. Read it from left to right. Each box is one kind of inventory, and the arrows show the order in which goods move. The first box is lightly shaded because its goods are untouched. The middle box has cross-hatching because work is in progress. The last box is dense, because its goods are complete. A customer stands at the far end.

Two different words in common use cause confusion, so one of them is set aside.

Stock. Word in use. In daily life, a supply of goods kept for later. In this book, stock means shares in a company; goods are called inventory.

Inventory moves from the balance sheet to the income statement at the moment of sale.

Not everything in a shop is inventory, because the test is the purpose for which the item is held.

A merchandising shop holds one kind of inventory; a factory holds three.

The first stage is the materials waiting at the start of the chain.

Raw materials. Materials a manufacturer holds to turn into products.

The second stage is the unfinished products in the middle of the chain.

Work in process. Goods in a factory that have been started but are not yet finished.

The third stage is the completed products waiting for a customer.

Finished goods. Completed products held for sale.

The figure shows the chain from materials to customer, and how each stage differs in hatching.

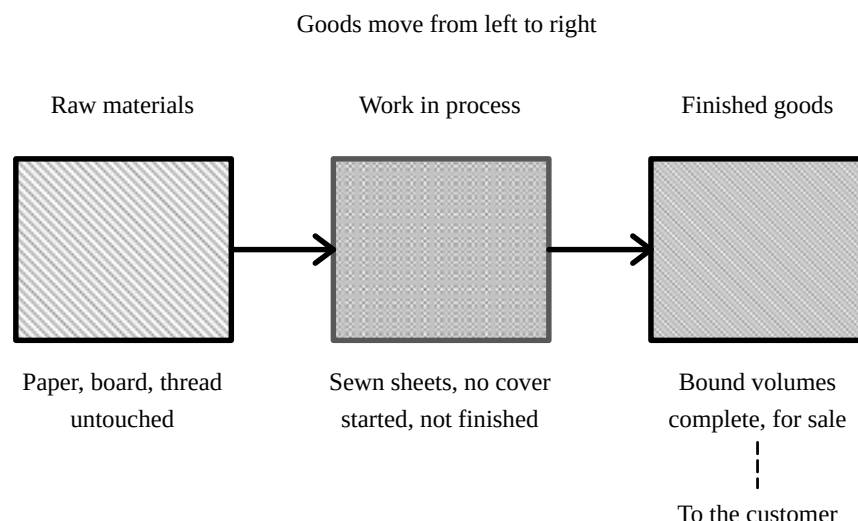


FIG. 13.1. *Three kinds of inventory in a factory chain*

From four viewpoints the kinds matter in different ways.

Before moving on, we collect the ideas of this section.

A business could try to hold nothing at all, and the reasons it does not form a framework.

Four reasons explain why every trading business holds some inventory.

Holding too little costs sales, goodwill and sometimes the customer for ever.

The kinds matter to different people. The owner wants to know how much money is tied up in each stage. The manager of the workshop watches the middle stage, because goods that stay long in the middle are a sign of delay. The employee who handles the materials needs a count of what is waiting. A lender looking at the balance sheet sees the total and asks how quickly it turns into cash. Finished goods are closest to cash, and raw materials are furthest from it.

Before moving on. Inventory is owned goods held for sale, or owned materials held to make goods for sale. It is a current asset until the sale, when its cost becomes cost of goods sold. A shop holds finished goods only. A factory holds raw materials, work in process and finished goods. You should now be able to say, for any item in a business, whether it is inventory and, if so, which kind.

13.2 WHY HOLD INVENTORY

Why does a business not hold nothing? It would save storage, it would never have unsold goods, and it would never have a loss from old books. The answer is that customers do not wait, and suppliers do not deliver at the moment of need. Section 13.1 asked what inventory is. This section asks why a business accepts the burden of holding it. There are four main reasons.

The first reason is the customer. A customer who finds the shelf empty usually goes elsewhere, so goods must be ready when the customer arrives. The second reason is delay in supply. A publisher may need two weeks to deliver, so the shop needs enough goods to last until the delivery comes. The third reason is price. Buying a large quantity at one time often earns a lower price per item, and fewer deliveries mean lower transport costs. The fourth reason is demand that changes with the season. A bookshop sells many more textbooks in the weeks before classes start, so it must prepare before that time. A factory has a fifth reason: the stages of making need materials and part-made goods to keep the workers busy.

Suppose the owner decides to hold very little. The first cost is the lost sale: the revenue and the gross profit on it are gone. The second cost is harder to see. A student who cannot find a textbook this term may never return next term. The third cost is the emergency purchase, for example paying extra to receive goods quickly. These costs do not appear on any ledger account, because the sale that

never happened leaves no record. They are real all the same, and that is why a shortage is easy to overlook.

Now suppose the owner decides to hold a great deal. The goods must be stored, insured and guarded. Paper can spoil in damp air. Fashions change, and a book may fall out of demand. Above all, the money spent on the goods could have been used elsewhere, for example to pay a debt or to buy goods that sell faster. The total of these burdens is called the holding cost, and it is usually stated as a percentage of the goods' cost for one year.

Case file, illustrative. On 31 December 2025, 800 titles had a total cost of Tk 380,000 and had not sold for 18 months. Meghna and Sons estimated its holding cost at 10% a year. The holding cost for one year is therefore $\text{Tk } 380,000 \times 10\% = \text{Tk } 38,000$. That is Tk 38,000 spent every year to keep goods that did not sell.

The table below compares the two mistakes. Introduce each row as a question: what is lost, and who sees it? Read across a row to compare the two cases.

THE COST OF TOO LITTLE AND THE COST OF TOO MUCH

QUESTION	TOO LITTLE INVENTORY	TOO MUCH INVENTORY
---	---	---
What is lost?	Sales and customers	Money tied up and goods that age
Is it recorded in the books?	No; the sale leaves no entry	Yes; storage and insurance appear as expenses
Who notices first?	The customer	The owner, when cash runs short
Typical remedy	Reorder sooner	Reduce purchases; sell at a lower price

The right amount cannot be set by a rule that fits every business. A grocer who sells fresh fish needs very little, because fish spoils in a day. A bookseller who sells textbooks needs more, because titles keep their value for a year or longer. There is also an exception to the idea that more is safer: if demand collapses suddenly, a large stock of goods becomes a large loss. We meet a famous case in the closing pack.

Before moving on. A business holds inventory because customers do not wait, suppliers need time, bulk buying saves money and demand follows the season. Holding too little costs sales. Holding too much costs storage, insurance, spoilage and the use of money. You should now be able to say one cost of each mistake and why the first is harder to see.

13.3 RECORDING INVENTORY

A business must know two things about its inventory: the quantity it holds and the value of that quantity. The value of each item is its cost, which Chapter 14 explains in detail. This section teaches how the quantity is found and recorded, and how an item whose value has fallen is treated. There are two systems for recording, and we begin with the more modern one.

Holding too much costs money in many small amounts that grow large together.

Cost. Word in use. In daily life, the price you pay for something. In accounting, the amount given up to acquire or hold an item; a holding cost is a different amount, spent to keep goods.

The running case gives a holding cost for a group of titles that did not sell.

A table sets the two mistakes side by side so that their different natures can be compared.

The right amount lies between the two mistakes and depends on the business.

Before moving on, we collect the ideas of this section.

To record inventory a business must answer two questions: how many, and what are they worth.

The perpetual system keeps a running record that changes after every purchase and every sale.

Perpetual inventory system. A system that updates the inventory account after every purchase and sale.

The periodic system leaves the account unchanged all year and finds the figure by counting at the end.

Periodic inventory system. A system that finds inventory only by a physical count at period end.

A small example shows that both systems reach the same cost of goods sold.

Under the perpetual system, every movement is recorded as it happens.

Under the periodic system, the answer comes from a formula applied after a count.

The two systems differ in what they reveal during the year, and a table makes the difference plain.

A count is needed under both systems, because records can differ from what is on the shelf.

The first system is the **perpetual inventory system**. A perpetual inventory system is a system that updates the inventory account after every purchase and sale. For example, when the shop buys five books at Tk 100 each, the Inventory account rises by Tk 500 at once. When it sells eight books, the account falls by their cost, and the cost of goods sold rises by the same amount. At any moment the account shows what the shop should be holding. The word *perpetual* means "going on without end".

The second system is the **periodic inventory system**. A periodic inventory system is a system that finds inventory only by a physical count at period end. For example, a small stall records every purchase in a Purchases account and every sale in a Sales account, but does not touch Inventory until the last day of the year. The stall then counts what remains, values it at cost, and calculates the cost of goods sold by subtraction. The word *periodic* means "at regular periods".

Take small, clean numbers so that the method is the only thing to learn. *Given.* A shop begins the year with 10 books at Tk 100 each. It buys 5 more at Tk 100. It sells 8. All books cost Tk 100, so no question arises about which book is sold. *Find.* The cost of goods sold, and the inventory at the end, under each system.

Perpetual. Start with $10 \times 100 = \text{Tk } 1,000$. The purchase adds $5 \times 100 = \text{Tk } 500$, so the account shows Tk 1,500. The sale of 8 books reduces it by $8 \times 100 = \text{Tk } 800$, so the account shows Tk 700. The cost of goods sold, recorded at each sale, is Tk 800.

Periodic. The year's records show the opening inventory of Tk 1,000 and purchases of Tk 500. A count at year end finds 7 books, which are valued at $7 \times 100 = \text{Tk } 700$. The cost of goods sold is the opening inventory plus purchases less the closing inventory: $1,000 + 500 - 700 = \text{Tk } 800$. *Answer.* Both systems give cost of goods sold of Tk 800 and closing inventory of Tk 700. *Check.* $10 + 5 - 8 = 7$ books, and $7 \times 100 = 700$. The two systems agree, as they should.

PAUSE AND CHECK 13.P1 A shop with a periodic system loses 3 books to theft during the year. Will the loss show as a separate item in its records, or will it be hidden? Give a reason.

If both systems give the same answer, why choose between them? The difference is in what they show along the way. Read each row of the table below as a question about the system.

PERPETUAL AND PERIODIC SYSTEMS COMPARED

QUESTION	PERPETUAL	PERIODIC
---	---	---
When is Inventory updated?	After every purchase and sale	Once, at period end
Can the owner see the quantity at any time?	Yes, from the records	No, only after a count
Is a loss from theft visible?	Yes, as a difference from the count	No; it hides inside cost of goods sold
Work needed during the year	More	Less

Neither system removes the need to count. In the periodic system the count is the only source of the closing figure. In the perpetual system the count is a check on the records. The traditional name for this count is the *stock count*, the one place where this book keeps the older word. We also call it a physical count. A team walks through the shelves and the store room and writes down each item

and its quantity, ideally with a second person who checks the first. Goods that belong to others are left out. Goods that have been sold but not yet collected are also left out, because they no longer belong to the business.

Under the perpetual system, suppose the records say 700 and the count finds 660. The difference of 40 may come from theft, damage, a mistake in recording, or a sale not entered. The business investigates first. If no cause is found, it reduces the Inventory account to the counted figure and charges the loss as an expense. The count is the final authority, because goods that cannot be found cannot be sold.

The last idea in this chapter is about value. A business usually reports inventory at its cost. But suppose an item can no longer be sold for as much as it cost. A business should not report an asset at more than the money it can bring in. The money it can bring in is the **net realisable value**. Net realisable value is the expected selling price of an item less the cost of selling it. For example, a textbook that cost Tk 100 may now sell for only Tk 150 after a discount, and selling it costs Tk 20 in packing and delivery. Its net realisable value is $150 - 20 = \text{Tk } 130$.

The rule is plain. For each item or group of the same item, compare the cost with the net realisable value. If cost is lower, report cost. If net realisable value is lower, report that figure and treat the difference as a loss in the period when the value fell. In the example above, cost Tk 100 is lower than Tk 130, so the book stays at Tk 100. If the discount were bigger and the net realisable value fell to Tk 70, the book would be reported at Tk 70, and a loss of Tk 30 would be recorded. We never raise an item above its cost because its value rises. Standards on this point differ between countries, so a business must follow the standards of the place where it reports.

Before moving on. A perpetual system updates Inventory after every purchase and sale. A periodic system finds it only by a count at period end. Both give the same cost of goods sold when the records are right. Inventory is reported at the lower of cost and net realisable value. You should now be able to say what each system shows during the year and how to test whether an item must be written down.

After a count, any difference between the records and the shelves must be explained and corrected.

Cost is not the only measure, because some goods will sell for less than they cost.

Net realisable value. The expected selling price of an item less the cost of selling it.

The rule: report the lower of cost and net realisable value, item by item.

Before moving on, we collect the ideas of this section.

WORKED EXAMPLE: EXCESS TITLES AND AN OLD EDITION

Case file, illustrative. Given for 31 December 2025: inventory at cost Tk 3,500,000. Excess inventory: 800 titles at cost Tk 380,000, unsold for 18 months, holding cost 10% a year. Old edition: 120 copies, cost Tk 300 each, net realisable value Tk 180 each.

Find. (a) the share of inventory in the excess titles, and the yearly holding cost; (b) the write-down for the old edition, and the value at which it is reported.

Part (a), steps. The excess titles are $380,000 \div 3,500,000 = 0.1086$, which is about 10.9% of the inventory. The holding cost is the cost times the rate: $380,000 \times 10\% = 380,000 \times 0.10 = \text{Tk } 38,000$ a year. *Answer.* About 11% of the inventory is excess, and keeping it costs Tk 38,000 a year. *Check.* 38,000 is one tenth of 380,000, which matches a rate of 10%.

Part (b), steps. The cost per copy is Tk 300 and the net realisable value is Tk 180. Since 180 is lower than 300, the copies are written down. The write-down per copy is $300 - 180 = \text{Tk } 120$. For 120 copies it is $120 \times 120 = \text{Tk } 14,400$. The old edition was carried at $120 \times 300 = \text{Tk } 36,000$, and is now reported at $120 \times$

The two groups from the case file are examined in full, with every line of arithmetic shown.

The task has two parts: the cost of holding the titles and the write-down of the old edition.

Part (a) uses a percentage of a number, which is the same step twice.

Part (b) compares cost with net realisable value, copy by copy, and multiplies by the number of copies.

The write-down is recorded as a loss and reduces the Inventory account by the same amount.

The common slips are in the units and in the direction of the comparison.

The chapter's ideas in a few lines.

Belief: any item the business owns and holds is inventory.

Belief: the periodic system is wrong because it does not update the account.

Belief: net realisable value is the selling price.

Belief: when the market value of an item rises, the item should be written up.

Belief: a count that differs from the records means the count is wrong.

A famous modern case shows how holding too much can become a very large loss.

180 = Tk 21,600. *Answer.* The write-down is Tk 14,400. *Check.* 36,000 – 21,600 = 14,400, which agrees.

The entry follows from the answer. Debit an expense account, here called Inventory Write-down Expense, with Tk 14,400, and credit Inventory with Tk 14,400. The income statement bears the loss in 2025, the year in which the value fell, and the balance sheet shows the lower figure.

Three slips are common. The first is to multiply by the wrong number of copies. The second is to use the selling price in place of net realisable value, forgetting to subtract the cost of selling. The third is to write up an item whose value has risen. A write-down is permitted; a write-up above cost is not.

SUMMARY

Inventory is owned goods held for sale, or owned materials held to make them. A shop holds finished goods; a factory holds raw materials, work in process and finished goods. Holding too little costs sales, and holding too much costs money and risk. A perpetual system updates Inventory at every purchase and sale; a periodic system finds it by a count at period end. Inventory is reported at the lower of cost and net realisable value.

THINK IT THROUGH

13.T1 The 800 unsold titles of Meghna and Sons cost Tk 380,000 and the holding cost is 10% a year. If they stay unsold for a further 12 months, what extra cost will the shop bear, and what other loss might it face?

13.T2 A bookseller says: "I never hold more than one week of goods, so I never lose money on inventory." Name two costs that this policy might create.

WHERE PEOPLE GO WRONG

The test is purpose. Equipment and office supplies are owned, but they are not held for sale, so they are not inventory.

Both systems are correct. They give the same cost of goods sold when the records are right. They differ in what they show during the year.

Net realisable value is the selling price less the cost of selling. Forgetting to subtract that cost raises the value wrongly.

Inventory is not reported above its cost. A fall below cost is recognised; a rise above cost is not.

The count is the final authority on quantity. A difference must be investigated, but if no cause is found the records are corrected to the count.

IN HISTORY

In April 2001 the American networking company Cisco Systems, based in San Jose, California, announced a large reduction in its inventory value. Demand for its products had fallen sharply, while it had ordered materials for much higher sales. In its third quarter of fiscal 2001 the company recorded an excess inventory charge of about USD 2.25 billion, and most of it related to materials it no longer expected to use.

The lesson fits this chapter. Goods that cannot be sold at cost must be reported at a lower value, and the loss falls in the period when the fall is recognised. Cisco's write-down was large in size, but the same rule applies to the 120 old copies of Meghna and Sons.

The case teaches that a count and a rule on value matter most when demand changes quickly.

REVIEW QUESTIONS

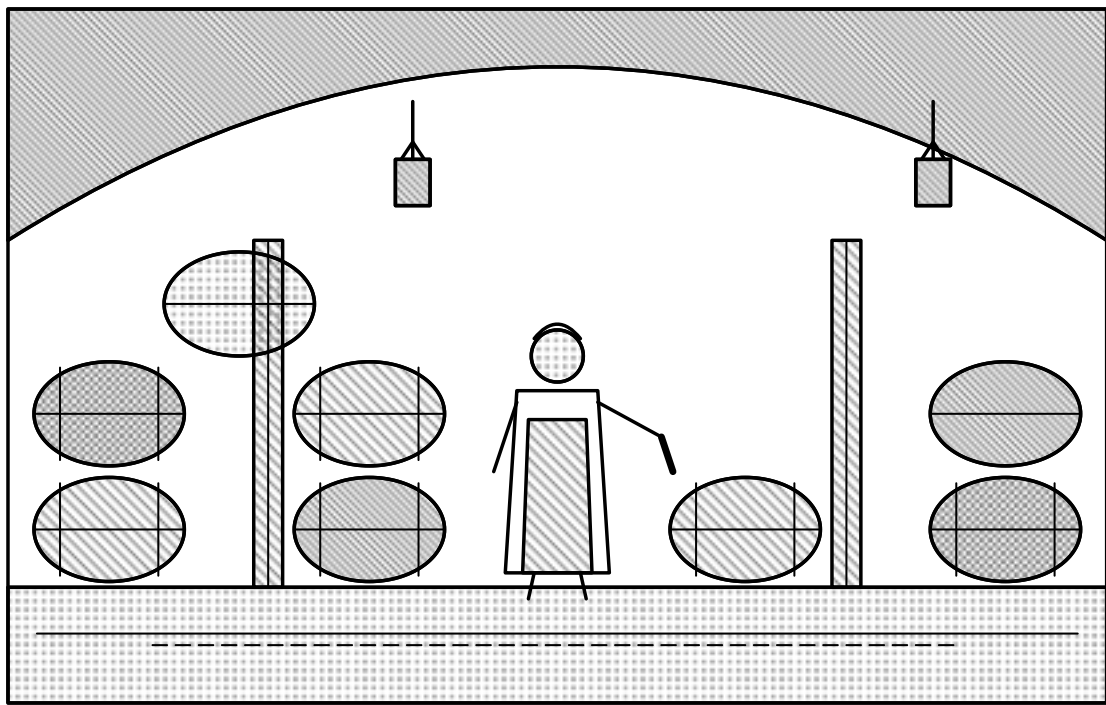
13.1 Classify each item as raw materials, work in process, finished goods, or not inventory: (a) reels of paper in a binding workshop; (b) sewn sheets with no cover; (c) a stapler on the owner's desk; (d) bound volumes ready for collection; (e) a supplier's books stored in the shop but still owned by the supplier.

13.2 A shop holds 200 copies of a guide book. Each cost Tk 450. Each can now be sold for Tk 500, and the cost of selling is Tk 90 a copy. Find the net realisable value per copy, and the write-down, if any, for all 200 copies.

13.3 A shop begins the year with inventory of Tk 40,000 at cost and buys goods for Tk 65,000. A count at year end shows goods with a cost of Tk 30,000. Find the cost of goods sold under a periodic system.

13.4 A perpetual record shows 500 units of a book. A count finds 470. Cost is Tk 60 a unit. Find the cost of the difference and say what the shop should do first.

13.5 An owner holds goods with a total cost of Tk 600,000 and a holding cost of 12% a year. Find the yearly holding cost. In the next year, 150 units that cost Tk 80 each have a net realisable value of Tk 55. Find the write-down, and state the new reported value of those units.



A cellar of casks in rows, with a cooper marking the oldest

CHAPTER XIV

Costing Inventory: FIFO, LIFO and Weighted Average

Two identical items were bought at different prices. Which cost did the customer take away?



This chapter teaches three rules for deciding which purchase costs leave inventory when goods are sold. You will apply each rule to one set of figures, compare the results, and see how the choice changes profit and the value of the goods still on hand.

YOU WILL NEED

You will need the difference between a perpetual record and a periodic record from Chapter 13 'Inventory: Meaning and Measurement', section 13.3 'Recording Inventory', and the idea of cost of goods sold from Chapter 10 'Accounts of a Merchandising Company', section 10.1 'Sales, Purchases and Returns'. Appendix A1 'Maths You Will Use' covers averages and rounding, which section 14.4 uses.

A shopkeeper holds two identical sacks of rice bought at different prices, and a customer buys one.

The profit on the sale is Tk 50 or Tk 30, depending on which cost we say the customer took away.

The choice looks small for one sack, but it grows large when thousands of units are sold.

This chapter teaches the three rules one at a time, using one set of figures throughout.

Picture a shopkeeper in a market. In the first week she buys one sack of rice for Tk 100. In the second week the price rises, and she buys a second sack for Tk 120. The two sacks stand side by side. They look the same, they weigh the same, and no label shows the week of purchase. A customer arrives and buys one sack for Tk 150.

Now the shopkeeper wants to know her profit. If the customer took the cheaper sack, the profit is Tk 150 less Tk 100, which is Tk 50. If the customer took the dearer sack, the profit is Tk 150 less Tk 120, which is Tk 30. Both answers are reasonable, and the sacks cannot tell us which is true. Many students find this puzzling at first, because they expect a fact to settle the matter. Here no physical fact can settle it. The business must choose a rule.

For one sack the difference is Tk 20. For a bookshop that buys thousands of books at changing prices, the difference can reach hundreds of thousands of taka. The owner, the tax collector and the bank that lends money to the business all read the profit figure. A rule that changes the profit changes what each of them sees. For that reason accountants name the possible rules, apply them carefully, and use one rule consistently.

Section 14.1 explains the problem and builds the vocabulary. Section 14.2 teaches the first rule, section 14.3 the second, and section 14.4 the third, with a comparison. A worked case then shows what happens when prices fall instead of rise. Go slowly. Each section finishes one idea before the next begins, and the same figures return every time so that only the rule changes.

* * *

14.1 THE PROBLEM OF CHANGING PRICES

GO SLOWLY HERE.

Prices of goods change, so units that look alike can carry different costs.

Recall: perpetual and periodic records, Chapter 13, 'Inventory: Meaning and Measurement', section 13.3 'Recording Inventory'.

A business rarely buys all its goods at one price. A supplier raises or lowers a price, a season ends, or a change in the exchange rate makes imported goods dearer. The business therefore holds goods that look the same but cost different amounts. Chapter 13 explained that the business records the goods it holds as inventory, and that it may count them at the end of the period or record every movement. In this chapter we count at the end of the period, and we apply each rule to the whole period at once. That choice keeps the arithmetic short while we learn the ideas.¹

¹ Where each unit can be told apart, for example a car with its own serial number, a business may use the actual cost of that unit. This chapter treats only goods that cannot be told apart. The same rules also apply to raw materials and work in process, which face the same problem.

Imagine the goods as a pile, built up as purchases arrive. Each purchase adds a **layer** of its own: a group of units that were bought at one time and at one unit cost. The opening inventory, which is the goods held when the period began, is the first layer. The sack bought for Tk 100 is one layer. The sack bought for Tk 120 is a second layer, placed on top of the first. When goods are sold, we must decide which layer the sold units are taken from.

A **cost flow assumption** is a rule that decides which purchase costs are treated as sold and which remain in inventory. Suppose a shop sells goods from a pile with two layers. The rule says which layer's cost goes to cost of goods sold first. The word flow refers to the order of costs, not to the movement of goods on the shelf. A shop may hand out its newest dictionary first, yet still assume that its oldest cost was sold first. We call it an assumption because we choose it, and no one proves it.

We now test the idea with the smallest numbers possible. *Given.* One unit bought at Tk 10 and one unit bought at Tk 12, both on hand. One unit is sold for Tk 20. *Find.* The cost of goods sold, the cost of the unit left, and the gross profit, under each of two rules: the oldest cost is sold first, or the newest cost is sold first. *Formula.* Gross profit equals net sales minus cost of goods sold. In symbols, $GP = S - C$, where GP is gross profit, S is net sales and C is cost of goods sold. *Steps.* If the oldest cost is sold first, $C = \text{Tk } 10$, the unit left costs Tk 12, and $GP = \text{Tk } 20 - \text{Tk } 10 = \text{Tk } 10$. If the newest cost is sold first, $C = \text{Tk } 12$, the unit left costs Tk 10, and $GP = \text{Tk } 20 - \text{Tk } 12 = \text{Tk } 8$. *Answer.* The profit is Tk 10 under the first rule and Tk 8 under the second. *Check.* In both cases the cost of goods sold plus the cost of the unit left equals Tk 22, the cost of both units.

The unit left in the tiny calculation has a name. **Ending inventory** is the cost of goods still on hand at the end of the period. The second sack in our opening puzzle, if the customer took the first, is the ending inventory, valued at Tk 120. Ending inventory appears as an asset in the balance sheet. It is also the opening inventory of the next period, so an error in one period carries into the next.

The tiny calculation shows a rule that never changes. The cost of goods available for sale is the opening inventory plus the purchases of the period. Each unit of that cost is either sold or left. In words: cost of goods available equals cost of goods sold plus ending inventory. In symbols, $A = C + E$, where A is the cost of goods available, C is the cost of goods sold and E is the ending inventory. A cost flow assumption does not change A. It decides only how A is divided between C and E.

Consider why the choice matters, from four viewpoints. The owner sees a higher or lower profit, and so a higher or lower reward for the year. The tax collector sees a larger or smaller amount on which tax is due, in countries where tax follows profit. The bank sees a higher or lower value for the goods that support its loan. The manager compares this year with last year, and a change of rule would spoil the comparison. A rule that gives a larger cost of goods sold gives a smaller gross profit and a smaller ending inventory. A rule that gives a smaller cost of goods sold does the opposite. Over the whole life of a business the total cost is the same; the rules differ in the timing of the cost.

Case file, illustrative. Meghna and Sons sells boxed dictionary sets. At the start of the period it held an opening inventory of 20 sets at Tk 300 a set, a cost

The word layer helps us picture goods bought at different times and prices.

Layer. Word in use. In daily life, a thickness of material lying over another. In accounting, a group of units bought at one time at one cost.

A cost flow assumption is a rule that names which layers are treated as sold.

Flow. Word in use. In daily life, water or traffic moving along a path. In accounting, the order in which costs are treated as leaving inventory, which need not match the movement of the goods.

Cost flow assumption. A rule that decides which purchase costs are treated as sold and which remain in inventory.

A tiny calculation with two sacks shows that each rule gives a different profit.

Ending inventory is the cost of the goods still on hand when the period closes.

Ending inventory. The cost of goods still on hand at the end of the period.

Every cost the business could sell ends up either in cost of goods sold or in ending inventory.

The method matters because it moves cost between the income statement and the balance sheet.

GO SLOWLY HERE.

The case file provides the figures used in the next three sections.

The table lists the three layers in order, with the cost of each layer.

of Tk 6,000. It then bought 30 sets at Tk 320 a set, a cost of Tk 9,600, and later bought 50 sets at Tk 340 a set, a cost of Tk 17,000. During the period it sold 70 sets at Tk 500 a set, net sales of Tk 35,000.

The table below shows the layers in the order in which they arrived. Read each row from left to right: the layer, the number of units, the unit cost and the total cost of that layer. The last row adds the units and the costs.

LAYERS OF BOXED DICTIONARY SETS AVAILABLE FOR SALE

LAYER	UNITS	UNIT COST (TK)	TOTAL COST (TK)
Opening inventory	20	300	6,000
First purchase	30	320	9,600
Second purchase	50	340	17,000
Available for sale	100		32,600

The cost of goods available is Tk 32,600, and 30 sets remain after 70 are sold.

Given. The three layers in the table. *Find.* The units and the cost available for sale, and the units left after the sale. *Formula.* Cost available equals the sum of the layer costs. In symbols, $A = L_1 + L_2 + L_3$, where A is the cost available and L_1 , L_2 and L_3 are the layer costs. *Steps.* Tk 6,000 + Tk 9,600 = Tk 15,600. Then Tk 15,600 + Tk 17,000 = Tk 32,600. Units: 20 + 30 + 50 = 100, and 100 – 70 = 30. *Answer.* Cost available is Tk 32,600, and 30 sets remain. *Check.* Each layer cost equals its units times its unit cost: 20×300 , 30×320 and 50×340 .

Recap: a cost flow assumption divides the cost available between cost of goods sold and ending inventory.

Before moving on. Goods bought at different prices form layers. A cost flow assumption decides which layers are treated as sold, and ending inventory is the cost that remains. Cost of goods sold plus ending inventory always equals the cost available. You should now be able to say why two businesses with identical goods can report different profits.

* * *

14.2 FIRST-IN FIRST-OUT

GO SLOWLY HERE.

A queue at a canteen shows the idea: the first person to arrive is the first to be served.

First-in first-out treats the oldest costs as sold first.

First-in first-out. A method that treats the oldest costs as sold first, leaving the newest costs in inventory.

We take 70 units from the oldest layers, and 30 units at Tk 340 remain.

Think of a queue at a canteen. The person who arrived first is served first, and the person who arrived last waits longest. Many shops follow the same order with their goods. They sell the oldest goods first, because old goods may spoil or fall out of fashion. A cellar of casks works in this way too: the cooper marks the oldest cask, and that cask is opened first.

The **first-in first-out** method, often written FIFO, is a method that treats the oldest costs as sold first, leaving the newest costs in inventory. In our pile, the opening layer is the oldest, so its cost leaves first. Then the cost of the first purchase leaves, and so on, until the number of units sold is reached. What remains is made of the newest layers.

We apply the method to the case. *Given.* The layers in the table of section 14.1, and a sale of 70 units for Tk 35,000. *Find.* The ending inventory, the cost of goods sold and the gross profit. *Formula.* Ending inventory equals the units left times the unit cost of the newest layer. Cost of goods sold equals cost available minus ending inventory. Gross profit equals net sales minus cost of goods sold. In symbols, $E = n \times u$, $C = A - E$ and $GP = S - C$, where n is the number of units left and u is the unit cost of the newest layer. *Steps.* The oldest 70 units are 20 + 30 + 20 from the second purchase. The 30 units left all come from the second purchase, so $E = 30 \times \text{Tk } 340 = \text{Tk } 10,200$. Then $C = \text{Tk } 32,600 - \text{Tk } 10,200 = \text{Tk } 22,400$. Then $GP = \text{Tk } 35,000 - \text{Tk } 22,400 = \text{Tk } 12,600$. *Answer.*

Under FIFO, ending inventory is Tk 10,200, cost of goods sold is Tk 22,400 and gross profit is Tk 12,600. *Check.* Directly: $20 \times 300 + 30 \times 320 + 20 \times 340 = 6,000 + 9,600 + 6,800 = \text{Tk } 22,400$.

PAUSE AND CHECK 14.P1 Suppose Meghna and Sons had sold only 60 sets. Using FIFO, find the ending inventory, the cost of goods sold, and check that they add to Tk 32,600.

Exceptions and limits need attention. FIFO is an assumption about costs, not a statement about which dictionary left the shelf. In our case the prices rise from Tk 300 to Tk 340. FIFO charges the cheap, old costs to the sale, so cost of goods sold is low and profit is high. The ending inventory carries the newest, dearest costs, so the asset in the balance sheet is close to what the goods cost now. When prices fall, the effects reverse. The worked case in this chapter shows this.

Before moving on. First-in first-out takes the oldest layers for cost of goods sold and keeps the newest layers as ending inventory. For the dictionary sets, it gives Tk 22,400 and Tk 10,200. You should now be able to say which layers remain after a sale under FIFO.

* * *

14.3 LAST-IN FIRST-OUT

Think of a stack of plates in a kitchen. A cook adds clean plates on top, and takes the top plate when one is needed. The plate placed last is taken first. The bottom plate may stay in the stack for months. A pile of coal or sand in a yard behaves in the same way, because the newest load lies on top and is taken first.

The **last-in first-out** method, often written LIFO, is a method that treats the newest costs as sold first, leaving the oldest costs in inventory. The method reverses the order of FIFO. In our pile, the newest layer is the second purchase, so its cost leaves first. Then the first purchase, and then the opening layer, if more units must be taken.

We apply the method to the same case. *Given.* The same layers and the same sale of 70 units for Tk 35,000. *Find.* The ending inventory, the cost of goods sold and the gross profit. *Formula.* The formulas are the same as for FIFO: $C = A - E$ and $GP = S - C$, but E is now built from the oldest layers. *Steps.* The newest 70 units are 50 from the second purchase and 20 from the first purchase. The 30 units left are 20 from the opening layer and 10 from the first purchase. So $E = 20 \times \text{Tk } 300 + 10 \times \text{Tk } 320 = \text{Tk } 6,000 + \text{Tk } 3,200 = \text{Tk } 9,200$. Then $C = \text{Tk } 32,600 - \text{Tk } 9,200 = \text{Tk } 23,400$. Then $GP = \text{Tk } 35,000 - \text{Tk } 23,400 = \text{Tk } 11,600$. *Answer.* Under LIFO, ending inventory is Tk 9,200, cost of goods sold is Tk 23,400 and gross profit is Tk 11,600. *Check.* Directly: $50 \times 340 + 20 \times 320 = 17,000 + 6,400 = \text{Tk } 23,400$.

Again we must state the limits. LIFO charges the newest, dearest costs to the sale, so cost of goods sold is high and gross profit is low. Ending inventory is valued at old prices, so the asset in the balance sheet may be far below what the goods would cost today. A business that uses LIFO in a time of rising prices reports less profit, and where tax follows profit it may owe less tax. Rules on this differ by country. The history section of this chapter describes the position in the United States and under international standards.

Before moving on. Last-in first-out takes the newest layers for cost of goods sold and keeps the oldest layers as ending inventory. For the dictionary sets, it gives Tk 23,400 and Tk 9,200. You

Pause and check that you can take costs from the oldest layers first.

When prices rise, FIFO gives the lowest cost of goods sold and the highest ending inventory.

Recap: FIFO sells the oldest costs first and leaves the newest costs in inventory.

GO SLOWLY HERE.

A stack of plates shows the idea: the plate placed last is the plate taken first.

Last-in first-out treats the newest costs as sold first.

Last-in first-out. A method that treats the newest costs as sold first, leaving the oldest costs in inventory.

We take 70 units from the newest layers, and 30 units of older cost remain.

When prices rise, LIFO gives the highest cost of goods sold and the lowest profit.

Recap: LIFO sells the newest costs first and leaves the oldest costs in inventory.

should now be able to say why LIFO gives a lower profit than FIFO when prices rise.

* * *

14.4 WEIGHTED AVERAGE COST

GO SLOWLY HERE.

Mixing tea of different prices in one jar shows the idea: the jar has one average price.

Weighted average cost gives every unit the average cost of all units available.

Weighted average cost. A method that gives every unit the average cost of all units available.

The unit cost is the total cost available divided by the total units available.

Pause and check that you can find the average cost of the units available.

The figure shows how each method takes cost from the pile of layers.

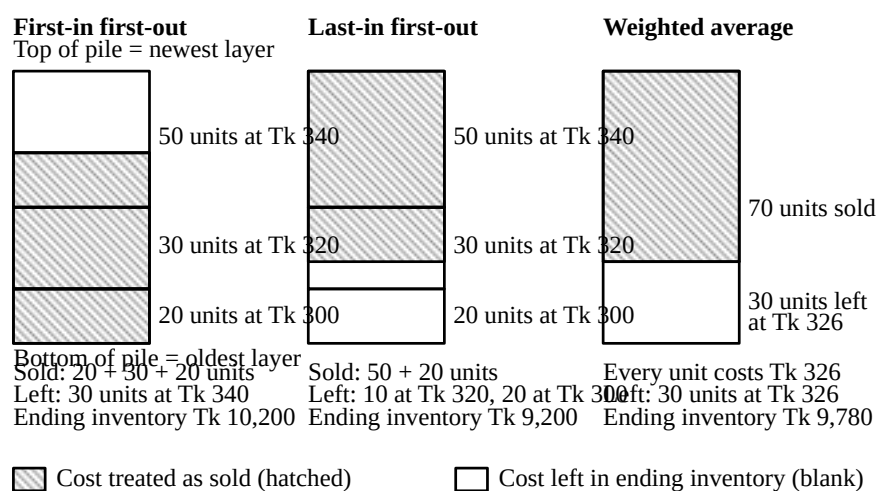
Think of a tea seller who pours tea bought at different prices into one large jar. After the pouring, no one can say which tea came from which purchase. Every spoonful of the mixture has the same average cost. The seller prices and counts the tea by that single figure. The third method treats the goods in the same way.

The **weighted average cost** method is a method that gives every unit the average cost of all units available. The word weighted means that each layer counts in proportion to its number of units. A layer of 50 units has more effect on the average than a layer of 20 units. The method needs no choice of oldest or newest layer, because every unit receives the same cost.

We apply the method to the case. *Given.* Cost available Tk 32,600 for 100 units, and a sale of 70 units for Tk 35,000. *Find.* The unit cost, the ending inventory, the cost of goods sold and the gross profit. *Formula.* Unit cost equals cost available divided by units available. In symbols, $u = A \div N$, where u is the unit cost, A is the cost available and N is the number of units available. Then $E = n \times u$ and $C = x \times u$, where n is the units left, x is the units sold, and $GP = S - C$. *Steps.* $u = \text{Tk } 32,600 \div 100 = \text{Tk } 326$ a unit. $E = 30 \times \text{Tk } 326 = \text{Tk } 9,780$. $C = 70 \times \text{Tk } 326 = \text{Tk } 22,820$. $GP = \text{Tk } 35,000 - \text{Tk } 22,820 = \text{Tk } 12,180$. *Answer.* Under weighted average cost, ending inventory is Tk 9,780, cost of goods sold is Tk 22,820 and gross profit is Tk 12,180. *Check.* $\text{Tk } 22,820 + \text{Tk } 9,780 = \text{Tk } 32,600$, the cost available. The unit cost Tk 326 lies between Tk 300 and Tk 340, as an average must.

PAUSE AND CHECK 14.P2 Using the unit cost found above, what is the cost of goods sold by weighted average if only 25 sets are sold from the 100 available?

The first figure shows the pile of Meghna and Sons as a column of three layers. The height of each block is in proportion to its number of units. Read a column from the bottom, where the oldest layer lies, to the top, where the newest layer lies. Hatched blocks show costs treated as sold. Blank blocks show costs that remain in ending inventory. In the third column all units are mixed, so the pile is one block at the average cost.



Each pile: 100 units available, cost Tk 32,600; 70 units sold.

FIG. 14.1. Three methods as layers of cost leaving a pile

The table collects the results. Read each column down for one method, and each row across to compare the methods. In every column the cost of goods sold plus the ending inventory equals Tk 32,600. Notice also that the weighted average figures lie between the FIFO and LIFO figures. This is no accident, because an average of the costs must lie between the lowest and the highest.

RESULTS OF THE THREE METHODS FOR THE DICTIONARY SETS

MEASURE	FIRST-IN FIRST-OUT	LAST-IN FIRST-OUT	WEIGHTED AVERAGE
Cost of goods sold (Tk)	22,400	23,400	22,820
Ending inventory (Tk)	10,200	9,200	9,780
Gross profit (Tk)	12,600	11,600	12,180

The second figure is a bar chart of the results for Meghna and Sons. Each group of bars belongs to one method. The vertical axis shows taka in thousands, and the height of a bar shows the size of that measure. In each group the three bars are cost of goods sold, ending inventory and gross profit, in that order. The bars are marked by different hatching, so the chart reads in grey as well as in colour. Notice how little the bars differ between methods: the choice moves a small share of the cost.

Comparing the three methods shows that weighted average cost lies between the other two.

The second figure shows the results of the three methods side by side.

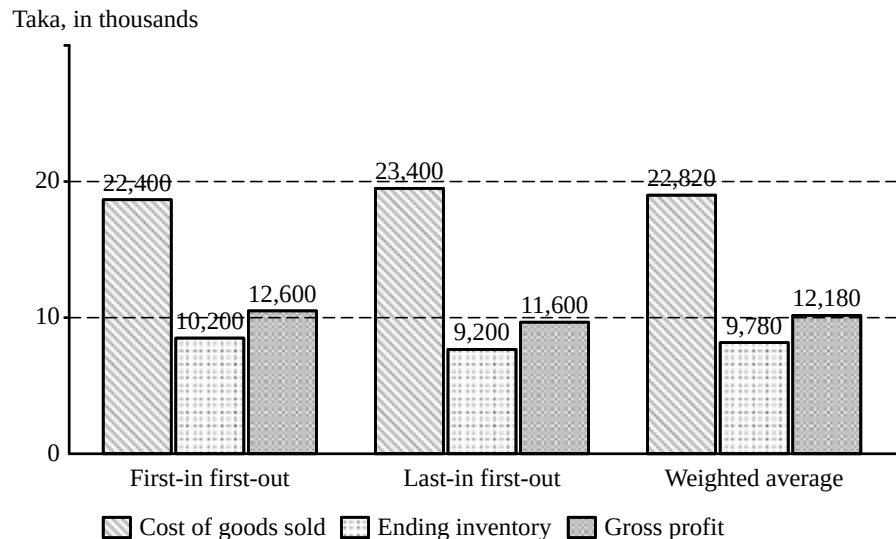


FIG. 14.2. Results of the three methods side by side

Recap: weighted average cost uses one unit cost, and its results lie between the other two methods.

Before moving on. Weighted average cost divides the cost available by the units available and gives every unit that cost. For the dictionary sets the unit cost is Tk 326. You should now be able to say why its profit lies between the profits of FIFO and LIFO.

* * *

WORKED CASE: PRICES THAT FALL

This worked case repeats the method with prices that fall, to show that the ranking reverses.

Situation. A different shop sells identical goods. It held 20 units at Tk 340, bought 30 units at Tk 320, and later bought 50 units at Tk 300. It sold 70 units at Tk 500, net sales Tk 35,000. These figures are for practice, and are not part of the case file. The prices fall from Tk 340 to Tk 300. **Question.** What are ending inventory, cost of goods sold and gross profit under each method?

The reasoning first finds the cost available, and then applies each rule in turn.

Reasoning. The cost available is $20 \times 340 + 30 \times 320 + 50 \times 300 = 6,800 + 9,600 + 15,000 = \text{Tk } 31,400$, for 100 units. Under FIFO, the 30 units left are the newest, all at Tk 300, so $E = \text{Tk } 9,000$, $C = \text{Tk } 22,400$ and $GP = \text{Tk } 12,600$. Under LIFO, the 30 units left are the oldest: 20 at Tk 340 and 10 at Tk 320, so $E = 6,800 + 3,200 = \text{Tk } 10,000$, $C = \text{Tk } 21,400$ and $GP = \text{Tk } 13,600$. Under weighted average, $u = 31,400 \div 100 = \text{Tk } 314$, so $E = 30 \times 314 = \text{Tk } 9,420$, $C = 70 \times 314 = \text{Tk } 21,980$ and $GP = \text{Tk } 13,020$.

The conclusion is that the ranking reverses when prices fall, while the cost available never changes.

Conclusion. When prices fell, LIFO gave the highest profit, Tk 13,600, and FIFO the lowest, Tk 12,600. When prices rose, as in the case file, the order was the reverse. Weighted average lay between the two in both situations. **Check.** In each method, cost of goods sold plus ending inventory is Tk 31,400: $22,400 + 9,000$, then $21,400 + 10,000$, then $21,980 + 9,420$.

Three slips occur often in this work, and each one can be prevented.

Common slips deserve a warning. The first is to take the sold units from the wrong end of the pile, which swaps the results of FIFO and LIFO. The second is to average the three unit costs, Tk 300, Tk 320 and Tk 340, instead of dividing the total cost by the total units. That error gives Tk 320, not Tk 326, because it ignores the number of units in each layer. The third is to forget that cost of goods sold and ending inventory must add to the cost available. Test every answer with that addition.

* * *

SUMMARY

Goods bought at different prices form layers, and a cost flow assumption decides which layers are treated as sold. First-in first-out sells the oldest costs first. Last-in first-out sells the newest costs first. Weighted average cost gives every unit the same average cost. For Meghna and Sons, with rising prices, gross profit was Tk 12,600, Tk 11,600 and Tk 12,180 under the three methods, and ending inventory was Tk 10,200, Tk 9,200 and Tk 9,780. In every method, cost of goods sold plus ending inventory equals the cost available, Tk 32,600.

The summary restates the whole chapter in a few lines.

THINK IT THROUGH

14.T1 Prices rise during the period. Rank the three methods from the highest to the lowest gross profit, and explain the ranking in a few sentences.

14.T2 Prices fall during the period. How does the ranking change? What stays the same for all three methods?

WHERE PEOPLE GO WRONG

Some students believe that FIFO describes which goods leave the shelf. It describes only the order of costs, and the goods may leave in any order. Others believe that LIFO always gives the lowest profit. That is true when prices rise, but when prices fall LIFO gives the highest profit. A third belief is that the weighted average is the plain average of the unit costs. It is the total cost divided by the total units, which gives larger layers more weight. A fourth belief is that a method can create profit. A method only moves cost between periods; the total cost over the life of the business does not change. Last, students sometimes change the method each year to suit the result. A business should use one method consistently, so that profits can be compared.

Five common misunderstandings are stated here, each with its correction.

IN HISTORY

The choice of cost flow assumption is a matter of law and standards as well as of arithmetic. In the United States, the Revenue Act of 1939 allowed LIFO for income tax. A rule known as LIFO conformity requires a business that uses LIFO for its tax return to use it in its financial reports as well.² In a time of rising prices, LIFO gives a higher cost of goods sold, a lower profit and a lower tax bill, which explains why many American businesses chose it.

The United States allowed LIFO for tax in 1939, while international standards now bar it.

Other countries follow the standards of the International Accounting Standards Board, a body based in London that issues International Financial Reporting Standards. Its International Accounting Standard 2, which deals with inventories, bars LIFO and allows FIFO and weighted average cost for goods that cannot be told apart. The board gave as one reason that tax considerations do not offer a sound basis for choosing an accounting method. A student must therefore ask which country and which standards apply before choosing a method.

Under international rules the position differs, because International Accounting Standard 2 bars LIFO.

² The Revenue Act of 1939 and the conformity rule are described in articles in *The Tax Adviser*, a journal of the American Institute of Certified Public Accountants.

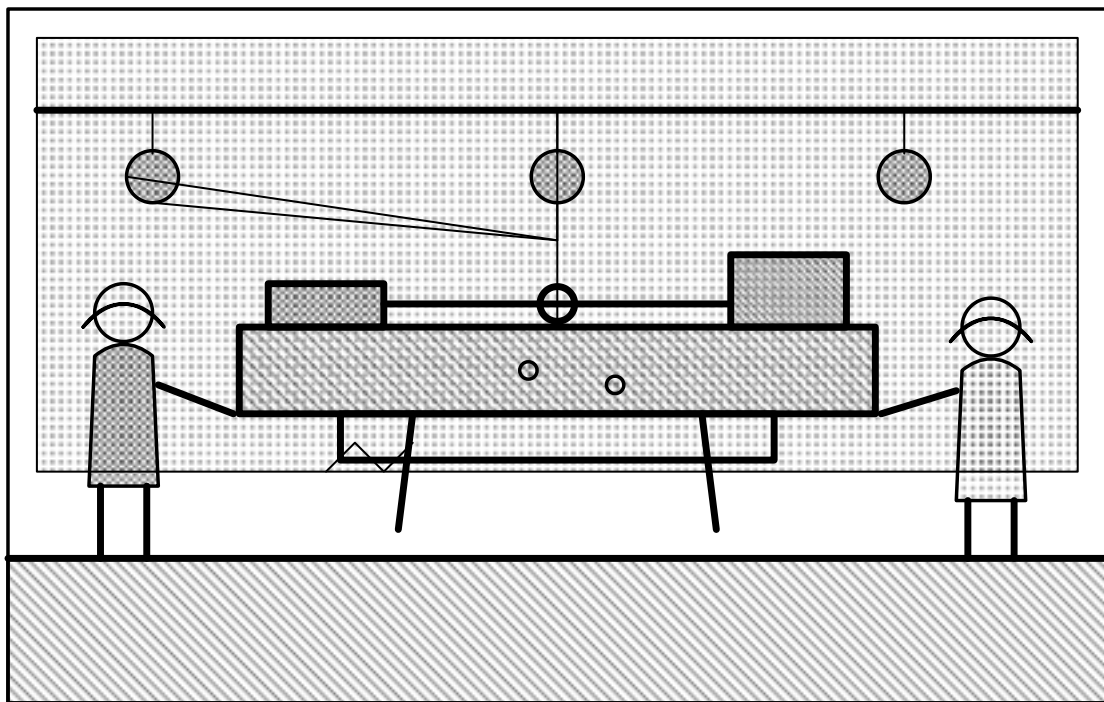
REVIEW QUESTIONS

14.1 A shop has an opening inventory of 10 units at Tk 10 and buys 10 units at Tk 12. It sells 10 units for Tk 20 each, net sales Tk 200. Find ending inventory, cost of goods sold and gross profit under FIFO, LIFO and weighted average cost.

14.2 A shop has an opening inventory of 40 units at Tk 45. It buys 60 units at Tk 48 and then 100 units at Tk 52. It sells 130 units at Tk 80 each, net sales Tk 10,400. Find ending inventory, cost of goods sold and gross profit under all three methods, and check each result.

14.3 Prices fall. A shop has an opening inventory of 25 boxes at Tk 80 and buys 25 boxes at Tk 70. It sells 30 boxes at Tk 100 each, net sales Tk 3,000. Find ending inventory, cost of goods sold and gross profit under all three methods, and state which method gives the highest profit.

14.4 Use your answers to question 14.2. Express each gross profit as a percentage of net sales, to one decimal place. Then state which method gives the highest ending inventory and explain why.



A machine shop of the iron age, with lathe, leather belt and workmen

CHAPTER XV

Plant Assets

Two bills of the same size, one for a new machine and one for a repair: why do they live in different parts of the accounts?



This chapter explains the long-lived assets a business uses to do its work, such as machines and buildings. You learn what counts as the cost of such an asset, which payments belong in that cost and which do not, and how to tell spending that builds an asset from spending that only keeps it working. The rules decide how large the profit and the balance sheet appear.

A small puzzle: two payments for the same machine, one that stays in the asset and one that does not.

The puzzle asks which payments become part of the machine in the accounts and which do not.

The running case supplies the payments for one machine, which the chapter follows to the end.

Imagine that the owner of a bookshop buys a machine to cut the edges of books. The seller sends an invoice. A lorry brings the machine, and the lorry driver wants payment. An electrician connects it, and a technician tests it on a few sheets of paper. Four months later a blade breaks, and a mechanic repairs the machine. Each of these events costs money, and the owner pays for all of them. Yet the accountant will treat them in two very different ways.

Why does the difference matter? Suppose the accountant adds every payment to the machine. The machine looks more valuable, and the profit for the year looks larger, because fewer costs are charged against it. Suppose instead that the accountant charges every payment as an expense at once. The profit for the year looks smaller, and the machine looks cheaper than it really was. Neither extreme is honest. The accountant needs a rule that decides, payment by payment, where each one belongs. This chapter builds that rule.

Case file, illustrative. Meghna and Sons bought a book-cutting machine on 1 April 2026. The invoice was Tk 300,000. Freight was Tk 10,000, installation was Tk 15,000 and a trial run cost Tk 5,000. On 14 August 2026 a breakdown required a repair that cost Tk 8,000.

YOU WILL NEED

This chapter uses the balance sheet and the income statement from Chapter 9 'Financial Statements and Closing Entries', section 9.1 'The Three Statements'. It also uses the idea of a cost paid now that benefits later periods, from Chapter 7 'Adjusting the Accounts', section 7.2 'Deferrals'. The arithmetic is addition and subtraction; Appendix A1 'Maths You Will Use' holds a short reference.

* * *

15.1 WHAT PLANT ASSETS ARE

A business owns some things to sell and other things to use, and this chapter is about the second group.

Plant asset is the name for long-lived physical things used in the business.

Plant asset. A long-lived physical asset used in running a business, such as a machine or a building.

Plant. Word in use. In daily life, a growing thing such as a tree, or a factory. In accounting, plant asset covers any long-lived physical asset used in the business, in a shop, an office or a factory alike.

Four features together mark an item as a plant asset.

Think about a family that runs a small food stall. The rice and vegetables on the table are for sale, and they are gone in a day. The stove, the large pot and the table itself are different. The family does not sell them. It uses them every day for years. Accountants separate these two groups, because they behave differently in the accounts. The first group is inventory, which Chapter 13 explains. The second group is the subject of this section.

The accounting name for the stove, the pot and the table is **plant asset**. A plant asset is a long-lived physical asset used in running a business, such as a machine or a building. For example, the book-cutting machine of Meghna and Sons is a plant asset. The word *plant* may mislead you. In daily speech it means a living plant or a factory. In accounting it applies to the equipment of any business, even a small shop with no factory at all.

How can you tell whether an item is a plant asset? Test it with four questions. First, is it physical, so that you can touch it? Second, does the business own it? Third, does the business use it to do its work, rather than hold it to sell? Fourth, will it last for more than one accounting period? If the answer to all four is yes, the item is a plant asset. A machine, a delivery van, a shelf unit, a computer and

a building all pass the test. A pack of paper fails the fourth question, because the business uses it up within weeks.

Think about what this means for the statements. When the business buys a plant asset, the money does not disappear as an expense. The business gives up cash and receives an asset of equal value. So the balance sheet shows the asset, and the income statement shows no expense on the day of purchase. The business then uses the asset over many periods. This is the same idea as the deferrals of Chapter 7, section 7.2 'Deferrals': a cost is paid now, and its benefit comes later. Chapter 16 'Depreciation' teaches how the cost is spread over the years of use.

The test depends on purpose, so the same object may fall into either group. A book-cutting machine on the floor of Meghna and Sons is a plant asset, because the shop uses it. The same machine in the showroom of a company that sells such machines is inventory, because that company holds it for sale. The question is always: for what purpose does this business hold the item?

Land deserves a short note. It is physical, owned and used, and it lasts. So it counts as a plant asset. But land does not wear out the way a machine does, and Chapter 16 'Depreciation' explains why this matters. For now, remember only that the class includes land, buildings, machines, vehicles and furniture.

Not every long-lived asset has a physical form. An **intangible asset** is a long-lived asset with no physical form, such as a patent or a trademark. For example, a patent is a legal right, given by the state, to be the only maker of a new invention for a number of years. A trademark is a name or sign that identifies a product. The owner cannot touch either one, yet each can earn money for years. The word *intangible* means "not able to be touched". Rules for intangible assets differ from country to country, so a business follows the standards of the place where it reports.

The table below compares the two kinds of long-lived asset with an ordinary item that is neither. Read each row as a question, and each column as an answer for one item.

PLANT ASSETS, INTANGIBLE ASSETS AND ORDINARY SUPPLIES

QUESTION	PLANT ASSET	INTANGIBLE ASSET	PACK OF PAPER
---	---	---	---
Physical form?	Yes	No	Yes
Used in the business, not held for sale?	Yes	Yes	Yes
Lasts more than one period?	Yes	Yes	No
Example	Book-cutting machine	Patent	Paper for printing

The class matters to different people. The owner wants to know how much money is tied up in equipment. A bank that lends money may accept the equipment as security, so it asks what the equipment cost. The manager decides when a machine must be replaced. The employee uses the machine every day and wants it kept in good order. All four depend on one reliable number: the cost of the asset.

Before moving on. A plant asset is a long-lived physical asset used in running a business. It is physical, owned, used and lasting. An intangible asset has no physical form but meets the other tests. The same object may be inventory in a

A plant asset moves from the business's cash to its balance sheet, and then it serves for years.

An exception: the same object can be a plant asset in one business and inventory in another.

Land is also a plant asset, but it behaves differently from the others.

Some valuable assets cannot be touched, and they form a separate group.

Intangible asset. A long-lived asset with no physical form, such as a patent or a trademark.

A table sets the two groups side by side so that you can compare them.

The viewpoints of owner, bank and manager show why the class matters.

Before moving on, we collect the ideas of this section.

different business. You should now be able to say, for any item, whether it is a plant asset, and give the reason.

15.2 COST OF A PLANT ASSET

The first task is to decide how much the asset cost, and the answer is larger than the invoice.

The rule says the cost covers everything spent to buy the asset and make it ready.

Acquisition cost. All the money spent to buy an asset and make it ready for use.

The items that belong in the cost can be listed, and each has a reason.

The items that do not belong are those that come after the asset is ready, or that do not add to it.

The case file lists four payments to the moment of readiness and one payment afterwards.

The figure shows how the cost builds up from its parts, one on top of the other.

How much did the machine cost? Many students say: the amount on the invoice. This answer is tempting, and it is incomplete. The owner paid for the lorry, the electrician and the test as well. The machine was not usable until all of these were done. An honest cost must include them.

The accounting name for this honest cost is **acquisition cost**. Acquisition cost is all the money spent to buy an asset and make it ready for use. For example, if a machine costs Tk 200,000 and the owner pays another Tk 10,000 to bring it and install it, the acquisition cost is Tk 210,000. The word *acquisition* means "getting". The test for each payment is simple to say: was it necessary to get the asset and to make it ready for use? If yes, it belongs in the cost.

Let us look at the usual items. The first is the price on the invoice, less any trade discount the seller gives. The second is freight, the cost of bringing the asset to the business, because the asset is of no use at the seller's door. The third is installation, which includes the electrician and any special base or wiring. The fourth is testing, because a machine is not ready until it has been tested. The fifth is any tax or duty on the purchase that the business cannot recover. Where a tax can be reclaimed from the government, it is not part of the cost; rules on this differ between countries, so the business checks the law of its own place.

Other payments are not part of the cost. Repairs after the asset is working are not part of it, because the asset was already ready. Training the staff to use the machine is not part of it, because the training does not make the machine itself ready. Insurance for the first year of use, and the running costs such as electricity, are not part of it either, because they relate to using the asset, not to getting it. Damage caused by careless handling during delivery is also excluded. It is a loss for the period, not a part of the asset. Many students find this list confusing at first. The test makes it easy: a payment up to the moment the asset is ready for use is in the cost, and a payment after that moment is out.

Case file, illustrative. For the book-cutting machine bought on 1 April 2026: invoice Tk 300,000; freight Tk 10,000; installation Tk 15,000; trial run Tk 5,000. These four payments were all made to bring the machine to a working state. The repair of Tk 8,000 on 14 August 2026 came after the machine was working.

The figure below shows the cost as a stack. Read it from the bottom upward. The widest block at the bottom is the invoice. Each block above it is one more payment added to the cost. The top block is the total, the acquisition cost. The blocks are not drawn to scale, because the smaller payments would be too thin to see.

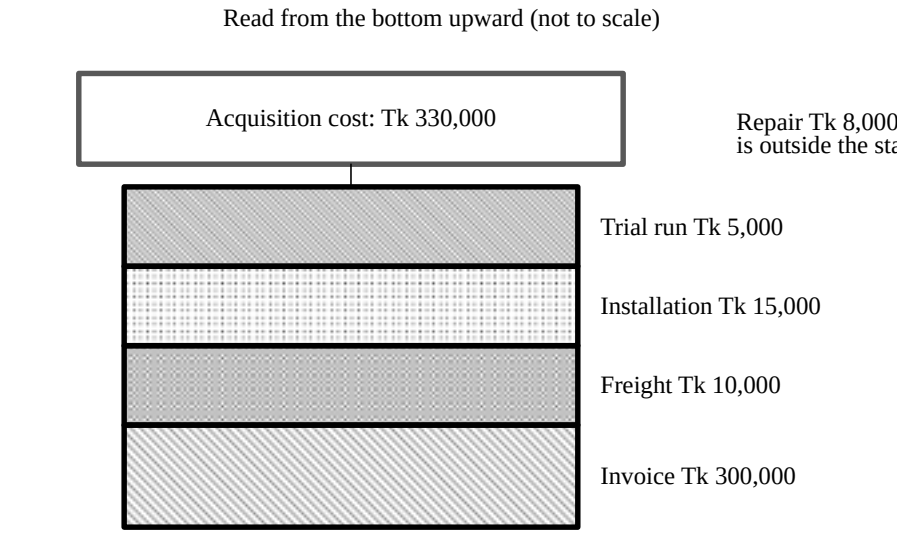


FIG. 15.1. Cost build-up of one machine as stacked parts

We now apply the method. *Given.* A shop buys a machine with an invoice price of Tk 200,000, less a trade discount of Tk 10,000. Freight is Tk 4,000. Installation is Tk 6,000. The staff training course costs Tk 3,000. *Find.* The acquisition cost.

Formula. Acquisition cost equals the invoice price less trade discount, plus freight, plus installation, plus testing and any tax that cannot be recovered. In symbols, let P be the invoice price, D the trade discount, F the freight, I the installation and T the other payments needed for readiness. Then the acquisition cost is $P - D + F + I + T$.

Steps. The price after discount is $200,000 - 10,000 = \text{Tk } 190,000$. Add freight: $190,000 + 4,000 = 194,000$. Add installation: $194,000 + 6,000 = 200,000$. There is no testing cost, so $T = 0$. The training course of Tk 3,000 is excluded, because it does not make the machine ready. *Answer.* The acquisition cost is Tk 200,000. *Check.* The total paid for the machine and its readiness is $190,000 + 4,000 + 6,000 = 200,000$, and the Tk 3,000 is treated as an expense.

PAUSE AND CHECK 15.P1 A delivery van is bought for Tk 900,000. The business also pays Tk 20,000 for the first year's road licence and Tk 12,000 for painting its name on the side before use. Which payments, if any, belong in the acquisition cost? Give a reason for each.

Now we apply the method to the machine of Meghna and Sons. The price is Tk 300,000, with no discount stated. Freight is Tk 10,000, installation is Tk 15,000 and the trial run is Tk 5,000. The acquisition cost is $300,000 + 10,000 + 15,000 + 5,000 = \text{Tk } 330,000$. The repair of Tk 8,000 is excluded, because it happened more than four months after the machine was working.

The entry follows. Suppose the shop paid by bank. The business debits Machinery with Tk 330,000 and credits Bank with Tk 330,000. The machine appears on the balance sheet at Tk 330,000, and no expense is recorded on the day of purchase. Why record one figure rather than five? The five payments together are the price of having a working machine, and the balance sheet should show that price in one place.

Before moving on. The acquisition cost of a plant asset is all the money spent to buy it and make it ready for use. It includes the invoice price less discounts, freight, installation and testing. It excludes payments after readiness, such as

A first example, with small and clean numbers, shows the method before the case.

The formula is stated in words first and then in symbols.

Each line of arithmetic is shown, and the training course is left out with a reason.

Now the case file is worked in the same order, with the same method.

The entry records the whole acquisition cost as one asset, not as five separate items.

Before moving on, we collect the ideas of this section.

repairs and training. You should now be able to list the payments for one asset and state which belong in the cost.

15.3 CAPITAL AND REVENUE SPENDING

Every payment a business makes either builds an asset or is used up, and the two kinds need names.

The first group is spending that creates or improves a lasting asset.

Capital expenditure. Spending that creates or improves an asset lasting more than one period.

Improvement. Word in use. In daily life, any change for the better. In accounting, a change that makes an asset last longer, work faster or do more, and so is added to its cost.

The second group is spending that keeps an asset in order and is used up in the period.

Revenue expenditure. Spending that keeps an asset in working order and is treated as an expense of the period.

The matching principle gives the reason for the split.

The hard cases are the repairs and improvements, and a single test sorts them.

A table compares the two kinds of spending question by question.

Sections 15.1 and 15.2 dealt with the cost of one asset. We now widen the question. Every day the business pays for things: wages, rent, paper, repairs, new machines. Some of these payments create something that lasts for years. Others are used up in the period. If the accountant puts a payment in the wrong group, the profit and the balance sheet are both wrong. We need two names for the two groups.

The first group is **capital expenditure**. Capital expenditure is spending that creates or improves an asset lasting more than one period. For example, the purchase of the book-cutting machine, at Tk 330,000, is capital expenditure. It is recorded as an asset on the balance sheet. The word *capital* here has nothing to do with the owner's money. It points to the long life of the thing that the spending creates.

The second group is **revenue expenditure**. Revenue expenditure is spending that keeps an asset in working order and is treated as an expense of the period. For example, the repair of Tk 8,000 on 14 August 2026 is revenue expenditure. It is recorded as an expense in the income statement of 2026. The name can confuse a new reader. It does not mean that the spending produces revenue. It means that the spending is matched against the revenue of the period in which it is made.

Why is the line drawn here? Think of the matching principle and the accrual basis of accounting, which you have met. An expense belongs to the period whose revenue it helps to earn. A repair keeps the machine running this year, so its cost belongs to this year. A new machine will help earn revenue for many years, so its cost is spread across those years. If the accountant charged the whole machine to the first year, that year would look poor and the later years would look rich, which is a false picture.

Some cases are not obvious. A mechanic works on a machine. Is that a repair or an improvement? The test asks what the spending does. If it restores the asset to the condition in which it was first bought, it is a repair: revenue expenditure. If it makes the asset last longer, work faster or do more than before, it is an improvement: capital expenditure. For example, replacing a broken blade with a new blade of the same kind is a repair. Fitting a motor that makes the machine cut twice as fast is an improvement, and its cost is added to the cost of the machine.

The table below sets the two kinds side by side. Read each row as a question and each column as an answer.

CAPITAL EXPENDITURE AND REVENUE EXPENDITURE COMPARED

QUESTION	CAPITAL EXPENDITURE	REVENUE EXPENDITURE
---	---	---
What does it do?	Creates or improves a lasting asset	Keeps the asset working
Where is it recorded first?	Balance sheet, as an asset	Income statement, as an expense
How long is the benefit?	More than one period	This period
Example from the case	Machine, Tk 330,000	Repair, Tk 8,000

There is one exception that avoids waste of effort. A business may buy a stapler that lasts five years. Strictly it is a plant asset. But recording it as an asset and tracking its cost for five years is not worth the effort. Most businesses therefore set a minimum amount, below which any such item is charged as an expense at once. The size of the minimum is a business decision and differs from one business to another.

Suppose the Tk 8,000 repair were wrongly added to the machine. The income statement would show Tk 8,000 less expense, so the profit would be Tk 8,000 higher. The balance sheet would show the machine at Tk 338,000, which is Tk 8,000 too much. Both statements would be wrong in the same direction. Now suppose the opposite error: a Tk 15,000 installation were charged as an expense. The profit would be Tk 15,000 too low and the machine Tk 15,000 too low in the balance sheet.

Before moving on. Capital expenditure creates or improves an asset lasting more than one period, and it is recorded as an asset. Revenue expenditure keeps an asset in working order, and it is an expense of the period. A repair restores; an improvement adds. A mistake in the choice moves profit and the asset by the same amount. You should now be able to classify any payment and say what happens if it is wrongly classified.

WORKED EXAMPLE: THE BOOK-CUTTING MACHINE

Case file, illustrative. Meghna and Sons paid the following for the book-cutting machine in 2026: invoice Tk 300,000; freight Tk 10,000; installation Tk 15,000; trial run Tk 5,000; repair after a breakdown on 14 August Tk 8,000.

Find. (a) the acquisition cost and the amount of revenue expenditure; (b) the balance sheet figure and the profit effect if the repair were wrongly added to the machine.

Part (a), steps. Payments up to readiness: $300,000 + 10,000 = 310,000$. Then $310,000 + 15,000 = 325,000$. Then $325,000 + 5,000 = 330,000$. The repair of Tk 8,000 is after readiness and is a revenue expenditure. *Answer.* The acquisition cost is Tk 330,000 and the revenue expenditure is Tk 8,000. *Check.* The total of all payments is $330,000 + 8,000 = 338,000$, which equals $300,000 + 10,000 + 15,000 + 5,000 + 8,000$.

Part (b), steps. If the repair were wrongly capitalised, the machine would appear at $330,000 + 8,000 = \text{Tk } 338,000$, which is Tk 8,000 too high. The expenses of 2026 would be Tk 8,000 too low, so profit would be Tk 8,000 too high. *Answer.* Balance sheet overstated by Tk 8,000; profit overstated by Tk 8,000.

The exception: a small payment may be treated as an expense even if the item lasts.

The effect of an error shows why the rule matters.

Before moving on, we collect the ideas of this section.

The whole bill for the machine is split into capital and revenue spending, with every line shown.

The question has two parts: the acquisition cost, and the effect on profit if the repair is wrongly capitalised.

Part (a) adds only the payments needed to make the machine ready.

Part (b) shows that a wrong choice moves profit and the asset by the same amount.

The entries record each kind of spending in its proper place.

Common slips: adding after-readiness payments, leaving out freight or installation, and confusing a repair with an improvement.

The chapter's ideas in a few lines.

Belief: the cost of an asset is the invoice price.

Belief: any payment connected with a machine is part of its cost.

Belief: capital expenditure means spending the owner's capital.

Belief: revenue expenditure produces revenue.

Belief: an improvement and a repair are the same thing.

A famous fraud began with the wrong choice between capital and revenue spending.

Check. The two errors are equal, because one payment is placed in the wrong statement.

The correct entries, if both were paid by bank, are these. For the purchase: debit Machinery Tk 330,000 and credit Bank Tk 330,000. For the repair: debit Repairs Expense Tk 8,000 and credit Bank Tk 8,000.

Three slips are common. The first is to add the repair to the cost because it concerns the machine. The test is timing and purpose, not the name of the asset. The second is to leave out freight or the trial run because they are not on the invoice. The third is to treat a restoring repair as an improvement, or the other way round.

SUMMARY

A plant asset is a long-lived physical asset used in running a business. An intangible asset has no physical form but lasts as well. The acquisition cost of a plant asset is all the money spent to buy it and make it ready for use. Capital expenditure creates or improves an asset lasting more than one period; revenue expenditure keeps an asset working and is an expense of the period. Wrong classification changes profit and the balance sheet together.

THINK IT THROUGH

15.T1 Suppose a business wrongly charges the Tk 15,000 installation of the machine as an expense in 2026. What is the effect on 2026 profit and on the balance sheet figure for the machine?

15.T2 A manager wishes to show a high profit this year. Explain why she might be tempted to capitalise repairs, and why this is dishonest even if it is not noticed at once.

WHERE PEOPLE GO WRONG

The cost includes freight, installation, testing and any tax that cannot be recovered. The invoice is only the first part.

Only payments up to the moment the asset is ready for use are part of the cost. Repairs, training and running costs come later and are expenses.

The word *capital* in this phrase refers to the lasting nature of the asset that the spending creates. It has no link to the owner's own money.

Revenue expenditure is an expense that is matched against the revenue of the same period. The name comes from that matching, not from producing revenue.

A repair restores the asset to its first condition and is an expense. An improvement raises its life, speed or output and is added to the asset.

IN HISTORY

In June 2002 the American telephone company WorldCom, one of the largest in the country, told the public that it had wrongly treated a large part of its costs as assets. The costs were line costs, the fees it paid to other companies to carry calls over their networks. On 25 June 2002 it announced that about USD 3.8 billion of such payments, made during 2001 and the first quarter of 2002, had been recorded as assets and not as expenses.

The effect was large. Reports at the time said that, without the transfers, WorldCom would have shown a loss for 2001 and for the first quarter of 2002. Its internal audit team, led by Cynthia Cooper, traced the entries. The company filed for bankruptcy in July 2002. The lesson is the one you have learned: spending that is used up in the period must be charged to the period, and only spending that creates a lasting asset can be added to the balance sheet.

The case shows why the choice of capital or revenue spending can decide profit or loss.

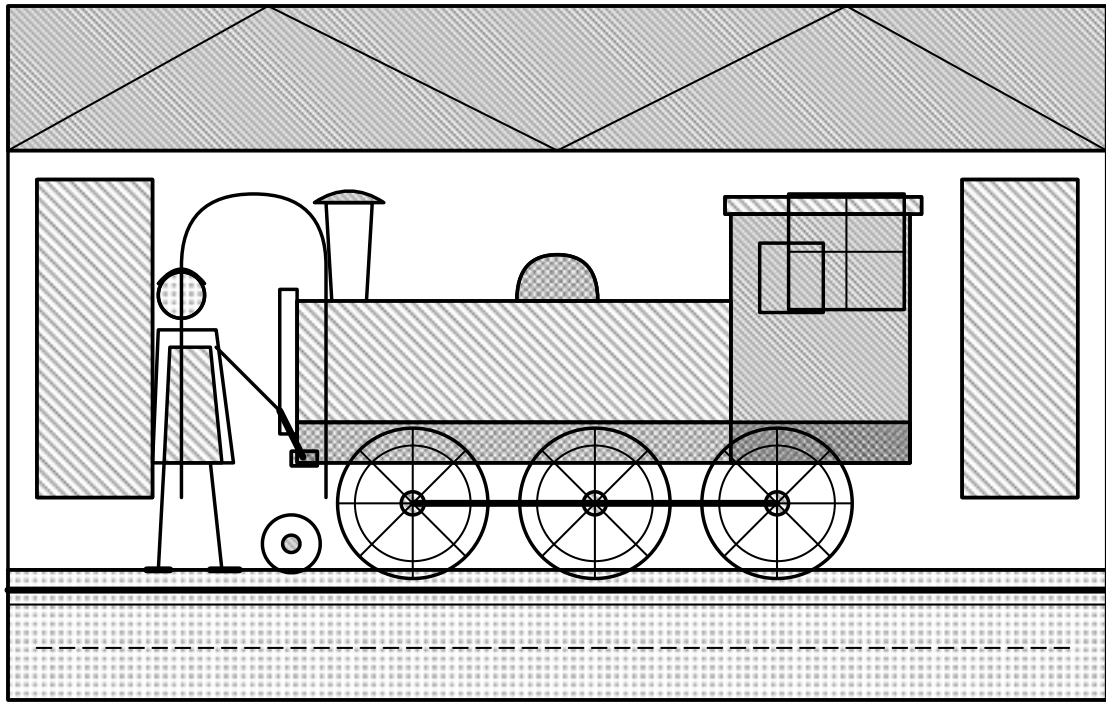
REVIEW QUESTIONS

15.1 A shop buys a printing machine. The invoice is Tk 450,000 and a trade discount of Tk 20,000 is allowed. Freight is Tk 8,000, installation Tk 12,000, a trial run Tk 4,000, and operator training Tk 6,000. Find the acquisition cost and the amount treated as an expense.

15.2 Classify each payment as capital or revenue expenditure: (a) new blades of the same kind, Tk 3,000; (b) a motor that raises output by half, Tk 40,000; (c) a concrete base laid for a new machine, Tk 9,000; (d) the yearly service of a van, Tk 7,500; (e) painting a building before first use, Tk 25,000.

15.3 A business treats a repair of Tk 12,000 as capital expenditure by mistake. State the effect on profit for the year and on the plant asset on the balance sheet.

15.4 A van costs Tk 600,000. The business pays Tk 30,000 to bring it from the port and Tk 18,000 to paint its name on it, both before use. In the first year it pays Tk 22,000 for repairs and Tk 9,000 for a licence. Find the acquisition cost, and the total revenue expenditure for the year.



A locomotive in a depot, with an engineer inspecting its wheels

CHAPTER XVI

Depreciation

*If a machine serves for five years, why should one year pay
for all of it?*



*This chapter teaches why the cost of a plant asset is spread over
the years in which it helps to earn revenue. You will learn three
methods of spreading cost, record the charge in the accounts, and
find the gain or loss when an asset is sold.*

YOU WILL NEED

You will need the cost of a plant asset from Chapter 15 'Plant Assets', section 15.2 'Cost of a Plant Asset', and the difference between capital and revenue spending from Chapter 15 'Plant Assets', section 15.3 'Capital and Revenue Spending'. Appendix A1 'Maths You Will Use' covers percentages and rounding, which this chapter uses throughout.

A bookshop buys a costly machine, and the question is which years should bear its cost.

Charging the whole cost to one year gives an unfair picture of every year.

The remedy is to charge a share of the cost to each year of use, and this chapter teaches how.

On 1 April 2026, Meghna and Sons pays Tk 330,000 for a book-cutting machine. The machine will cut books for about five years. In the year of purchase the shop spends Tk 330,000, and its profit for that year is far smaller than usual. In the next four years the machine cuts thousands of books, yet the shop records no cost at all for it. Those years look more profitable than they really are.

Consider what the owner would conclude. In the first year, the bookshop seems to lose money, and a lender may refuse a loan. In the later years, it seems to earn large profits, and the owner may spend too much. Both pictures are false, because the machine helps to earn revenue in all five years. Many students find this point difficult at first, because they remember that the cash left the business on one day. The accounts, however, must follow the use of the machine and not the day of payment.

The remedy has a name, and this chapter teaches it. Section 16.1 explains the idea and the words that belong to it. Section 16.2 and section 16.3 teach three methods of choosing the share for each year. Section 16.4 shows how to record the share and what happens when the asset is sold. Go slowly. The same machine returns in every section, so only the method changes.

* * *

16.1 WHY DEPRECIATE

GO SLOWLY HERE.

A bicycle bought for a delivery job shows why cost should be shared over the years of use.

Depreciation is the spreading of an asset's cost over the years of its use.

Depreciation. The spreading of a plant asset's cost over the years it is expected to be useful.

Suppose you begin a part-time delivery job and buy a bicycle for Tk 10,000. You expect to use it for four years, and then to sell it for about Tk 2,000. Would it be fair to say that all your earnings in the first year were reduced by Tk 10,000? The bicycle helps you earn in each of the four years. A fair view charges a share of its cost to each year. You pay once, but you use it four times.

Depreciation is the spreading of a plant asset's cost over the years it is expected to be useful. The bicycle shows the idea: a share of Tk 10,000 is charged in each of the four years. Depreciation is a method of allocation. It does not measure the fall in the market price of the asset, and it does not mean that cash is paid each year. The cash was paid on the day of purchase.

Before we continue, notice that two ordinary words carry special meanings here. The word *value* in accounting often means a figure in the records, and the word *life* means the period of expected service. The two notes beside this paragraph give both meanings. We shall use each word in its accounting sense from now on.

The **useful life** of an asset is the period, or the amount of use, over which the asset is expected to serve the business. The bicycle has a useful life of four years. A delivery van might have a useful life of 200,000 kilometres. Notice that useful life is an expectation, not a fact, and it can be shorter than the time the asset physically lasts. A business may sell a machine after five years even though the machine could run for ten.

The **residual value** of an asset is the amount the business expects to receive when it disposes of the asset at the end of its useful life. The bicycle has a residual value of Tk 2,000. If the business expects to receive nothing, the residual value is zero. Only the part of the cost that will be used up needs to be spread, so the residual value is not charged to the years of use.

We now use the smallest numbers possible. *Given.* A bicycle costs Tk 10,000, has a residual value of Tk 2,000, and has a useful life of four years. *Find.* The amount to be spread, and an equal share for each year. *Formula.* The amount to be spread equals cost minus residual value. An equal yearly share equals the amount to be spread divided by the number of years. In symbols, $S = C - R$ and $D = S \div n$, where C is the cost, R is the residual value, S is the amount to be spread, n is the useful life in years and D is the yearly share. *Steps.* $S = \text{Tk } 10,000 - \text{Tk } 2,000 = \text{Tk } 8,000$. $D = \text{Tk } 8,000 \div 4 = \text{Tk } 2,000$. *Answer.* Tk 8,000 is spread, at Tk 2,000 a year. *Check.* Four years at Tk 2,000 make Tk 8,000, and Tk 10,000 less Tk 8,000 leaves the residual value of Tk 2,000.

Several factors lead a business to choose a life and a residual value. Wear from use reduces the usefulness of a machine. The passing of time rots wood and rusts iron, even when the machine stands idle. Changes in the market can make an asset out of date: a faster machine appears, or the books the shop sells change. A manager considers all three. The owner then chooses an estimate. Different people may reach different estimates, and tax rules in some countries use their own lives and rates, which may differ from those used in the accounts.¹

Exceptions need care. Land is a plant asset, but it is not depreciated, because it does not wear out or become out of date in the way that buildings and machines do. An estimate of life or of residual value may also prove wrong. If so, the business revises the estimate and spreads the remaining cost over the remaining years. It does not go back and alter the past years.

Before moving on. Depreciation spreads the cost of a plant asset over its useful life. The amount to be spread is the cost less the residual value. You should now be able to say

Two everyday words, value and life, have special meanings when used with assets.

Value. Word in use. In daily life, what something is worth to a person. In accounting, a figure in the records, such as cost less depreciation, which need not equal the price in the market.

Life. Word in use. In daily life, the time a person lives. In accounting, the period over which an asset is expected to serve the business, which need not equal its physical age.

Useful life is the period, or the amount of use, over which an asset is expected to serve.

Useful life. The period, or the amount of use, over which an asset is expected to serve the business.

Residual value is the amount the business expects to receive when the asset is disposed of.

Residual value. The amount a business expects to receive when it disposes of an asset at the end of its useful life.

A first calculation with small numbers shows the amount to spread and the share for each year.

Three factors affect the yearly charge: wear from use, passing of time, and change in the market.

Land is the usual exception, and estimates may be revised as experience grows.

Recap: depreciation spreads cost; life and residual value are estimates that set the amount and the period.

¹ Intangible assets, such as a licence, are also spread over their useful lives, but this chapter treats plant assets only.

why a machine bought in one year should not reduce the profit of that year alone.

* * *

16.2 STRAIGHT-LINE AND UNITS-OF-PRODUCTION

GO SLOWLY HERE.

Equal shares are the easiest rule, and the case file gives the figures for the machine.

The straight-line method charges the same amount in every year.

Straight-line method. A depreciation method that charges an equal amount each year.

The machine's yearly charge is Tk 60,000, and the nine months of 2026 carry Tk 45,000.

Pause and check that you can calculate the straight-line charge.

The units-of-production method links the charge to how much the machine is used.

Units-of-production method. A depreciation method that charges an amount based on how much the asset is used in the period.

The machine's charge is Tk 0.60 for each book, so 90,000 books give Tk 54,000.

Each method has limits: straight-line ignores use, and units-of-production needs a good estimate of use.

Case file, illustrative. Meghna and Sons bought a book-cutting machine on 1 April 2026 at a cost of Tk 330,000. The residual value is Tk 30,000, the useful life is five years, and the capacity is 500,000 books. The business closes its books on 31 December each year.

The **straight-line method** is a depreciation method that charges an equal amount each year. The bicycle in section 16.1 used it, at Tk 2,000 a year. The word straight describes the picture of book value falling by equal steps, which forms a straight line on a graph. The method is the most common because it is easy to calculate and easy to explain.

Given. Cost Tk 330,000, residual value Tk 30,000, useful life five years, bought on 1 April 2026. *Find.* The charge for a full year, and the charge for 2026. *Formula.* Yearly charge equals cost minus residual value, divided by useful life in years. In symbols, $D = (C - R) \div n$. For a part of a year, multiply by the fraction of the year, $m \div 12$, where m is the number of months of use. *Steps.* $D = (Tk\ 330,000 - Tk\ 30,000) \div 5 = Tk\ 60,000$. From 1 April to 31 December is 9 months, so the 2026 charge is $Tk\ 60,000 \times 9 \div 12 = Tk\ 45,000$. *Answer.* The full-year charge is Tk 60,000, and the charge for 2026 is Tk 45,000. *Check.* $Tk\ 60,000 \times 5 = Tk\ 300,000$, which is the cost less the residual value.

PAUSE AND CHECK 16.P1 A van costs Tk 120,000, has a residual value of Tk 20,000 and a useful life of four years. What is the straight-line charge for a full year?

Some assets wear out by use more than by time. The **units-of-production method** is a depreciation method that charges an amount based on how much the asset is used in the period. A cutting machine is used more in a busy year than in a quiet year, and its wear follows the number of books cut. The method first finds a charge for each unit of use, and then multiplies by the units used.

Given. The machine can cut 500,000 books in its life, and it cut 90,000 books in 2026. *Find.* The charge for each book, and the charge for 2026. *Formula.* The charge for each unit equals cost minus residual value, divided by the total units expected. In symbols, $k = (C - R) \div U$ and $D = k \times u$, where U is the total units expected, k is the charge for each unit and u is the units used in the period. *Steps.* $k = Tk\ 300,000 \div 500,000 = Tk\ 0.60$ a book. $D = 90,000 \times Tk\ 0.60 = Tk\ 54,000$. *Answer.* The 2026 charge is Tk 54,000. *Check.* If all 500,000 books were cut, the total would be $500,000 \times Tk\ 0.60 = Tk\ 300,000$, which equals the amount to be spread.

Exceptions apply to both methods. The straight-line method charges the same amount in a busy year and in a quiet year, so it ignores use. The units-of-production method needs a reliable estimate of total units, which is hard for some assets, and it charges nothing in a year when the machine stands idle, even though time still passes. The 2026 results also differ: Tk 45,000 by straight line and Tk 54,000 by units of production. Both are correct under their own rule.

Before moving on. The straight-line method charges Tk 60,000 a year for the machine, and Tk 45,000 for 2026. The units-of-production method charges Tk 0.60 a book, and Tk 54,000 for 90,000 books. You should now be able to say which method suits an asset that wears by use.

* * *

Recap: straight-line gives equal shares; units-of-production follows use.

16.3 DECLINING-BALANCE

Some assets lose most of their usefulness early. A new machine cuts faster and breaks down less than an old one, and a new vehicle is worth more in its first year than in its fifth. For such assets a larger charge in the early years is a fairer picture. The third method provides it.

The **declining-balance method** is a depreciation method that charges a fixed percentage of the asset's remaining cost each year. We call the remaining cost the *book value*, a term that section 16.4 defines fully: it is the cost less all depreciation charged so far. Because the base falls each year, the charge falls each year. The balance that declines gives the method its name. The method uses the cost in the first year, and the residual value is not subtracted when the percentage is applied.

Given. Cost Tk 330,000, residual value Tk 30,000, a rate of 40% a year. *Find.* The charge and the remaining cost at the end of year 1 and year 2. *Formula.* The charge equals the rate times the remaining cost at the start of the year. The remaining cost at the end equals the start less the charge. In symbols, $D = r \times B$ and $B' = B - D$, where r is the rate, B is the remaining cost at the start and B' is the remaining cost at the end. *Steps.* Year 1: $D = 40\% \times \text{Tk } 330,000 = \text{Tk } 132,000$, and $B' = \text{Tk } 330,000 - \text{Tk } 132,000 = \text{Tk } 198,000$. Year 2: $D = 40\% \times \text{Tk } 198,000 = \text{Tk } 79,200$, and $B' = \text{Tk } 198,000 - \text{Tk } 79,200 = \text{Tk } 118,800$. *Answer.* The charges are Tk 132,000 and Tk 79,200, and the remaining cost after two years is Tk 118,800. *Check.* $\text{Tk } 132,000 + \text{Tk } 79,200 = \text{Tk } 211,200$, and $\text{Tk } 330,000 - \text{Tk } 211,200 = \text{Tk } 118,800$.

PAUSE AND CHECK 16.P2 An asset costs Tk 80,000 and is depreciated at 25% by the declining-balance method. Find the charge and the remaining cost at the end of year 1 and year 2.

The method has a limit that students often miss. The remaining cost must never fall below the residual value. The machine's remaining cost after year 3 is Tk 71,280, and after year 4 it is Tk 42,768. A charge of 40% in year 5 would be Tk 17,107, which would leave Tk 25,661, below the residual value of Tk 30,000. The year 5 charge is therefore cut to $\text{Tk } 42,768 - \text{Tk } 30,000 = \text{Tk } 12,768$.

The table below compares the yearly charges by the straight-line method and by the declining-balance method. Read each row across to compare the two methods for one year. The last row shows that both methods charge the same total, Tk 300,000, over five years. Only the timing differs.

GO SLOWLY HERE.

A larger charge in early years suits an asset that loses usefulness quickly.

The declining-balance method charges a fixed percentage of the remaining cost each year.

Declining-balance method. A depreciation method that charges a fixed percentage of the asset's remaining book value each year.

The machine's charge at 40% is Tk 132,000 in year 1 and Tk 79,200 in year 2.

Pause and check that you can apply a percentage to the remaining cost.

The method must stop at the residual value, so the last charge is cut to fit.

The table compares the yearly charges, and both methods spread the same Tk 300,000.

YEARLY DEPRECIATION CHARGES FOR THE MACHINE OVER FIVE YEARS (TK)

YEAR	STRAIGHT-LINE	DECLINING-BALANCE
1	60,000	132,000
2	60,000	79,200
3	60,000	47,520
4	60,000	28,512
5	60,000	12,768
Total	300,000	300,000

The figure shows the remaining cost falling faster under declining-balance than under straight-line.

The first figure draws the remaining cost of the machine against the years. The horizontal axis shows the years since purchase, and the vertical axis shows taka. A point shows the remaining cost at the end of that year. The solid line, for straight-line, falls by equal steps. The dashed line, for declining-balance, falls steeply at first and then flattens. Both lines end at the residual value of Tk 30,000.

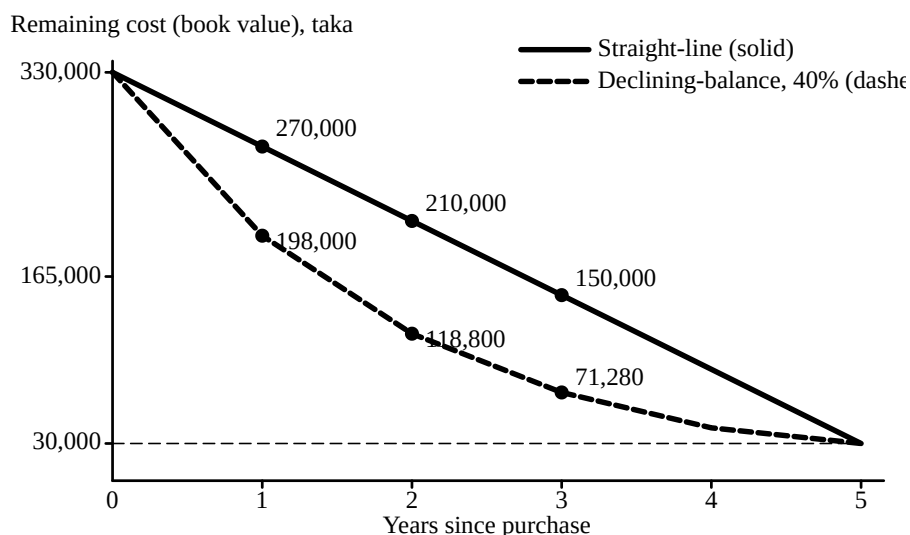


FIG. 16.1. Straight-line against declining-balance as two descending lines

Recap: declining-balance charges more early, less later, and stops at the residual value.

Before moving on. The declining-balance method applies a fixed rate to the remaining cost. For the machine at 40%, the charges are Tk 132,000 and Tk 79,200 in the first two years. You should now be able to say why the charge falls each year, and why the last charge may be cut.

* * *

16.4 RECORDING AND DISPOSAL

Each year the charge is recorded by debiting an expense and crediting a separate account.

At the end of each year the business records the charge in an adjusting entry, of the kind taught in Chapter 7 'Adjusting the Accounts'. The entry debits an expense account, called depreciation expense, which reduces profit. It credits a second account, which collects the charges and is shown as a deduction from the cost of the asset. The cost account itself is left unchanged.

Accumulated depreciation is the total depreciation charged to date on an asset, shown as a deduction from its cost. After one year it equals one charge, after two years two charges, and so on. The business keeps the original cost visible, so that a reader can see both what the asset cost and how much has been charged. For the machine, the 2026 charge by straight line is Tk 45,000, and the entry is shown below.

ENTRY ON 31 DECEMBER 2026 FOR THE STRAIGHT-LINE CHARGE

ACCOUNT	DEBIT (TK)	CREDIT (TK)
Depreciation expense, machine	45,000	
Accumulated depreciation, machine		45,000

The **book value** of an asset is its cost less its accumulated depreciation. At 31 December 2026 the book value of the machine is Tk 330,000 – Tk 45,000 = Tk 285,000. The balance sheet shows the three figures together: cost, accumulated depreciation and book value. Book value is a figure in the records. It is not the price at which the machine could be sold, which may be higher or lower.

THE MACHINE IN THE BALANCE SHEET AT 31 DECEMBER 2026 (TK)

ITEM	AMOUNT
Machine, at cost	330,000
Less accumulated depreciation	45,000
Book value	285,000

The second figure shows the book value of the machine at the end of each year. The horizontal axis shows the years, and the vertical axis shows taka. Each horizontal step shows the book value that holds through a year, and each drop shows one yearly charge. After year 3 the book value is Tk 150,000, the amount used in the next calculation.

Book value at the end of each year, taka

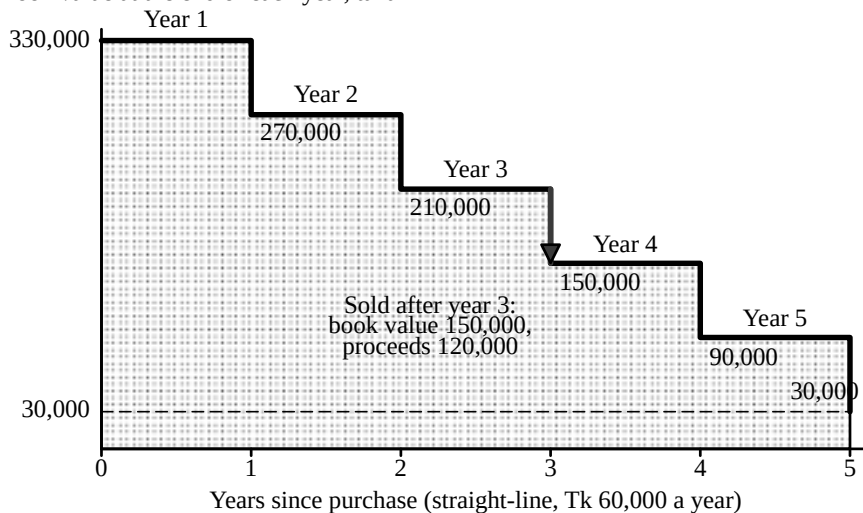


FIG. 16.2. Book value of the machine stepping down year by year

Case file, illustrative. After three years of straight-line depreciation, Meghna and Sons sells the book-cutting machine for Tk 120,000. The machine cost Tk 330,000 and the straight-line charge is Tk 60,000 a year.

Accumulated depreciation is the running total of the charges made on one asset.

Accumulated depreciation. The total depreciation charged to date on an asset, shown as a deduction from its cost.

Book value is cost less accumulated depreciation, and it shows the cost not yet charged.

Book value. An asset's cost less its accumulated depreciation.

The figure shows book value stepping down once a year under straight-line.

A sale after three years brings a gain or a loss, found by comparing proceeds with book value.

The calculation first finds the book value at the date of sale and then compares it with the proceeds.

The entry removes the asset and its accumulated depreciation and records the cash and the loss.

Depreciation must be brought up to the date of sale first, and a sale above book value gives a gain.

Recap: accumulated depreciation and book value are linked, and a sale compares proceeds with book value.

This worked case shows that the method chosen changes the gain or loss on sale.

The reasoning finds the book value under each method and compares it with the proceeds.

The conclusion is a loss of Tk 90,000 under straight-line and a gain of Tk 14,800 under declining-balance.

Given. Cost Tk 330,000, charge Tk 60,000 a year for three years, proceeds Tk 120,000. *Find.* The gain or loss on sale. *Formula.* Gain or loss equals proceeds minus book value. A positive result is a gain and a negative result is a loss. In symbols, $G = P - B$, where P is the proceeds and B is the book value. *Steps.* Accumulated depreciation = $3 \times \text{Tk } 60,000 = \text{Tk } 180,000$. $B = \text{Tk } 330,000 - \text{Tk } 180,000 = \text{Tk } 150,000$. $G = \text{Tk } 120,000 - \text{Tk } 150,000 = -\text{Tk } 30,000$. *Answer.* The sale gives a loss of Tk 30,000. *Check.* The machine was carried at Tk 150,000 but brought only Tk 120,000, a shortfall of Tk 30,000.

The entry for the sale removes the machine from the records. It credits the machine for its full cost. It debits accumulated depreciation, so that the charges made so far are cleared. It debits cash for the proceeds, and it debits a loss account for the loss. The debits and credits must be equal, and the table shows that they are.

ENTRY FOR THE SALE OF THE MACHINE AFTER THREE YEARS

ACCOUNT	DEBIT (TK)	CREDIT (TK)
Cash	120,000	
Accumulated depreciation, machine	180,000	
Loss on sale of machine	30,000	
Machine, at cost		330,000

Two exceptions complete the picture. If the sale falls during a year, the charge for the part of the year must be recorded first, so that the book value is correct on the date of sale. If the machine had sold for Tk 170,000, the result would be $\text{Tk } 170,000 - \text{Tk } 150,000 = \text{Tk } 20,000$, a gain, and the entry would credit a gain account instead of debiting a loss account.

Before moving on. Accumulated depreciation is the total charged so far, and book value is cost less that total. At sale, gain or loss equals proceeds minus book value. You should now be able to say why the machine's sale gave a loss of Tk 30,000.

* * *

WORKED CASE: A VAN SOLD AFTER TWO YEARS

Situation. A different shop bought a delivery van on 1 January 2026 for Tk 480,000. It estimated a residual value of Tk 60,000 and a useful life of six years. At 31 December 2027, after two years, it sold the van for Tk 250,000. These figures are for practice and are not part of the case file. *Question.* What is the gain or loss on sale if the shop used straight-line depreciation, and what is it if the shop used declining-balance at 30%?

Reasoning. Straight line: the yearly charge is $(480,000 - 60,000) \div 6 = \text{Tk } 70,000$. After two years the accumulated depreciation is Tk 140,000, so the book value is $\text{Tk } 480,000 - \text{Tk } 140,000 = \text{Tk } 340,000$. Declining balance: year 1 is $30\% \times 480,000 = \text{Tk } 144,000$, leaving Tk 336,000. Year 2 is $30\% \times 336,000 = \text{Tk } 100,800$, leaving Tk 235,200. The accumulated depreciation is Tk 244,800. The proceeds are Tk 250,000 in both cases.

Conclusion. Straight-line: $\text{Tk } 250,000 - \text{Tk } 340,000 = -\text{Tk } 90,000$, a loss. Declining balance: $\text{Tk } 250,000 - \text{Tk } 235,200 = \text{Tk } 14,800$, a gain. The same van, sold at the same price, gives a loss under one method and a gain under the other. The faster method had already charged more cost, so less remained to set against

the proceeds. *Check.* Under straight-line, the debits are Tk 250,000 + Tk 140,000 + Tk 90,000 = Tk 480,000, which equals the credit to the van. Under declining balance, the debits are Tk 250,000 + Tk 244,800 = Tk 494,800, and the credits are Tk 480,000 + Tk 14,800 = Tk 494,800.

Common slips deserve a warning. The first is to forget the residual value in the straight-line formula, which makes every charge too large. The second is to subtract the residual value from the cost before applying the declining-balance rate, instead of applying the rate to the full remaining cost. The third is to forget the fraction of the year in the year of purchase. Test every answer by asking whether the total of the charges equals the cost less the residual value.

* * *

Three slips occur often in this work, and each one can be prevented.

SUMMARY

Depreciation spreads the cost of a plant asset over its useful life, after subtracting the residual value. The straight-line method charges an equal amount each year, the units-of-production method charges by use, and the declining-balance method charges a fixed percentage of the remaining cost. The charge is recorded as an expense and as accumulated depreciation. Book value is cost less accumulated depreciation, and the gain or loss on a sale is the proceeds minus the book value. For Meghna and Sons, the machine's straight-line charge is Tk 60,000 a year, and a sale after three years for Tk 120,000 gives a loss of Tk 30,000.

The summary restates the whole chapter in a few lines.

THINK IT THROUGH

16.T1 A manager raises the estimated life of the book-cutting machine from five years to eight years, with the same residual value. What happens to the yearly straight-line charge, to the profit of each year, and to the total charged over the whole life?

16.T2 A student says: "Depreciation sets aside cash for buying a new machine." Explain what is wrong with this statement.

WHERE PEOPLE GO WRONG

Some students believe that depreciation measures the fall in the market price of an asset. It does not: it spreads the cost over the years of use, by a rule. Others believe that depreciation is a payment of cash. It is not: the cash was paid when the asset was bought, and the yearly entry moves no cash. A third belief is that book value equals the price at which the asset could be sold. Book value is a figure in the records, and the market price may be higher or lower. A fourth belief is that a faster method gives a larger total charge. Over the whole life all methods charge the same total, and only the timing differs. Last, students sometimes depreciate land. Land is usually not depreciated, because it does not wear out in the same way.

Five common misunderstandings are stated here, each with its correction.

IN HISTORY

In 1850, Dionysius Lardner published *Railway Economy* in London. Lardner was an Irish-born writer on science and engineering, who lived from 1793 to

A book of 1850 about railways discussed the wearing out of locomotives and track.

The railways showed that heavy equipment does not last for ever, so its cost belongs to many years.

1859.² The railways were then about twenty years old, and their owners had to decide how to treat the wearing out of locomotives, carriages and track. Lardner discussed how parts wear with use and must be replaced, and he described the cost of keeping a railway in repair.

Railway companies owned costly equipment that served for many years, which is the same situation as that of the book-cutting machine. Lardner's discussion is an early case for the idea of this chapter: when equipment wears out, a fair record charges a part of its cost to each year of service. The methods in this chapter, straight-line, declining-balance and units-of-production, are later ways of making that idea exact.

REVIEW QUESTIONS

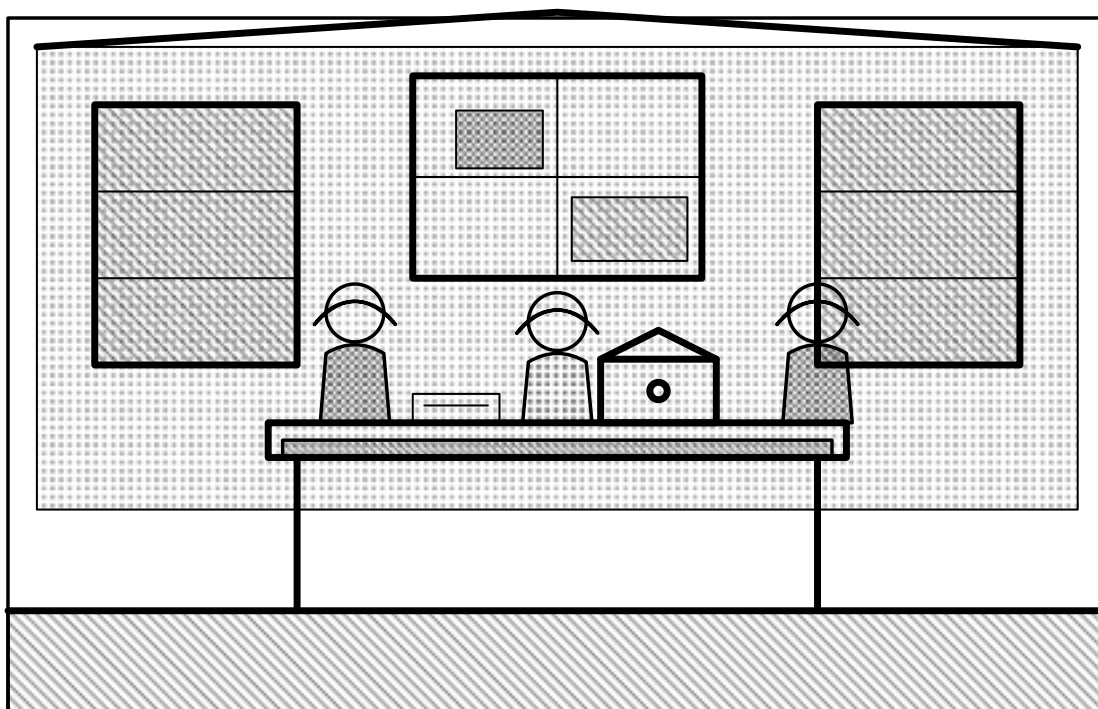
16.1 A machine costs Tk 50,000, has a residual value of Tk 10,000 and a useful life of four years. Find the straight-line charge for a year and the book value after two years.

16.2 Equipment costs Tk 275,000 on 1 October 2026. Its residual value is Tk 25,000, its useful life is eight years, and its capacity is 400,000 units. The business closes its books on 31 December. Find the straight-line charge for 2026, and the units-of-production charge for 2026 if 36,000 units were produced.

16.3 An asset costs Tk 150,000 and is depreciated at 20% by the declining-balance method. Find the charge and the remaining cost for each of the first three years. After three years it is sold for Tk 80,000. Find the gain or loss.

16.4 A press costs Tk 240,000 on 1 January 2026, with a residual value of Tk 40,000 and a useful life of five years. Using straight-line depreciation, write the entry for 2026. At 31 December 2028 the press is sold for Tk 100,000. Find the gain or loss and write the entry for the sale.

² Accounts of the book and its author are given in articles on the history of accounting and railways. This chapter relies on those summaries and not on the text of the book itself.



A village reading room, with the committee table and the treasurer's box

CHAPTER XVII

Accounts of Clubs and Sole Traders

The treasurer says the box holds more money than last year: is the club better off?



This chapter shows how organisations that do not aim at profit keep their accounts, and how their yearly cash record is turned into a fair statement of surplus or deficit. It then turns to a small trader who has kept no full books, and shows how to find the year's profit from the change in owner's equity. The same principles of Chapters 7 and 9 apply throughout.

A small puzzle: the cash grew, but the members are not sure the club is better off.

The puzzle asks what the cash figure leaves out.

The chapter has two settings, and both use the principles you know.

Imagine a village reading room, run by its members. They pay a yearly fee, and with the money they rent a room, buy newspapers and heat the room in winter. At the end of the year the treasurer opens the cash box and announces good news. A year ago it held Tk 4,000. Today it holds Tk 8,000. The cash has doubled.

A member asks three questions. Some members have not yet paid this year's fee; does that not count? Some members paid next year's fee early; is that money really this year's? The club bought a new table for Tk 10,000; is that a cost of this year only, or something that will serve for years? The treasurer's cash figure answers none of these questions. It records money moving in and out. It does not say whether the club has done well. This chapter shows how to answer with the tools you already know, in a new setting where nobody takes a profit.

There is a second puzzle, in a different setting. A small trader has kept no full books for a year. She has no journal, no ledger and no income statement. She knows only what she owns and owes at the start and at the end. Can she find her profit? She can, and the method is a pleasant surprise: it needs only two lists and some subtraction.

YOU WILL NEED

This chapter uses the three statements from Chapter 9 'Financial Statements and Closing Entries', section 9.1 'The Three Statements'. It also uses the accrual basis from Chapter 7 'Adjusting the Accounts', section 7.1 'Why Adjust'. The arithmetic is addition and subtraction; Appendix A1 'Maths You Will Use' holds a short reference.

* * *

17.1 NON-TRADING ORGANISATIONS

Most of this book has described businesses that exist to earn a profit, and some organisations do not.

The accounting name for this group describes its purpose.

Non-trading organisation. An organisation, such as a club or charity, that exists to serve its members or a cause rather than to earn profit.

The accounts follow the same principles, but some names change to suit the purpose.

Surplus. Word in use. In daily life, an amount left over or more than needed. In a non-trading organisation, the amount by which income exceeds expenditure in a period; if expenditure is larger, the result is a deficit.

Until now this book has described businesses that are run to earn profit for an owner. Many organisations exist for another reason. A village reading room, a sports club, a school parents' association and a charity are examples. Nobody puts money in to take profit out. The members join to enjoy something together, or the donors give to help others.

The accounting name for this group is **non-trading organisation**. A non-trading organisation is an organisation, such as a club or charity, that exists to serve its members or a cause rather than to earn profit. For example, the village reading room serves its members by providing newspapers and a quiet place to read. The name does not mean that the organisation never sells anything. A club may sell tea or tickets. It means that the main aim is not profit.

Because no owner takes profit, some words change. The result of a year is not called profit but *surplus* when income is larger than expenditure, and *deficit* when it is smaller. A surplus is not a gain for an owner. It is money kept for the use of the members in the future. The owner's equity of a business has a counterpart in the club: the *accumulated fund*, which is the total of all past surpluses plus any starting fund. It is the equity of the members as a group. The principles are the same as for any business: assets, liabilities, income, expenditure and the matching of the two.

The main income of most clubs is the **subscription**. A subscription is a regular payment made by a member for belonging to an organisation. For example, each member of the reading room pays Tk 500 a year. Other income comes from donations, from the entry fees of new members and from events such as an annual dinner. A subscription needs special care because members may pay late, on time or early. We shall return to this in section 17.2.

Small clubs usually keep little more than a cash book, and the treasurer's first report is a summary of it. This summary is called the **receipts and payments account**. A receipts and payments account is a summary of all cash received and paid in a period, with opening and closing balances. For example, the account of the reading room lists every taka that came into the cash box and every taka that went out during the year. The cash balance at the start is added on the receipts side, and the balance at the end is found by subtraction.

The account has two sides. The left side lists the receipts, beginning with the opening cash balance. The right side lists the payments, ending with the closing cash balance, which makes both sides equal. We now build a small account with clean numbers, so that the method is the only thing to learn. *Given*. The reading room had cash of Tk 4,000 on 1 January 2025. During 2025 it received subscriptions of Tk 30,000, donations of Tk 5,000 and Tk 8,000 from sales of tickets to an annual dinner. It paid rent of Tk 12,000, newspapers of Tk 9,000, dinner costs of Tk 6,000, repairs of Tk 2,000 and Tk 10,000 for a new table. *Find*. The closing cash balance.

Steps. Receipts: $4,000 + 30,000 + 5,000 + 8,000 = \text{Tk } 47,000$. Payments: $12,000 + 9,000 + 6,000 + 2,000 + 10,000 = \text{Tk } 39,000$. Closing cash: $47,000 - 39,000 = \text{Tk } 8,000$. *Answer*. The cash on 31 December 2025 is Tk 8,000. *Check*. The treasurer said Tk 8,000, so the account agrees with the box.

RECEIPTS AND PAYMENTS ACCOUNT FOR THE YEAR ENDED 31 DECEMBER 2025

RECEIPTS	TK	PAYMENTS	TK
---	---	---	---
Opening cash	4,000	Rent	12,000
Subscriptions	30,000	Newspapers	9,000
Donations	5,000	Dinner costs	6,000
Dinner tickets	8,000	Repairs	2,000
		New table	10,000
		Closing cash	8,000
Total	47,000	Total	47,000

The receipts and payments account is useful. It is easy to prepare, and it proves that the cash is in the box. But it has three limits, and they match the member's questions in the puzzle. It shows cash only, so it ignores fees that are owed but not yet paid. It does not separate this year's money from money paid in advance for the next year. And it mixes everyday costs with the purchase of a table that will last for years. A profit-seeking business could not rely on such a record, and a club should not either.

Before moving on. A non-trading organisation exists to serve members or a cause, not to earn profit. Its result is a surplus or a deficit, and its equity is the

The main source of money is the regular fee paid by members.

Subscription. A regular payment made by a member for belonging to an organisation.

Subscription. Word in use. In daily life, a payment to receive a newspaper or a service for a period. In a club, the yearly or monthly fee that a member pays for belonging to it.

A club keeps simple records, and its first summary is of cash only.

Receipts and payments account.

A summary of all cash received and paid in a period, with opening and closing balances.

The account has the same two-sided form as a cash book.

The steps add each side and subtract to find the closing balance.

The account has strengths, and also limits that explain the rest of the chapter.

Before moving on, we collect the ideas of this section.

accumulated fund. A subscription is a regular payment by a member. The receipts and payments account summarises the cash only, so it cannot say whether the year went well. You should now be able to say what the account shows and the three things it leaves out.

17.2 INCOME AND EXPENDITURE ACCOUNT

The remedy is to apply the accrual basis, and the result has its own name.

Income and expenditure account. The non-trading organisation's equivalent of an income statement, showing surplus or deficit on an accrual basis.

The figure shows how the cash record is turned into the accrual record, step by step.

The remedy for the three limits is a different statement, the **income and expenditure account**. An income and expenditure account is the non-trading organisation's equivalent of an income statement, showing surplus or deficit on an accrual basis. For example, it records the fees that belong to 2025, whether they were paid in 2025 or not. The accrual basis, which you met in Chapter 7, section 7.1 'Why Adjust', counts income and expense in the period to which they belong. The matching principle then puts the two together.

The figure below shows the change. Read it from left to right. The left box holds the receipts and payments account, which counts cash. The arrows mark the three changes you make: adjust for fees owed or paid early, adjust for costs owed or paid early, and take out the purchase of lasting assets. The right box holds the income and expenditure account, which counts what belongs to the year.

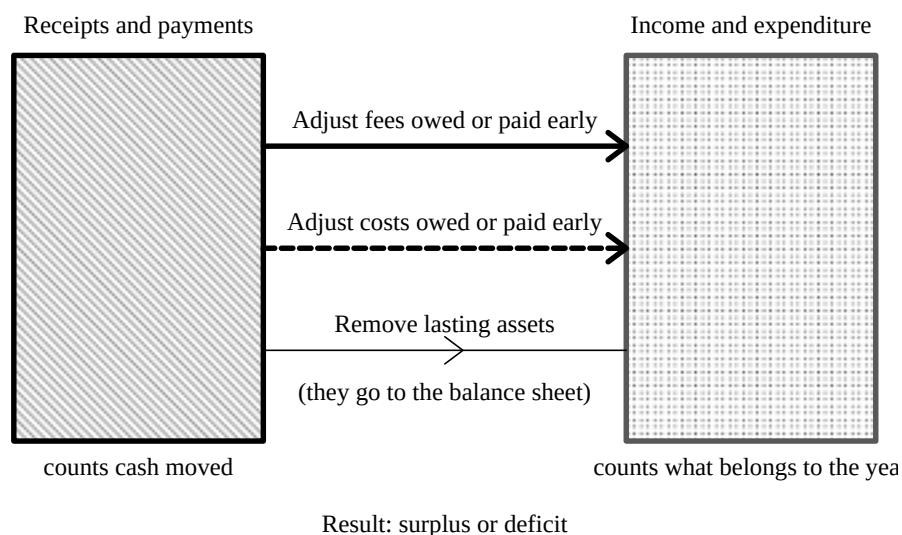


FIG. 17.1. Receipts and payments account turned into income and expenditure

The first change adjusts items that belong to a different period from the cash.

The first change deals with accrued and prepaid items. An accrued item belongs to this period but has not been paid in cash. A prepaid item has been paid in cash but belongs to a later period. Take subscriptions as the main example. Suppose some members owe fees for the year; those fees are income of the year, though no cash has come in. Suppose some members paid next year's fee; that cash is not income of this year. Suppose some members paid last year's arrears this year; that cash belongs to the earlier year. A single rule covers all three. In words: income for the year equals cash received, less arrears from earlier years received this year, less amounts received in advance for later years, plus amounts still owed for this year.

The same idea applies to costs such as rent and newspapers.

The same idea applies to costs. Rent paid in advance for January of the next year is not an expense of this year. A newspaper bill that is owed at the year end is an expense of this year, though it has not been paid. Expense for the year equals cash paid, less amounts paid in advance, plus

amounts still owed. This is exactly the method of Chapter 7, now applied to a club.

The second change deals with the table. A table bought for Tk 10,000 will serve for years, so its cost is not an expense of this year alone. It is an asset, and it appears on the balance sheet. Chapter 15 'Plant Assets' deals with such assets, and Chapter 16 'Depreciation' teaches how a share of the cost is charged to each year of use. In this chapter we do not charge depreciation, so that the principle of conversion stands out.

A third matter concerns events. When the club holds a dinner, it receives money from tickets and pays for food. It is clearer to show only the difference, the surplus or deficit on the dinner, than to list both amounts among the other items. For the dinner of the example, tickets of Tk 8,000 less costs of Tk 6,000 give a surplus of Tk 2,000.

The club also prepares a balance sheet at the year end. It lists the assets: the table, the cash, the fees still owed by members and the rent paid in advance. It lists the liabilities: the fees paid in advance by members and the unpaid newspaper bill. It ends with the accumulated fund, which equals assets less liabilities. The fund at the end equals the fund at the start plus the surplus for the year, or less the deficit.

PAUSE AND CHECK 17.P1 A club receives Tk 1,000 in January for the year that begins in the following April. Is it income of the year in which it is received? Give a reason.

Before moving on. The income and expenditure account applies the accrual basis to a non-trading organisation. It adjusts the cash record for amounts owed and amounts paid early, removes the purchase of lasting assets and shows the result as a surplus or a deficit. The balance sheet ends with the accumulated fund. You should now be able to say how cash received for subscriptions becomes income for the year.

The second change removes the purchase of a lasting asset.

Events that earn money are shown by their net result.

The club also draws up a balance sheet, which uses the same form as any other.

Before moving on, we collect the ideas of this section.

17.3 THE SOLE TRADER FROM INCOMPLETE RECORDS

Now we turn to the second puzzle. A sole trader is one person who owns and runs a business alone. Many small traders do not keep a journal and a ledger. They keep a cash box, a few bills and a memory. At the year end they, or a bank, or the tax office, want to know the profit. Records are incomplete, but two facts can be found by counting: what the business owns and owes at the start, and what it owns and owes at the end.

That list is called the **statement of affairs**. A statement of affairs is a list of a business's assets and liabilities at one date, used to find capital when records are incomplete. For example, the trader counts her tools, her goods, the money customers owe her and the cash in her box, and then lists what she owes to suppliers. The assets less the liabilities give the owner's equity at that date. The statement of affairs looks like a balance sheet, but it is built from counting and memory, not from ledger accounts.

How can profit come from two lists? Think of the owner's equity as a tank of water. Profit pours water in. The owner's own withdrawals for private use pour water out. Money the owner brings into the business from outside also pours water in. At the end of the year the level of the tank is known, because the statement of affairs gives it. So the profit is the change in the level, corrected for the two other flows. In words: profit equals closing owner's equity, less opening

Some small traders keep no full books, yet they still need to know their profit.

A list of assets and liabilities at one date is the tool, and it has a name.

Statement of affairs. A list of a business's assets and liabilities at one date, used to find capital when records are incomplete.

The profit is found from the change in owner's equity, after two corrections.

A first example with small numbers shows the method.

The steps build the two statements of affairs and then apply the formula.

The final accounts are then built with the profit as the balancing figure.

The method has limits, and an honest accountant states them.

A deficit is not always a failure, and a surplus is not always a success.

Before moving on, we collect the ideas of this section.

The reading room's cash record is converted in full, with each adjustment shown.

owner's equity, plus withdrawals, less new money brought in by the owner. In symbols, with E_2 for closing equity, E_1 for opening equity, W for withdrawals and N for new money: $\text{profit} = E_2 - E_1 + W - N$.

Given. At 1 January the trader owned equipment Tk 20,000, goods Tk 12,000, money owed by customers Tk 6,000 and cash Tk 2,000, and she owed suppliers Tk 8,000. At 31 December she owned equipment Tk 18,000, goods Tk 15,000, money owed by customers Tk 7,000 and cash Tk 5,000, and she owed suppliers Tk 6,000. During the year she took Tk 10,000 for private use and brought in Tk 5,000 of her own savings. *Find.* The profit for the year.

Steps. Opening assets: $20,000 + 12,000 + 6,000 + 2,000 = \text{Tk } 40,000$. Opening owner's equity: $40,000 - 8,000 = \text{Tk } 32,000$. Closing assets: $18,000 + 15,000 + 7,000 + 5,000 = \text{Tk } 45,000$. Closing owner's equity: $45,000 - 6,000 = \text{Tk } 39,000$. Profit = $39,000 - 32,000 + 10,000 - 5,000 = \text{Tk } 12,000$. *Answer.* The profit for the year is Tk 12,000. *Check.* Start from Tk 32,000, add the profit of Tk 12,000 and the new money of Tk 5,000, and subtract withdrawals of Tk 10,000; the result is Tk 39,000, which equals the closing owner's equity.

The trader can now prepare her final accounts. The balance sheet at 31 December lists the closing assets and liabilities from the statement of affairs, and its owner's equity is the opening owner's equity plus profit and new money, less withdrawals. The income statement shows the profit of Tk 12,000 as one figure. If she also has records of cash received and paid, she can show sales and expenses in more detail, by the methods of this chapter and Chapter 7. When she does not, the profit alone is the best result available.

The method has limits. It gives the amount of profit but not how it arose: you cannot see sales, expenses or gross profit. It depends on honest counting, since a missed asset or an unrecorded debt changes the answer. It also depends on correct figures for withdrawals, which small traders often do not record. For these reasons the result is an estimate that rests on the lists, and a trader should keep fuller records as the business grows. Reports that rely on such figures should state how they were prepared.

A word of judgement is useful here. A deficit in one year may be planned, for example when a club spends savings from earlier years on repairs. A surplus may be a warning, for example when a club saves money by delaying repairs that must be made later. The members of a club, the donors to a charity and the officials who inspect them read the accounts for these reasons. The figure matters less than the story behind it, and the accounts help the reader to find that story.

Before moving on. A statement of affairs lists the assets and liabilities at one date, and the difference is the owner's equity. Profit equals the change in owner's equity, plus withdrawals, less new money brought in. The method gives profit but not its parts. You should now be able to find a trader's profit from two statements of affairs and two further facts.

WORKED EXAMPLE: FROM CASH RECORD TO ACCRUAL ACCOUNTS

Given. The receipts and payments account of the reading room for 2025 is in the table of section 17.1. Four further facts are known. First, members owed Tk 1,500 for 2024 at 1 January 2025, and this was paid in 2025. Second, Tk 2,000 of the subscriptions received in 2025 was paid in advance for 2026. Third, at 31 December 2025 members owed Tk 3,000 for 2025. Fourth, the rent paid includes

	Tk	1,000	for
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January 2026, and a bill of Tk 500 for newspapers was unpaid on 31 December 2025.

Find. The income and expenditure account for 2025, the surplus, and the balance sheet at 31 December 2025.

Formula. In words: subscriptions for the year equal cash received, less arrears of the earlier year, less advance receipts for the next year, plus arrears at the end of this year. In symbols, with R for cash received, A_1 for earlier arrears, P for advance receipts and A_2 for closing arrears: $\text{income} = R - A_1 - P + A_2$.

Steps. Subscriptions: $30,000 - 1,500 - 2,000 + 3,000 = \text{Tk } 29,500$. Rent: $12,000 - 1,000 = \text{Tk } 11,000$. Newspapers: $9,000 + 500 = \text{Tk } 9,500$. Dinner surplus: $8,000 - 6,000 = \text{Tk } 2,000$. The table of Tk 10,000 is an asset and is not an expense. Repairs stay at Tk 2,000.

INCOME AND EXPENDITURE ACCOUNT FOR THE YEAR ENDED 31 DECEMBER 2025

INCOME	TK	EXPENDITURE	TK
---	---	---	---
Subscriptions	29,500	Rent	11,000
Donations	5,000	Newspapers	9,500
Surplus on annual dinner	2,000	Repairs	2,000
		Surplus for the year	14,000
Total	36,500	Total	36,500

At 1 January 2025 the club held cash of Tk 4,000 and was owed Tk 1,500, and it owed nothing, so its accumulated fund was Tk 5,500. The year added a surplus of Tk 14,000, so the fund at the end is $5,500 + 14,000 = \text{Tk } 19,500$. The assets are the table, Tk 10,000; cash, Tk 8,000; fees owed by members, Tk 3,000; and rent paid in advance, Tk 1,000. The total is Tk 22,000. The liabilities are fees paid in advance, Tk 2,000, and the unpaid newspaper bill, Tk 500, a total of Tk 2,500.

Answer. The surplus for 2025 is Tk 14,000 and the accumulated fund at the end is Tk 19,500. *Check.* Assets less liabilities are $22,000 - 2,500 = \text{Tk } 19,500$, which equals the fund. The agreement proves that the adjustments are consistent. The surplus of Tk 14,000 is larger than the rise in cash of Tk 4,000 because of the new table (Tk 10,000 of cash that is not an expense) and the other adjustments.

Three slips are common. The first is to use the cash figure for subscriptions without adjusting. The second is to charge the table as an expense, which would turn a healthy surplus into a small one. The third is to add arrears of the earlier year instead of subtracting them. The formula protects you: earlier arrears and advance receipts are subtracted, and closing arrears are added.

SUMMARY

A non-trading organisation exists to serve members or a cause. Its first summary is the receipts and payments account, which shows cash only. The income and expenditure account applies the accrual basis, adjusts subscriptions and costs for amounts owed and paid early, leaves out lasting assets and ends in a surplus or a deficit. A sole trader with incomplete records builds a statement of affairs at the start and the end, and finds profit from the change in owner's equity, plus withdrawals, less new money.

The task is to find the income and expenditure for the year and the accumulated fund at the end.

The subscriptions come first, because they need the longest calculation.

The steps are shown one line at a time.

The balance sheet shows what the club owns and owes at the end of the year.

The answer and the check show that the two statements agree.

Common slips: using cash for subscriptions, treating the table as an expense, and the wrong sign on an adjustment.

The chapter's ideas in a few lines.

THINK IT THROUGH

17.T1 A club reports a deficit in a year in which its cash balance rose. Give two reasons, using the ideas of this chapter, why both can be true.

17.T2 A trader's owner's equity fell from Tk 50,000 to Tk 46,000 during a year in which she took Tk 9,000 for private use and brought in no new money. Was her business profitable? Explain.

WHERE PEOPLE GO WRONG

Belief: the closing cash balance shows how well the club has done.

Belief: a surplus means the club has made a profit for an owner.

Belief: all payments in the receipts and payments account become expenditure.

Belief: the profit of a sole trader equals the rise in owner's equity.

Belief: a statement of affairs is a balance sheet from the ledger.

Cash shows only money that moved. It leaves out fees owed, fees paid early and the cost of lasting assets, so it can mislead in either direction.

A surplus belongs to the members as a group. It is added to the accumulated fund and used for the purposes of the club.

Payments for lasting assets, such as a table, become assets in the balance sheet. Only the costs that belong to the year appear in the income and expenditure account.

The rise must be corrected. Withdrawals reduced the equity and new money raised it, so both must be put back or taken out before profit appears.

It is built by counting and by memory when no ledger exists. It looks like a balance sheet, but it is less exact, because nothing proves it.

IN HISTORY

A famous charity began with a committee that needed to account for money received and spent.

The case shows why a charity measures income and spending and not profit.

On 5 October 1942, during the Second World War, a group of citizens met in the Old Library of the University Church in Oxford, England, and formed the Oxford Committee for Famine Relief. Its aim was to help civilians starving in Greece under occupation. The committee later became known as Oxfam. Its first big appeal, held in Oxford in 1943, raised £12,700 for the Greek Red Cross, and £2,300 of that came from a temporary gift shop.

A charity such as this one has no owner who takes profit. Its accounts show money received from donors and money spent on its cause, which is the pattern of the receipts and payments account and the income and expenditure account of this chapter. Its test of success is not profit but how much of the money reaches the people it was raised for.

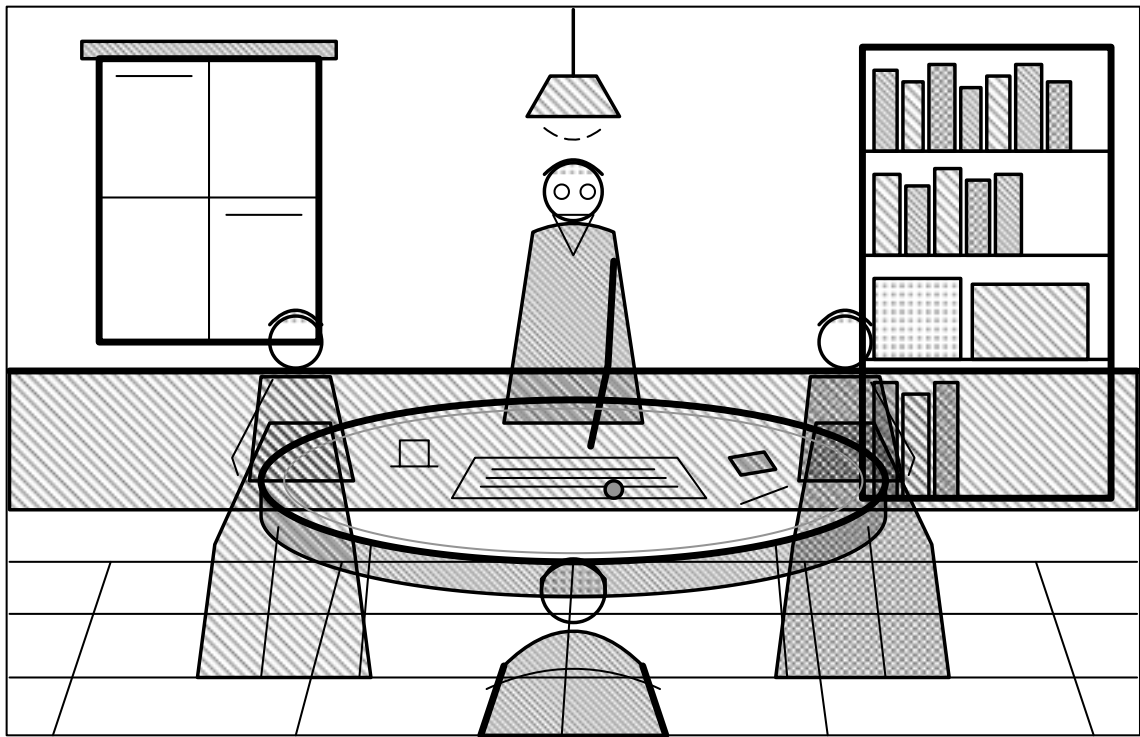
REVIEW QUESTIONS

17.1 A club began 2026 with cash of Tk 6,000. It received subscriptions of Tk 40,000 and donations of Tk 3,000, and paid rent of Tk 15,000, wages of Tk 12,000 and Tk 8,000 for a new cupboard. Prepare its receipts and payments account and find the closing cash.

17.2 In question 17.1 members owed Tk 2,000 of fees for 2025 at 1 January 2026, and these were paid in 2026. Tk 1,500 was received in 2026 for 2027. At 31 December 2026 members owed Tk 3,500 for 2026. Wages of Tk 1,000 were unpaid at the end of 2026. Find subscriptions income, wages expenditure and the surplus for 2026.

17.3 A trader had assets of Tk 60,000 and liabilities of Tk 15,000 at the start of a year, and assets of Tk 72,000 and liabilities of Tk 18,000 at the end. She withdrew Tk 20,000 for private use and brought in Tk 8,000 of savings. Find the profit.

17.4 A club bought a piano for Tk 30,000 in the year. Explain where this appears in the receipts and payments account, and in the income and expenditure account and balance sheet.



A solicitor and three partners study the deed that governs their business

CHAPTER XVIII

Partnership Accounts

Who receives what when three people share one business?



This chapter shows how the books of a business with several owners are kept. It explains each partner's capital, the rules for sharing profit, the admission of a new partner, and the ending of the business by liquidation. Each method is worked by hand, from small numbers to a realistic case.

YOU WILL NEED

Chapter 9, 'Financial Statements and Closing Entries', section 9.1 'The Three Statements'; Chapter 5, 'The Journal and the Ledger', section 5.2 'The Journal'; Appendix A1, 'Maths You Will Use'.

Three friends earn one profit from one stall, and each believes a different split is fair.

Equal shares and shares by money both seem unfair, so a better rule is needed.

This chapter answers four questions about people who own one business together.

Three friends run a market stall together. Anwar paid for most of the first goods, but he visits only on Sundays. Babita paid very little, yet she works at the stall every day. Chandra paid a middle sum and keeps the accounts. At the end of the first month the stall has earned a profit of Tk 30,000. How should the three friends divide it?

Divide it into three equal parts, and Anwar feels cheated, because his money carried most of the risk. Divide it by the money each person paid in, and Babita feels cheated, because her daily work counts for nothing. Both friends have a fair point. Neither rule is wrong in itself; each is incomplete. Friends who sit together can argue until they agree. A business that lasts for years cannot rely on memory and goodwill, so its owners write the agreement down, and the accountant keeps books that follow it.

Accountants have a standard answer to the puzzle, and this chapter teaches it step by step. You will learn how the money each owner contributes is recorded, how profit is divided in layers so that both money and work are rewarded, what changes in the books when a new owner joins, and how the business is closed and the owners are paid when it ends. We begin, as the friends must, with the question of who owns what.

* * *

18.1 FORMING A PARTNERSHIP

A business with several owners raises three questions that a single owner never faces.

A business with one owner is easy to describe. The owner puts money in, takes profit out, and the books show one amount, the owner's equity. That amount is also called capital, net worth or proprietorship, and this book uses the term owner's equity. When two or more people own a business together, three questions arise at once. How much has each person put in? How will the profit be divided? Who answers for the debts? The owners must agree on these answers before the first sale is recorded.

Partnership. A business owned by two or more people who agree to share its profit and loss.

A partnership is a business with two or more owners who share profit and loss.

The law gives a name to such an arrangement. A **partnership** is a business owned by two or more people who agree to share its profit and loss. Each owner is called a partner. Three friends who open a bookshop together, and agree to share what it earns and loses, form a partnership. In many countries the law holds every partner personally responsible for the firm's debts, so the choice of partners is a serious one.¹

Partnership deed. A written agreement setting out each partner's capital, duties and share of profit.

A deed puts the partners' agreement in writing so that nobody must rely on memory.

Partners protect themselves by writing their agreement down. A **partnership deed** is a written agreement setting out each partner's capital, duties and share of profit. A deed might say, for example, that one partner pays in money while another pays in nothing but works full time in the shop. If the partners make no agreement on a point, the law of the place decides it. In the United Kingdom, the Partnership Act 1890 divides profit equally between partners, whatever each one

¹ Partnership law differs by country. In the United Kingdom the Partnership Act 1890 supplies rules for partners who have not agreed their own; in Bangladesh the Partnership Act 1932 has the same purpose. Read the law of your own country before applying any rule from this chapter.

paid in. Partners who contribute unequally usually want a different rule, which is the reason deeds exist.

Now consider the books. When one person owns a business, one account records the owner's equity. When several people own it, the accountant keeps one account for each person. This account is called the partner's capital. It shows the amount that partner has put into the firm and left there, as a credit balance, which is a balance on the right-hand side of the account. When a partner pays in cash, we debit Cash and credit that partner's capital. When a partner brings an asset other than cash, such as a delivery van, we debit that asset instead of Cash. The capital of all the partners together makes up the owners' equity of the firm.

Different readers use the same records in different ways. A partner reads the capital accounts to see what share of the firm is his or her own. A bank that lends to the firm reads them to learn how much owners' money stands behind the loan. An employee has less interest in them, but may still ask whether the owners are able to pay wages in a poor year.

Case file, illustrative. A partnership deed is dated 1 July 2026. It records capital of Tk 2,500,000 from Nurul, Tk 1,000,000 from Rafiq and Tk 600,000 from Tariq. The three capital accounts together total Tk 4,100,000. The deed also fixes how profit is shared; section 18.2, 'Sharing Profit', explains the rule. On the balance sheet each of the three amounts appears on its own line.

Before moving on. A partnership is a business owned by two or more people who share profit and loss. The partnership deed records their agreement; without a deed, the law of the place decides. Each partner has a separate capital account, and the accounts together form the owners' equity of the firm.

* * *

18.2 SHARING PROFIT

The three friends of the opening can settle their puzzle in two layers. In the first layer, each friend is paid a reward for the money put in, so the largest lender gains most. In the second layer, the profit that remains is divided in a fixed proportion that reflects the work each friend does. Accountants use the same two layers, and each layer has a name.

We begin with the second layer. A **profit-sharing ratio** is the fixed proportion in which partners divide profit and loss. A ratio of 5:3:2 cuts the amount into ten equal parts: five go to the first partner, three to the second and two to the third. Of Tk 100, those partners receive Tk 50, Tk 30 and Tk 20. In symbols, a partner's share is $S = (p \div P) \times R$, where p is that partner's number of parts, P is the total number of parts, and R is the amount to be divided.

Many students expect the ratio to match the capital. It need not. A partner may hold the smallest capital and the largest ratio, because the ratio rewards work and responsibility as well as money. Losses are shared in the profit-sharing ratio too, unless the deed says otherwise.

Capital. Word in use. In daily life, a sum of money, or the main city of a country. In accounting, the amount owners have put into the business and left there.

Each partner's capital account records what that one partner has put into the firm.

A lender reads the deed to learn who owns the firm and how much capital stands behind it.

Recall: balance sheet, Chapter 9, section 9.1 'The Three Statements'.

The case file records three partners and the capital each one contributed.

Three ideas carry forward: the partnership, its deed and the capital accounts.

The two friends' puzzle is solved by dividing profit in two layers.

Share. Word in use. In daily life, a part of something divided among people. In accounting, a partner's part of the profit; in a company, a unit of ownership, taught in Chapter 19.

Profit-sharing ratio. The fixed proportion in which partners divide profit and loss.

A ratio cuts profit into equal parts and gives each partner a fixed number of them.

The ratio need not match the capital, and it also shares losses.

Interest on capital. A payment credited to partners for their capital before the remaining profit is divided.

Interest on capital rewards money in proportion to the capital each partner contributed.

Interest on capital divides profit; it is not a cost paid to an outsider.

The appropriation account shows profit being divided, layer by layer, in one place.

Example 1 uses small round numbers so that the method is the only thing to learn.

The first layer rewards money. **Interest on capital** is a payment credited to partners for their capital before the remaining profit is divided. A partner with capital of Tk 100,000 and a rate of 10% a year is credited Tk 10,000 for a full year. In words, interest equals capital times rate times time. In symbols, $I = C \times r \times t$, where C is the capital, r is the yearly rate and t is the time in years. Six months is half a year, so $t = 6 \div 12$.

Interest on capital is a way of dividing profit. It is not a cost paid to an outsider, and no cash need leave the firm. It exists only if the deed provides for it. We also assume that the profit is large enough to cover all the interest; a deed must say what happens if it is not.

PAUSE AND CHECK 18.P1 A partner has capital of Tk 200,000. The deed allows interest at 5% a year. How much interest is credited for six months?

Both layers are recorded in an appropriation account. The word *appropriation* means setting money aside for a stated use. This account begins with the profit from the income statement. It takes out interest on capital, divides the remainder by the profit-sharing ratio, and ends with the total credited to each partner. In this way the account shows who received what, and why, in one place.

Example 1 uses small, round numbers. Work through each row of the table in order; every line of arithmetic is shown.

EXAMPLE 1: DIVIDING ONE YEAR'S PROFIT IN TWO LAYERS

PART	WORKING
Given	Capital: A Tk 100,000, B Tk 50,000. Interest 10% a year. Ratio 2:1. Profit for the year Tk 30,000.
Find	The total profit credited to A and to B.
Formula	In words: interest = capital $\times$ rate $\times$ time; share = (parts $\div$ total parts) $\times$ remainder. In symbols: $I = C \times r \times t$ and $S = (p \div P) \times R$.
Step 1	Interest for A = $100,000 \times 10\% \times 1 = \text{Tk } 10,000$.
Step 2	Interest for B = $50,000 \times 10\% \times 1 = \text{Tk } 5,000$.
Step 3	Remainder $R = 30,000 - (10,000 + 5,000) = \text{Tk } 15,000$.
Step 4	A's share = $(2 \div 3) \times 15,000 = \text{Tk } 10,000$. B's share = $(1 \div 3) \times 15,000 = \text{Tk } 5,000$.
Answer	A is credited Tk 20,000 (10,000 + 10,000). B is credited Tk 10,000 (5,000 + 5,000).
Check	$20,000 + 10,000 = \text{Tk } 30,000$, which equals the profit.

Three common slips are a wrong period, a wrong order of layers and a wrong divisor.

Two layers, interest first and ratio second, are the core of this section.

Three slips are common. First, students forget that the rate is yearly and credit a full year of interest for a shorter period. Second, they divide the profit by the ratio before taking out interest, which reverses the layers. Third, they divide by one part of the ratio instead of its total: with a ratio of 2:1 the divisor is 3, not 2.

Before moving on. A profit-sharing ratio divides profit and loss in fixed parts. Interest on capital pays each partner for money put in, and it comes out of profit first. The appropriation account records both layers and ends with each partner's total.

* * *

18.3 ADMISSION OF A PARTNER

What must be agreed, and what must be recorded, when a fourth person asks to join a firm of three? The two questions have different answers, and many students mix them up. The first belongs to the partners and their deed. The second belongs to the accountant and the journal.

When a new partner is admitted, the people who own the firm change. In many places the law treats the firm before and after the change as two different partnerships. For this reason the partners agree new terms: the date, the new partner's capital and the profit-sharing ratio that applies from then on. The deed is rewritten or added to. The accountant records the new capital only when it actually arrives.

Case file, illustrative. On 1 January 2027, Nusrat Meghna is admitted as a partner. She pays in cash of Tk 700,000, which is credited as her capital. The case file gives no new profit-sharing ratio, so none is used in this chapter.

Recording the new capital uses the rule from section 18.1. Cash comes in, so we debit Cash. The new partner's claim on the firm grows, so we credit her capital. Each journal entry has a date, the accounts, the amounts and a short narration. The table shows the entry for the case.

JOURNAL ENTRY FOR THE ADMISSION OF NUSRAT MEGHNA

DATE	ACCOUNT	DEBIT (TK)	CREDIT (TK)
1 January 2027	Cash	700,000	
	Nusrat Meghna, Capital		700,000
	Cash paid in by the new partner on admission		

Check the entry in two ways. The debit of Tk 700,000 equals the credit of Tk 700,000, as every entry requires. The capital accounts of Nurul, Rafiq and Tariq are untouched, because their claims did not change on that day. If the new partner brings a delivery van instead of cash, the debit goes to the van, and the credit to her capital stays the same.

Before moving on. Admission of a partner changes the owners, so the partners agree new terms. The accountant records only the new capital: debit Cash, or the asset brought, and credit the new partner's capital. Existing capital accounts stay as they were.

* * *

18.4 LIQUIDATION

Every business ends one day. How does an owner recover money from a shop full of shelves, counters and unsold books? The answer is to sell everything, pay what is owed and divide what remains. Accountants give a name to this process and to its first step.

A new partner changes who owns the firm, so the agreement and the records must both change.

Admission of a partner means a new person becomes an owner, usually after new terms are agreed.

The case file records Nusrat Meghna joining on 1 January 2027 with cash.

Recall: journal, Chapter 5, section 5.2 'The Journal'.

Recording a new partner's capital follows the rule for any cash paid in: debit Cash, credit capital.

Debits equal credits, and the existing partners' accounts do not change.

New terms are agreed by the partners; new capital is recorded by the accountant.

Ending a business means turning assets into cash, paying debts and sharing what is left.

Liquidation. The process of selling a business's assets, paying its debts and sharing what remains among the owners.

Liquidation is the whole process of closing a business and paying everyone.

Realisation. The sale of assets, in liquidation, and the gain or loss on that sale.

Realisation is the sale of the assets and the gain or loss that results.

The case file records the sale of all assets at a loss and the payment of debts and partners.

Paying the partners follows four steps in a fixed order: sell, share the loss, pay debts, pay partners.

Example 2 works the liquidation arithmetic in the standard order.

Liquidation is the process of selling a business's assets, paying its debts and sharing what remains among the owners. A shop that sells its books and fittings, repays its suppliers and bank, and gives the balance to its owners is in liquidation. Debts to outsiders come first. The owners receive only what is left, because their claim comes after every other claim.

The first step has its own name. **Realisation** is the sale of assets, in liquidation, and the gain or loss on that sale. If an asset with a book amount of Tk 100 sells for Tk 80, the realisation loss is Tk 20. The book amount is the amount at which the asset stands in the accounts. The partners share a loss on realisation in the profit-sharing ratio, as section 18.2 explained, unless the deed says otherwise.

Case file, illustrative. The liquidation takes place on 31 December 2027. The assets, at a book amount of Tk 5,300,000, are sold for Tk 4,400,000. The liabilities are Tk 1,200,000. The loss of Tk 900,000 is shared as Tk 450,000, Tk 270,000 and Tk 180,000. Cash of Tk 3,200,000 remains after the debts and is paid to the partners as Tk 2,050,000, Tk 730,000 and Tk 420,000.

Paying the partners comes last, and the order matters. First, sell the assets and find the loss. Second, share the loss in the ratio 5:3:2 by reducing each partner's capital. Third, pay the liabilities to outsiders from the cash received. Fourth, pay each partner the capital that remains. A realisation account collects the assets sold and the cash received, so that its balance shows the loss. The table records the four steps.

ENTRIES FOR THE LIQUIDATION, 31 DECEMBER 2027

STEP	DEBIT	CREDIT	TK
1	Cash	Realisation	4,400,000
1	Realisation	Assets	5,300,000
2	Capital: Nurul 450,000, Rafiq 270,000, Tariq 180,000	Realisation	900,000
3	Liabilities	Cash	1,200,000
4	Capital: Nurul 2,050,000, Rafiq 730,000, Tariq 420,000	Cash	3,200,000

Example 2 works the arithmetic. The case file implies that each partner's capital on 31 December 2027 equals the amount in the deed, and the working uses those amounts.

EXAMPLE 2: PAYING THE PARTNERS AT LIQUIDATION

PART	WORKING
Given	Capital: Nurul Tk 2,500,000, Rafiq Tk 1,000,000, Tariq Tk 600,000. Ratio 5:3:2. Assets, book amount Tk 5,300,000, sold for Tk 4,400,000. Liabilities Tk 1,200,000.
Find	The loss on realisation, each partner's share of it, and the cash paid to each partner.
Formula	In words: loss = book amount – amount received; payment = capital – share of loss. In symbols: $L = B - A$ and $X = C - D$, where B is the book amount, A is the amount received, C is the capital and D is the partner's share of the loss.
Step 1	Loss $L = 5,300,000 - 4,400,000 = \text{Tk } 900,000$.
Step 2	Shares of loss: Nurul $(5 \div 10) \times 900,000 = \text{Tk } 450,000$. Rafiq $(3 \div 10) \times 900,000 = \text{Tk } 270,000$. Tariq $(2 \div 10) \times 900,000 = \text{Tk } 180,000$.
Step 3	Payments X: Nurul $2,500,000 - 450,000 = \text{Tk } 2,050,000$. Rafiq $1,000,000 - 270,000 = \text{Tk } 730,000$. Tariq $600,000 - 180,000 = \text{Tk } 420,000$.
Step 4	Cash after debts $= 4,400,000 - 1,200,000 = \text{Tk } 3,200,000$.
Answer	The loss is Tk 900,000. Nurul receives Tk 2,050,000, Rafiq Tk 730,000 and Tariq Tk 420,000.
Check	$2,050,000 + 730,000 + 420,000 = \text{Tk } 3,200,000$, which equals the cash after debts.

Two slips are usual. Some students pay the partners before the outside debts, which is wrong, because outsiders have the first claim. Others share the loss in proportion to capital instead of the profit-sharing ratio. Capital and ratio are different things, and the deed names the ratio.

Occasionally a loss is larger than a partner's capital. That partner's capital account then has a debit balance, which means the partner owes the firm the shortage. The partner must pay it in before the final payments are made. The case file has no such partner, because each capital is larger than that partner's share of the loss.

Before moving on. Liquidation sells the assets, pays the debts and shares what remains. Realisation is the sale and its gain or loss, which the partners share in the ratio. Outsiders are paid before partners, and each partner receives capital less his or her share of the loss.

* * *

Paying partners before debts, or sharing the loss by capital, are the two usual slips.

A partner whose capital falls below zero must pay in the shortage.

Sell, share the loss, pay debts, then pay the partners: this order is the technique of liquidation.

WORKED CASE: DIVIDING SIX MONTHS OF PROFIT

Section 18.2 used a full year. The case file is messier, because the profit covers only six months. Read the figure below from top to bottom. The top bar is the whole profit. The dashed layer shows interest on capital, and the dotted layer shows the remainder divided by the ratio. The bottom layer gives each partner's total. The words in each layer carry the same meaning as the shading, so the figure reads in grey as well.

The first six months of the case give a realistic division over a period shorter than a year.

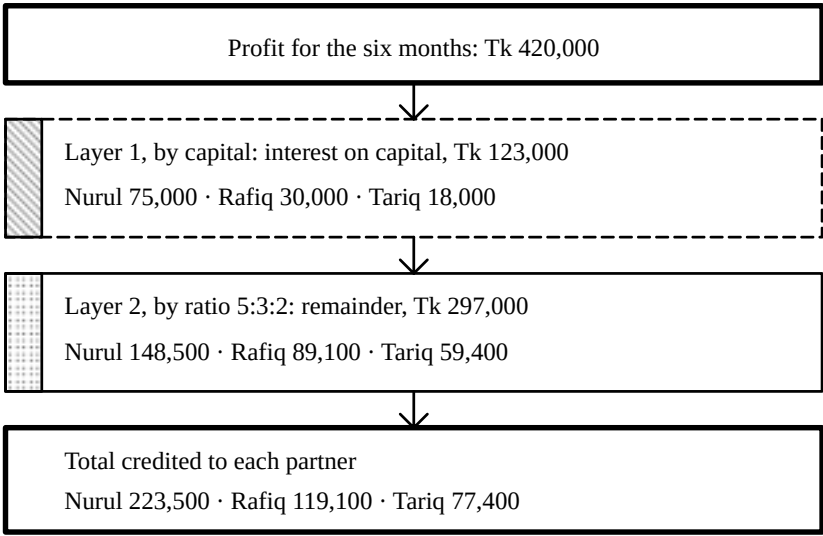


FIG. 18.1. Profit shared in two layers: interest on capital first, then the ratio

The case file records the profit for the first six months and the rules that divide it.

Example 3 divides the case-file profit in standard order, including the change of period.

Case file, illustrative. The deed is dated 1 July 2026. Capital: Nurul Tk 2,500,000, Rafiq Tk 1,000,000 and Tariq Tk 600,000. The profit-sharing ratio is 5:3:2 and interest on capital is 6% a year. The profit from 1 July to 31 December 2026 is Tk 420,000.

Example 3 divides this profit. Notice the period: 1 July to 31 December is six months, so $t = 6 \div 12 = 0.5$ year. Many students find this step confusing, because the rate is stated for a year while the profit covers half of one. The table keeps both facts in view.

EXAMPLE 3: DIVIDING SIX MONTHS OF PROFIT

PART	WORKING
Given	Capital: Nurul Tk 2,500,000, Rafiq Tk 1,000,000, Tariq Tk 600,000. Interest 6% a year. Ratio 5:3:2. Profit for six months Tk 420,000.
Find	The total profit credited to each partner.
Formula	In words: interest = capital $\times$ rate $\times$ time; share = (parts $\div$ total parts) $\times$ remainder. In symbols: $I = C \times r \times t$ and $S = (p \div P) \times R$.
Step 1	Interest: Nurul $2,500,000 \times 6\% \times 0.5 = \text{Tk } 75,000$. Rafiq $1,000,000 \times 6\% \times 0.5 = \text{Tk } 30,000$. Tariq $600,000 \times 6\% \times 0.5 = \text{Tk } 18,000$.
Step 2	Total interest = $75,000 + 30,000 + 18,000 = \text{Tk } 123,000$.
Step 3	Remainder $R = 420,000 - 123,000 = \text{Tk } 297,000$.
Step 4	Shares of R: Nurul $(5 \div 10) \times 297,000 = \text{Tk } 148,500$. Rafiq $(3 \div 10) \times 297,000 = \text{Tk } 89,100$. Tariq $(2 \div 10) \times 297,000 = \text{Tk } 59,400$.
Step 5	Totals: Nurul $75,000 + 148,500 = \text{Tk } 223,500$. Rafiq $30,000 + 89,100 = \text{Tk } 119,100$. Tariq $18,000 + 59,400 = \text{Tk } 77,400$.
Answer	Nurul is credited Tk 223,500, Rafiq Tk 119,100 and Tariq Tk 77,400.
Check	$223,500 + 119,100 + 77,400 = \text{Tk } 420,000$, which equals the profit.

The appropriation account sets out the same figures in the form accountants use.

The appropriation account shows the same figures in the form that accountants use. Read each column downward: a partner's interest, then the share of the remainder, then the total.

APPROPRIATION ACCOUNT FOR THE SIX MONTHS ENDED 31 DECEMBER 2026 (TK)

ITEM	NURUL	RAFIQ	TARIQ	TOTAL
Profit for the period				420,000
Interest on capital	75,000	30,000	18,000	123,000
Remainder, shared 5:3:2	148,500	89,100	59,400	297,000
Total credited to partners	223,500	119,100	77,400	420,000

A common slip is to charge a full year of interest. The interest would then be 150,000, 60,000 and 36,000, a total of Tk 246,000, and the remainder would fall to Tk 174,000. Every total would be wrong. Always ask first: how long is the period?

Charging a full year of interest gives wrong totals, even though the arithmetic looks neat.

Tariq holds the smallest capital. Interest gives him Tk 18,000, and the ratio gives him a fifth of the remainder. Nurul holds the largest capital, and for him the interest layer protects the reward for his money. The ratio then gives him half of what remains, as the deed provides.

Each partner sees the two layers differently, and both layers serve a purpose.

* * *

SUMMARY

A partnership is a business owned by two or more people who share profit and loss under a deed. Each partner has a capital account. Profit is divided in two layers, interest on capital and then the remainder by the profit-sharing ratio, and both are recorded in the appropriation account. A new partner's capital is recorded by debiting Cash and crediting that partner's capital. In liquidation, the assets are realised, the loss is shared by the ratio, debts are paid first, and each partner receives capital less his or her share of the loss.

A partnership shares profit and loss by agreement and records each partner's capital separately.

THINK IT THROUGH

18.T1 Tariq adds Tk 400,000 to his capital, and the ratio stays 5:3:2. What changes in his share of the remainder, and what changes in his interest at 6% a year?

18.T2 After a firm sells all its assets, its cash is smaller than its debts. Who is paid first, and what happens to the partners' capital?

WHERE PEOPLE GO WRONG

Belief: the partner with the most capital must have the biggest share of profit. **Correction:** the ratio is fixed by the deed and need not match the capital. Interest on capital rewards money separately, in the first layer.

A larger capital does not give a larger ratio; the deed sets the ratio.

Belief: interest on capital is a cost of the firm, like rent. **Correction:** it is a way of dividing profit among the partners, so it appears in the appropriation account and not in the income statement.

Interest on capital divides profit; it is not an expense paid to an outsider.

Belief: the owners are paid first, because they own the firm. **Correction:** suppliers, banks and other outsiders are paid before the partners receive anything.

Owners are paid last in liquidation, because outsiders hold the first claim.

Belief: a loss follows the size of each partner's capital. **Correction:** unless the deed says otherwise, the loss follows the profit-sharing ratio, exactly as profit does.

A loss on realisation is shared by the ratio, not by capital.

Goldman Sachs, a New York firm founded in 1869, was owned by its partners for 130 years.

On 4 May 1999 the firm sold shares to the public and its partners became shareholders.

IN HISTORY

Marcus Goldman, a German immigrant, founded Goldman Sachs in New York City in 1869. The firm grew into one of the best-known financial firms in the United States, and for 130 years its owners were partners. In the 1990s the partners debated for years whether to change this. Some feared the loss of the partnership's special culture; others wanted permanent capital to fund the firm's growth.

The partners voted to make the change. On 4 May 1999 the firm's shares began to trade, in what was then the second-largest first public sale of shares in United States history. The lesson for this chapter is that a partner's capital belongs to named people, who may take it back when they leave, while money raised from shares stays in the business. Chapter 19, 'Company Accounts: Shares', section 19.1 'Share Capital', teaches that second form.

REVIEW QUESTIONS

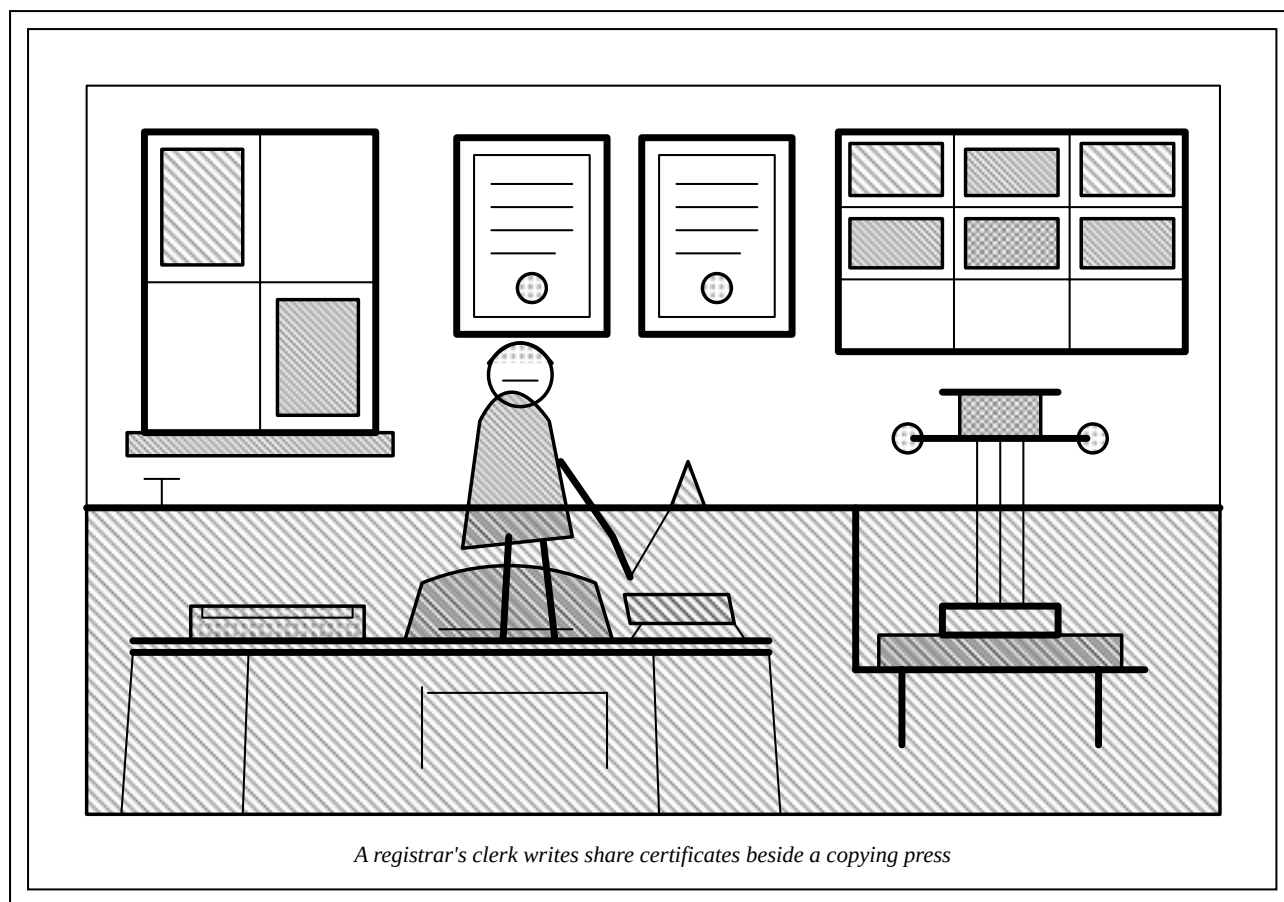
18.1 Two partners share profit in the ratio 3:1 and the deed gives no interest on capital. The profit is Tk 80,000. How much does each partner receive?

18.2 A new deed gives Mira capital of Tk 450,000 and Naveed capital of Tk 300,000. Interest on capital is 8% a year, and the ratio is 3:2. The profit for the year is Tk 211,000. Find the total credited to each partner.

18.3 Under the same deed as question 18.2, the profit of Tk 180,000 is for nine months. Find the total credited to each partner.

18.4 On 1 April 2028 a new partner, Priya, pays Tk 300,000 by cheque into the firm's bank account, and the amount is credited as her capital. Journalise the admission.

18.5 Partners P and Q have capital of Tk 400,000 and Tk 200,000 and share profit and loss 2:1. Their assets, at a book amount of Tk 850,000, are sold for Tk 700,000, and the liabilities are Tk 250,000. Find the loss, each partner's share of it, and the payment to each partner.



CHAPTER XIX

Company Accounts: Shares

How can a thousand strangers own one business together?



This chapter explains how a company raises money by selling shares. It shows what share capital is, how ordinary and preference shares differ, and how an issue of shares is recorded from application to allotment, including the share premium. Every entry is worked by hand, from small numbers to a realistic case.

YOU WILL NEED

Chapter 9, 'Financial Statements and Closing Entries', section 9.3 'Reversing Entries'; Chapter 18, 'Partnership Accounts', section 18.1 'Forming a Partnership'; Appendix A1, 'Maths You Will Use'.

A bookshop needs far more money than its four partners own, so it looks to many small contributors.

The puzzle is to give each contributor a small, equal piece of the business and a firm limit on the risk.

This chapter takes three steps: what share capital is, which kinds of share exist, and how shares are issued.

A company is owned by many people through shares, and a share is one equal part of that ownership.

Recall: balance sheet, Chapter 9, section 9.3 'Reversing Entries'.

Stock. Word in use. In everyday English, a supply of goods kept ready for sale. In the United States and some other places, stock also means a company's shares. This book uses inventory for goods and stock only for company shares.

Share capital. The money a company receives from shareholders in exchange for shares.

Share capital is the money shareholders pay to the company for their shares.

Suppose a bookshop wishes to open ten new branches. The cost is Tk 5,000,000. The four partners who own it have already put in all the money they can spare. A thousand neighbours would each be willing to give Tk 5,000, if they could be sure of a fair return. Yet the partners cannot ask a thousand people to become partners. A deed with a thousand names would be impossible to write, and in many countries every partner is personally responsible for the firm's debts. No sensible neighbour would accept that risk for Tk 5,000.

What arrangement would be attractive to the neighbours and safe for the partners? Each neighbour needs a small piece of the business, a paper to prove it, a claim on profit, and a promise that the loss can never be more than the money paid. The partners need a way to keep control of the shop while they raise the money. Merchants found an answer long ago, and it is the subject of this chapter. The money raised becomes the capital of a company, and each small piece of ownership is called a share.

We shall proceed in three steps. First, we see what the money raised is called and how its size is limited. Second, we meet the two chief kinds of share. Third, we record an issue of shares in the books, from the first application to the last entry. Chapter 18 described a business with a few owners who know each other. Here the owners may be many, and the records change to suit them.

* * *

19.1 SHARE CAPITAL

Ten families in a village decide to dig a well. The well will cost Tk 100,000, so each family pays Tk 10,000 and receives a written note of its part. Nobody owns the well alone, and nobody's part is larger than another's. A company works in a similar way, on a larger scale. Its owners are called shareholders. A shareholder is a person who owns one or more shares, and a share is one of the equal parts into which the ownership of a company is divided.

The accountant needs a name for the money that the shareholders pay in. **Share capital** is the money a company receives from shareholders in exchange for shares. If a shareholder pays Tk 1,000 for 100 shares, then Tk 1,000 enters the share capital. In a partnership, each partner has a separate capital account, as Chapter 18 showed. A company keeps one share capital account for all its shareholders together. On the balance sheet it appears among the owners' equity. In the United States, share capital is often called capital stock.

Every share is created with an amount printed on it. This amount is called the face value, and it is the same for every share of the same kind. A company might create shares with a face value of Tk 10. The face value is a fixed label. It is not the price at which people trade the shares afterwards, and it is not what the company's books record when one shareholder sells to another. Those later sales happen between shareholders, so the company's own accounts do not change.

A company cannot create unlimited shares. The people who found it choose a ceiling, and write that ceiling into its founding documents.¹ Think of a theatre. The building has a fixed number of seats, whatever the size of tonight's audience. The ceiling of a company works as the number of seats does.

Authorised capital is the largest amount of share capital a company is allowed to issue under its founding documents. A company with authorised capital of Tk 50,000 may sell shares worth up to Tk 50,000 and no more. If the founders later want to sell shares worth more, they must first change the founding documents by the legal method of their country.

Most companies do not sell all their authorised shares at once. **Issued capital** is the part of authorised capital that has actually been sold to shareholders. If the company above has sold shares worth Tk 30,000, then its issued capital is Tk 30,000. The remaining Tk 20,000 is authorised but not issued, and the company may sell it later. Issued capital can never be larger than authorised capital.

Example 1 applies these ideas to a small company. The first line of each step names the rule, and every line of arithmetic follows it.

EXAMPLE 1: AUTHORISED CAPITAL AND ISSUED CAPITAL

PART	WORKING
Given	Authorised capital Tk 50,000, in shares of face value Tk 5. The company has issued 6,000 shares.
Find	The number of shares authorised, the issued capital, and the capital not yet issued.
Formula	In words: shares authorised = authorised capital ÷ face value; issued capital = shares issued × face value; capital not yet issued = authorised capital – issued capital. In symbols: $N = A \div F$, $I = n \times F$ and $U = A - I$, where A is authorised capital, F is face value and n is the number of shares issued.
Step 1	$N = 50,000 \div 5 = 10,000$ shares.
Step 2	$I = 6,000 \times 5 = \text{Tk } 30,000$.
Step 3	$U = 50,000 - 30,000 = \text{Tk } 20,000$, which is $20,000 \div 5 = 4,000$ shares.
Answer	The company may issue 10,000 shares. It has issued capital of Tk 30,000 and may still issue shares worth Tk 20,000.
Check	$30,000 + 20,000 = \text{Tk } 50,000$, which equals the authorised capital.

Each share carries a face value, the amount printed on it, which is fixed when the share is created.

A company may not create shares without limit; its founding documents set a ceiling.

Authorised capital. The largest amount of share capital a company is allowed to issue under its founding documents.

Authorised capital is the ceiling on the shares a company may sell.

Issue. Word in use. In daily life, a topic of concern, or a copy of a magazine. In accounting, to issue shares means to sell and deliver them to the people who apply for them.

Issued capital. The part of authorised capital that has actually been sold to shareholders.

Issued capital is the part of the ceiling that has actually been sold.

Example 1 uses small round numbers, so that the method is the only thing to learn.

¹ The rule differs by country. Under the Companies Act 1994 of Bangladesh, the founding documents state a ceiling of share capital. In the United Kingdom, the Companies Act 2006 ended the need for most companies to state one. Read the law of your own country before applying any rule from this chapter.

The case file gives a proposed company a ceiling of Tk 10,000,000 and a first issue to its partners.

Shares of face value Tk 10 given for net assets of Tk 4,100,000 are recorded as Tk 4,100,000 of share capital.

The entry is shown in short form; the full entry debits each asset and credits each liability.

Different readers want different things from the same figures: owners, managers, lenders and employees.

Authorised capital is a ceiling and issued capital is what has been sold; the two are not interchangeable.

Three terms carry forward: share capital, authorised capital and issued capital.

Investors differ: one wants steady income and another wants a voice and a chance of higher reward.

Case file, illustrative. Meghna and Sons Limited is a proposal to turn the business into a company. Its authorised capital is Tk 10,000,000, in 1,000,000 shares of face value Tk 10. The first issue is 410,000 shares to the partners, in exchange for net assets of Tk 4,100,000. Net assets are the assets of the business less its liabilities. The partners' own capital accounts, which Chapter 18 described, are replaced by the shares they receive.

Now we record the first issue. The partners hand over net assets, and the company gives them shares in return. The company debits the net assets it receives. Debit means the left-hand side of an account. It credits Share Capital with the face value of the shares given. Here 410,000 shares $\times$ Tk 10 = Tk 4,100,000, which equals the value of the net assets, so no further entry is needed. The table shows the entry.

JOURNAL ENTRY FOR THE FIRST ISSUE OF SHARES

ACCOUNT	DEBIT (TK)	CREDIT (TK)
Net assets taken over	4,100,000	
Share Capital		4,100,000
Shares issued to the partners in exchange for net assets		

This entry is shown in short form. In the full entry, every asset is debited by its own amount and every liability is credited by its own amount, and the difference is Tk 4,100,000. After this entry, the issued capital is Tk 4,100,000. The capital not yet issued is Tk 10,000,000 – Tk 4,100,000 = Tk 5,900,000.

Each reader looks at share capital for a different reason. A shareholder asks what part of the company he or she owns. A manager asks how much more may be raised without changing the founding documents. A bank asks how much owners' money stands behind its loan. An employee, a supplier or a customer asks mainly whether the company is stable.

Many students find these two terms confusing, because both are amounts of capital in the founding documents' language. The test is simple. If shareholders have paid for it, it is issued. If the company has only permission to sell it, it is authorised and not yet issued.

Before moving on. Share capital is the money a company receives from shareholders in exchange for shares. Authorised capital is the ceiling set by the founding documents. Issued capital is the part of the ceiling that has been sold, and it is never larger than the ceiling.

* * *

19.2 TYPES OF SHARE

Two neighbours each have Tk 100,000 to invest in the bookshop. Anwar is retired and wants a steady, predictable income. Babita is young and is willing to wait for a larger reward, and she wishes to have a say in how the shop is run. One kind of share cannot suit both of them equally. For this reason, companies create more than one kind.

We begin with the common kind. An **ordinary share** is a share that gives a holder a vote and a claim on profit after other claims are paid. The vote lets the holder help to choose the people who direct the company. The claim on profit has no fixed amount, so it is large when the company does well and may be nothing when it does badly. Babita would choose ordinary shares.

The second kind suits Anwar. A **preference share** is a share that gets a fixed dividend before ordinary shareholders, usually without a vote. A dividend is a share of profit that the company pays out to its shareholders. If Anwar holds preference shares with a fixed rate of 8% a year, he receives 8% of their face value first, before any ordinary shareholder is paid, but the company has not promised that it will earn that profit.

We now have the rights of each kind. The table below sets them side by side. To read a row, find the right in the left-hand column and move across to see how each kind treats it.

THE RIGHTS OF THE TWO KINDS OF SHARE

RIGHT	ORDINARY SHARE	PREFERENCE SHARE
Vote at meetings	Yes	Usually no
Dividend	Not fixed; paid from profit that remains	Fixed rate; paid first
Order of payment	After the preference dividend	Before the ordinary dividend
Gain when profit is large	Large, because the claim is on what remains	Limited to the fixed rate
Loss when profit is small	May receive nothing	Paid first, if profit is enough

The table shows a trade between safety and reward. The preference holder gains safety and gives up the vote and the chance of a large dividend. The ordinary holder accepts risk and gains a vote and a larger claim when profit is large. Neither kind is better for everyone, because the right choice depends on what the investor needs.

Example 2 uses clean figures, so that the method is clear. The company has 100,000 preference shares and 200,000 ordinary shares, all with a face value of Tk 10. These numbers are chosen to teach the method and do not belong to the case file.

Ordinary share. A share that gives a holder a vote and a claim on profit after other claims are paid.

An ordinary share gives a vote and a claim on whatever profit remains.

Preference share. A share that gets a fixed dividend before ordinary shareholders, usually without a vote.

A preference share gives a fixed dividend, paid before the ordinary shareholders, usually without a vote.

The two kinds differ in their rights, and a table makes the differences easy to compare.

In the table, the fixed rate protects the preference holder, and the remainder rewards the ordinary holder.

Example 2 divides one year's dividend between the two kinds of share.

EXAMPLE 2: PAYING A DIVIDEND TO THE TWO KINDS OF SHARE

PART	WORKING
Given	Preference capital Tk 1,000,000 at a fixed rate of 8% a year. Ordinary capital Tk 2,000,000. The company decides to pay Tk 400,000 as dividends for the year.
Find	The dividend to the preference shareholders, the dividend to the ordinary shareholders, and the amount for each ordinary share.
Formula	In words: preference dividend = preference capital $\times$ rate; ordinary dividend = total dividend – preference dividend. In symbols: $P = C \times r$ and $O = T - P$, where C is preference capital, r is the rate, and T is the total dividend.
Step 1	$P = 1,000,000 \times 8\% = \text{Tk } 80,000$.
Step 2	$O = 400,000 - 80,000 = \text{Tk } 320,000$.
Step 3	There are $2,000,000 \div 10 = 200,000$ ordinary shares, so each receives $320,000 \div 200,000 = \text{Tk } 1.60$.
Answer	The preference shareholders receive Tk 80,000. The ordinary shareholders receive Tk 320,000, which is Tk 1.60 for each share.
Check	$80,000 + 320,000 = \text{Tk } 400,000$, which equals the total dividend.

A dividend is paid out of profit, so it is a division of the company's profit and not an expense like rent.

The case file names no preference shares, so every share in the case has the same rights.

Rights vary by country and by company; the founding documents of each company state its own rules.

Two kinds of share, ordinary and preference, trade vote and reward against fixed income and priority.

A common slip is to treat a dividend as an expense. It is a division of profit among the owners, as the appropriation account of Chapter 18 was. Another slip is to pay the ordinary shareholders first. The preference dividend always comes first, as the table showed.

Case file, illustrative. The proposal for Meghna and Sons Limited states one face value, Tk 10, for all its shares. The case file names no preference shares, so this chapter treats every share in the case as an ordinary share.

The rights described here are the common pattern. The exact rights of each kind of share depend on the law of the country and on the founding documents of each company. Some preference shares carry a vote in certain circumstances, and some fixed dividends may be carried forward when profit is too small to pay them. Read the documents of the company concerned.

Before moving on. An ordinary share carries a vote and a claim on profit that remains after other claims are paid. A preference share carries a fixed dividend, paid before the ordinary dividend, usually without a vote. The right choice for an investor depends on whether safety or reward matters more.

* * *

19.3 ISSUING SHARES

A company that wants outsiders' money must invite applications, choose among them and then deliver the shares.

Share application. A request to buy shares, sent with money, to a company that is offering them.

A share application is a request to buy shares, sent with money.

Money held before allotment is recorded in a temporary account named Share Application.

Suppose a company offers shares to the public. How does it know who wants them, how much money has arrived, and who has actually become a shareholder? The process has two formal steps, and each step has its own name and its own entry in the books.

The first step belongs to the buyer. A **share application** is a request to buy shares, sent with money, to a company that is offering them. A neighbour who sends Tk 5,000 and a form asking for 500 shares has made a share application. At this point the neighbour is not yet a shareholder. The company holds the money, but it has not yet agreed to deliver any shares.

The company records the money received from applicants. We debit Bank, because cash has arrived, and credit an account named Share Application. This

account holds the money until the company decides who receives shares. It is a temporary account, closed at the next step.

The second step belongs to the company. **Allotment** is the company's formal decision to give shares to applicants. When the company allots 500 shares to the neighbour, he or she becomes a shareholder. From this moment the company may credit its share capital, because the shares now exist in the hands of their owners. The Share Application account is closed, and its balance moves into the permanent accounts.

Some companies sell shares for more than their face value. **Share premium** is the amount paid for a share above its face value. A share with a face value of Tk 10 sold for Tk 25 carries a premium of Tk 15. The premium is not part of the share capital, which holds only the face value. It is credited to a separate account named Share Premium. In many countries the law limits what a company may do with that account.

Why would anyone pay more than the face value? Because the business may already be worth more than the amounts printed on its shares. A shop with a good name, loyal customers and years of trading is worth more than a new shop with the same number of shares. In a partnership, a new partner may pay for such a gain, as Chapter 18 showed. In a company, the share premium records the same idea.

PAUSE AND CHECK 19.P1 A share of face value Tk 10 is issued at Tk 18. What is the premium on each share, and what is the total premium on 1,000 shares?

The entries follow the order of events, and each has a debit total equal to its credit total. When the application money is received, we debit Bank and credit Share Application. When the shares are allotted, we debit Share Application, and we debit Bank for any further money received at that time. We credit Share Capital with the face value of the shares and credit Share Premium with the premium. This chapter assumes that all money is paid when it is due. A real company must also record money that arrives late.

Example 3 uses small numbers. The company issues 1,000 shares of face value Tk 10 at Tk 12. Applicants pay Tk 12 a share with the application, and all 1,000 shares are allotted.

EXAMPLE 3: RECORDING A SMALL ISSUE WITH A PREMIUM

PART	WORKING
Given	1,000 shares, face value Tk 10, issued at Tk 12. All the money is paid with the application, and all 1,000 shares are allotted.
Find	The share capital, the share premium, and the two journal entries.
Formula	In words: share capital = shares $\times$ face value; share premium = shares $\times$ (issue price – face value). In symbols: $K = n \times F$ and $M = n \times (P - F)$, where n is the number of shares, F is the face value and P is the issue price.
Step 1	Application money = $1,000 \times 12 = \text{Tk } 12,000$. Debit Bank 12,000; credit Share Application 12,000.
Step 2	$K = 1,000 \times 10 = \text{Tk } 10,000$.
Step 3	$M = 1,000 \times (12 - 10) = \text{Tk } 2,000$.
Step 4	On allotment: debit Share Application 12,000; credit Share Capital 10,000 and Share Premium 2,000.
Answer	Share capital is Tk 10,000 and share premium is Tk 2,000.
Check	$10,000 + 2,000 = \text{Tk } 12,000$, which equals the application money received.

Allotment. The company's formal decision to give shares to applicants.

Allotment is the company's decision to give shares to the applicants.

Share premium. The amount paid for a share above its face value.

Share premium is the amount paid above face value, and it is kept in its own account.

Recall: partnership, Chapter 18, section 18.1 'Forming a Partnership'.

A company may charge a premium because the business is already worth more than the face value of its shares.

Two journal entries record an issue: one when applications arrive and one on allotment.

Example 3 records a small issue with a premium, so that the method is the only thing to learn.

Three slips are common: credit capital with the whole price, forget the premium, or mix the two steps.

Application comes first, then allotment, then premium: this is the order of the technique.

The case file offers 200,000 shares at Tk 25, so the same method is now applied to a realistic issue.

Three slips are common. The first is to credit Share Capital with the whole issue price, which hides the premium. The second is to record the issue before the allotment, so that Share Capital appears before anybody owns a share. The third is to credit the application money to Share Capital directly, which skips the temporary account.

Before moving on. A share application is a request to buy shares, sent with money. Allotment is the company's decision to give shares to applicants. Share premium is the amount paid above face value, and it is credited to its own account. The entries follow the order of events: application, allotment, premium.

* * *

WORKED CASE: THE PUBLIC ISSUE OF MEGHNA AND SONS LIMITED

Example 3 used small numbers. The case file now gives a realistic issue. Read the figure below from left to right. The bar is the issue price of Tk 25 for one share. The hatched part is the face value of Tk 10, and the dotted part is the premium of Tk 15. The words under the bar name the money asked at each step.

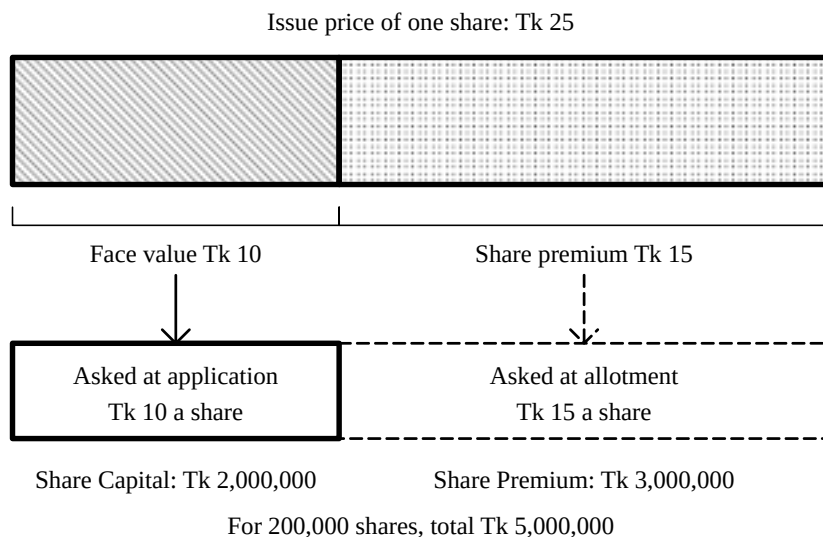


FIG. 19.1. The issue price of one share, split into face value and share premium

The case file records the public issue, the two money stages, and the totals credited.

Example 4 works the public issue in the standard order, with the journal entries at the end.

Case file, illustrative. Meghna and Sons Limited makes a public issue of 200,000 shares at Tk 25, which is a face value of Tk 10 and a premium of Tk 15. Application money is Tk 10 a share, which is Tk 2,000,000. Allotment money is Tk 15 a share, which is Tk 3,000,000. The total is Tk 5,000,000. Share capital credited is Tk 2,000,000, and share premium credited is Tk 3,000,000.

Example 4 works the case in full. Many students find this example confusing, because the application money equals the face value and the allotment money equals the premium. This is a feature of the case, and it is not a rule: other issues divide the money in other ways, as question 19.2 shows.

EXAMPLE 4: THE PUBLIC ISSUE OF MEGHNA AND SONS LIMITED

PART	WORKING
Given	200,000 shares, face value Tk 10, issue price Tk 25. Application money Tk 10 a share. Allotment money Tk 15 a share. All money is paid when due.
Find	The money at each step, the share capital, the share premium, and the journal entries.
Formula	In words: money at a step = shares $\times$ amount a share; share capital = shares $\times$ face value; share premium = shares $\times$ (issue price – face value). In symbols: $K = n \times F$ and $M = n \times (P - F)$.
Step 1	Application money = $200,000 \times 10 = \text{Tk } 2,000,000$.
Step 2	Allotment money = $200,000 \times 15 = \text{Tk } 3,000,000$.
Step 3	Total money = $2,000,000 + 3,000,000 = \text{Tk } 5,000,000$, which also equals $200,000 \times 25$.
Step 4	Share capital $K = 200,000 \times 10 = \text{Tk } 2,000,000$.
Step 5	Share premium $M = 200,000 \times (25 - 10) = \text{Tk } 3,000,000$.
Answer	Share capital credited is Tk 2,000,000 and share premium credited is Tk 3,000,000.
Check	$2,000,000 + 3,000,000 = \text{Tk } 5,000,000$, which equals the total money received.

The journal entries follow. The first records the money received with the applications. The second records the allotment and the money received at that stage.

The two journal entries put the money into Bank first and into capital and premium second.

JOURNAL ENTRIES FOR THE PUBLIC ISSUE

STEP	ACCOUNT	DEBIT (TK)	CREDIT (TK)
Application	Bank	2,000,000	
	Share Application		2,000,000
Allotment	Share Application	2,000,000	
	Bank	3,000,000	
	Share Capital		2,000,000
	Share Premium		3,000,000

Now consider the effect on the company. Before this issue, its issued capital was Tk 4,100,000. After the public issue it is $\text{Tk } 4,100,000 + \text{Tk } 2,000,000 = \text{Tk } 6,100,000$, which is still below the authorised capital of Tk 10,000,000. The share premium of Tk 3,000,000 is held in its own account, outside the share capital. The company has raised Tk 5,000,000 in cash in all. The balance sheet shows both accounts among the owners' equity, in the way that Chapter 9 described.

After the issue, the company has issued capital of Tk 6,100,000 and a share premium account of Tk 3,000,000.

* * *

SUMMARY

A company is owned by shareholders through shares. Share capital is the money it receives for them. Authorised capital is the ceiling set by the founding documents, and issued capital is the part sold. Ordinary shares carry a vote and a claim on remaining profit; preference shares carry a fixed dividend paid first,

A company raises share capital by selling shares, and the entries follow application, allotment and premium.

usually without a vote. In an issue, a share application is recorded in a temporary account, allotment moves it into Share Capital at face value, and share premium is credited to its own account.

THINK IT THROUGH

19.T1 The company offers 200,000 shares, but applications arrive for 300,000. Suggest two ways the company could treat the extra applications, and say what must happen to the money of any applicant who receives no shares.

19.T2 A company with shares of face value Tk 10 sells new shares for Tk 40. Explain, in two or three sentences, why the buyers may be willing to pay Tk 30 more than the face value.

WHERE PEOPLE GO WRONG

Authorised capital is a permission and not money; only issued capital has been paid for.

The premium is not part of share capital, because share capital holds only face value.

A preference dividend is fixed in rate, but the company must still earn profit to pay it.

A shareholder who sells shares to another person does not change the company's share capital.

Belief: authorised capital is money the company already has. **Correction:** authorised capital is a ceiling. Only the part that has been issued, and paid for, has become share capital.

Belief: the whole issue price goes to Share Capital. **Correction:** only the face value is credited to Share Capital. The amount above face value is credited to Share Premium.

Belief: a preference dividend is a promise that is always paid. **Correction:** it is a fixed rate and the first claim on profit. The company can pay it only if there is profit, and the rules of each company decide what happens when there is none.

Belief: when shares are sold between shareholders, the company's share capital changes. **Correction:** the company's books change only when the company itself issues shares and receives money for them. Later sales are between shareholders.

IN HISTORY

The Joint Stock Companies Act 1856 let any seven people register a company, and the registered company could limit its owners' liability.

The 1856 Act spread share ownership by making small investments safe, and the neighbours of our puzzle would have understood it.

In the nineteenth century, a person who invested in a business could, in many cases, lose more than the money paid in. The United Kingdom Parliament changed this. The Joint Stock Companies Act 1856, which received royal assent on 14 July 1856, allowed seven or more people to register a company by signing a memorandum of association. The company could be registered with limited liability, which means that a shareholder could lose no more than the amount he or she had agreed to pay for shares.

A company with limited liability had to carry the word "Limited" at the end of its name, so that everybody who dealt with it knew the limit. The neighbours in our opening puzzle would have understood the value of this rule. A person could put in Tk 5,000, lose at most Tk 5,000, and share in the profit. This is the reason that share ownership spread, as the Act made small investments safer for people who did not run the business themselves.

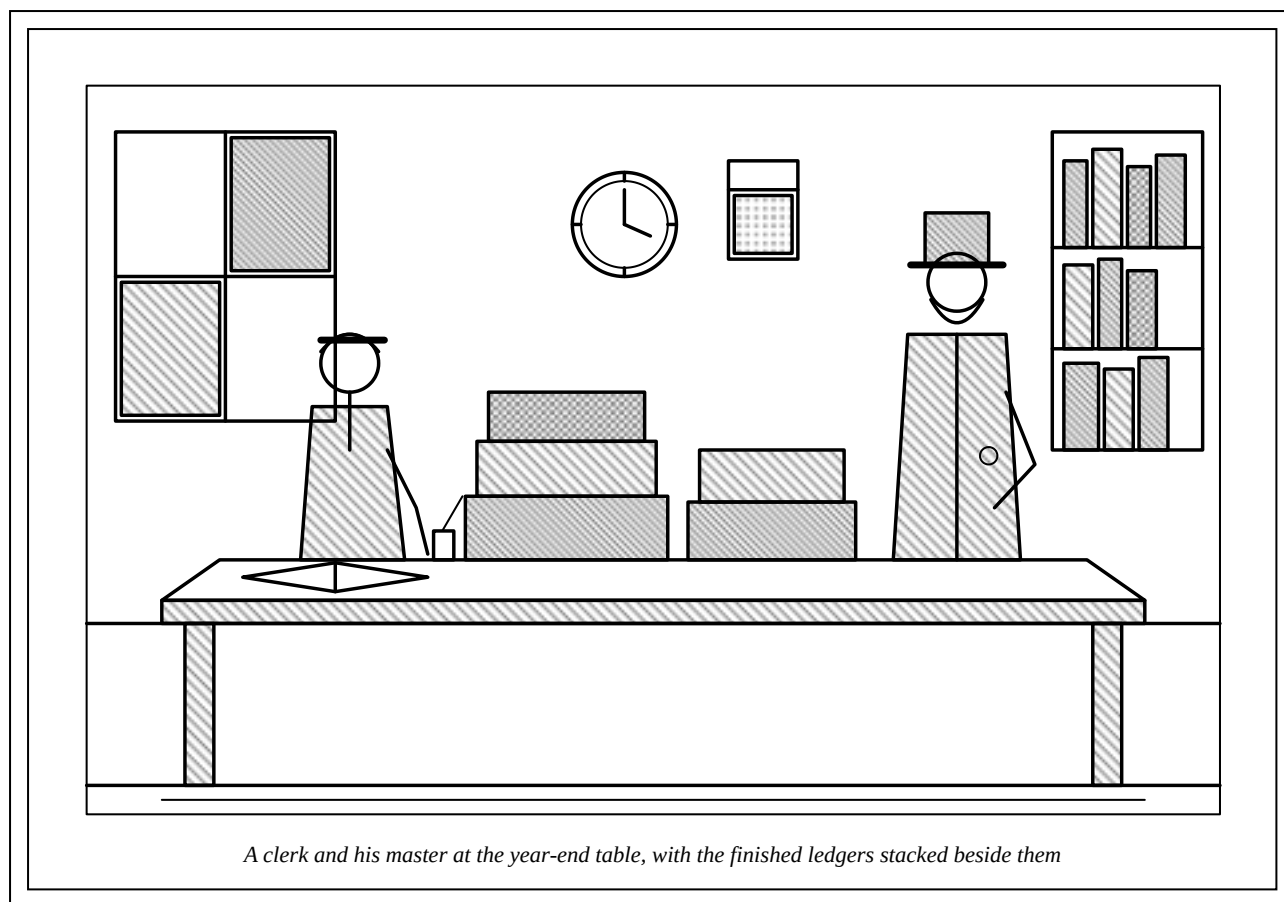
REVIEW QUESTIONS

19.1 A company issues 1,000 shares of face value Tk 10 at Tk 10. Applicants pay the whole Tk 10 with the application, and all 1,000 shares are allotted. Journalise the application and the allotment.

19.2 A company issues 50,000 shares of face value Tk 10 at Tk 16. Applicants pay Tk 8 a share with the application and Tk 8 a share on allotment. Find the share capital and the share premium, and journalise the application and the allotment.

19.3 A company has authorised capital of Tk 2,000,000 in shares of face value Tk 5. It has issued 120,000 shares. Find the number of shares authorised, the issued capital, and the number of shares still unissued.

19.4 Mira and Naveed, former partners with capital of Tk 450,000 and Tk 300,000, form a company. They receive shares of face value Tk 10, equal in value to their capital, in exchange for net assets of Tk 750,000. Find the number of shares each receives, and journalise the issue.



CHAPTER XX

The Whole Cycle and Reporting Periods

Which twelve months are we reporting, and which costs belong to them?



This chapter completes your first pass through the accounting cycle for a trading business. It explains why a business reports for a chosen year and for shorter periods inside it, and why some costs wait in inventory while others reach the income statement at once. It then links every report you have met into one connected whole.

A friend's question about a shop can be answered with three different stretches of time.

A second puzzle: two payments that look alike are treated in different ways.

A third puzzle: the many records and reports of earlier chapters must join into one system.

The chapter rests on earlier work on the balance sheet, the classified income statement and cost flow.

Imagine that you own a small bookshop. On the evening of 30 September, a friend who has lent you money asks how the shop is doing. You open your records. You could describe the whole of last year. You could describe the three months that ended today. Or you could describe the time since 1 July, when the government's own twelve-month period began. All three answers are true, and each gives different figures. Which one should you give, and who decides?

Now consider a second puzzle. During those three months the shop paid Tk 2,450,000 for new books and Tk 180,000 for rent. Both payments are money that left the business, yet an accountant handles them differently. The books are not charged against profit until a customer buys them; until then they wait on the shelf as inventory. The rent belongs to the three months in which the shop used the building, whether or not a single book is sold. Why the difference?

The third puzzle is larger. By now you have met the journal, the ledger, the trial balance, adjusting entries, the worksheet, the financial statements and closing entries. Each has its own chapter. This chapter asks how they fit together. If one number changes in the ledger, which reports must change with it, and in what order? When you can answer that, you hold the whole accounting cycle in your mind, and not only its parts.

You will need: Chapter 9, 'Financial Statements and Closing Entries', section 9.3 'Reversing Entries'; Chapter 10, 'Accounts of a Merchandising Company', section 10.2 'Classified Income Statement'; Chapter 14, 'Costing Inventory: FIFO, LIFO and Weighted Average', section 14.1 'The Problem of Changing Prices'; and Appendix A1, 'Maths You Will Use'.

* * *

GO SLOWLY HERE.

A business never stops, so accountants cut its life into equal slices of time and measure each slice.

A fiscal year is any twelve-month reporting period that a business chooses; it may start in any month.

Fiscal year. Any twelve-month period a business chooses for its yearly reports, which may start in any month.

Year. Word in use. In daily life, a year is the twelve months from 1 January to 31 December. In accounting, a year is whichever twelve months a business chooses for its reports.

The word year has two meanings, and the accounting meaning is the chosen one.

The case file shows a business that has chosen the calendar year.

20.1 FISCAL YEAR AND INTERIM PERIODS

Think first of a family budget. A family that earns a monthly salary does not wait until old age to check its spending. It looks at one month, sees what came in and what went out, and plans the next month. A school does the same with its school year, which begins when classes begin and not on 1 January. A business needs the same habit. It may trade for fifty years, but its owner cannot wait fifty years to learn whether it earns a profit. So the owner chooses a slice of time and measures it.

The usual name for this slice is the **fiscal year**. A fiscal year means any twelve-month period that a business chooses for its yearly reports, and it may start in any month. A shop might choose 1 April to 31 March, or 1 July to 30 June. The word *fiscal* comes from the Latin *fiscus*, a basket for money, and it now means "to do with money and accounts".

Notice that the word *year* now has two meanings. In daily life it usually means 1 January to 31 December. In accounting it means whichever twelve months the business has chosen. The most familiar choice is the calendar year, which runs from 1 January to 31 December. A business that picks the calendar year has a fiscal year that matches the calendar. A business that picks another start month does not.

Case file, illustrative. Meghna and Sons, the family-run bookshop in Dhanmondi, Dhaka, reports for a calendar year ending 31 December. Its yearly reports therefore cover 1 January to 31

December. Bangladesh's government fiscal year is different: it runs from 1 July to 30 June.

The case file shows an important point. A government and a business in the same country can use different twelve-month periods, and neither one is wrong. Each chooses the period that suits its own work. Many students find this confusing, because they expect one year for everyone. The remedy is to ask of every report the same question: from which date to which date does it run?

Why would a business choose a year that is not the calendar year? One common reason is the pattern of trade. A seller of school uniforms might sell most items in the weeks before term begins. If the year ended in the middle of that rush, the shelves would be in constant movement and inventory would be hard to count. Ending the year when trade is quiet gives a calmer count. Rules differ from country to country, and tax law or company law can limit the choice, so a business must check the rules of its own country.

A fiscal year answers the question "How did we do?" only once a year. A lender who must decide this week whether to lend more money cannot wait. An owner who sees losses building would like to know in the third month, and not after the twelfth. For this reason businesses also prepare reports for shorter slices of time.

An **interim period** is a reporting period shorter than a year, such as a quarter. A quarter is three months; four quarters make a year.¹ Months, quarters and half-years are all interim periods.

Case file, illustrative. Meghna and Sons prepared a report for the interim period of the quarter ended 30 September 2026. Net sales in that quarter were Tk 3,900,000. The quarter began on 1 July 2026 and is the third quarter of the shop's calendar year.

The table below sets four periods side by side. Read each row from left to right: the name, the dates, and the length. The dates for Meghna and Sons come from the case file; the other two come from the facts you have already read.

FOUR REPORTING PERIODS COMPARED

PERIOD	DATES	LENGTH
Meghna and Sons, calendar year	1 January to 31 December	12 months
Meghna and Sons, interim period	1 July to 30 September 2026	3 months
Bangladesh government fiscal year	1 July to 30 June	12 months
United States federal fiscal year (from 1976)	1 October to 30 September	12 months

Interim reports use the same accounts and the same method as yearly reports. They are, however, less final. Some adjusting entries, such as the use of equipment during the quarter, must be estimated, and the full-year count of inventory may _____ not _____ yet _____ have _____

¹ A quarter may begin in any month. For a business with a calendar year, the quarters end on 31 March, 30 June, 30 September and 31 December.

Two bodies in one country may use two different twelve-month periods.

Some businesses end their year when trade is quiet, because counting inventory is then easier.

A year gives news only once, so lenders and owners also want reports for shorter slices.

Interim period. A reporting period shorter than a year, such as a quarter.

An interim period is a reporting period shorter than a year, such as a quarter.

Period. Word in use. In daily life, a period is any stretch of time, such as a school lesson. In accounting, a period is the span of time that one set of reports covers, such as a month, quarter or year.

The case file records one interim period of three months.

Different bodies measure different periods, and a table lays the dates side by side.

Interim reports follow the same methods as yearly reports but rely more on estimates.

Recap of fiscal year and interim period.

The reader should now be able to state the difference between the two kinds of period.

A shop owner pays for books and for rent, yet only one of these payments waits on the shelf.

Product cost. A cost of buying or making goods held for sale, such as the price paid for books and freight in.

A product cost is the cost of buying goods that are held for sale, including freight in.

Cost. Word in use. In daily life, cost is the price you pay for something. In accounting, a cost is an amount given up to obtain goods or services, and it is later either held as an asset or charged as an expense.

The word cost has an everyday meaning and a narrower accounting meaning.

Period cost. A cost charged to the period in which it is incurred, such as rent or wages.

A period cost belongs to the time in which it is incurred, not to any particular goods.

The case file lists the product costs and period costs of one quarter.

The two classes of cost differ in what they are tied to and in when they reach the income statement.

taken place. Treat interim figures as a useful early signal, and not as the last word.

Before moving on. A fiscal year is any twelve months a business chooses, and the calendar year is one such choice. An interim period is shorter than a year, and a quarter is the usual example. Always ask of a report: from which date to which date?

You should now be able to say: "A fiscal year is a chosen twelve months, and an interim period is a shorter slice inside it."

20.2 PRODUCT COSTS AND PERIOD COSTS

Return to the second puzzle. The owner of Meghna and Sons has paid for books, for freight in, for rent and for wages. All four are costs, and all four reduce cash. But the first two are tied to goods, and the last two are tied to time. That difference decides when each cost reaches the income statement.

A **product cost** is a cost of buying or making goods held for sale, such as the price paid for books and freight in. Freight in means the carrier's charge for bringing the goods to the shop. A product cost first becomes part of inventory. It stays there until the goods are sold, and only then does it move to cost of goods sold in the income statement.²

The word *cost* needs care. In daily life it means the price paid. In accounting it means an amount given up to get goods or services, and the later treatment depends on what was bought. A cost for goods held for sale is held as inventory until the sale. A cost for something used up with time is charged at once.

A **period cost** is a cost charged to the period in which it is incurred, such as rent or wages. The shop must pay rent for July, August and September whether it sells ten books or ten thousand. The cost belongs to the time, and not to a particular book. So the income statement for the quarter carries the rent in full, and nothing waits on the shelf.

Case file, illustrative. For the quarter ended 30 September 2026, Meghna and Sons recorded these product costs: books bought Tk 2,450,000 and freight in Tk 40,000. It recorded these period costs: rent Tk 180,000 and wages Tk 735,000.

The table below compares the two classes. Read down each column to see how a cost behaves. The first row shows what the cost is tied to, the second shows where it first appears, and the third shows when it reaches the income statement.

² Which cost is attached to which unit sold depends on the cost flow assumption. Chapter 14 teaches this in section 14.1, 'The Problem of Changing Prices'. This chapter assumes first-in first-out, but any method would classify the costs in the same way.

PRODUCT COSTS AND PERIOD COSTS COMPARED

FEATURE	PRODUCT COST	PERIOD COST
Tied to	Goods held for sale	A stretch of time
First appears	In inventory, on the balance sheet	Straight in the income statement
Reaches income statement	When the goods are sold	In the period incurred
Examples from the case	Books bought, freight in	Rent, wages

PAUSE AND CHECK 20.P1 Meghna and Sons pays Tk 40,000 to a carrier to bring books into the shop, and Tk 180,000 as rent. Which payment is a product cost, and which is a period cost?

Chapter 10 shows how a classified income statement groups costs under headings.³ The product costs sit inside cost of goods sold, which is deducted from net sales to find gross profit. The period costs sit below gross profit as operating expenses. You can think of the income statement as a staircase: product costs come off first, then period costs.

Some costs are easy to misplace. Delivery to a customer is freight out, and it is tied to the act of selling, not to the buying of goods, so it is a period cost. Freight in, which brings goods to the shop, is a product cost. The name "freight" appears in both, but the question that sorts them is the same: is the cost tied to goods held for sale, or to time and selling? A second exception concerns unsold goods. If the shop buys books in September and sells none before the quarter ends, the whole product cost is still in inventory on 30 September, and none of it has reached the income statement.

Each reader of the reports uses this split differently. The owner wants to know whether the books bought are selling, which the product costs show. The manager wants to control rent and wages, which belong to the period and do not change with each sale. A lender wants to know whether sales cover both classes of cost. If the two classes were mixed, none of these readers could separate the cost of the goods from the cost of running the shop.

Before moving on. A product cost belongs to goods held for sale and waits in inventory until the sale. A period cost belongs to the time in which it is incurred and reaches the income statement at once. Freight in is a product cost; rent and wages are period costs.

You should now be able to say: "A product cost follows the goods, and a period cost follows the calendar."

A classified income statement groups these costs under separate headings.

A cost can look the same on paper yet belong to the other class, so ask what it is tied to.

The viewpoint of the owner, the manager and the lender explains why the split matters.

Recap of product cost and period cost.

The reader should now be able to state what separates the two classes of cost.

20.3 THE LINKED STATEMENTS

Think of a water pipe that feeds three taps in a house. If the pipe is blocked, all three taps fail. The accounts work the same way. Every figure in the ledger feeds the reports, and a change at the source changes every tap it feeds. Learn the pipe first and the taps follow.

The cycle has a fixed order. First, transactions are recorded in the journal and posted to the ledger. Second, the accounts are listed in a trial balance to test that debits equal credits. Third, adjusting entries are made for items such as accrued wages and used supplies, and the adjusted accounts are listed again. Fourth, the

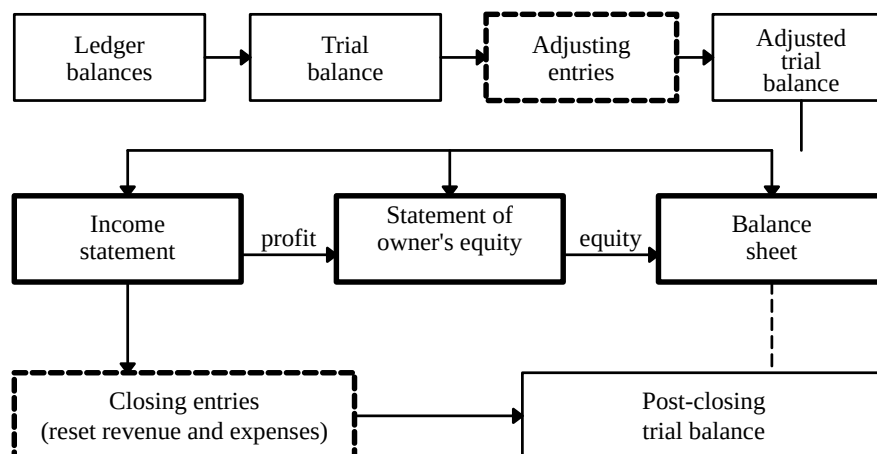
GO SLOWLY HERE.

A single number in the ledger flows through every report, much as one water pipe feeds several taps.

The cycle begins with transactions and ends with a post-closing check on the accounts.

³ Chapter 10, 'Accounts of a Merchandising Company', section 10.2 'Classified Income Statement'.

income statement, the statement of owner's equity and the balance sheet are prepared. Fifth, closing entries reset the revenue and expense accounts to zero for the next period.



Dashed boxes are the steps that change account balances.

FIG. 20.1. How the trial balance, the adjustments, the statements and the closing entries connect

The figure shows the cycle as boxes joined by arrows, with dashed boxes for the steps that change balances.

How the three statements connect is the heart of the cycle.

An account payable is an amount we owe; an account receivable is an amount owed to us.

The trial balance and the balance sheet look alike but serve different purposes.

The complete cycle can be reviewed by following one quarter of Meghna and Sons from start to end.

To read the figure, follow the arrows. Each box is a step or a report, and each arrow means that the figures pass on to the next box. The dashed boxes mark the two steps that change account balances: the adjusting entries and the closing entries. The solid boxes only collect and report the balances that already exist.

The three statements pass figures along a chain. The income statement finds the profit for the period. That profit goes into the statement of owner's equity, where it is added to the opening owner's equity, after any owner's withdrawals are subtracted. The closing owner's equity then goes into the balance sheet. If any link is wrong, the balance sheet will not balance.⁴

A common confusion lies between two accounts. An account payable means an amount the business owes to a supplier. An account receivable means an amount a customer owes to the business. Some books call these creditors and debtors, but this book uses payable and receivable. Both appear on the balance sheet: the payable among the liabilities, and the receivable among the assets. A payable is money going out later; a receivable is money coming in later.

The trial balance and the balance sheet also look alike. Both list accounts with amounts. The trial balance lists every ledger account, including revenue and expense accounts, to test that total debits equal total credits. It is a working test, and nobody outside the business needs to see it. The balance sheet lists only assets, liabilities and owner's equity at one date, and it is a report for readers. A trial balance can balance and still contain an error that affects the profit, such as a cost put in the wrong account.

Follow one quarter through the whole cycle. In July, August and September, each purchase of books, payment of rent and sale is recorded in the journal and posted to the ledger. On 30 September the trial balance lists the account balances. Adjusting entries then bring in costs that belong to the quarter but are not yet recorded, such as wages earned but not yet paid. The adjusted figures feed the income statement, which shows the profit found in the worked example

⁴ Chapter 9, 'Financial Statements and Closing Entries', section 9.3 'Reversing Entries', describes the balance sheet and the entries that follow the statements.

below. At the end of a full year, closing entries move the revenue and expense balances into owner's equity. For an interim report, many businesses prepare the statements without closing the accounts, because the year has not ended.

IN SHORT

The cycle has six verbs: record, post, test, adjust, report and close. Record and post fill the ledger. Test and adjust prepare it. Report tells the story. Close resets the temporary accounts for the next period.

PAUSE AND CHECK 20.P2 A business pays Tk 5,000 in cash to settle an account payable of the same amount. Which two totals on the balance sheet change, and does the balance sheet still balance?

Before moving on. The ledger feeds the trial balance; adjustments correct it; the statements report it; closing entries reset the temporary accounts. A payable is owed by us; a receivable is owed to us. The trial balance tests; the balance sheet reports.

You should now be able to say: "The statements are linked, so profit flows from the income statement through owner's equity to the balance sheet."

Recap of the linked statements.

The reader should now be able to state how the statements are linked.

WORKED EXAMPLE: ONE QUARTER OF MEGHNA AND SONS

This example applies the chapter to the case file. We first sort the costs, and then we find the profit for the interim period. Take each step slowly and notice where each figure comes from.

Given. Net sales for the quarter ended 30 September 2026 are Tk 3,900,000. Product costs are books bought Tk 2,450,000 and freight in Tk 40,000. Period costs are rent Tk 180,000 and wages Tk 735,000. To keep the arithmetic clear, assume that all the books bought were sold in the quarter and that inventory did not change. *Find.* The total product cost, the total period cost, the gross profit and the profit for the quarter.

Formula. Product cost total equals books bought plus freight in, or $P = B + F$. Period cost total equals rent plus wages, or $C = R + W$. Gross profit equals net sales minus product cost total, or $G = S - P$. Profit equals gross profit minus period cost total, or $N = G - C$. Here S stands for net sales, and N stands for profit.

Steps. First, $P = \text{Tk } 2,450,000 + \text{Tk } 40,000 = \text{Tk } 2,490,000$. Second, $C = \text{Tk } 180,000 + \text{Tk } 735,000 = \text{Tk } 915,000$. Third, $G = \text{Tk } 3,900,000 - \text{Tk } 2,490,000 = \text{Tk } 1,410,000$. Fourth, $N = \text{Tk } 1,410,000 - \text{Tk } 915,000 = \text{Tk } 495,000$.

A worked example sorts the costs and finds the profit for the quarter ended 30 September 2026.

The given figures and the question to be answered.

The formulas are stated in words first and then in symbols.

The steps show every line of arithmetic with units.

INCOME STATEMENT FOR THE QUARTER ENDED 30 SEPTEMBER 2026 (TK)

LINE	AMOUNT
Net sales	3,900,000
Less product costs (books bought and freight in)	2,490,000
Gross profit	1,410,000
Less period costs (rent and wages)	915,000
Profit	495,000

Answer. Product costs are Tk 2,490,000 and period costs are Tk 915,000. Gross profit is Tk 1,410,000, and profit for the quarter is Tk 495,000. *Check.* Take all

The answer and a check of the result against another route.

A second, messier example uses new figures and one cost that is easy to misplace.

Common slips in this calculation arise from mixing classes, periods or signs.

A worked case applies the ideas to a decision instead of a calculation.

the costs together: Tk 2,490,000 + Tk 915,000 = Tk 3,405,000. Then Tk 3,900,000 – Tk 3,405,000 = Tk 495,000, which equals the profit found above.

Now a second example with new figures. A small bakery has net sales of Tk 800,000 for a quarter. It bought flour and other goods for Tk 310,000 and paid freight in of Tk 12,000. It paid rent Tk 90,000, wages Tk 160,000 and delivery to customers Tk 18,000. Product costs are Tk 310,000 + Tk 12,000 = Tk 322,000. Delivery to customers is freight out, so it is a period cost; period costs are Tk 90,000 + Tk 160,000 + Tk 18,000 = Tk 268,000. Gross profit is Tk 800,000 – Tk 322,000 = Tk 478,000, and profit is Tk 478,000 – Tk 268,000 = Tk 210,000. Check: Tk 800,000 – Tk 322,000 – Tk 268,000 = Tk 210,000.

Watch for three slips. The first is to place freight in with the period costs, which would understate gross profit. The second is to mix periods, for example by comparing a quarter's profit with a full year's profit. The third is to add the profit instead of subtracting the costs, which gives a figure larger than sales.

Situation. The owner of Meghna and Sons wants to know whether the quarter's profit of Tk 495,000 can stand for a full year. **Question.** Is it safe to multiply the quarter's profit by four? **Reasoning.** The quarter is one interim period of a calendar year. Trade may be higher or lower in the other quarters, and some adjustments in an interim report are estimates. **Conclusion.** The figure is a useful signal but not a forecast of the year. **Check.** The case file does not give figures for the other quarters, so no firm yearly figure can be drawn from this one.

* * *

SUMMARY

A fiscal year is a chosen twelve-month reporting period, and an interim period is a shorter one. A product cost is tied to goods and waits in inventory until the sale, while a period cost is charged to the time in which it is incurred. The statements are linked: profit flows from the income statement to owner's equity to the balance sheet. A payable is owed by the business and a receivable is owed to it. The trial balance tests the accounts, and the balance sheet reports them.

THINK IT THROUGH

20.T1 Suppose Meghna and Sons changed its fiscal year to run from 1 July to 30 June, the same as the Bangladesh government. Name two things that would change in its reports, and one thing that would stay the same.

20.T2 A seller of school uniforms trades most heavily in the weeks before term starts. Which twelve-month period would you suggest for this business, and why?

WHERE PEOPLE GO WRONG

Many students believe that a fiscal year must follow the calendar. It need not: it is any twelve months the business chooses. Some believe that every payment is an expense at once. A product cost waits in inventory until the sale. Others treat freight in and freight out alike, but the first is a product cost and the second is a period cost. A further error is to think that a balanced trial balance proves every account is right. It proves only that debits equal credits.

A short summary of the whole chapter.

Common misunderstandings and their corrections.

IN HISTORY

Until 1976, the United States federal government counted its fiscal year from 1 July to 30 June. The Congressional Budget and Impoundment Control Act of 1974, signed into law on 12 July 1974, changed this. Under section 501, the fiscal year began on 1 October from 1976. To make the change, the period from 1 July to 30 September 1976 was treated as a separate three-month transition quarter, between fiscal years 1976 and 1977. The case shows that a fiscal year is a choice, and that a change of choice needs a bridging period.

*A true account of a government
that chose a new fiscal year.*

REVIEW QUESTIONS

- 20.1 Classify each cost of a bookshop as a product cost or a period cost: books bought, freight in, shop rent, wages of the shop assistants, delivery to a customer.
- 20.2 A business records net sales of Tk 600,000, product costs of Tk 380,000 and period costs of Tk 150,000 for a quarter. Find the gross profit and the profit.
- 20.3 Trace the cycle from one transaction to the post-closing check. Name each step in order and say which steps change account balances.
- 20.4 Explain in two sentences the difference between an account payable and an account receivable, and say where each appears on the balance sheet.
- 20.5 A business has a fiscal year from 1 April to 31 March. Name the dates of its second quarter, and say which of these is an interim period: the year, the second quarter, or the full cycle.

GLOSSARY

Account. A record that collects all the increases and decreases of one kind of item. The Cash account, for example, shows every receipt and payment of cash, so its balance tells the business how much cash it holds. (*Chapter IV*)

Accounting. The process of recording, sorting and reporting the money effects of a business's events to people who need them. When a shop sells a novel, accounting records the sale, sorts it with other sales, and reports the total to the owner, the bank and others. (*Chapter II*)

Accounting cycle. The series of steps repeated each period, from recording transactions to preparing closing records. It begins with source documents, passes through the journal and the ledger, and ends with reports and closing records, after which the next period starts the same steps again. (*Chapter V*)

Accounting equation. The rule that assets always equal liabilities plus owner's equity, written $A = L + E$. It holds because every asset was provided either by a lender or by the owner. (*Chapter III*)

Accounting period. The span of time, such as a month, a quarter or a year, for which a business measures and reports its results. For example, Meghna Binding Service reports for the calendar year 2026. (*Chapter VII*)

Accounting standard. A written rule about how to record and report items, so that different businesses report in the same way. For example, a standard may say how goods held for sale must be measured. In daily life, a standard means a level of quality. (*Chapter II*)

Accrual basis of accounting. A method that records revenue when it is earned and an expense when it is incurred, whether or not cash has moved. For example, wages earned in December belong to December even if paid in January. (*Chapter VII*)

Accrued expense. A cost that has already been used up but not yet paid, shown as a liability. For example, wages earned by workers in the last days of December but paid in January. (*Chapter VII*)

Accrued revenue. Revenue that has already been earned but not yet received or billed, shown as an asset. For example, binding work finished in December for which the invoice is sent in January. (*Chapter VII*)

Accumulated depreciation. The total depreciation charged to date on an asset, shown as a deduction from its cost. It grows by one charge each year. For example, after three years of Tk 60,000 it equals Tk 180,000. (*Chapter XVI*)

Acquisition cost. All the money spent to buy an asset and make it ready for use. It includes the invoice price less discounts, freight, installation and testing. Payments after the asset is ready, such as repairs, are not part of it. (*Chapter XV*)

Adjusted trial balance. A trial balance prepared after the adjusting entries have been taken into account. It shows the correct balance of each account, and its debit total equals its credit total. Example: Supplies of Tk 1,000 less an adjustment of Tk 600 appear as Tk 400. (*Chapter VIII*)

Adjusting entry. A journal entry made on the last day of a period to bring an account to its correct balance for that period. For example, debiting Insurance Expense and crediting Prepaid Insurance for the part used. (*Chapter VII*)

Allotment. The company's formal decision to give shares to applicants. From this moment the applicants become shareholders, and the company credits Share

Capital and, where there is a premium, Share Premium. The Share Application account is then closed. (*Chapter XIX*)

Asset. Something of value that a business owns or controls, such as cash, goods held for sale or equipment. For example, the shelving bought by a workshop is an asset, and so is the cash in its drawer. (*Chapter III*)

Authorised capital. The largest amount of share capital a company is allowed to issue under its founding documents. It is a ceiling and not money received. A company with authorised capital of Tk 50,000 may sell shares worth up to Tk 50,000 and no more. (*Chapter XIX*)

Balance sheet. A report of assets, liabilities and owner's equity at one date. Its two sides must agree, because assets equal liabilities plus owner's equity. Its heading says *At* and then the date. Some standards call it the statement of financial position. (*Chapter IX*)

Bank reconciliation statement. A statement that explains the difference between the bank balance in the cash book and the balance on the bank's statement. It has two halves, each ending in an adjusted balance, and the two must be equal. It is a working paper, not a ledger account. (*Chapter XII*)

Book value. An asset's cost less its accumulated depreciation. It is a figure in the records and need not equal the price at which the asset could be sold. For example, Tk 330,000 less Tk 180,000 gives a book value of Tk 150,000. (*Chapter XVI*)

Bookkeeping. The recording part of accounting: writing down each transaction in the right book. A bookkeeper writes each sale and payment as it happens, while an accountant also sorts the records, checks them and prepares reports for readers who need them. (*Chapter II*)

Business entity. The rule that a business is treated as separate from its owner when records are kept. The owner's household spending stays out of the business records, and two businesses of one owner keep separate books, even when the law treats the owner and business as one person. (*Chapter II*)

Capital expenditure. Spending that creates or improves an asset lasting more than one period. It is recorded as an asset on the balance sheet. The purchase of a machine is capital expenditure, and so is a motor fitted to make it work faster. (*Chapter XV*)

Cash book. A book that serves as both journal and ledger for cash and bank receipts and payments. It has a debit side for money received and a credit side for money paid, usually with a Cash column and a Bank column. A customer's Tk 500 in notes is written once, in the Cash column. (*Chapter XII*)

Chart of accounts. A list of every account a business uses, with a code for each. It stops the business from opening duplicate accounts or forgetting one. Meghna Binding Service, for example, uses eight accounts, so its chart is short. (*Chapter IV*)

Closing entry. A journal entry made at the end of a period that moves the balance of a temporary account to capital, usually through the income summary. After it, the temporary account stands at zero. Example: debit Revenue Tk 788,000, credit Income Summary Tk 788,000. (*Chapter IX*)

Company. A business that the law treats as a person separate from its owners. Its ownership is divided into equal parts called shares, so it can have many owners. It can own goods and owe money in its own name. (*Chapter I*)

Compensating error. Two or more mistakes that cancel each other so that the trial balance still agrees. Example: Wages debited Tk 300 too much and Rent debited Tk 300 too little, so the debit total is still correct. (*Chapter X*)

Compound entry. A journal entry that has more than one debit or more than one credit. Total debits still equal total credits. Buying a press for Tk 30,000, part paid in cash and part owed, debits one account and credits two. (*Chapter V*)

Control account. A main ledger account that sums up the balances of all the accounts in a subsidiary ledger. If both are correct, the control account balance equals the total of the individual balances. A difference shows that an error exists somewhere. (*Chapter XI*)

Cost flow assumption. A rule that decides which purchase costs are treated as sold and which remain in inventory. It concerns the order of costs, not the movement of goods. For example, a shop may sell its newest dictionary first but still assume that the oldest cost was sold first. (*Chapter XIV*)

Cost of goods sold. The cost of the goods that were sold during the period. It equals opening inventory plus purchases less purchase returns less closing inventory. Example: Tk 2,000 + Tk 6,000 – Tk 300 – Tk 1,700 = Tk 6,000. (*Chapter X*)

Credit. An entry on the right side of an account. It increases liabilities, owner's capital and revenue, and it decreases assets, drawings and expenses. Like debit, the word names a side only and carries no judgement. (*Chapter IV*)

Credit memo. A bank's notice that it has added money to the account, for example a collection made for the business or interest paid. The name follows the bank's books, where more owed to the customer is a credit. It is added on the book side of the reconciliation. (*Chapter XII*)

Current asset. An asset expected to be turned into cash or used up within one year. Examples are cash, accounts receivable, inventory and prepaid expenses. Current assets are listed in the order in which they come closest to cash. (*Chapter X*)

Current liability. A debt that must be paid within one year. Accounts payable are the usual example. The part of a long loan that falls due within one year is also current. Example: a loan instalment due in six months. (*Chapter X*)

Debit. An entry on the left side of an account. It increases assets, drawings and expenses, and it decreases liabilities, owner's capital and revenue. The word names a side; it does not mean good, bad, more or less. (*Chapter IV*)

Debit memo. A bank's notice that it has deducted money from the account, for example a service charge. The name follows the bank's books, where less owed to the customer is a debit. It is subtracted on the book side of the reconciliation. (*Chapter XII*)

Declining-balance method. A depreciation method that charges a fixed percentage of the asset's remaining book value each year. The charge is largest in the first year and falls after that. For example, 40% of Tk 330,000 gives Tk 132,000 in year 1. (*Chapter XVI*)

Deposit in transit. Money recorded as deposited in the books but not yet shown on the bank statement, because the bank has not yet processed it. For example, a deposit made after the bank closes on the last day of the month. It is added on the bank side of the reconciliation. (*Chapter XII*)

Depreciation. The spreading of a plant asset's cost over the years it is expected to be useful. It is a method of allocation, not a measure of the fall in market price and not a payment of cash. For example, a Tk 10,000 bicycle may carry a charge of Tk 2,000 a year. (*Chapter XVI*)

Dishonoured cheque. A cheque that the bank refuses to pay, usually because the writer's account lacks funds. The bank takes back any amount it credited, so the business subtracts it on the book side and restores the customer's account receivable. For example, a customer's cheque for Tk 800 returned unpaid. (*Chapter XII*)

Double entry. The rule that every transaction is recorded twice, once as a debit and once as a credit of equal amount. It keeps the accounting equation in balance and exposes many mistakes, though equal totals do not prove that every entry is right. (*Chapter IV*)

Drawings. Money or goods that the owner withdraws from the business for personal use. Drawings lower the owner's equity, but they are not an expense, because they do not help to earn revenue. (*Chapter III*)

Ending inventory. The cost of goods still on hand at the end of the period. It appears as an asset in the balance sheet and becomes the opening inventory of the next period. Cost of goods sold plus ending inventory equals the cost of goods available. (*Chapter XIV*)

Error of commission. A mistake in which an entry is made in the right class of account but the wrong account, or with a wrong amount. Example: electricity of Tk 700 debited to Telephone Expense, when both are expenses. (*Chapter X*)

Error of omission. A mistake in which a transaction is left out of the books completely. Neither side of the entry is made, so the trial balance still agrees. Example: a credit sale of Tk 800 that is never entered. (*Chapter X*)

Error of principle. A mistake in which an entry is made in the wrong class of account, such as an expense recorded as an asset. Example: Tk 40,000 spent on a delivery van and debited to Repairs Expense. (*Chapter X*)

Expense. A cost used up in earning revenue, such as rent or wages. For example, rent paid for a month in which a workshop has used the room is an expense; buying shelving is not. (*Chapter III*)

Financial accounting. Accounting that prepares reports for outsiders, such as lenders, owners and tax authorities. Its reports follow written rules, so that outsiders can compare one business with another. A bank's yearly report on a borrower is an example. (*Chapter II*)

Financial statement. A formal report of a business's results or position, prepared at the end of a period. A period may be a month, a quarter or a year. The income statement, which shows profit, and the balance sheet are examples taught in a later chapter. (*Chapter II*)

Finished goods. Completed products held for sale. They are closest to being turned into cash. Bound volumes ready for collection are finished goods, and a bookshop's inventory consists wholly of them. (*Chapter XIII*)

First-in first-out. A method that treats the oldest costs as sold first, leaving the newest costs in inventory. Often written FIFO. When prices rise it gives the lowest cost of goods sold, the highest gross profit and an ending inventory close to current prices. (*Chapter XIV*)

Fiscal year. Any twelve-month period a business chooses for its yearly reports. It may start in any month, and the calendar year is one choice among many. Meghna and Sons uses 1 January to 31 December, while the Bangladesh government uses 1 July to 30 June. (*Chapter XX*)

Going concern. The assumption that a business will continue to operate for the foreseeable future. Because of it, a shop's shelves are treated as items used for years. The assumption fails when the business is about to close, and records must then show sale values. (*Chapter II*)

Gross profit. Net sales less cost of goods sold, before operating expenses are deducted. It shows what the business earns from buying and selling goods. Example: Tk 14,400,000 – Tk 9,800,000 = Tk 4,600,000. (*Chapter X*)

Historical cost. The rule of recording an asset at the price paid for it, not at its present value. The price paid can be proved by a receipt, while a present value is an opinion. A shelf bought for a price stays at that price in the records. (*Chapter II*)

Income and expenditure account. The non-trading organisation's equivalent of an income statement, showing surplus or deficit on an accrual basis. It records income and costs in the period to which they belong, and leaves out the purchase of lasting assets. (*Chapter XVII*)

Income statement. A report of revenue and expenses for one period, such as a year, ending in net income or net loss. Its heading says *For the year ended*. Example: revenue Tk 788,000 less expenses Tk 604,000 gives net income of Tk 184,000. (*Chapter IX*)

Income summary. A temporary account used to collect revenue and expenses at period end. Its balance equals net income or net loss, which is then moved to capital. After that move, the account stands at zero. It exists only during closing. (*Chapter IX*)

Intangible asset. A long-lived asset with no physical form, such as a patent or a trademark. The business cannot touch it, yet it can earn money for years. A patent is a legal right to be the only maker of an invention for a number of years. (*Chapter XV*)

Interest on capital. A payment credited to partners for their capital before the remaining profit is divided. It rewards money put in. For a period shorter than a year, it is capital times the yearly rate times the fraction of the year. It divides profit; it is not an outside cost. (*Chapter XVIII*)

Interim period. A reporting period shorter than a year. A quarter of three months is the usual example, and months and half-years are others. Interim reports use the same methods as yearly reports but depend more on estimates. (*Chapter XX*)

Inventory. Goods held for sale, or materials held to make goods for sale, in the ordinary course of business. The business must own them. It is a current asset until sale, when its cost becomes cost of goods sold. Books on a bookshop's shelves are inventory. (*Chapter XIII*)

Issued capital. The part of authorised capital that has actually been sold to shareholders. It can never exceed authorised capital. If authorised capital is Tk 50,000 and shares worth Tk 30,000 have been sold, issued capital is Tk 30,000. (*Chapter XIX*)

Journal. The book of first entry, in which transactions are recorded in date order with debits and credits. Each entry shows the date, the debited account, the credited account, the amounts and a narration. A purchase on credit is written here before anywhere else. (*Chapter V*)

Last-in first-out. A method that treats the newest costs as sold first, leaving the oldest costs in inventory. Often written LIFO. When prices rise it gives the highest cost of goods sold and the lowest gross profit. International Accounting Standard 2 bars it. (*Chapter XIV*)

Ledger. The book containing all of a business's accounts, where the effects of journal entries are collected. Each account has its own page, so all the Cash entries or all the Rent entries sit together and can be balanced. (*Chapter V*)

Liability. An amount that a business owes to others, such as a bank loan or an unpaid supplier bill. For example, Tk 200,000 owed to a publisher for books is a liability until it is paid. (*Chapter III*)

Liquidation. The process of selling a business's assets, paying its debts and sharing what remains among the owners. Outsiders are paid first. Each partner then receives capital less his or her share of any loss, so the partners are the last to be paid. (*Chapter XVIII*)

Management accounting. Accounting that prepares reports inside the business to help managers plan and control. No outside rule governs these reports, and the manager designs them to answer a question, such as which books sell fastest each week. (*Chapter II*)

Manufacturing business. A business that makes goods from materials and then sells them. It holds materials, goods half finished and goods ready to sell. A bakery that turns flour and fuel into bread is an example. (*Chapter I*)

Matching principle. The rule that expenses are recorded in the same period as the revenue they helped to earn. For example, insurance for October to December is placed beside the revenue of those months. (*Chapter VII*)

Merchandising business. A business that buys finished goods and sells them again at a higher price. It does not change the goods. A bookshop that buys books from publishers and sells the same books to readers is an example. (*Chapter I*)

Narration. A short note under a journal entry explaining the transaction. A good narration names the event, not the accounts. For example, "Bought shelving for cash" tells a later reader why Shelving was debited and Cash credited. (*Chapter V*)

Net income. The amount by which total revenue exceeds total expenses for a period. It is found by subtracting total expenses from total revenue, and it increases owner's equity. Example: revenue of Tk 6,000 less expenses of Tk 3,400 gives net income of Tk 2,600. (*Chapter VIII*)

Net loss. The amount by which total expenses exceed total revenue for a period. It is found by subtracting total revenue from total expenses, and it reduces owner's equity. Example: expenses of Tk 3,400 and revenue of Tk 3,000 give a net loss of Tk 400. (*Chapter VIII*)

Net realisable value. The expected selling price of an item less the cost of selling it. If it is lower than cost, the item is reported at net realisable value. An item that sells for Tk 150 with Tk 20 of selling cost has a value of Tk 130. (*Chapter XIII*)

Net sales. Sales less sales returns and allowances. It is the revenue a merchandising business really keeps. Example: sales of Tk 10,000, less returns of Tk 500 and allowances of Tk 200, give net sales of Tk 9,300. (*Chapter X*)

Non-trading organisation. An organisation, such as a club or charity, that exists to serve its members or a cause rather than to earn profit. Its result is called a surplus or a deficit, and its equity is the accumulated fund. A village reading room is an example. (*Chapter XVII*)

Normal balance. The side of an account, debit or credit, on which its increases are recorded and where its balance usually ends. Cash has a debit normal balance, and Accounts Payable has a credit normal balance. A balance on the other side is a signal to look closer. (*Chapter IV*)

Operating expense. A cost of running the business, such as wages or rent, apart from the cost of goods sold. Operating expenses are grouped as selling expenses and administrative expenses. Example: advertising and office rent. (*Chapter X*)

Ordinary share. A share that gives a holder a vote and a claim on profit after other claims are paid. The dividend is not fixed, so it is large when profit is large and may be nothing when profit is small. It carries more risk and more possible reward. (*Chapter XIX*)

Outstanding cheque. A cheque written and recorded in the books but not yet paid by the bank, because the payee has not yet presented it or it has not yet cleared. For example, a cheque for Tk 900 written on 30 June and paid on 4 July. It is subtracted on the bank side. (*Chapter XII*)

Owner's equity. The owner's claim on the business: what is left of the assets after all liabilities are paid. For example, assets of Tk 565,000 less liabilities of Tk 120,000 leave owner's equity of Tk 445,000. (*Chapter III*)

Partnership. A business owned by two or more people who agree to share its profit and loss. The owners are called partners. Three friends who open a bookshop and agree to share what it earns and loses form a partnership. The law of each country adds its own rules. (*Chapter XVIII*)

Partnership deed. A written agreement setting out each partner's capital, duties and share of profit. It prevents disputes by recording what was agreed. Where the deed is silent on a point, the law of the place supplies a rule, which partners often wish to replace. (*Chapter XVIII*)

Period cost. A cost charged to the period in which it is incurred, because it belongs to time and not to particular goods. It reaches the income statement at once. Rent and wages are examples. (*Chapter XX*)

Periodic inventory system. A system that finds inventory only by a physical count at period end. During the year only purchases are recorded, and cost of goods sold is found by subtraction: opening inventory plus purchases less closing inventory. (*Chapter XIII*)

Permanent account. An account for assets, liabilities or capital. It shows the position of the business, so it is never closed. Its balance at the end of one period becomes the opening balance of the next. Example: the Cash account. (*Chapter IX*)

Perpetual inventory system. A system that updates the inventory account after every purchase and sale, so that the account always shows what the business should hold. A count is still needed, as a check on the records. (*Chapter XIII*)

Plant asset. A long-lived physical asset used in running a business, such as a machine or a building. It is owned, used in the work of the business, and lasts more than one period. A book-cutting machine in a bookshop is a plant asset. (*Chapter XV*)

Posting. Copying the debits and credits from the journal to the right accounts in the ledger. Posting changes no amount. Each journal line is copied once, and the folio columns record that it has been done. (*Chapter V*)

Preference share. A share that gets a fixed dividend before ordinary shareholders, usually without a vote. For example, a holder of preference shares at 8% receives 8% of their face value before any ordinary dividend, if the company has enough profit. (*Chapter XIX*)

Prepaid expense. A cost paid in advance that is an asset until it is used up, after which it becomes an expense. For example, a year of insurance paid on 1 October is mostly still an asset on 31 December. (*Chapter VII*)

Product cost. A cost of buying or making goods held for sale. It waits in inventory until the goods are sold and then moves to cost of goods sold. The price paid for books and freight in are examples. (*Chapter XX*)

Profit-sharing ratio. The fixed proportion in which partners divide profit and loss. A ratio of 5:3:2 cuts the amount into ten equal parts, giving five to the first partner, three to the second and two to the third. Unless the deed says otherwise, it also divides losses. (*Chapter XVIII*)

Purchases journal. A special journal that records only purchases of goods on credit. Each line comes from an invoice received from a supplier and shows the date, supplier and amount. Its total is posted once to the main ledger. Cash purchases and purchases of equipment are not recorded in it. (*Chapter XI*)

Raw materials. Materials a manufacturer holds to turn into products. They have not yet been changed by the factory's work. In a binding workshop, reels of paper, board, thread and glue are raw materials. (*Chapter XIII*)

Realisation. The sale of assets, in liquidation, and the gain or loss on that sale. The loss is the book amount less the amount received. For example, an asset with a book amount of Tk 100 sold for Tk 80 gives a realisation loss of Tk 20. (*Chapter XVIII*)

Receipts and payments account. A summary of all cash received and paid in a period, with opening and closing balances. It has the form of a cash book. It shows cash only, so it ignores amounts owed, amounts paid early and the difference between lasting assets and everyday costs. (*Chapter XVII*)

Residual value. The amount a business expects to receive when it disposes of an asset at the end of its useful life. It may be zero. Only the cost less the residual value is spread over the years. For example, a machine costing Tk 330,000 with a residual value of Tk 30,000 has Tk 300,000 to spread. (*Chapter XVI*)

Revenue. Money earned, or becoming due, from selling goods or doing services. For example, a binding fee of Tk 40,000 earned by a workshop is revenue, whether the customer pays now or later. (*Chapter III*)

Revenue expenditure. Spending that keeps an asset in working order and is treated as an expense of the period. It is recorded in the income statement. A repair that restores a machine to its first condition is revenue expenditure. (*Chapter XV*)

Reversing entry. An entry made on the first day of a new period that cancels an earlier accrual. It lets the later payment be recorded in the usual way. Example: debit Wages Payable and credit Wages Expense for Tk 14,000 on 1 January 2027. (*Chapter IX*)

Sales journal. A special journal that records only sales of goods on credit. Each line comes from a copy of a sales invoice and shows the date, invoice number, customer and amount. Its total is posted once to the main ledger. Cash sales are not recorded in it. (*Chapter XI*)

Service business. A business that earns money by doing work for customers rather than by selling goods. The customer pays for the work performed, such as a repaired book, a haircut or a lesson, and takes home no goods that the business bought or made. (*Chapter I*)

Share application. A request to buy shares, sent with money, to a company that is offering them. The sender is not yet a shareholder. The company holds the money in a temporary account called Share Application until it decides who receives shares. (*Chapter XIX*)

Share capital. The money a company receives from shareholders in exchange for shares. It records the face value of the shares issued. If shareholders pay Tk 1,000 for 100 shares of face value Tk 10, share capital is Tk 1,000. It appears among the owners' equity. (*Chapter XIX*)

Share premium. The amount paid for a share above its face value. A share of face value Tk 10 sold for Tk 25 carries a premium of Tk 15. The premium is credited to its own account and is not part of share capital. (*Chapter XIX*)

Shareholder. A person who owns one or more shares of a company. A person who holds one tenth of the shares owns one tenth of the company. More shares give a larger part of the ownership. (*Chapter I*)

Source document. Paper or electronic proof of a transaction, such as an invoice, receipt or cheque stub. Records begin from the document, not from memory, because a bank, a tax authority or an auditor can inspect it. A publisher's invoice for books is an example. (*Chapter V*)

Special journal. A journal used for only one kind of repeated transaction, such as credit sales. It names the two accounts once in its heading and records only the changing parts, such as date, name and amount, on each line. Using one saves writing and reduces postings. (*Chapter XI*)

Statement of affairs. A list of a business's assets and liabilities at one date, used to find capital when records are incomplete. The assets less the liabilities give the owner's equity. It is built by counting, so it is less exact than a balance sheet from a ledger. (*Chapter XVII*)

Statement of owner's equity. A report showing how the owner's capital changed during one period. It starts with capital at the beginning, adds net income, subtracts drawings, and ends with capital at the end. Example: Tk 400,000 + Tk 184,000 – Tk 80,000 = Tk 504,000. (*Chapter IX*)

Straight-line method. A depreciation method that charges an equal amount each year. The yearly charge equals cost less residual value, divided by the useful life in years. For example, $(\text{Tk } 330,000 - \text{Tk } 30,000) \div 5$ gives Tk 60,000 a year. (*Chapter XVI*)

Subscription. A regular payment made by a member for belonging to an organisation. It is the main income of most clubs. Fees may be paid late, on time or

early, so the cash received in a year is adjusted to find the income that belongs to the year. (*Chapter XVII*)

Subsidiary ledger. A ledger holding the individual accounts of customers or suppliers, supporting a main ledger account. A business may keep one for customers and another for suppliers. Each line of a special journal is posted to the matching account in it. (*Chapter XI*)

Suspense account. A temporary account that holds a difference until its cause is found. It takes the side that makes the two column totals agree, and it must be cleared by a correcting entry before reports are prepared. (*Chapter VI*)

T-account. A simple account drawn as a letter T, with the name of the account across the top. The left side holds the debit entries and the right side holds the credit entries. It is a quick way to show an account's entries and balance. (*Chapter IV*)

Temporary account. An account for revenue, expense or drawings that measures one period only. It is closed to zero at the end of the period, so the next period starts afresh. Example: the Revenue account of one year. (*Chapter IX*)

Transaction. An event that changes the money position of a business and must be recorded. Buying shelving for cash is a transaction; a customer saying she may buy next month is not, because no money has yet changed. (*Chapter III*)

Transposition error. A mistake in which two digits of a number are written in the wrong order, such as 53 for 35. Posted on one side only, it makes the trial balance totals differ by a multiple of 9, here 18. (*Chapter VI*)

Trial balance. A list of all ledger account balances, with debits and credits totalled, to test that they are equal. It is taken at one date. If the two totals agree, the test is passed; if they differ, a slip must be found. (*Chapter VI*)

Unadjusted trial balance. A trial balance taken before the period-end adjustments have been recorded. The January list of Meghna Binding Service is unadjusted, because no adjustment has yet been made to the accounts. (*Chapter VI*)

Unearned revenue. Money received before the work is done, held as a liability until the work is performed. For example, a school pays Tk 24,000 in advance; the part not yet worked remains a liability. (*Chapter VII*)

Units-of-production method. A depreciation method that charges an amount based on how much the asset is used in the period. A charge for each unit of use is found first and then multiplied by the units used. For example, Tk 0.60 a book times 90,000 books gives Tk 54,000. (*Chapter XVI*)

Useful life. The period, or the amount of use, over which an asset is expected to serve the business. It may be given in years, kilometres or units made. It is an estimate and may be shorter than the time the asset would physically last. (*Chapter XVI*)

Weighted average cost. A method that gives every unit the average cost of all units available. The unit cost is the cost available divided by the units available, so a larger layer counts for more. For example, Tk 32,600 divided by 100 units gives Tk 326 a unit. (*Chapter XIV*)

Work in process. Goods in a factory that have been started but are not yet finished. Their cost includes the materials and labour used so far. Sewn sheets that have no cover yet are work in process. (*Chapter XIII*)

Worksheet. A large ruled working paper with columns. An accountant uses it to move the figures of the trial balance through the adjustments towards the

statements. It is private, it is not a financial statement, and it changes no ledger account. Example: Supplies show Tk 1,000 before an adjustment and Tk 400 after it. (*Chapter VIII*)

ANSWERS TO QUESTIONS

CHAPTER I · BUSINESS ACTIVITY AND COMPANIES

1.p1 It is a merchandising business, because it buys finished goods (rice) and sells them again without changing them.

1.p2 One quarter, because 25 is one quarter of 100.

1.t1 Its customers pay for work, so it holds no goods bought for resale. It still records fees, wages, rent and any money owed, and the materials it uses are a cost of the work.

1.t2 The salon combines a service business (hair work) with a merchandising business (shampoo bought and resold). If shampoo is a small part of its income, the service kind brings in most of its money.

1.t3 Not in the sense of this chapter, because the club does not supply goods or services to earn a profit. It still needs records of the fees it collects and the money it spends.

1.t4 It combines manufacturing (bread baked from materials) and merchandising (tea bought and resold). Ownership has the shape of a partnership, with three owners. Any one of the three friends, a lender or the tax authority would need its records.

1.1 The tailor is a service business, the grocer is a merchandising business, and the furniture workshop is a manufacturing business.

1.2 Examples: the owner (whether the bakery earns a profit), a lender (whether a loan can be repaid), the tax authority (how much tax is due), and an employee (whether wages will be paid).

1.3 $410,000 \text{ shares} \times \text{Tk } 10 = \text{Tk } 4,100,000$. One tenth of 410,000 is 41,000 shares.

1.4 A shareholder is a person who owns one or more shares of a company. A merchandising business buys finished goods and resells them, while a manufacturing business makes goods from materials and then sells them.

1.5 It is a service business, because it repairs and binds books for schools. Separate records let the owner see the result of each business alone and judge each one fairly.

CHAPTER II · WHAT ACCOUNTING IS

2.1 Nurul asks whether the shop earned enough profit (Tk 750,000). The bank asks whether the loan (Tk 600,000) can be repaid. The publishers ask whether the Tk 1,150,000 owed to them will be paid. The tax authority asks whether the profit return is honest. The staff ask whether wages (Tk 2,940,000) and jobs are safe.

2.t1 A household bill is not a cost of the shop. If it appears in the shop's records, the shop's profit looks too low and the bank forms a false picture of the shop's ability to repay. The business entity rule keeps the two sets of money apart.

2.p1 The book leaves the shop's inventory, so the shop's records must show that event. The rule says the shop is separate from Nurul, so the shop's goods are not Nurul's to take without a record.

2.p2 It is bookkeeping, because the clerk only records each payment in the right book. Accounting would also sort the payments and report totals to readers.

2.2 The weekly book-sales report is management accounting, because it is made for the manager inside the business and follows no outside rule. It is not enough for the bank, which is an external user and needs a formal report that follows written standards.

2.3 The business entity rule is broken, because family costs are mixed with the business. Anyone who reads her records, such as a lender or a tax authority, is misled about how well the business is doing.

2.4 The going concern assumption no longer holds, because the shop will not use its shelves for years. The records must then show what the shelves would fetch when sold, instead of the price paid under historical cost.

CHAPTER III · THE ACCOUNTING EQUATION

3.p1 Owner's equity = 80,000 – 25,000 = Tk 55,000. Check: liabilities plus equity, 25,000 + 55,000 = 80,000, which equals the assets.

3.p2 Owner's equity = 40,000 + 15,000 – 9,000 – 2,000 = Tk 44,000.

3.t1 Buying shelving for cash: one asset (shelving) rises by Tk 120,000 and another (cash) falls by Tk 120,000. Total assets are unchanged, so liabilities and owner's equity need no change and the equation stays equal.

3.t2 She is not right. Paying the publisher lowered cash by Tk 80,000 and lowered the account payable by Tk 80,000. Owner's equity did not change, so no expense arose.

3.1 (a) asset; (b) liability; (c) asset; (d) liability; (e) owner's equity.

3.2 Liabilities = 240,000 – 150,000 = Tk 90,000. Check: 90,000 + 150,000 = 240,000.

3.3 Owner's equity at the end = 300,000 + 410,000 – 365,000 – 50,000 = Tk 295,000.

3.4 Equity = 1,250 – 480 = 770 thousand, which is Tk 770,000.

3.5 Assets Tk 252,000 (cash 162,000 and equipment 90,000); liabilities Tk 50,000 (90,000 – 40,000); owner's equity Tk 202,000 (200,000 + 30,000 – 18,000 – 10,000). Check: 50,000 + 202,000 = 252,000.

CHAPTER IV · ACCOUNTS, DEBITS AND CREDITS

4.p1 Debit. Rent is an expense, and expenses increase by debit, so its balance usually falls on the debit side.

4.p2 Shelving is an asset that increases, so it is debited. Cash is an asset that decreases, so it is credited.

4.t1 Cash is an asset, and assets stand on the left of the equation $A + D + E = L + C + R$. Increases of left-side items are recorded on the left, which is the debit side.

4.t2 Debit Drawings and credit Cash, both for Tk 10,000. Drawings is its own type because the owner's personal use is not a cost of earning revenue, so it is not an expense.

4.1 Debit Cash 2,000 and credit Owner's Capital 2,000; debit Shelf 500 and credit Cash 500; debit Cash 300 and credit Sales Revenue 300. Cash: debits 2,300 less credits 500 gives a debit balance of Tk 1,800.

4.2 Debit Inventory (paper) 18,000, credit Accounts Payable 18,000; debit Accounts Payable 7,500, credit Cash 7,500; debit Cash 26,400, credit Service Revenue 26,400; debit Wages 9,200, credit Cash 9,200; debit Drawings 4,000,

credit Cash 4,000. Cash: $50,000 + 26,400 - 7,500 - 9,200 - 4,000 = 55,700$
debit. Accounts Payable: $18,000 - 7,500 = 10,500$ credit.

4.3 Borrowing: debit Cash 100,000, credit Bank Loan 100,000. Repaying: debit Bank Loan 40,000, credit Cash 40,000. Bank Loan has a credit balance of Tk 60,000.

4.4 An asset normally has a debit balance, so a credit balance means more cash was recorded as paid out than received, through an error or an overdraft. Check each entry against its source document.

4.5 Inventory: asset, debit, debit. Accounts Payable: liability, credit, credit. Drawings: drawings, debit, debit. Service Revenue: revenue, credit, credit. Wages: expense, debit, debit.

CHAPTER V · THE JOURNAL AND THE LEDGER

5.p1 Debit Electricity (an expense) Tk 10,000 and credit Cash Tk 10,000. Narration: Paid electricity bill in cash.

5.t1 Publisher Payable would show a credit balance of Tk 200,000 instead of Tk 120,000. The credit balances would then total Tk 740,000 against debit balances of Tk 660,000, so credits would exceed debits by Tk 80,000.

5.1 Debit Cash 100,000, credit Owner's Equity 100,000 (narration: Owner invested cash). Then debit Shelving 20,000, credit Cash 20,000 (narration: Bought shelving for cash).

5.2 Cash debit side: 100,000. Cash credit side: 20,000. Balance = $100,000 - 20,000 = \text{Tk } 80,000$ debit.

5.3 Debit total = $125,500 + 48,250 = 173,750$. Credit total = $30,750 + 61,400 = 92,150$. Balance = $173,750 - 92,150 = \text{Tk } 81,600$, on the debit side.

5.4 Debit side: $245,000 + 52,000 = 297,000$. Credit side: 118,500. Balance = $297,000 - 118,500 = \text{Tk } 178,500$ debit.

5.5 Debit Equipment 30,000; credit Cash 10,000; credit Equipment Dealer Payable 20,000. Debits 30,000 equal credits 30,000. Balances: Equipment Tk 30,000 debit; Cash Tk 10,000 credit on these entries alone; Equipment Dealer Payable Tk 20,000 credit.

CHAPTER VI · THE TRIAL BALANCE

6.p1 Every entry has equal debits and credits, and posting copies both sides, so the ledger holds equal totals of debits and credits. Balancing removes the same amount from both totals, so the debit balances still equal the credit balances.

6.t1 Two examples: the 15 January rent entry left out entirely, which leaves Cash at Tk 305,000 and Rent at nil but both totals at Tk 660,000; and wages debited to Rent instead of Wages, which leaves the totals equal but Rent wrongly at Tk 95,000 and Wages at nil.

6.1 Debits: $50,000 + 30,000 = \text{Tk } 80,000$. Credits: Tk 80,000. The totals are equal.

6.2 Debits: $187,450 + 96,300 + 150,000 + 42,000 + 78,500 = \text{Tk } 554,250$. Credits: $64,250 + 400,000 + 90,000 = \text{Tk } 554,250$. The trial balance is balanced.

6.3 Difference = $465,500 - 458,900 = \text{Tk } 6,600$, with credits larger. $6,600 \div 9$ is not a whole number, so a transposition is unlikely. Half is Tk 3,300, which suggests a debit balance of Tk 3,300 listed in the Credit column.

6.4 The debits fall short by Tk 7,200, so a suspense account with a debit balance of Tk 7,200 is added in the Debit column.

6.5 Difference = $81,000 - 18,000 = \text{Tk } 63,000$, with debits larger ($63,000 \div 9 = 7,000$). The credits are short, so the suspense balance of Tk 63,000 goes on the credit side. Correcting entry: debit Suspense 63,000, credit Rent 63,000. Narration: Corrected rent posted as 81,000 instead of 18,000, and cleared the suspense account.

CHAPTER VII · ADJUSTING THE ACCOUNTS

7.p1 Monthly rent = $24,000 \div 12 = \text{Tk } 2,000$. November and December are two months, so Tk 4,000 belongs to the year. The matching principle (with the accrual basis) decides it: the cost is placed in the period in which it is used.

7.p2 The Tk 6,500 owed to workers is an accrued expense, a liability. The Tk 11,000 of unbilled work is accrued revenue, an asset.

7.t1 The records would show a profit of Tk 280,000 instead of Tk 184,000, which is Tk 96,000 too high. The owner might withdraw too much, and a lender or tax authority would judge the business on a wrong figure. Assets would also be overstated and liabilities understated.

7.t2 The Tk 36,000 was paid in cash on 1 October. The adjustment on 31 December only moves the used part, Tk 9,000, from the asset Prepaid Insurance to the expense Insurance Expense. No money moves on that date.

7.1 $6,000 \div 6 = \text{Tk } 1,000$ a month. March to June is 4 months, so the expense is Tk 4,000 and the prepaid expense is $6,000 - 4,000 = \text{Tk } 2,000$.

7.2 Supplies expense = $52,500 - 18,200 = \text{Tk } 34,300$. Entry: debit Supplies Expense 34,300; credit Supplies 34,300.

7.3 $18,000 \div 6 = \text{Tk } 3,000$ a month. November and December are 2 months, so revenue earned is Tk 6,000 and unearned revenue is $18,000 - 6,000 = \text{Tk } 12,000$.

7.4 Depreciation = $240,000 \div 8 = \text{Tk } 30,000$: debit Depreciation Expense, credit Accumulated Depreciation. Wages: debit Wages 9,000, credit Wages Payable 9,000. Unbilled work: debit Accounts Receivable 15,000, credit Service Revenue 15,000. Expenses rise by 39,000 and revenue by 15,000, so profit falls by Tk 24,000.

CHAPTER VIII · THE WORKSHEET

8.p1 Tk 400, a debit: the debit of Tk 1,000 less the credit of Tk 600 leaves a debit of Tk 400.

8.p2 The debit side, because revenue (Tk 6,000) exceeds expenses (Tk 3,400), so the debit column is short by Tk 2,600 and the balancing figure makes the two totals equal.

8.t1 The credit total is $\text{Tk } 8,500 + \text{Tk } 2,600 = \text{Tk } 11,100$, so the difference is Tk 300 (debit larger). Wages Payable is Tk 300, so check first whether that balance was extended to the balance sheet credit column.

8.t2 The difference is Tk 270, and $270 \div 9 = 30$, so it is divisible by 9. This suggests a transposition error. Wages of Tk 2,300 written as Tk 2,030 would give exactly this shortfall.

8.1 Adjusted balances: Cash 3,000; Prepaid Insurance 900; Wages 3,000; Insurance Expense 300 (debits, total 7,200); Capital 3,200; Fees Revenue 4,000 (credits, total 7,200). Income columns: debit 3,300, credit

4,000, so net income is Tk 700. Balance sheet: debit 3,900; credit $3,200 + 700 = 3,900$.

8.2 Adjustments: Supplies Expense 2,700 (Supplies 4,200 less 1,500 on hand); Wages 1,300 with Wages Payable 1,300; Depreciation Expense 4,800 with Accumulated Depreciation 4,800. Adjusted totals are 131,100 each side. Expenses: $18,500 + 4,800 + 2,000 + 2,700 + 4,800 = 32,800$. Net income = $61,400 - 32,800 = \text{Tk } 28,600$. Balance sheet columns: debit 98,300; credit $69,700 + 28,600 = 98,300$.

8.3 Net loss of 3,500 thousand ($21,500 - 18,000$). It is entered in the credit column of the income pair ($18,000 + 3,500 = 21,500$) and the debit column of the balance sheet pair ($40,000 + 3,500 = 43,500$). In full taka, the net loss is Tk 3,500,000.

8.4 Adjusted balances: Cash 9,000; Equipment 12,000; Prepaid Insurance 900; Wages 3,400; Insurance Expense 900; Depreciation Expense 1,200 (debits, total 27,400); Accumulated Depreciation 1,200; Wages Payable 400; Unearned Revenue 900; Capital 17,400; Fees Revenue 7,500 (credits, total 27,400). Expenses 5,500; net income = $7,500 - 5,500 = \text{Tk } 2,000$. Balance sheet: debit 21,900; credit $19,900 + 2,000 = 21,900$.

8.5 A worksheet gathers all balances on one page, tests the adjustments before they are recorded, and checks the arithmetic with equal totals. It changes no account because it is working paper; the ledger changes only when the adjusting entries are journalised and posted.

CHAPTER IX · FINANCIAL STATEMENTS AND CLOSING ENTRIES

9.p1 A credit of Tk 300: credit Tk 800 less debit Tk 500. This balance is the net income.

9.t1 $\text{Tk } 788,000 + \text{Tk } 900,000 = \text{Tk } 1,688,000$. The figure mixes two years, so the 2027 income statement would overstate revenue by Tk 788,000.

9.t2 The balance sheet needs ending capital. Ending capital comes from the statement of owner's equity (starting capital plus net income minus drawings), so that statement must come first.

9.1 Income statement: what did the business earn? Statement of owner's equity: how did capital change? Balance sheet: what does the business own and owe on one date? The first two headings end *For the year ended ...*; the balance sheet heading ends *At*

9.2 Net income is $\text{Tk } 12,000 - \text{Tk } 7,000 = \text{Tk } 5,000$. Ending capital is $\text{Tk } 10,000 + \text{Tk } 5,000 - \text{Tk } 1,000 = \text{Tk } 14,000$. Assets are $\text{Tk } 6,000 + \text{Tk } 2,000 + \text{Tk } 9,000 = \text{Tk } 17,000$; liabilities Tk 3,000 plus capital Tk 14,000 also give Tk 17,000.

9.3 Step 1: debit Revenue 12,000, credit Income Summary 12,000. Step 2: debit Income Summary 7,000, credit Expenses 7,000. Step 3: debit Income Summary 5,000, credit Owner's Capital 5,000. Step 4: debit Owner's Capital 1,000, credit Drawings 1,000.

9.4 Temporary: Revenue, Wages Expense, Drawings. Permanent: Prepaid Insurance, Wages Payable, Owner's Capital.

9.5 1 January 2027: debit Wages Payable 6,000, credit Wages Expense 6,000. 4 January 2027: debit Wages Expense 8,000, credit Cash 8,000. Wages expense of 2027 is $\text{Tk } 8,000 - \text{Tk } 6,000 = \text{Tk } 2,000$.

9.6 The net loss is $\text{Tk } 11,000 - \text{Tk } 9,000 = \text{Tk } 2,000$, so Income Summary has a debit balance. Step 3:

debit Owner's Capital 2,000, credit Income Summary 2,000. Capital falls by Tk 2,000.

CHAPTER X · ACCOUNTS OF A MERCHANDISING COMPANY

10.p1 Tk 500 + Tk 2,000 – Tk 700 = Tk 1,800.

10.p2 Tk 22,500, the first instalment, because it falls due within one year. The other Tk 67,500 is non-current.

10.t1 The two mistakes are equal and opposite on the same side, so total debits still equal total credits. Example: Wages debited Tk 300 too much and Rent debited Tk 300 too little.

10.t2 Cost of goods sold is Tk 2,000 too large, gross profit is Tk 2,000 too small, and inventory is missing from current assets, so the balance sheet understates assets.

10.1 Tk 5,000 – Tk 300 – Tk 100 = Tk 4,600.

10.2 Net sales Tk 78,000. Cost of goods sold = 12,000 + 50,000 – 1,000 – 14,000 = Tk 47,000. Gross profit Tk 31,000. Operating expenses 6,000 + 7,000 = Tk 13,000. Profit Tk 18,000.

10.3 Current assets Tk 22,000 (3,000 + 4,000 + 14,000 + 1,000); equipment Tk 20,000; total assets Tk 42,000. Current liabilities Tk 7,000 (5,000 + 2,000); long-term loan Tk 10,000; owner's equity Tk 25,000; total Tk 42,000.

10.4 (a) omission; (b) principle; (c) commission; (d) compensating error.

10.5 (b) Debit Vehicles Tk 40,000; credit Repairs Expense Tk 40,000. (c) Debit Electricity Expense Tk 700; credit Telephone Expense Tk 700.

10.6 Inventory: current asset. Shop fittings: non-current asset. Accounts payable: current liability. Loan due in four years: non-current liability. Loan due in six months: current liability.

CHAPTER XI · SPECIAL JOURNALS

11.p1 The subsidiary ledger holds the balance of one named customer. The accounts receivable control account holds the total owed by all customers together.

11.t1 Five credit sales make five entries and ten postings, so a special journal saves very little. The answer changes when credit sales become many, for example dozens each month, or when more than one clerk must write at the same time.

11.t2 Every sale is posted twice to the control account, so it is too high by the total of the lines. The customer accounts are correct, so the two totals disagree by exactly the amount posted twice, and the test reveals the slip.

11.t3 The agreement test reveals it. The control account, posted from the wrong total, is Tk 1,800 lower than the sum of the customer accounts, since Tk 54,200 – Tk 52,400 = Tk 1,800.

11.1 Lines of Tk 100, Tk 200 and Tk 300 give a total of Tk 600. Debit the accounts receivable account Tk 600 and credit Sales Tk 600, each posted once.

11.2 Total: Tk 42,500 + Tk 18,750 + Tk 31,200 = Tk 92,450. Closing balance: Tk 150,000 + Tk 92,450 = Tk 242,450. Both Greenfield School lines go to one customer account.

11.3 They do not agree. The control account should be Tk 310,000 + Tk 126,400 = Tk 436,400. The customer accounts total Tk

426,400. The difference is Tk 10,000, which suggests a posting is missing or wrong.

11.4 Total: Tk 36,000 + Tk 54,500 + Tk 29,500 = Tk 120,000. Debit Purchases Tk 120,000 and credit the accounts payable control account Tk 120,000. Closing balance: Tk 200,000 + Tk 120,000 = Tk 320,000.

CHAPTER XII · THE CASH BOOK AND BANK RECONCILIATION

12.p1 A difference of Tk 1,000 means an item is missing, wrongly placed or wrongly added. Tick the items again and check the arithmetic; make no entry until the two adjusted balances agree.

12.t1 Entries are needed for the service charge (Tk 1,500), the dishonoured cheque (Tk 8,000), the interest (Tk 2,500) and the school payment (Tk 40,000), because the bank recorded them first. No entry is needed for the deposit in transit or the two cheques, because the business recorded them first and the bank will catch up.

12.t2 The business should contact the payee to learn why the cheque has not been presented, because rules on old cheques differ between banks and countries. Until it clears, the cheque stays on the reconciliation as an outstanding cheque and is subtracted on the bank side each month.

12.1 Bank side: $3,400 + 700 - 300 = 3,800$. Book side: $3,000 + 900 - 100 = 3,800$. The adjusted balance is Tk 3,800 on both sides.

12.2 Bank side: $91,250 + 12,000 + 6,500 - 9,800 - 4,150 = 95,800$. Book side: $95,050 + 5,000 - 750 - 3,500 = 95,800$. The adjusted balance is Tk 95,800.

12.3 The cash book is higher by Tk 1,750 before adjusting. Bank side: $18,400 + 5,000 - 3,200 - 1,100 = 19,100$. Book side: $20,150 - 250 - 800 = 19,100$. The adjusted balance is Tk 19,100.

12.4 Bank side: $47,000 + 4,000 - 9,000 = 42,000$. Book side: $40,000 + 3,000 + 500 - 1,500 = 42,000$. Entries: debit Bank, credit Account Receivable Tk 3,000; debit Bank, credit Interest Revenue Tk 500; debit Bank Charges, credit Bank Tk 1,500. Bank on the balance sheet is Tk 42,000.

CHAPTER XIII · INVENTORY: MEANING AND MEASUREMENT

13.p1 It will be hidden. A periodic system calculates cost of goods sold by subtraction, so the cost of the 3 stolen books is included in cost of goods sold with no separate sign.

13.t1 The extra holding cost is $380,000 \times 10\% = \text{Tk } 38,000$. The shop may also face a write-down if the titles can only be sold for less than their cost.

13.t2 Two possible costs: lost sales when demand rises or a delivery is late, and the loss of customers who do not return. More frequent orders may also raise transport costs and remove bulk-buying discounts.

13.1 (a) raw materials; (b) work in process; (c) not inventory, because it is not held for sale; (d) finished goods; (e) not inventory, because the shop does not own it.

13.2 Net realisable value = $500 - 90 = \text{Tk } 410$ a copy. Cost is Tk 450, so each copy is written down by $450 - 410 = \text{Tk } 40$. For 200 copies the write-down is Tk 8,000.

13.3 Cost of goods sold = $40,000 + 65,000 - 30,000 = \text{Tk } 75,000$.

13.4 The difference is $500 - 470 = 30$ units, and $30 \times 60 = \text{Tk } 1,800$. The shop should first investigate the cause. If none

is found, it reduces Inventory to the counted figure and charges Tk 1,800 as a loss.

13.5 Holding cost is $600,000 \times 12\% = \text{Tk } 72,000$ a year. The write-down per unit is $80 - 55 = \text{Tk } 25$, so for 150 units it is Tk 3,750. The units are reported at $150 \times 55 = \text{Tk } 8,250$.

CHAPTER XIV · COSTING INVENTORY: FIFO, LIFO AND WEIGHTED AVERAGE

14.p1 The oldest 60 units are 20 at Tk 300, 30 at Tk 320 and 10 at Tk 340, so the cost of goods sold is $\text{Tk } 6,000 + \text{Tk } 9,600 + \text{Tk } 3,400 = \text{Tk } 19,000$. The 40 units left are $40 \times \text{Tk } 340 = \text{Tk } 13,600$. $\text{Tk } 19,000 + \text{Tk } 13,600 = \text{Tk } 32,600$.

14.p2 The unit cost is Tk 326, so 25 sets cost $25 \times \text{Tk } 326 = \text{Tk } 8,150$.

14.t1 FIFO is highest at Tk 12,600, weighted average is next at Tk 12,180, and LIFO is lowest at Tk 11,600 in the case. When prices rise, FIFO charges the oldest, cheapest costs to the sale and LIFO charges the newest, dearest costs. Weighted average lies between them.

14.t2 The ranking reverses: LIFO gives the highest gross profit, FIFO the lowest, and weighted average lies between. For all three methods, cost of goods sold plus ending inventory equals the cost available.

14.1 Cost available is $\text{Tk } 100 + \text{Tk } 120 = \text{Tk } 220$. FIFO: ending Tk 120, cost of goods sold Tk 100, gross profit Tk 100. LIFO: ending Tk 100, cost of goods sold Tk 120, gross profit Tk 80. Weighted average: unit cost Tk 11, ending Tk 110, cost of goods sold Tk 110, gross profit Tk 90.

14.2 Cost available is $\text{Tk } 1,800 + \text{Tk } 2,880 + \text{Tk } 5,200 = \text{Tk } 9,880$ for 200 units; 70 remain. FIFO: ending Tk 3,640, cost of goods sold Tk 6,240, gross profit Tk 4,160. LIFO: ending Tk 3,240, cost of goods sold Tk 6,640, gross profit Tk 3,760. Weighted average: unit cost Tk 49.40, ending Tk 3,458, cost of goods sold Tk 6,422, gross profit Tk 3,978. Each pair adds to Tk 9,880.

14.3 Cost available is $\text{Tk } 2,000 + \text{Tk } 1,750 = \text{Tk } 3,750$ for 50 boxes; 20 remain. FIFO: ending Tk 1,400, cost of goods sold Tk 2,350, gross profit Tk 650. LIFO: ending Tk 1,600, cost of goods sold Tk 2,150, gross profit Tk 850. Weighted average: unit cost Tk 75, ending Tk 1,500, cost of goods sold Tk 2,250, gross profit Tk 750. LIFO gives the highest profit because prices fall.

14.4 FIFO: $4,160 \div 10,400 = 40.0\%$. LIFO: $3,760 \div 10,400 = 36.2\%$. Weighted average: $3,978 \div 10,400 = 38.3\%$. FIFO gives the highest ending inventory, Tk 3,640, because prices rise and FIFO leaves the newest, dearest costs in inventory.

CHAPTER XV · PLANT ASSETS

15.p1 The van price of Tk 900,000 and the painting of Tk 12,000, which is needed before use, belong in the cost. The road licence of Tk 20,000 is a cost of using the van during the first year, so it is an expense, not part of the cost.

15.t1 Profit for 2026 would be Tk 15,000 too low, and the machine on the balance sheet would be Tk 15,000 too low at Tk 315,000 instead of Tk 330,000.

15.t2 Capitalising a repair removes it from expenses, so profit rises and the asset grows. This is dishonest because the repair does not create a lasting asset, so the profit and the balance sheet both give a false picture, even if the entry is not noticed at first.

15.1 Acquisition cost = $450,000 - 20,000 + 8,000 + 12,000 + 4,000 = \text{Tk } 454,000$. The training of Tk 6,000 is an expense.

15.2 (a) revenue; (b) capital; (c) capital; (d) revenue; (e) capital, because it is done before first use.

15.3 Profit is Tk 12,000 too high and the plant asset is Tk 12,000 too high on the balance sheet.

15.4 Acquisition cost = 600,000 + 30,000 + 18,000 = Tk 648,000. Revenue expenditure for the year = 22,000 + 9,000 = Tk 31,000.

CHAPTER XVI · DEPRECIATION

16.p1 (Tk 120,000 – Tk 20,000) ÷ 4 = Tk 25,000 a year.

16.p2 Year 1: 25% × Tk 80,000 = Tk 20,000, leaving Tk 60,000. Year 2: 25% × Tk 60,000 = Tk 15,000, leaving Tk 45,000.

16.t1 The yearly charge falls from Tk 60,000 to (Tk 330,000 – Tk 30,000) ÷ 8 = Tk 37,500, so each year's profit rises by Tk 22,500. The total charged over the whole life is unchanged at Tk 300,000, because only the timing changes.

16.t2 Depreciation is an allocation of cost, not a transfer of cash. The cash was paid on the day of purchase, and the yearly entry moves no cash. Saving for a new machine is a separate decision about cash.

16.1 Charge: (Tk 50,000 – Tk 10,000) ÷ 4 = Tk 10,000 a year. After two years the accumulated depreciation is Tk 20,000, so the book value is Tk 30,000.

16.2 Straight-line: (Tk 275,000 – Tk 25,000) ÷ 8 = Tk 31,250 a year; for 3 months, Tk 31,250 × 3 ÷ 12 = Tk 7,812.50. Units-of-production: Tk 250,000 ÷ 400,000 = Tk 0.625 a unit; 36,000 × Tk 0.625 = Tk 22,500.

16.3 Year 1: Tk 30,000, remaining Tk 120,000. Year 2: Tk 24,000, remaining Tk 96,000. Year 3: Tk 19,200, remaining Tk 76,800. Gain = Tk 80,000 – Tk 76,800 = Tk 3,200.

16.4 Yearly charge: (Tk 240,000 – Tk 40,000) ÷ 5 = Tk 40,000. Entry for 2026: debit depreciation expense Tk 40,000, credit accumulated depreciation Tk 40,000. After three years the accumulated depreciation is Tk 120,000 and the book value is Tk 120,000, so the sale gives a loss of Tk 20,000. Sale entry: debit cash Tk 100,000, debit accumulated depreciation Tk 120,000, debit loss on sale Tk 20,000; credit press at cost Tk 240,000.

CHAPTER XVII · ACCOUNTS OF CLUBS AND SOLE TRADERS

17.p1 No. The money is for a later year, so it belongs to the period that it pays for. It is received early and is shown as a liability, fees paid in advance, until that period begins.

17.t1 Cash can rise because of money not charged against the year, such as a loan or the sale of an asset, or because the club bought no lasting assets. A deficit arises when expenditure on the accrual basis, including amounts owed but unpaid, exceeds income that belongs to the year, such as when many fees are still owed.

17.t2 Profit = 46,000 – 50,000 + 9,000 – 0 = Tk 5,000. The equity fell by Tk 4,000 only because she withdrew Tk 9,000, so the business was profitable.

17.1 Receipts: 6,000 + 40,000 + 3,000 = Tk 49,000. Payments: 15,000 + 12,000 + 8,000 = Tk 35,000. Closing cash = 49,000 – 35,000 = Tk 14,000.

17.2 Subscriptions = 40,000 – 2,000 – 1,500 + 3,500 = Tk 40,000. Wages = 12,000 + 1,000 = Tk 13,000. Income = 40,000 + 3,000 = 43,000. Expenditure = rent 15,000 +

wages 13,000 = 28,000. Surplus = Tk 15,000. The cupboard is an asset, not an expense.

17.3 Opening owner's equity = 60,000 – 15,000 = 45,000. Closing = 72,000 – 18,000 = 54,000. Profit = 54,000 – 45,000 + 20,000 – 8,000 = Tk 21,000.

17.4 In the receipts and payments account the Tk 30,000 is a payment. In the income and expenditure account it does not appear as expenditure, because it is a lasting asset. In the balance sheet it appears as an asset at its cost.

CHAPTER XVIII · PARTNERSHIP ACCOUNTS

18.p1 Interest for a year is $200,000 \times 5\% = \text{Tk } 10,000$. For six months, $10,000 \times 6 \div 12 = \text{Tk } 5,000$.

18.t1 His share of the remainder does not change, because the ratio stays 5:3:2 and he still receives two parts in ten. His interest rises by $400,000 \times 6\% = \text{Tk } 24,000$ a year, so his total grows through the first layer only.

18.t2 Outsiders such as suppliers and banks are paid first, from all the cash available. If the cash cannot cover the debts, nothing is left for the partners, whose capital is used up by the loss.

18.1 Four parts in all. The first partner receives $(3 \div 4) \times 80,000 = \text{Tk } 60,000$ and the second receives $(1 \div 4) \times 80,000 = \text{Tk } 20,000$. Check: $60,000 + 20,000 = 80,000$.

18.2 Interest: Mira $450,000 \times 8\% = 36,000$; Naveed $300,000 \times 8\% = 24,000$; total 60,000. Remainder $211,000 - 60,000 = 151,000$. Shares: Mira $(3 \div 5) \times 151,000 = 90,600$; Naveed $(2 \div 5) \times 151,000 = 60,400$. Totals: Mira Tk 126,600 and Naveed Tk 84,400, which add to 211,000.

18.3 Time is $9 \div 12 = 0.75$ year. Interest: Mira $450,000 \times 8\% \times 0.75 = 27,000$; Naveed $300,000 \times 8\% \times 0.75 = 18,000$; total 45,000. Remainder $180,000 - 45,000 = 135,000$. Shares: Mira 81,000; Naveed 54,000. Totals: Mira Tk 108,000 and Naveed Tk 72,000, which add to 180,000.

18.4 Debit Bank Tk 300,000 and credit Priya, Capital Tk 300,000, dated 1 April 2028, with a narration that the amount was paid in by the new partner on admission. Debits equal credits.

18.5 Loss = $850,000 - 700,000 = \text{Tk } 150,000$. Shares: P $(2 \div 3) \times 150,000 = \text{Tk } 100,000$; Q $(1 \div 3) \times 150,000 = \text{Tk } 50,000$. Payments: P $400,000 - 100,000 = \text{Tk } 300,000$; Q $200,000 - 50,000 = \text{Tk } 150,000$. Check: cash after debts is $700,000 - 250,000 = 450,000$, which equals $300,000 + 150,000$.

CHAPTER XIX · COMPANY ACCOUNTS: SHARES

19.p1 The premium on each share is $18 - 10 = \text{Tk } 8$. On 1,000 shares the total premium is $1,000 \times 8 = \text{Tk } 8,000$.

19.t1 The company could allot shares to every applicant in proportion, for example 2 shares for every 3 applied for, or accept some applications in full and refuse the rest. Money for shares not allotted must be returned to the applicant, or used for the later amount due if the applicant agrees.

19.t2 The business may already be worth more than the face value of its shares, because of its name, its customers and the profit it has kept. Buyers pay the extra Tk 30 for a share of that value, and the company records it as share premium.

19.1 Application: debit Bank 10,000; credit Share Application 10,000. Allotment: debit Share Application 10,000; credit Share Capital 10,000. There is no premium, because the issue price equals the face value.

19.2 Share capital = $50,000 \times 10 = \text{Tk } 500,000$. Share premium = $50,000 \times 6 = \text{Tk } 300,000$. Application: debit Bank 400,000; credit Share Application 400,000. Allotment: debit Share Application 400,000 and Bank 400,000; credit Share Capital 500,000 and Share Premium 300,000. Check: $500,000 + 300,000 = 800,000 = 400,000 + 400,000$.

19.3 Shares authorised = $2,000,000 \div 5 = 400,000$. Issued capital = $120,000 \times 5 = \text{Tk } 600,000$. Shares still unissued = $400,000 - 120,000 = 280,000$, worth Tk 1,400,000.

19.4 Total shares = $750,000 \div 10 = 75,000$. Mira receives $450,000 \div 10 = 45,000$ shares and Naveed receives $300,000 \div 10 = 30,000$ shares. Entry: debit Net assets taken over 750,000; credit Share Capital 750,000.

CHAPTER XX · THE WHOLE CYCLE AND REPORTING PERIODS

20.p1 Freight in (Tk 40,000) is a product cost, because it is part of bringing goods to the shop. Rent (Tk 180,000) is a period cost, because it belongs to the time the building is used.

20.p2 Cash, an asset, falls by Tk 5,000, and accounts payable, a liability, falls by Tk 5,000. Both sides fall by the same amount, so the balance sheet still balances.

20.t1 Two changes: every report would cover 1 July to 30 June, and the quarters would end on 30 September, 31 December, 31 March and 30 June, so the quarter ended 30 September 2026 would become a first quarter. Moving from a December year-end also needs a bridging period of six months, 1 January to 30 June. The accounts, the debit and credit rules and the split between product costs and period costs stay the same.

20.t2 A year ending when trade is quiet, such as soon after the term-start rush, is a good choice. Inventory is low and easy to count then, and the whole busy season falls inside one year.

20.1 Product costs: books bought and freight in. Period costs: shop rent, wages of the shop assistants and delivery to a customer (freight out).

20.2 Gross profit is $\text{Tk } 600,000 - \text{Tk } 380,000 = \text{Tk } 220,000$. Profit is $\text{Tk } 220,000 - \text{Tk } 150,000 = \text{Tk } 70,000$.

20.3 The order is: journal, posting to the ledger, trial balance, adjusting entries, adjusted trial balance, the three statements, closing entries, post-closing trial balance. The adjusting entries and the closing entries change account balances.

20.4 An account payable is an amount the business owes, and it appears among the liabilities. An account receivable is an amount owed to the business, and it appears among the assets.

20.5 The second quarter runs from 1 July to 30 September. The second quarter is an interim period; the year is the fiscal year, and the cycle is a set of steps and not a period of time.

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What is accountancy, and how are the books of a business kept? This treatise answers from the beginning. It sets out the nature and purpose of accounting, then follows the work through recording transactions, classifying them in ledgers, summarising them in a trial balance, and reporting the result in final accounts. Each chapter opens with a puzzle, closes with review questions, and needs no earlier knowledge.